MILITARY ENGINEER SERVICES
HQ CHIEF ENGINEER (AF) SHILLONG ZONE

NAME OF WORK : PROVN OF CENTRAL SEWAGE TREATMENT PLANT AT AF STN SALUA

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SIGNATURE OF CONTRACTOR
DATED:_____

AAD (CONTRACTS)
FOR ACCEPTING OFFICER

HQ Chief Engineer (AF)
Shillong Zone
Military Engineer Services
Elephant Falls Camp
Nonglyer : P.O
Shillong – 793009

88853/KKD/107/E8 27 Aug 2025

M/S_____

_____

_____

NAME OF WORK : PROVN OF CENTRAL SEWAGE TREATMENT PLANT AT AF STN SALUA

Dear Sir (S)

1. Tender documents in respect of above work are uploaded on the sitedefproc.gov.in. The tender is on single stage two cover e-tendering system. The contents of Cover I & Cover II are specified in NOTICE OF TENDER.

2. Bids will be received online by ACCEPTING OFFICER upto the date & time mentioned in the NOTICE INVITING TENDER (NIT). No tender/bid will be received in physical form and any tender/bid received in such manner will be treated as non bonafide tender/bid.

3. Bid will be opened on due date and time fixed for opening in the presence of tenderer/bidders or their authorized representatives, who have uploaded their quotation bid and who wish to be present at the time of opening the bids.

4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents and/or depute your technical representative for discussion on tender/drawing and to clarify doubts, if any, on or before the stipulated clarification end date. You are requested not to write piece meal points and forward your points duly consolidated before due date viz the stipulated clarification end date.

5. Unenlisted contractors are required to submit the scanned copies (in pdf file) of documents required as per eligibility criteria mentioned in instructions for filling the tender documents and appendix 'A' to NIT alongwith EARNEST MONEY DEPOSIT (EMD) and tender fee on e-procurement portal and submit the physical documents in the office of**HQ CHIEF ENGINEER (AF), SHILLONG ZONE, MILITARY ENGINEER SERVICES,ELEPHANT FALLS CAMP, PO : NONGLYER, SHILLONG – 793009**within time limit specified in NIT. Inadequacy/deficiency of documents shall make the bid liable for rejection resulting in disqualification for opening of finance bid.

6. Enlisted contractors of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appx 'A' to NIT on eprocurement portal and submit physical documents in the office of **HQ CHIEF ENGINEER (AF), SHILLONG ZONE, MILITARY ENGINEER SERVICES, ELEPHANT FALLS CAMP, PO : NONGLYER, SHILLONG – 793009**before date & time fixed for this purpose.

7. The contractor must ensure that the tender/bid on the proper form is uploaded in time as the accepting officer will take no cognizance of any quotations/offer received in any other electronic or physical form like email/fax/by hand/through post from tenderer/bidder even if they are received in time.

8. In view of delays due to system failure or other communication related failures, it is suggested that the tender/bid be uploaded, if necessary sufficiently in advance of the last due date and time fixed.

9. General conditions of Contracts (IAFW-2249) (1989 Print) and errata and amendments thereto, schedule of minimum fair wages and MES SSR (Part-I and Part-II) are not enclosed with these documents. These are available for perusal in the Office of GE concerned and this office.

10. **ANY TENDERER, WHICH PROPOSES ALTERATIONS TO ANY OF THE CONDITION, SPECIFICATIONS LAID DOWN IN THE TENDER DOCUMENTS OR ANY NEW CONDITION, WHATSOVER, IS LIABLE TO BE REJECTED**.

Yours faithfully,

Encls: (As above)

AAD(CONTRACTS)
For Accepting Officer

Signature of contractor

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER

1. **EARNEST MONEY DEPOSIT (EMD):**
Contractor(s) who are not enlisted with MES/who are enlisted, but have not executed the Standing Security Bond shall submit Earnest Money Deposit as detailed in Notice of Tender in any one of the following forms, along with their tender/bid:-

(a) Deposit at Call Receipt from a Scheduled Bank in favour of concerned **GE/GE(I)/AGE(I)** .

(b) Receipted Treasury Challan, the amount being credited to the Revenue Deposit of concerned **GE/GE(I)/AGE(I)** .

It is advisable that Earnest Money is deposited in the form of deposit call receipt from an approved Scheduled Bank for easy refund. In case the tenderer/bidder wants to lodge 'EARNEST MONEY DEPOSIT' in any other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money Deposit shall be submitted in the name of concerned GE/AGE(I).

NOTES:- (i) Earnest Money Deposit (EMD) in the form of Cheque/Bank Guarantee etc., will not be accepted. NON SUBMISSION OF EARNEST MONEY DEPOSIT (EMD) (scanned copy alongwith Technical Bid & hard copy before the date & time fixed for opening of BOQ) WILL RENDER THE BID DISQUALIFIED FOR OPENING OF COVER-II (FINANCE BID).

2. **PERFORMANCE SECURITY DEPOSIT (APPLICABLE FOR BOTH UNENLISTED AS WELL AS ENLISTED CONTRACTORS):**
The tenderers shall note that the successful tenderer is required to lodge '**Performance Security Deposit**' for an amount equivalent to **5% of the Contract Sum/ lumpsum** within 28 Days of receipt of letter of Acceptance in the form of '**Bank Guarantee' /FDR/ Insurance Surety Bond or any other Government Instruments** in favour of the Accepting Officer in terms of Condition – 19 of IAFW - 2249, General Conditions of Contracts. **The Performance Security shall be valid till the completion of PH-III (i.e. AMC and Manning & Operation of STP) here-in-after PLUS one months**. Irrespective of whatever is mentioned in condition 19.3 of IAFW 2249 with regard to suspension of tenders on account of non-submission of Performance Security, issue of tenders to such tenderers shall remain suspended for a period of six months from the date of cancellation of contract under condition 19.3 of IAFW 2249 in case of unenlisted Contractors. In case of MES enlisted contractor, issue of tenders shall remain suspended till deposit of EMD or six months from date of cancellation whichever is later. The contractor shall not be entitled for any compensation whatsoever on this account. The Work Order for commencing the work shall be placed by the GE only after receipt of above Performance Security by the Accepting Officer within the stipulated time period.

3. **GENERAL INSTRUCTIONS FOR COMPLIANCE:**

a. The bids received only in the electronic form will be considered. All bids shall be submitted on defproc.gov.in portal. Documents should be scanned and forwarded in 'pdf' format and 'xls' form as indicated. Bids shall be uploaded on 'defproc.gov.in' portal on or before the bid closing date mentioned in the tender. No tender/bid in any other electronic or physical form like date email/fax/by hand/through post will be considered.

b. The bid shall be DIGITALLY signed using authorised DSC. All pages of tender documents, drawings, corrections/alterations shall be signed/initialed by the lowest bidder after acceptance of tender for making original and CTC of the contract.

c. Digital signatory of the bid/tender shall ensure that he is competent to bind the contractor (through partnership deed, general power of attorney or Memorandum and Article of Association of the Company) in all the matters pertaining to the contract with Union of India including arbitration clause. A scanned copy of the documents in confirmation of such authority shall be attached with the tender/bid in 'pdf' form in Cover 1. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Article of Association.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER (CONTD…)

d. Drawings, if issued in physical form, must be returned duly initialled by the tenderer/bidder in separate envelope indicating his name and address.

e. The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initialled. The Contractor shall initial every page of tender and shall sign all drawings forming part of the tender. Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

4.0 **BOQ :-**

4.1 The tenderer shall quote his rates on the BOQ EXCEL FILE only as per guide line of e-procure web portal. No alteration to the format will be accepted and such bid will be disqualified.

4.2 In case any tenderer wishes to revise/modify the rates quoted in the BOQ file, he can do so only in the BOQ files before uploading the tender through https://defproc.gov.in site only before closing date & time.

4.3 After the uploading of tender, Department may upload the errata/ amendment through corrigendum. The tenderer/bidder should submit their offer considering the errata/amendment carried out through corrigendum issued from time to time.

4.4 While uploading the bid, the tenderers/bidders should specifically check whether any revised BOQ has been uploaded by department through corrigendum prior to Bid submission start date. Tenderers/bidders attention is specifically drawn to the fact that they should submit their offer on revised BOQ only. In case any tenderer/bidder submits offer on pre-revised BOQ in lieu of Revised BOQ, it will be considered as a willful negligence by the tenderer/bidder and quotation shall be considered non-bonafide.

5. **REVOCATION/ UPWARD REVISION OF OFFER/VOLUNTARY REDUCTION BY THE LOWEST TENDERER AFTER OPENING OF COVER 2:**

Revoking the offer or revising the rates upward/offering voluntary reduction by the lowest tenderer, after opening of Cover 2, shall be considered as a wilful default. For this default a penalty of an amount equal to earnest money shall be levied. In case of an un-enlisted tenderer, earnest money deposited by him shall be forfeited. In case of MES enlisted tenderer, an amount equal to the earnest money stipulated in the NIT shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the aforementioned amount is deposited in the Government Treasury. No other disciplinary/administrative action shall be taken against such tenderer. In such a situation, the next lowest offer shall not be considered for acceptance. Instead retendering shall be resorted to in a transparent and fair manner and the defaulting tenderer and his related firm if any, shall not be eligible for the tender in second call or subsequent calls. Reduction offered by the tenderer in respect of the freakishly high rates shall not be treated as voluntary reduction.

6. Department may issue amendments/errata in form of CORRIGENDUM to tender/revised BOQ to the tender documents. The tenderer/bidder is requested to read the tender documents in conjunction with the errata/amendment/corrigendum, if any, issued by the department.

7. Bidders should check the website periodically for any modified/Revised BOQ uploaded by the Department through Corrigendum and it is bidder's responsibility to ensure uploading of correct BOQ. The bidder whose bid is not on correct BOQ shall be treated as NON-BONAFIDE.

8. Tenders /bidders who uploaded their priced tenders/bids and are desirous of being present at the time of opening of the tenders/bids, may do so at the appointed time.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER (CONTD…)

9. Hard copies of all above documents shall be sent by the contractor to the Tender issuing authority well in advance to be received before the date & time fixed for the same.

10. The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to relevant provisions in Special Condition referred here-in-after and also Conditions 24 & 25 of IAFW-2249 (General Conditions of Contract).

11. These instructions shall form part of the contract documents.

SIGNATURE OF CONTRACTOR
DATED:_____

AAD(CONTRACTS)
FOR ACCEPTING OFFICER

MILITARY ENGINEER SERVICES
NOTICE INVITING TENDER (NIT)

1. A TENDER IS INVITED FOR THE WORK AS MENTIONED IN Appendix 'A' to this (NOTICE INVITING TENDER (NIT))

2. The work is estimated to cost as indicated in aforesaid Appendix 'A'. This estimate, however, is not a guarantee and is merely given as rough guide and if the work cost more or less, a tenderer/bidder will have no claim on that account.

3. The work to be completed within the period as indicated in aforesaid Appendix 'A' in accordance with the phasing, if any, indicated in the tender from the date of handing over site, which will be on or about two weeks after the date of Acceptance of tender.

4. Contractors whose names are on the MES approved list and within whose financial category the estimated amount would fall and un-enlisted contractors may submit tender/bid subject to other criteria mentioned in Appendix A. However in case of term contracts, enlisted contractors of Class SS to E may submit tender. Not more than one tender shall be submitted/uploaded by one contractor/ firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. Two firms shall be deemed to have business dealing if any of the partners/proprietor/director is common among both of them. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

5. **PERFORMANCE SECURITY**: - The tenderers shall note that the successful tenderer is required to lodge **'Performance Security Deposit'** for an amount equivalent to **5% of the Contract Sum/ lumpsum** within 28 Days of receipt of letter of Acceptance in the form of **Bank Guarantee' /FDR/ Insurance Surety Bond or any other Government Instruments**in favour of the **Accepting Officer** in terms of Condition – 19 of IAFW - 2249, General Conditions of Contracts. **The Performance Security shall be valid till the completion of PH-III (i.e. AMC and Manning & Operation of STP) here-in-after PLUS one months.**

6. The office of HQ CHIEF ENGINEER (AF), SHILLONG ZONE, MILITARY ENGINEER SERVICES, ELEPHANT FALLS CAMP, PO : NONGLYER, SHILLONG – 793009 will be the Accepting Officer here-in-after referred to as such for purpose of the contract.

7. Not more than one tender/bid shall be submitted/uploaded by one bidder firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same tender as separate competitors. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

8. The Technical Bid and Financial Bid (Cover-1 and Cover-2) shall be uploaded by the contractor on or before the date & time mentioned in **NIT**. A scanned copy of DD with enlistment details and other documents as specified in Appendix A shall be uploaded as Cover-1 (Technical bid) of the tender on e-tendering portal. DD is refundable in case the contractor is not considered eligible in technical evaluation of Cover 1 resulting in non-opening of Cover1. The applicant contractor shall bear the cost of bank charges for procuring and encashing the DD including revalidation of DDs and shall not have any claim from Government whatsoever on this account.

9. Tender form and conditions of contract and other necessary documents shall be available on website defproc.gov.in for download and shall form part of contract agreement in case the tender/bid is accepted.

10. In case of MES enlisted contractor who has not executed the Standing Security Bond and un enlisted contractor, the Cover-I shall be accompanied by Earnest Money for the amount mentioned in Appendix 'A' in the form of deposit at call receipt in favour of concerned CCE/GE/GE (I)/AGE (I) (see Appendix 'A') by a Scheduled Bank or in received treasury Challan the amount being credited to the revenue deposit of the concerned CCE/GE/GE(I)/AGE(I) (see Appendix 'A'). The CCE/GE/GE (I)/AGE (I) will return the Earnest Money, wherever applicable, to all unsuccessful tenderers/bidders by endorsing an authority on the deposit at call receipt for it's refund, on receipt of intimation from the Accepting Officer to do that..

11. In case of successful contractor i e the lowest contractor having submitted EMD, he shall have the option of converting the EMD instrument into part of the Performance Security to be deposited by him within 28 days from the receipt of intimation of acceptance of tender from Accepting Officer

12. Sample of materials and stores to be supplied by the contractor will also be available for inspection by the bidder at the office of concerned GE/GE (I)/AGE (I)/Project Manager during working hours. The bidder is advised to visit the site of work by making prior appointment with GE/GE (I)/AGE (I)/CCE/Project Manager, who is the Executing Agency of the work (see Appendix 'A'). The bidder shall be deemed to have full knowledge of all relevant documents, samples, site etc whether he has inspected them or not.

13. Any bid which proposes any alteration to any of the conditions laid down or proposes any other new condition whatsoever, is liable to be rejected.

14. The uploading of bid by a bidder implies that bidder has read this notice and the conditions of contract and has made himself aware of the scope and specification of work to be done and of the conditions and rates at which stores (as applicable) etc will be issued to him and local conditions and other factors having bearing on the execution of the work.

15. The bidder must be in possession of a copy of the MES Schedule (SSR) (Part-I & Part-II of latest edition) including amendments and errata thereto.

16. Accepting Officer does not bind himself to accept the lowest or any tender/bid or to give any reason for not doing so.

17. The Accepting Officer reserves the right to accept a tender submitted by a Public Undertaking/Small & Medium Enterprises (SMEs), giving a price preference/purchase preference over other tender(s)/bids which may be lower, as are admissible under the Government Policy. No claim for any compensation or otherwise shall be admissible for such tenderer/bidder whose tender/bid is rejected.

18. Court of the place from where tender has been issued shall alone have jurisdiction to decide any dispute out of or in respect of this tender. After acceptance of tender, Condition 72 (Jurisdiction of Courts) of IAFW-2249 shall be applicable.

19. This Notice Inviting Tender (NIT) including Appendix 'A' shall form part of the contract.

SIGNATURE OF CONTRACTOR **AAD(CONTRACTS)**
DATED:_____ **FOR ACCEPTING OFFICER**

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

1.	Name of work	**PROVN OF CENTRAL SEWAGE TREATMENT PLANTAT AF STATION SALUA**
2.	Estimated Cost	**Rs. 690.00 Lakhs** (At Par Market)
3.	Period of completion	Phase I- 365days Phase II- 365 days (Arboriculture work) Phase III- 1800 days (AMC and Manning & Operation of STP)
4.	Cost of tender documents	Rs.3000/- in the shape of DD/Bankers cheque from any schedule Bank in favour **GE (P) (AF) Kalaikunda** and payable at **Kalaikunda** (Note: In case of retendering, the contractor who had quoted in the previous call is not required to submit the cost of tender.)
5.	Website/portal address	www.defproc.gov.in
6.	Type of contract	The Lump sum tender shall be based on drawing and specifications as per IAFW: 2249 & GCC (IAFW: 2159) with Schedule 'A' (list of items of work) to be priced by contractor/ pre priced by MES. The contractor is required to quote the lump sum amounts for parts of Schedule 'A' and quote rates against items of other parts of Schedule 'A'.
7.	Timeline Details :	
	(a) Bid submission start date	Refer critical dates on the website.
	(b) Bid submission end date	
	(c) Date of bid opening	
8.	ELIGIBILITY CRITERIA	
	(a) For MES enlisted contractors	Contractor shall satisfy the following: (i) Enlistment in Class 'A'and above subject to satisfactoryremarks wrt performance in respect of works in hand as reflectedin Work Load Return (WLR) or any other report circulated by thecompetent engineer authority. (ii) Prequalification Criteria as in Note No. 1 below. (iii) They should submit requisite documents listed here-in-after.
	(b) Forcontractors not Enlisted with MES	(i) Contractors shall meet the enlistment criteria of Class 'A' and above with regards to satisfactory completion of requisite value works with Central/State Government/ Central/ State PSUs/ AWHO/ AFNHB/ CGEWHO / DGMAP, annual turnover, bank solvency, working capital and other requirements given in Para 1.4 & 1.5 of Section 1 of MES Manual of Contracts 2020 as available in all MES formations as well as MES website (www.mes.gov.in). (ii) Not carrying adverse remarks in Work Load Report (WLR) or any other similar report circulated by any competent authority, if already working in MES. (iii) Not suspended/debarred/blacklisted (either permanently or temporarily) from participating in any bid or for business dealings by any Central/State Government Department or any Central/State Government PSU or any Autonomous Body under Central/State Government or any Local Body as on the bid submission end date. (iv) Details of works completed and under progress in MES be submitted in the following format :-

Srl No.	CA No. & Name of Work	Value of CA	Date of Commencement	Date of Completion	Extended Date of Completion

(v) Un-enlisted contractor who have secured two works in MES should get themselves registered in the appropriate designated class with any Registering Authority, else the firm will not be eligible for participation in the tender unless until the firm is enlisted with the MES.
(vi) Prequalification Criteria as in Note No. 1 below.
(vii) They should submit requisite documents listed here-in-after.

APPENDIX 'A' TO NOTICE INVITING TENDER (CONTD)

9.	Tender issuing and Accepting Officer		**HQ CHIEF ENGINEER (AF), SHILLONG ZONE, MILITARY ENGINEER SERVICES, ELEPHANT FALLS CAMP, PO : NONGLYER, SHILLONG – 793009**
10.	Executing agency		**GE (P) (AF) Kalaikunda**
11.	Earnest money		**Rs. 6,70,000.00**/-in favour of **GE (P) (AF) Kalaikunda**in the form of Deposit at call receipt, FDR not acceptable.
12.	Performance Security		The tenderers shall note that the successful tenderer is required to lodge '**Performance Security Deposit**' for an amount equivalent to **5% of the Contract Sum/ lumpsum** within 28 Days of receipt of letter of Acceptance in the form of **Bank Guarantee' /FDR/ Insurance Surety Bond or any other Government Instruments**in favour of the Accepting Officer in terms of Condition – 19 of IAFW - 2249, General Conditions of Contracts. **The Performance Security shall be valid till the completion of PH-III (i.e. AMC and Manning & Operation of STP) here-in-after PLUS one months.** Irrespective of whatever is mentioned in condition 19.3 of IAFW 2249 with regard to suspension of tenders on account of non-submission of Performance Security, issue of tenders to such tenderers shall remain suspended for a period of six months from the date of cancellation of contract under condition 19.3 of IAFW 2249 in case of unenlisted Contractors. In case of MES enlisted contractor, issue of tenders shall remain suspended till deposit of EMD or six months from date of cancellation whichever is later. The contractor shall not be entitled for any compensation whatsoever on this account. The Work Order for commencing the work shall be placed by the GE only after receipt of above Performance Security by the Accepting Officer within the stipulated time period.
13.	Integrity Pact		(a) Scan copy of the Integrity Pact duly signed by the Accepting Officer is enclosed as per Annexure – II to Appendix – 'A' to Notice Inviting Tender (NIT) here-in-after. Scanned copy of Integrity pact duly signed on each page by the bidder(s) shall be uploaded as part of technical bid (Cover-1) and original Integrity Pact duly signed on each page shall be forwarded by post along with demand draft. Bidder(s) who do not upload scanned copy of Integrity Pact duly signed, shall be intimated of the same alongwith intimation of other such deficient documents, through option of 'Short Fall Documents' (in e-tendering portal) before 'T' Bid evaluation. Any bidder who fails to forward the copy of IP duly signed even after this communication, shall be disqualified in the Technical Bid (Cover-1). (b) Integrity pact is deemed as part of contract and both the parties are bound by its provisions.

APPENDIX 'A' TO NOTICE INVITING TENDER (CONTD…)

NOTES:-

1. **PRE-QUALIFICATION CRITERIA (PQC) :-**

The bids of Enlisted/Un-enlisted contractors meeting the following criteria only shall qualify for further consideration:-

(a) Average Annual financial turnover during the last three years, ending 31st March 2025, shall be at least **2.10 Cr** of the estimated cost.

(b) Experience of having successfully completed similar works during last seven years, ending last day of month previous to the month of bid submission start date, should be either of the following :

Three **similar completed works** costing not less than equal to **2.80 Cr** or each having quantity of specific items of work not less than 40% of estimated quantities of tender for which PQC is being given.

OR

Two **similar completed works** costing not less than equal to **3.45 Cr** or each having quantity of specific items of work not less than 50% of estimated quantities of tender for which PQC is being given.

OR

One **similar completed work** costing not less than equal to **5.52 Cr** or each having quantity of specific items of work not less than 80% of estimated quantities of tender for which PQC is being given.

Note :- Contractor shall be uploaded detailed calculation of qualifying work ("Sewage Treatment Works" or "items related to sewage treatment works") on their letter head along with BOQ in Technical bid.

(c) 'Similar works' means construction of **Sewage Treatment Works**(sewage treatment work include both laying of sewer network and construction of STP) with any existing technology & connected civil engineering work.

(d) For the purpose of value of completed works, the value of previously completed works shall be enhanced @ 5% per year to bring them at par with present cost.(Only completed numbers of years shal be considered & part/fraction of year shall not be considered)

(e) For purpose of evaluation, if O&M of the work is in progress, the value of capital expenditure (CAPEX) i.e value of main capital work of STP and the date of completion of the corresponding work shall be considered. However, if the O&M of the work is completed, then the entire value of work & the date of completion of the entire work shall be considered for the purpose of evaluation.

2. In Case after opening of Cover 1, the number of MES enlisted contractors of eligible class as well as eligible un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT is less than 07 (Seven), applications in respect of contractors one class or two classes (in case of remote and difficult areas to be decided as per list circulated by CE Command/ ADG) below the eligible class shall also be considered subject to fulfillment of **Pre-Qualification Criteria given Note No. 1** and other eligibility criteria given in the NIT. Therefore MES contractor's one class below (two classes below in case of remote and difficult areas) may also bid for the tender. Such contractors (Contractors of one/ two classes below the eligible class) shall not be considered in case their present residual work in hand is more than TWO TIMES their present tendering limit. However in case such Contractors fulfill the criteria of up-gradation to the stipulated eligible class bases on past experience of completed works (individual work experience and/ or average annual turnover, as applicable) and financial soundness (solvency/ financial soundness and working capital), the ceiling of present residual work will not apply and they will be considered for issue of tender. Such bidders shall upload in their Cover 1 bid details of works in hand showing names of work, names of Accepting Officers, Contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date. Such contractors, if claim to fulfill the criteria of up-gradation shall also upload the requisite information/ documents in support of up-gradation. These details shall be verified by the Tender Issuing Authority from concerned formations in case bids of such contractors are considered for evaluation.

APPENDIX 'A' TO NOTICE INVITING TENDER (CONTD…..)

3. In case after opening of Cover 1, the number of MES enlisted contractors of eligible class as well as unenlisted contractor (if any), fulfilling the other eligibility criteria given in NIT, are 7 (seven) or more, applications of only those one class below the eligible class bidders shall be considered, who fulfill the **Pre-Qualification Criteria given in Note No. 1 above** and are meeting the criteria of upgradation in respect of past experience of completed works (individual work experience and/ or average annual turnover as applicable) and financial soundness (solvency/ financial soundness and working capital) as per details given in Manual on Contracts provided the value of tendered work is less than twice the tendering limit of such contractor. Therefore such contractors may upload the requisite information/ documents in Cover-I.

4. Un-enlisted contractor shall be considered provided he meets the criteria. Foreign firms shall not be eligible for this tender. However Indian Firms having foreign national/ Indian National staying abroad / Indian national having take foreign citizenship, as director(s) shall be considered subject to security clearance from the concerned authorities.

5. Contractors enlisted with MES will upload following documents in Cover-I for checking eligibility :-

(a) Application for tender on Firm's letterhead.

(b) Enlistment letter issued by the Registering Authority duly renewed for cycle period invogue.

(c) Scanned copy of DD/Bankers Cheque toward cost of tender and EMD instrument in case SSD bond is not signed at the time of registration.

(d) Documents required in respect of PQC criteria.

(e) GST Registration

(f) Any other documents required as described in this Appendix.

6. Contractors not enlisted with MES will be required to upload following documents in Cover I for checking eligibility : -

(a) Application for tender on Firm's letterhead.

(b) Scanned copy of DD/Bankers cheque toward cost of tender and Earnest Money Deposit (EMD) instrument.

(c) Copy of police Verification Report/Police Clearance Certificate/Character Certificate from the police Authority of the area where the registered office of the firm is located notarized copy of valid passport of Proprietor/each Partner/ each Director.

(d) All documents required for enlistment in MES for the class mentioned in Para 8(b) above as per Para 1.5 of Section 1 of MES Manual of Contractors 2020.

(e) Details of works executed and completed in MES or any other Govt agency in past Seven years with completion certificate(**only those works completed in last seven years ending last day of month previous to the month of bid submission start date shall be considered**).

(f) Details of work being executed/under progress in MES. Details of work for which tender/bid has been accepted but work order has not been placed shall also be shown.

(g) Documents required in respect of PQC criteria.

(h) GST Registration

(j) Any other documents required as described in this Appendix.

7. Tender not accompanied by scanned copies of requisite DD /Bankers Cheque towards cost of tender and earnest money (as applicable) in Cover I shall not be considered for validation of 'T' bid and their Financial Bids will not be opened.

8. Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD instruments (as applicable) reach the office of Accepting Officer within 05 (five) days of bid submission end date failing which following action shall be taken.

(a) In case of tenders from an enlisted contractor of MES, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will be opened. However non-submission of physical copies of cost of tender shall be considered as willful negligence of the tenderer with ulterior motives and such tenderer shall be banned from bidding for a period of six months commencing from the date of opening of Financial bid (Cover 2).

(b)In case of tenders from unenlisted contractor, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover2) will not be opened. Name of such contractors along with complete address shall be circulated for not opening of their bids for a period of six months commencing from the date opening of Financial bid (Cover 2).Contd…

APPENDIX 'A' TO NOTICE INVITING TENDER (CONTD…)

(c) In case of tender from enlisted and unenlisted contractors, where scanned copies of instruments of Earnest Money Deposit (as applicable) have been uploaded in Cover 1 but the same are not received in physical form within stipulated period, such tenders shall not qualify for opening of financial bid (Cover 2).

9. Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/another firm on his behalf. However a contractor can execute the work through power of attorney to sons/daughters /spouse of Proprietor/Partner/Director and firm's own employees, director, project manager **provided they are not having a separate enlisted firm in MES in their name as Proprietor/Partner/Director**.

10. After opening of Cover I and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderers as per NIT, a communication in the form of e-mail/SMS/Speed Post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from date of communication failing which their financial bid (Cover 2) shall not be opened and contractor shall not any claim on the same.

11. Invitation for e-tender does not constitute any guarantee for validation of Technical bid and subsequent opening of financial bid of any applicant/bidder merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the Technical bid and not to open the financial bid of any applicant/ bidder. Technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given in this Appendix. Tender /bidder will be informed regarding non-validation of his Technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the next higher Engineer Authority (NHEA) viz **ADG NEI Guwahati** on email id **adgnei-mes@gov.in** with copy of the Accepting Officer on email before the scheduled date of opening of Cover 9. NHEA shall decide the matter within a period of seven working days from the date of receipt of appeal. The decision of NHEA shall be final and binding. The tenderer/ bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

12. In case an unenlisted contractor is already executing work in MES, he shall not be considered eligible for the subject tender if the bidder is already having secured two works in MES. For this purpose, details of the works being executed by such a contractor shall be uploaded in the cover 1 of the bid and shall be checked /verified by the Accepting Officer.

13. In case the BOQ is revised through the corrigendum and the bidder has failed to quote on revised BOQ (i.e he has quoted on pre revised BOQ), such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non-bonafide. In such cases the lowest tender shall be determined from amongst the valid bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance.

14. Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tenderer after opening of Cover 2 shall be considered as a willful default. For this default a penalty of an amount equal to Earnest Money shall be levied. In case of an unenlisted tenderer, Earnest Money deposited by him shall be forfeited. In case of MES enlisted tenderer having deposited the Standing Security Bond, an amount to the earnest money stipulated in the NIT, shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the aforementioned amount is deposited in the Government Treasury. No other disciplinary/administrative action shall be taken against such tenderers. In such a situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be restored to in a transparent and fair manner and the defaulting tenderer and his related firm if any, shall not be eligible for this tender in second call or subsequent calls.

14A. Irrespective of whatever is mentioned in condition 19.3 of IAFW 2249 with regard to suspension of tenders on account of non-submission of Performance Security, issue of tenders to such tenderers shall remain suspended for a period of six months from the date of cancellation of contract under condition 19.3 of IAFW 2249 in case of unenlisted Contractors. In case of MES enlisted contractor, issue of tenders shall remain suspended till deposit of EMD or six months from date of cancellation whichever is later.

15. Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/one or more Partners/Directors are common. Decision of Accepting Officer on issue/ deny the tender to any one of the related forms shall be final and binding.

AAD(Contracts)

SIGNATURE OF CONTRACTOR **FOR ACCEPTING OFFICER**

ANNEXURE-II TO NOTICE INVITING TENDER(NIT)

INTEGRITY PACT

GENERAL

1. Whereas **THE PRESIDENT OF INDIA**, represented by **Chief Engineer (AF) Shillong Zone** here-in-after referred to as Principal / Owner and the first part, has floated the Tender and intends to award, under laid down organizational procedure, contract for **PROVISION OF CENTRAL SEWAGE TREATMENT PLANT AT AF STN SALUA** here-in-after referred to as works/Services and M/S_____ represented by_____(which term unless expressly indicated by the contract, shall be deemed to include its successors and its assignees), here-in-after referred to as the Bidder/Contractor and the second part is willing to carryout the works / services.

2. Whereas the Bidder is a Proprietorship Concern / Partnership Firm / Limited Liability Firm / Private Limited Company / Limited Company / Joint Venture constituted in accordance with the relevant law in the matter and the Principal / Owner is Chief Engineer (CE) performing its functions on behalf of the President of India.

OBJECTIVES

3. Now, therefore, the Principal / Owner and the Bidder agree to enter into this pre-contract agreement, referred to as INTEGRITY PACT (IP), to avoid all forms of corruption by following a system that is fair, transparent and free from any influence/prejudiced dealings prior to, during and subsequent to the conclusion of the contract to be entered into with a view to:-

   3.1. Enabling the Principal / Owner to get the desired works / services at a competitive price in conformity with the defined specifications of the Services by avoiding high cost and the distortionary impact of corruption on public procurement.

   3.2. Enabling Bidders to abstain from bribing or any corrupt practice in order to secure the contract by providing assurance to them that their competitors will also refrain from bribing and other corrupt practices and the Principal / Owner will commit to prevent corruption, in any form, by their officials by following transparent procedures.

COMMITMENTS OF THE PRINCIPAL/ OWNER

4. The Principal / Owner commits itself to the following:-

   4.1. The Principal / Owner undertakes that, no official of the Principal / Owner, connected directly or indirectly with the contract will demand, take a promise for or accept, directly or through intermediaries, any bribe, consideration, gift, reward, favour or any material or immaterial benefit or any other advantage from the Bidder, either for themselves or for any person, organisation or third party related to the contract; in exchange for an advantage; in the bidding process, bid evaluation, contracting or implementation process related to the Contract.

   4.2. The Principal / Owner will, during the pre-contract stage, treat all Bidders alike and will provide to all Bidders the same information and will not provide any such information to any particular Bidder which could afford an advantage to that particular Bidder in comparison to other Bidders.

   4.3. All the officials of the Principal / Owner will report to the appropriate Government office any attempted or completed breach(s) of the above commitments as well as any substantial suspicion of such a breach.

5. In case of any such preceding misconduct on the part of such official(s) is reported by the Bidder to the Principal / Owner willful and verifiable facts and the same is prima facie found to be correct by the Principal / Owner, necessary disciplinary proceedings, or any other action as deemed fit, including criminal proceedings may be initiated by the Principal / Owner and such a person shall be debarred from further dealing related to the tender / contract process. In such a case while an Inquiry is being conducted by the Principal / Owner the tender process / proceedings under the contract would not be stalled.

COMMITMENTS OF BIDDERS

6. The Bidder commits himself to take all measures necessary to prevent corrupt practices, unfair means and illegal activities during any stage of his bid or during any pre-contract or post-contract stage in order to secure the contract or in furtherance to secure it and in particular commits himself to the following-.

6.1. Bidder will not offer, directly or through intermediaries, any bribe, gift, consideration, reward, favour any material or non-material benefit or other advantage, commission, fee, brokerage or inducement to any official of the Principal / Owner, connected directly or indirectly with the bidding process, or to any person, organization or third party related to the contract in exchange for any advantage in the bidding, evaluation, contracting and implementation of the Contract.

6.2. The Bidder further undertakes that he has not given, offered or promised to give, directly or indirectly any bribe, gift, consideration, reward, favour any material or non-material benefits or other advantage, commission, fees, brokerage or inducement to any official of the Principal / Owner or otherwise in procuring the Contract or forbearing to do or having done any act in relation to the obtaining or execution of the contract or any other Contract with the Government for showing or forbearing to show favour or disfavour to any person in relation to the Contract or any other Contract with the Government.

6.3. The Bidder will not collude with other parties interested in the contract to impair the transparency, fairness and progress of the bidding process, bid evaluation, contracting and implementation of the contract.

6.4. The Bidder will not accept any advantage in exchange for any corrupt practice, unfair means and illegal activities.

6.5. The Bidder would not enter into conditional contract with any Agent(s), broker(s) or any other intermediaries wherein payment is made or penalty is levied, directly or indirectly, on success or failure of the award of the contract.

6.6. The Bidder commits to refrain from giving any complaint directly or through any other manner without supporting it with full and verifiable facts. Complaint will be processed as per **Guidelines for Handling of Complaints** in vogue. In case the complaint is found to be vexatious, frivolous or malicious in nature, it would be construed as a violation of Integrity pact.

7. PREVIOUS TRANSGRESSION

7.1. The Bidder declares that no previous transgression occurred in the last three years immediately before signing of this Integrity Pact with any other company in respect of any corrupt practices envisaged hereunder or with any Public Sector Enterprise in India or any Government Department in India.

7.2. If the Bidder makes incorrect statement on this subject, Bidder can be disqualified from tender process or the contract and if already awarded, same can be terminated for such reason.

INTEGRITY PACT (CONTD…)

8. COMPANY CODE OF CONDUCT

8.1. Bidders are advised to have a company code of conduct (clearly rejecting the use of bribes and other unethical behaviour) and a compliance program for the implementation of the code of conduct throughout the country.

9. SANCTION FOR VIOLATION

9.1. Any breach of the aforesaid provisions by the Bidder or any one employed by him or acting on his behalf (whether with or without the knowledge of the Bidder) or the commission of any offence by the Bidder or any one employed by him or acting on his behalf, as defined in Chapter IX of the Indian Penal Code, 1860 or the Prevention of Corruption Act 1988 or any other act enacted for the prevention of corruption shall entitle the Principal / Owner to take all or any one of the following actions, wherever required:-

(i) Technical bid of the Bidder will not be opened. Bidder will not be entitled to or given any compensation. However, the proceedings with the other Bidder(s) would continue.

(ii) Financial bid of the Bidder will not be opened. Bidder will not be entitled to or given any compensation. However, the proceedings with the other Bidder(s) would continue.

(iii) The Earnest Money Deposit shall stand forfeited either fully or partially, as decided by the Principal / Owner, in case contract is not awarded to the Bidder and the Principal / Owner shall not be required to assign any reason therefor. For enlisted contractors an amount less than or equal to Earnest Money Deposit as decided by the Principal/ Owner shall be deducted from any amount held with the Department / any payment due.

(iv) To immediately cancel the contract, if already concluded / awarded without any compensation to the Bidder.

(v) To encash the Performance Security furnished by the Bidder.

(vi) To cancel all or any other Contract(s) with the Bidder.

(vii) To temporarily suspend or temporarily debar / permanently debar the bidder as per the extant policy.

(viii) If adequate amount is not available in the present tender / contract, the deficient amount can be recovered from any outstanding payment due to the Bidder from the Principal / Owner in connection with any other contract for any other works/ services.

(ix) If the Bidder or any employee of the Bidder or any person acting on behalf of the Bidder, either directly or indirectly, is closely related to any of the officers of the Principal / Owner, or alternatively if any close relative of an officer of the Principal / Owner has financial interest/stake in the Bidder's firm, the same shall be disclosed by the Bidder at the time of submission of tender. Any failure to disclose the interest involved shall entitle the Principal / Owner to debar the Bidder from the bid process or rescind the contract without payment of any compensation to the Bidder. The term 'close relative' for this purpose would mean spouse whether residing with the Government servant or not, but does not include a spouse separated from the Government servant by a decree or order of a competent Court; son or daughter or step son or step daughter and wholly dependent upon Government servant, but does not include a child or step child who is no longer in any way dependent upon the Government servant or of whose custody the Government servant has been deprived of by or under any law; any other person related, whether by blood or marriage, to the Govt servant or to the Government servant's wife or husband and wholly dependent upon Government servant.

(x) The Bidder shall not lend to or borrow any money from or enter into any monetary dealings or transactions, directly or indirectly, with any employee of the Principal / Owner and if he does so, the Principal / Owner shall be entitled forthwith to cancel the contract and all other contracts with the Bidder.

9.2. The decision of the Principal / Owner to the effect that a breach of the provisions of this Integrity Pact has been committed by the Bidder shall be final and binding on the Bidder. However, the Bidder can approach the Independent External Monitor(s) (IEMs) appointed for the purposes of this Pact.

10. INDEPENDENT EXTERNAL MONITORS (IEMS)

10.1. MOD has appointed the following Independent External Monitors for this pact in consultation with the Central Vigilance Commission:-

Sl No	Name of IEM	e-mail id
1	Shri Narayan Murty Ganapathy, IFos (Retd)	gana_narayan@yahoo.com
2	Shri Lalatendu Mohanti, IPS (Retd)	L.mohanti@gmail.com

10.2. Details of Nodal officer nominated by E-in-C's Branch are as follows:-

Srl No.	Name of the Nodal Officer of MES	E-mail ID	Mobile No.
(a)	Shri P K S Sengar Director (Contracts)	dircont1einc-mes@nic.in	Tel No. (Office); 011-23019154 Mobile No. - 9131948501

10.3. In case of any complaint with regard to violation of Integrity Pact, either party can approach IEMs with copy to the Nodal Officer and the other party. If any such complaint from bidder is received by the Principal / Owner, the Principal / Owner shall refer the complaint to the Independent External Monitors for their recommendations / inquiry report.

10.4. If the IEMs need to peruse the relevant records of the Principal/ Owner and/or of the Bidder / Contractor in connection with the complaint sent to them, the Principal/ Owner and/ or the Bidder/ Contractor shall make arrangement for such perusal of records by the IEMs as demanded by them including unrestricted and unconditional access to the project documentation and minutes of meeting. If records / documents Of Sub-Contractor(s) are also required to be perused by the IEMs, the Bidder shall make arrangement for such perusal of records by the IEMs as demanded by them, IEMs are under obligation to treat the information and documents of the Principal/Owner and Bidder/ contractor/sub-contractors with confidentiality.

10.5. The task of the IEMs, is to review independently and objectively, any complaint received with regard to violation Integrity Pact and offer recommendations or carry out inquiry as deemed fit. The IEMs are not subject to any instructions by the representatives of the parties and shall perform their functions neutrally and independently. The report of inquiry, if any, made by the IEMs shall be submitted to either of the following for a final and appropriate decision in the matter keeping in view the provision of this Pact:-

(a) Engineer-in-Chief in normal cases

(b) CVO(MES & BRO) /MoD in cases involving vigilance angle

INTEGRITY PACT (CONTD…)

11. EXAMINATION OF BOOKS OF ACCOUNTS

In case of any allegation of violation of any provisions of this Integrity Pact or payment of commission, the Principal / Owner or its agencies shall be entitled to examine the Books of Account of the Bidder and the Bidder shall provide necessary information of the relevant financial documents in English and shall extend all possible help for the purpose of such examination.

12. LAW AND PLACE OF JURISDICTION

This Pact is subject to Indian Law. The place of performance and jurisdiction is the seat of the Principal / Owner.

13. OTHER LEGAL ACTIONS

The actions stipulated in this Integrity pact are without prejudice to any other legal action that may follow in accordance with the provisions of the extant law in force relating to any civil or criminal proceedings.

14. SIGNING OF INTEGRITY PACT ON BEHALF OF BIDDER

(a) Proprietorship Concern — The Integrity Pact must be signed by the proprietor or by an authorised signatory holding power of attorney signed by the proprietor.

(b) Partnership firm - The Integrity Pact must be signed by all partners or by one or more partner holding power of attorney signed by all partners.

(c) Limited Liability firm - The Integrity Pact must be signed by all partners or by one or more partner holding power of attorney signed by all partners.

(d) Private Limited / Limited Company - The Integrity Pact must be signed by a representative duly authorized by Board resolution.

(e) Joint Venture - The Integrity Pact must be signed by all partners and members to Joint Venture or by one or more partner holding power of attorney signed by all partners and members to the Joint Venture.

15. VALIDITY

15.1. The validity of this Integrity Pact shall be from date of its signing. It expires for the Contractor after the final payment under the contract has been made or till the continuation of Defect liability period, whichever is later and for all other bidders, till the Contract has been awarded.

15.2. Should one or several provisions of this Pact turn out to be invalid; the remainder of this Pact remains valid. In this case, the parties will strive to come to an agreement to their original intentions.

INTEGRITY PACT

To

_____

_____

_____

NAME OF WORK : PROVISION OF CENTRAL SEWAGE TREATMENT PLANT AT AF STN SALUA

Dear Sir,

It is hereby declared that MES is committed to follow the principle of transparency, equity and competitiveness in public procurement.

The subject Notice Inviting Tender (NIT) is an invitation to offer made on the condition that the Bidder will sign the Integrity Pact, which is an integral part of tender/bid documents, failing which the tender/bidder will stand disqualified from the tendering process and the bid of the bidder would be summarily rejected.

This declaration shall form part and parcel of the Integrity Pact and signing of the same shall be deemed as acceptance and signing of the Integrity Pact on behalf of MES.

Yours faithfully,

for Chief Engineer (AF) Shillong Zone

CONTD….

INTEGRITY PACT

To,
The Chief Engineer
HQ Chief Engineer (AF) Shillong Zone
Military Engineer Services
Elephant Falls Camp
P.O. :Nonglyer
Shillong – 793009

NAME OF WORK : PROVISION OF CENTRAL SEWAGE TREATMENT PLANT AT AF STN SALUA

Dear Sir,

I/We acknowledge that MES is committed to follow the principles thereof as enumerated in the Integrity Pact enclosed with the tender/bid document.

I/We agree that the Notice Inviting Tender (NIT) is an invitation to offer made on the condition that I/We will sign the Integrity Pact, which is an integral part of tender document, failing which I/We will stand disqualified from the tendering process I/We acknowledge that THE MAKING OF THE BID SHALL BE REGARDED AS AN UNCONDITIONAL AND ABSOLUTE ACCEPTANCE of the conditions of the NIT.

I/We confirm acceptance and compliance with the Integrity Pact in letter and spirit and further agree that execution of the said Integrity Pact shall be separate and distinct from the main contract, which will come into existence when tender/bid is finally accepted by MES. I/We acknowledge and accept the validity of the Integrity Pact, which shall be in line with Para 15 of the enclosed Integrity Pact.

I/We acknowledge that in the event of my/our failure to sign and accept the Integrity Pact, while submitting the tender/bid, MES shall have unqualified, absolute and unfettered right to disqualify the tender/bidder and reject the tender/bid in accordance with terms and conditions of the tender/bid.

Yours faithfully,

(Duly authorized signatory of the Bidder)

CONTD….

IN LIEU OF IAFW-2159

(TO BE READ IN CONJUNCTION WITH
GENERAL CONDITIONS OF CONTRACT: IAFW-2249 (1989 PRINT)
MILITARY ENGINEER SERVICES

HQ Chief Engineer (AF)
Shillong Zone
Military Engineer Services
Elephant Falls Camp
Nonglyer : P.O
Shillong – 793009

88853/KKD/ 107/E8 27 Aug 2025

NAME OF WORK : PROVISION OF CENTRAL SEWAGE TREATMENT PLANT AT AF STN SALUA

1. Shri/M/s_____of_____ is/are hereby authorised to bid for the above work. This bid is to be submitted on- line as per critical dates mentioned in web site "www.defproc.gov.in".

2. All documents must be submitted as per NIT and instructions on "www.defproc.gov.in".

3. All correspondence concerning to this tender shall be addressed as indicated at the top of this sheet quoting the reference given.

PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANYTENDER.

Signature of the Officer
Issuing the Tender Documents
Appointment: AAD (CONTRACTS)

Signature of Contractor
Dated: _____

CONTD….

SCHEDULE `A' NOTES

NAME OF WORK:-PROVISION OF CENTRAL SEWAGE TREATMENT PLANTAT AF STN SALUA

A. NOTES:

1. **(GENERAL) – APPLICABLE TO ALL PARTS**

This schedule is divided into 11 (Eleven) parts as detailed below:-

(a)	PART- I	BUILDING/STRUCTURESS WORKS (To be quoted by contractor)
(b)	PART- II	SITE CLEARANCE AND DEVELOPEMENT (Pre-priced)
(c)	PART- III	INTERNAL ELECTRIFICATION(Pre-priced)
(d)	PART- IV	INTERNAL WATER SUPPLY (Pre-priced)
(e)	PART- V	ROADS (Pre-priced)
(f)	PART- VI	AREA DRAINAGE (Pre-priced)
(g)	PART- VII	SEWAGE DISPOSAL (Pre-priced)
(h)	PART- VIII	EXTERNAL WATER SUPPLY (Pre-priced)
(j)	PART- IX	EXTERNAL ELECTRIC SUPPLY (Pre-priced)
(k)	PART- X	COMPOUND WALL(Pre-priced)
(l)	PART- XI	MISCELLANEOUS ITEMS (To be quoted by contractor)

2. All quantities indicated in BOQ of all parts of Schedule 'A' except Part-I are provisional.

3. The description of building works and services given in various parts of BOQ/Schedule `A' are in brief. These are deemed to be amplified and read in conjunction with special conditions, particular specifications, drawing (including notes thereon), specifications for materials and workmanship given in MES Schedule Part-I, 2009 (Specification) and preambles of items and special condition given in relevant section of MES Schedule Part-II, 2020 (Rates).

4. **PERIOD OF COMPLETION:** The entire work covered under this contract shall be completed within a period of **2165(Two Thousand One Hundred Sixty Five) days (PHASE-I, PHASE-II & PHASE-III)** from the date of commencement of the work as mentioned in the Work Order which shall be issued by GE after acceptance of the tender as under:-

(a) **PHASE- I** : Total work under Phase I shall be completed within **365 (Three Hundred and Sixty Five) days** from the date of commencement of work, except Serial Item No **38.00 to 53.00** of BOQ. In case, it is not possible for the department, to make the entire site available on the date of handing over of site, the contractor shall have to arrange his programmed accordingly. No claim whatsoever for not giving the entire site on the date of handing over of site and for giving site gradually will be tenable. However suitable extension of time will be granted under condition 11 of IAFW-2249 (General Conditions of the contract) in case of delay in completion of work due to gradual handing over of site.. The completion certificate for Phase I shall be issued by the GE after completion of items listed in BOQ except Srl Item No **38.00 to 53.00** of BOQ.

(b) **PHASE II** : Arboriculture against Srl Item No **38.00 to 45.00** of BOQ shall commence on very next day after successful completion of Phase I and will run for 1 Yr (365 days). The completion certificate for Phase II shall be issued by the GE after completion of items listed in BOQ as Srl Item No **38.00 to 45.00** of BOQ.

(c) **PHASE III** : AMC, Manning and operation of Sewage Treatment Plant against Srl Item No **46.00 to 53.00** of BOQ shall commence on very next day after successful completion of Phase I and will run for 5 Yrs (1800 days). The completion certificate for Phase III shall be issued by the GE after completion of items listed in BOQ except Srl Item No **46.00 to 53.00** of BOQ.

Notes:-

(i) Completion for each phase shall be issued separately by the GE in terms of condition 49 of IAFW-2249.

(ii) Final bill for the work covered under Phase-I, Phase-II & Phase-III shall be prepared separately and paid to the contractor after satisfactory completion of each Phase.

(iii) The contractor shall plan the work in such a phased manner so as to hand over the complete work.

CONTD….

SCHEDULE 'A' NOTES (CONTD…)

(iv) Irrespective of the variation in prices of material and labour during period of Phase-III i.e. AMC, Manning & Operation and supplying consumable materials, escalation shall not given to the contractor, The bidder/firms should quote their rates accordingly and no claim whatsoever on this account shall be admissible at a later stage.

(v) The defects liability period for Phase I shall commence after completion of Phase I as certified by GE. AMC during defect liability period (24 Months) shall be included in quoted rate of STP.

5. The tenderer are advised to visit the work site by taking prior appointment with GE before submission of tender. The tenderer shall have no claim what-so-ever on this account at a later stage whether he has actually inspected the work site or not.

6. Unit rate of buildings (as listed in Schedule `A` Part I under BOQ) shall also include the cost of numbering on both sides as per details shown on drawings.

7. The `Unit Rates' inserted by MES under column 6 of Schedule `A' Part II to Part X have been worked out at par MES Standard Schedule of Rates 2020 including amendments/ errata as indicated in tender page here-in-after or at the rates analogous thereto. The accuracy of the 'Unit Rates' is however not guaranteed. The tenderer's attention is invited to Condition 6-A sub para (B) of IAFW-2249 (General Conditions of Contracts) and the lump sum price shall be worked out by him independently of the prices or rates inserted by MES in the tender and irrespective of any errors or inaccuracies therein. **In respect of Sch 'A' Part XI, the rate shall be worked out by the tenderer and shall be inserted in BOQ.**

7A. In case details in respect of items shown on main drawings are not given in the drawings referred to in the main drawings, then the same shall be followed from any other drawings included in the list of drawings. Any drawings mentioned in the contract/contract drawings but inadvertently not included in the list of drawings shall also be deemed to form part of the contract.

8. Probable distribution of various items of internal/external services are indicated on drawings. These are tentative and may be varied as per site requirement, where necessary at the discretion of the Engineer-in-Charge. The effect of such changes/variations in quantities due to realignment or re-siting shall deemed to be included in the rates quoted and no price adjustment on this account shall be allowed.

9. For structural details, structural drawings only to be referred. If there is any discrepancy between Architectural drawings and structural drawings with regard to structural details, detail shown on structural drawings shall prevail and with regard to Architectural details, detail shown in Architectural drgs shall prevail. The decision of Accepting Officer as to what constituted structural or Architectural details shall be final, conclusive and binding.

10. Layout of buildings indicated in the site plan are tentative and no price adjustment shall be made on account of any change in final approved layout within the site of work.

11. The tenderer are advised to visit the work site by taking prior appointment with GE before submission of tender. The tenderer shall have no claim what-so-ever on this account at a later stage whether he has actually inspected the work site or not.

12. The lump sum quoted by the tenderer against each part of Sch 'A' will be deemed to allow for all minor extras and constructional details, which are not specifically shown on drawings and specified in particular specifications but are essential for execution of work services in workman like manner and sound construction. Some of the minor details/items which are deemed essential for execution and entire completion of work are detailed below for guidance:-

(a) Reinforcement for any RCC member not indicated in the drawings but required due to codel requirement.

(b) Dwarf wall in situations like verandah, passage, ramp etc not indicated in drawings.

(c) Lintels over doors, windows, openings and electric meter box not shown in drawings.

In all the above and other similar cases the details indicated elsewhere in the drawings which are similar or near similar to the missed out items of work shall be followed. In the absence of any other similar or near similar details, minimum essential requirement for completion of the work from structural and utility point of view shall be deemed to be included in the lump sum quoted. In case of difference of opinion as to whether or not certain items of work constitute minor extras and constructional details, the decision of the Accepting Officer in this regard shall be final, conclusive and buildings.

CONTD….

SCHEDULE `A' NOTES (Contd..)

13. Cost of testing including transportation of materials or equipments or items, provision of all facilities for testing in accordance with specifications and/or IS shall be borne by the contractor unless otherwise mentioned.

13A. **TAXES, DUTIES & LEVIES ETC. (REFER SPECIAL CONDITION FOR REIMBURSEMENT/ REFUND ON VARIATION IN TAXES DIRECTLY RELATED TO CONTRACT VALUE):** It is pertinent to mention that contractor's quoted rates shall be deemed to include for all statutory levies, duties, taxes including GST, labour welfare cess/tax etc as prevailing on the last date of bid submission. It may be noted that conditional tender on this aspect and/or on any other account will not be considered and shall be liable to be REJECTED."

B. SCHEDULE `A' PART-I.

14. The rate quoted against item of Schedule `A' Part-I shall be deemed to include for all items of works complete as specified in Particular Specifications and shown on drawings, notes thereon for completion of work and all minor details though not specifically shown on drawings or specified in particular specification except works covered under Schedule 'A' Part-II to Part-X or unless specifically stated elsewhere in the tender documents.

15. Cutting and forming chases in masonry/concrete work etc. including cutting and or leaving holes/recesses, sinking etc. where required and as directed by the Engineer-in-charge shall be done as far as possible while the work is in progress. The cost of materials and labour for cutting and/or forming chases, cutting or leaving holes/recesses, sinking and making good in cement mortar (1:3) for filling up to 20mm and in PCC (1:2:4) type B-1 concrete for filling more than 20mm shall deemed to be included in the lump sum quoted by the contractor. In addition cutting or forming chases, cutting or leaving holes/recesses, sinking and making good in cement mortar 1:3/PCC 1:2:4 type B-1 in connection with the work required in plumbing, internal water supply, internal electrification etc. shall deemed to be included in the unit rates of respective building works in Sch 'A' part-I. However no price adjustment shall be made for any variation in the items/quantities of chases/holes/recesses etc. consequent upon changes in quantities of plumbing, internal water supply, internal electrification works etc.

16 The buildings under this contract have been designed for SBC of soil as indicated in structural drawings. Variation in SBC of soil if found at site and required redesigning of the foundation, the same shall be regularized through proper deviation order.

16.1 **Deviation percentage** :- for pricing deviations in items under **BOQ serial No.1.00under Schedule A Part-I (Buildings/Structures works)**, the percentage addition over the rates in "MES STANDARD SCHEDULES-2020, Part II (Rates)" as per condition 62(c) of IAFW 2249 shall be **(+) 10% (Plus Ten Percentage)**.

17. Lump sum quoted by the contractor for items under Schedule `A' Part- I shall also be deemed to include for the following items complete:-

(i) BLANK.

(ii) APP roof treatment in the Building as mentioned at Ser Item No. 1.00 of Schedule 'A' Part I of BOQ and as shown in drawing.

(iii) RCC Jalli and RCC/PCC platform at the locations as shown in drawings.

(iv) Blank

(v) Plinth protection with drain, RCC/PCC cill, Ramps, platforms, verandahs, brick steps, PCC flooring, Granite flooring, tiling, plastering, Cement Concrete tiles on roof, kota stone etc. all as shown on drawing.

(vi) Excavation & earth work complete required for all items of Sch 'A' part-I. Surface dressing of the premises around 3.00 metre perimeter of the building from edge of plinth protection on completion of work. Surplus earth retrieved from foundation of proposed buildings (against schedule 'A' part-I). Extra earth, if required for filling shall be taken from **outside MD Land.** Necessary arrangement to this effect shall be made by the bidder/tenderer and cost of the same shall be deemed to be included in the cost of the building. No extra cost to be given by the department.

(vii) Cutting and forming chases in masonry/concrete work etc. including cutting and or leaving holes/recesses, sinking etc. where required and as directed by the Engineer-in-charge and making good disturbed surface with PCC 1:2:4 type B1 or in CM 1:3.

(viii) HDPE Tanks where ever shown on drawings Capacity of each tank shall be as shown in drawing.

(ix) All type of doors, windows and ventilators with guard bar all as shown on drawings.

CONTD….

SCHEDULE `A' NOTES (Contd..)

(x) Seismic strengthening.

(xi) All other items./fittings/fixtures as indicated in the drawing whether of specifically covered in the specification.

(xii) All the Civil units of STP i.e. screen chamber, oil & grease trap, equlisation tank, moving bed biofilm reactor(MBBR)- I&II, Tube settler tank, Chlorine contact tank, Sludge holding tank, Treated water tank, Micro filteration tank, Plant room etc. shall be as per shown in drawings.

(xiii) All the Mechanical units of STP i.e. Bar screen, Twin lobe air blower, Filter Feed Pump, Multigrade filter, Activated Carban filter, Sludge recycling pump, chlorine dosing pump, EQT sump pump, screw pump, filter press, micro filteration feed pump, Treated water transfer Pump etc. shall be as per shown in drawings.

(xiv) Contractor shall give satisfactory test results of plant to GE, which shall be produced to the Department to **Concern State Pollution Control Board** for getting their consent for discharge of Sewage for proceeding with treatment of sewage. Representative of contractor shall accompany GE to **Concern State Pollution Control Board** for discussions to clarify any other relevant inputs / data desired by **Concern State Pollution Control Board**. After **State** Pollution Control Board gives aforesaid consent for discharge of Sewage, Rate to be included in quoted rate and nothing shall be admissible for this.

Notes for STP:-

(i) All RCC work for STP plant shall be in RCC grade **M-30 (Design mix)** as specified.

(ii) Reinforcement steel TMT bars of grades **Fe 500(D) and shall be CRS (Corrosion Resistant Steel)**.

(iii) Please note that amount quoted for Sewage Treatment Plant shall deemed to be include for full and final completion of work as per Schedule 'A' and BOQ read in conjunction with notes, particular specifications, special conditions as forming part of tender.

(iv) In case any deficiency is noticed in STP plants during testing / commissioning the same will be removed and the plant will be made functional by contractor without any extra cost to govt.

(v) If any deficiency is noticed by any bidder in specification or BOQ the same shall be intimated by the bidder to Deptt. for clarification for such deficiency.

(vii) Plinth protection shall be provided as per location as shown in the drawing.

(viii) All Inlet pipe and outlet pipe with connections as shown in the drawings or otherwise upto 1.00 meter length from faces of surfaces shall deemed to be included in lump sum amount quoted by the contractor.

(ix) Cement for the work shall be as follows:

(f) **For works of STP: Sulphate Resistant cement conforming to IS 12330: 1998 shall be used.**

(g) For other structural works: OPC 43 Grade conforming to IS 269: 2015 shall be used.

(h) For all other works: PPC conforming to IS 1489: 1991 satisfying the strength criteria of OPC 43 Grade shall be used

NOTES:- The list enumerated above is not an exhaustive list, non inclusion of anything /items in the above list does not mean that thing/items is not included in the lump-sum quoted for Sch 'A' Part-I, however please note that lump-sum quoted for Sch 'A' Part-I includes for all the aspects as enumerated in Sch 'A' Note No 17 here-in-before.

17A. The tenderer shall calculate his own rates and insert the same in the BOQ against Schedule 'A' Part-I. The unit rate shall be based on the description of items in Schedule –'A', drawings and specifications enclosed herewith and other applicable provisions of the contract. Amounts quoted against these items by the bidder shall be the lump sum quoted for the respective part of Sch A. Percentage above SSR Pre-Priced amount shall be drived in the excel sheet automatically by the system.

C. **SCHEDULE 'A' PART-II TO PART- X**

18. All quantities are PROVISIONAL. Any variation in quantities if considered necessary due to site requirement /any other technical reasons as decided by the GE may be executed to the extent as specified in condition 7 of General Conditions of Contracts (IAFW-2249).

19. Unless otherwise specified the unit rate of each item of work inserted by MES or quoted by tenderer shall be deemed to include for "Materials and labours" or "Supplying and fixing" or "Supplying, laying, jointing and testing" etc complete unless otherwise specifically mentioned.

20. Excavation and earthwork for Schedule 'A' Part – V to X shall be measured and paid under respective items of Schedule 'A' Part – II unless otherwise specifically specified against Schedule 'A' Part V to X.

21. Blank

CONTD….

SCHEDULE `A' NOTES (Contd..)

22. Special condition, condition in MES standard schedule Part-I and the preambles to the items given in MES Standard schedule of rates Part-II under respective trades shall be applicable. If any provision in the description of items of Schedule 'A' and particular specification is at variance with the provision laid down in the condition of MES Schedule and preambles to MES schedule item, the provision in description of items of Sch 'A' and provision in particular specification shall take precedence.

23. The contractor shall provide digital photographs taken at various stages as decided by GE during the progress of work and submit the same with RAR. Two hard copies (A4 size) and two soft copies (CD) of each digital photograph shall be submitted by contractor to GE" The unit rates against Sch 'A' Part I shall be deemed include the cost of above provision and nothing extra payable on this account. In the event of failure by contractor to submit the same within 30 days of issuance of completion certificate, an amount of Rs. 50,000/- (Rupees Fifty thousand only) will be recovered from the contractor.

24. The contractor shall submit Road Register, PSMBs for all buildings, record drawings of the completed buildings showing B/R & E/M services on A1 size polyester film and one blue print along with a CD showing all changes shall be submitted by the contractor within 03 months of issue of completion certificate failing which an amount of Rs.50,000/- (Rupees fifty thousand only) will be recovered towards it from the dues of the contractor.

25. **ANTIQUITIES AND USEFUL MATERIALS**

25.1 All findings of antiquities and useful materials of archaeological interest such as relics of antiquity, coins, fossils or other valuable articles shall become the property of the Government and accordingly shall be delivered to the Engineer-in-Charge without any extra cost.

26. Contractor shall provide structure plate in each building to be constructed in the above tender. Building plate shall be provided to any suitable location at external wall surface of the respective building. 18 to 20 mm thick granite of size 75 cm x 60 cm shall be provided over 20 mm thick cement mortar (1:3). The following information shall be engraved over granite.

(a) Job No.
(b) CA No and Year.
(c) Name of work.
(d) Name of Contractor.
(e) Name of GE.
(f) Name of Engineer-in-Charge.
(g) Date of Commencement.
(h) Details of completion phase wise.

27. **The tenderer shall satisfy himself by visiting the site with prior appointment with GE/AGE(I) regarding working conditions, local conditions, nature of site/ground, climatic conditions, nature of work and means to access to site and other issues/circumstances which may effect the execution of work at a later date. No extra payment with regard to afore said reasons shall be admissible under any circumstances.**

28. Retrieved materials (due to dismantling/demolition/ taking down etc) of serviceable nature shall be removed from site of work after deducting credit amount from RAR. Unserviceable retrieved materials and other debris shall be removed off from site to a distance exc. 50 mtr and not exc. 100 mtr and deposited as directed by Engineer-in-charge and the unit rate quoted for the dismantling/demolition/taking down etc shall be deemed to be inclusive for above provision.

29. All the Code of Practice, Standards and Specifications applicable shall be the latest editions with upto date correction slip etc.

30. Please note that No foreign exchange or import license facility will be arranged by the department for any material included in the subject tender.

CONTD….

SCHEDULE `A' NOTES (Contd..)

31. The following abbreviation wherever occurs in tender documents shall have the meaning as indicated against each :-

(a)	RM, M, m, Rm	-	Running Metre
(b)	CM, cm	-	Cement mortar/Centimetre
(c)	MM, mm	-	Millimetre
(d)	CUM, cum, cuM	-	Cubic Metre
(e)	SQM, Sqm, sqm	-	Square Metre
(f)	M/L, M & L	-	Material & Labour
(g)	S/F, S&F	-	Supplying and fixing
(h)	C/O, CO	-	Carried over
(j)	B/F, BF	-	Brought forward
(k)	Kg	-	Kilogram
(l)	x Sqm/ X Sqm /10 Sqm	-	10 Square Metre
(m)	x RM/ X RM/ 10RM/ 10 Rm	-	Ten Running Metre
(n)	N.B.	-	Nominal Bore
(o)	Drg/Drgs/drg/drgs	-	Drawing/Drawings/drawing/ drawings
(p)	Cu cm/CU CM	-	Cubic Centimetre
(q)	PQC	-	Pavement quality concrete
(r)	Pbw	-	Part by Weight
(s)	Qty/qty	-	Quantity
(t)	Ltr/ltr	-	Litre
(u)	Qtl	-	Quintal
(v)	GE	-	Garrison Engineer

32. Security/ Restricted Area. The contractor, his agents, work people etc shall adhere to this security instruction as laid down in the Contract Conditions strictly. No claim whatsoever for observance of restrictions imposed at site shall be admissible at any stage.

33. **PROJECT MANAGEMENT**: A Project Schedule will be jointly prepared by contractor and GE using MS Project/Primavera within two weeks of acceptance of the contract and shall be kept on record of GE/AGE(I) for further reference.

(i) The details in the Project Schedule Network will include:-

(a) All activities in detail with assigned T&P labour and duration with facility to follow progress.
(b) Material Procurement planning and utilization (quantity and time of availability) details to ensure actual availability prior to the commencement of the activity/task.
(c) Plant deployment details (activity wise).
(d) Activity Vs Labour employment details.

(ii) The Project Schedule will be updated daily with all necessary details and the "work done report" will be signed by the contractor and included as part of the "Work Diary" by the JE and checked by the Engineer-in-Charge. Weekly review of the work progress will be done between the Engineer-in-Charge and the contractor to monitor the progress made during the week vis-à-vis total progress. The weekly report will also include forecast of resources to include plant, stores, labour, etc for the next week. A detailed monthly report will be prepared and submitted to the GE by the contractor and review done jointly to examine increasing resources to ensure completion within the laid down time period.

(iii) All RAR's submitted by the contractor will be accompanied with the "Weekly Review of Work Progress Report" as mentioned above for the intervening period between last RAR paid and the present RAR being prepared.

34. Integrity Pect (IP) is deemed as part of contract and both are bound bby its provisions.

35. INSPECTION OF THE WORKS (DEFECTS LIABILITY PERIOD)(Refer Condition 46 of IAFW – 2249 (General Conditions of Contracts)):

The **Defects liability period for the scope of work(s) covered under the contract is TWENTY FOUR CALENDAR** months. Any defects in the work noticed during this period shall be rectified by the contractor forthwith, on demand in writing from the GE specifying the area affected not withstanding that the work executed might have been inadvertently passed, certified and paid. The contractor shall carry out the rectification of the defects notified at his own expense during the defects liability period and in event of his failure to do within a period to be specified by the GE in his demand, the GE shall undertake such rectification work at the risk and expense of the contractor.

CONTD….

SCHEDULE `A' NOTES (Contd..)

36. The tenderers shall note that the successful tenderer is required to lodge '**Performance Security Deposit**' for an amount equivalent to **5% of the Contract Sum/ lumpsum** within 28 Days of receipt of letter of Acceptance in the form of '**Bank Guarantee' /FDR/ Insurance Surety Bond or any other Government Instruments** in favour of the Accepting Officer in terms of Condition – 19 of IAFW - 2249, General Conditions of Contracts. **The Performance Security shall be valid till the completion of PH-III (i.e. AMC and Manning & Operation of STP) here-in-before PLUS one months**. Irrespective of whatever is mentioned in condition 19.3 of IAFW 2249 with regard to suspension of tenders on account of non-submission of Performance Security, issue of tenders to such tenderers shall remain suspended for a period of six months from the date of cancellation of contract under condition 19.3 of IAFW 2249 in case of unenlisted Contractors. In case of MES enlisted contractor, issue of tenders shall remain suspended till deposit of EMD or six months from date of cancellation whichever is later. The contractor shall not be entitled for any compensation whatsoever on this account. The Work Order for commencing the work shall be placed by the GE only after receipt of above Performance Security by the Accepting Officer within the stipulated time period.

37. **INSTRUCTIONS ON APPROVAL OF SPECIAL (STAR) RATES.**

37.1 In case of any deviation, mode of pricing shall be decided by Accepting Officer in terms of Condition 62 of IAFW-2249.

37.2 In the event of a deviation order involving fixation of Special (Star) Rate, Draft Rate shall be prepared by GE (within a maximum period of 30 days) while initiating the proposal for deviation seeking approval of the Accepting Officer and notified to contractor. While notifying the Draft Rate, it will be clearly stipulated that the same is merely an estimated rate and firm rate shall be fixed based on actuals and receipt of supporting documents from contractor such as vouchers/literature of product/test certificates etc (as applicable) on completion of the work involving Star Rate. Any objection to the method of fixing Star Rate will be dealt as per Condition 7 of IAFW-2249.

37.3 Draft Star Rate shall be made based on market enquiry through telephonic enquiry/quotations/email/rate lists/internet based sources, material & labour constants available in various publications and record available in respect of Star Rates approved in the past for similar items of work etc. Contractor may also assist GEs office in preparation of draft Star Rate

37.4 The Draft Star Rate shall be purely a draft rate and shall not be used for claiming final payment during execution of work. However GE shall allow part payment to the tune of 80% during execution to avoid any financial hardship to the contractor.

37.5 After completion of the item of work involving Star Rate, the contractor shall submit the vouchers/literature of product/test certificates (as applicable, decision of GE being final in case of any disagreement) for finalization of Star Rate. The Star Rate shall be technically checked by DCWE (C)/Director (C) depending upon the financial effect & approved by competent authority within a period of one month from submission of the relevant documents by contractor as mentioned above.

37.6 The star rate as approved by competent authority after technical check by DCWE (C)/Director (C) depending upon the financial effect shall be referred as "the rate decided by GE" under Para 62 (G) of IAFW-2249.

38. **PERFORMANCEEVALUATIONANDMONITORINGOF WORKS**

38.1 **PERFORMANCEEVALUATION**

Performance evaluation of the work as executed by the contractor shall be carried out at specified, distinct and laid down stages of the work. The contractor shall give on site presentation in these evaluations to the Accepting officer in presence of CWE, GE, AGE, staff officers of CE's Office [E8, E2 (Design), E6, E2 (Plg) and E4]. These shall also be attended by representatives of CFA, PMG, users, audit and references quoted in the minutes of meeting (MoM), which shall be issued by the CE. All evaluations will be carried out on site.

FIRST EVALUATION

First evaluation shall be carried out at the end of mobilization but not later than two months from date of commencement of work indicated in the Work Order No. 1 The contractor and GE shall finalise the Works Programme (CPM or PERT or any other method) which shall be discussed threadbare, during the meeting.

The minutes of the evaluation shall be recorded and signed by the Accepting Officer, CWE and GE as well as the contractor attending the meeting. This shall become the action plan agreed to by all parties and will be contractually binding.

CONTD….

SCHEDULE `A' NOTES (Contd..)

SECOND / INTERMEDIATE EVALUATION

Contracts having period of completion more than twelve months shall have these meetings at intervals of every six months from the date of commencement indicated in Work Order No. 1. These shall be held at the work site on a date fixed well in advance to ensure presence of all concerned. The present progress of work and reasons for any delay shall be analysed and time bound action plan to remove any encumbrances / bottlenecks shall be discussed in detail.

EVALUATION ONE MONTH PRIOR TO ORIGINAL/EXTENDED DATE OF COMPLETION

Like six monthly meeting mentioned above, this meeting will evaluate progress achieved one month prior to original date of completion vis-à-vis extended date of completion. In case progress is not satisfactory on account of the contractor's deficiencies, he will be informed of decision of compensation leviable from original or extended date of completion.

FINAL EVALUATION

Final evaluation shall be held at work site after one month of actual date of completion. The meeting shall be attended by the Accepting Officer, CWE, GE, AGE and the Contractor. The progress of pending issues if any, shall be analysed and timelines arrived at in detail for ensuring preparation and submission of final bill by the contractor within 03 months of the date of completion.

SIGNATURE OF CONTRACTOR
DATED : _____

AAD (CONTRACTS)
FOR ACCEPTING OFFICER

CONTD….

SCHEDULE 'A' PART -I

BUILDINGS/STRUCTURES WORK

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9

Rate To be quoted in BOQ

Signature of Contractor
Date :

AAD (Contracts)
for Accepting Officer

SCHEDULE 'A' PART-II
SITE CLEARANCE AND DEVELOPMENT
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9

{Note:-Necessary Earthwork and Excavation as required for different services (except Schedule 'A' Part-I) also shall be measured under this Schedule unless otherwise specifically excluded}

1.00	Surface dressing not exceeding 15cm deep in hard/dense soil, and removing surplus soil to a distance n. exc. 50 m and making good surface complete all as specified and directed.		Sqm	8940.00	33.50	299490.00		
2.00	M&L for surface excavation n exc 30 cm deep and averaging 15cm deep and getting out soft/loose soil , complete, all as specified and directed.		Sqm	3560.00	53.10	189036.00		
3.00	Removing excavated soil to a distance not exceeding 50 m and depositing where directed at a level not exceeding 1.5m above the starting point complete all as specified and as directed.		Cum	352.00	320.40	112780.80		
4.00	Removing excavated soil to a distance exceeding 250m but not exceeding 500m and depositing where directed at a level, not exceeding 1.5 metre above the starting point complete all as specified and as directed.		Cum	1106.30	582.60	644530.38		
5.00	Removing excavated soft/disintegrated rock to a distance exceeding 250m but not exceeding 500m and depositing where directed at a level, not exceeding 1.5 metre above the starting point complete all as specified and as directed.		Cum	55.00	745.80	41019.00		
6.00	Removing excavated soil to a distance exceeding 500m but not exceeding 1.5 Km and depositing where directed at a level, not exceeding 1.5 metre above the starting point complete all as specified and as directed.		Cum	534.00	643.80	343789.20		
7.00	Excavating in trenches, n exc 1.5 m wide and exc 1.5 m but n exc 3.0m in depth; for foundation, etc. or for shafts, wells, cesspits, manholes, pier holes, etc. n exc 10 sqm on plan and n exc 1.5 m in depth and getting out in soft/loose soil complete all as specified and as directed.		Cum	1.08	542.40	585.79		
8.00	Excavating in trenches, n exc 1.5 m wide and n exc 1.5 m in depth; for foundation, etc. or for shafts, wells, cesspits, manholes, pier holes, etc. n exc 10 sqm on plan and n exc 1.5 m in depth and getting out in hard/dense soil complete all as specified and as directed.		Cum	4691.23	675.00	3166580.25		
9.00	Excavating in trenches, n. exc. 1.5 m wide and exc 1.5 m but n. exc. 3.0m in depth, for foundation, etc. or for shafts, wells, cesspits, manholes, pier holes, etc. n exc 10 sqm on plan and n exc 1.5 m in depth and getting out in hard/dense soil complete all as specified and as directed.		Cum	625.00	833.70	521062.50		
10.00	Excavating in trenches, n. exc. 1.5 m wide and exc 3.0 m but n. exc. 4.5m in depth, for foundation, etc. or for shafts, wells, cesspits, manholes, pier holes, etc. n exc 10 sqm on plan and n exc 1.5 m in depth and getting out in hard/dense soil complete all as specified and as directed.		Cum	312.00	992.40	309628.80		
11.00	Excavating in trenches, n. exc. 1.5 m wide and exc 4.5 m but n. exc. 6.0m in depth, for foundation, etc. or for shafts, wells, cesspits, manholes, pier holes, etc. n exc 10 sqm on plan and n exc 1.5 m in depth and getting out in hard/dense soil complete all as specified and as directed.		Cum	116.00	1151.10	133527.60		
12.00	Excavating in trenches, n exc 1.5 m wide and n exc 1.5 m in depth; for foundation, etc. or for shafts, wells, cesspits, manholes, pier holes, etc. n exc 10 sqm on plan and n exc 1.5 m in depth and getting out in soft / disintegrated rock complete all as specified and as directed.		Cum	55.00	1542.20	84821.00		
13.00	Returning, filling in with soil including spreading, levelling, watering and well ramming in layers not exc 25 cm thick complete all as specified and as directed.		Cum	4285.92	139.10	596171.47		
	Total carried over to BOQ					**64,43,022.79**		

Signature of the Contractor
Date :

AAD (Contracts)
for Accepting Officer

SCHEDULE 'A' PART-III
INTERNAL ELECTRIFICATION
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5.00	6	7	8	9
Note:- Cost of cutting chases in wall and making good for concealing conduit/ terminal boxes etc., for point wiring/sub main wiring/ Telephone cables and cutting opening for fixing exhaust fans are included in the lump sum price of Building under Schedule 'A' Part-I								
1.00	M & L for point wiring with 1.5 Sqmm FRLS & PVC insulated unsheathed single core multi stranded copper conductor concealed type 1100 volt grade drawn in and including non-metalic PVC concealed regid conduit of minimum 20mm medium grade size for one light/ fan point controlled by one switch including one 1.5 Sqmm PVC insulated unsheathed copper conductor for earth continuity conductor and connected to earth dolly and to be terminated to new sunken /pressed flush metal boxes of thickness 1.2mm covered with 3mm thick hylum sheet as per requirement at site for mounting switches , sockets , fan regulators etc including testing on complete all as specified and as directed. (Note: Cost of metal flush box and cover plate shall be included in the item.)		Per Point	22.00	992.75	21840.50		
2.00	All as per item No. 1 here-in-before but for One 3 pin 5 amps socket out let point on independent board complete all as specified and as directed.		Per Point	2.00	1070.40	2140.80		
3.00	All as per item No. 1 here-in-before but for one 3/6 pin switch socket outlet 15 Amp point on independent board with 4.0 Sq mm PVC insulatede unsheathed FRLS multistranded copper conductor and 2.5 Sq mm PVC insulatede unsheathed FRLS multistranded copper earth wire complete, all as specified and as directed.		Per Point	5.00	1131.39	5656.95		
4.00	S&F piano flush type switch, 5/6 Amp, 230V, single pole, 1 way complete all as specified and as directed.		Each	24.00	120.80	2899.20		
5.00	S&F piano flush type switch, 15 Amp, 230V, single pole, 1 way complete all as specified and as directed.		Each	2.00	218.40	436.80		
6.00	S&F socket outlet 2 in 1, 3 pin, 6A complete all as specified and as directed.		Each	2.00	130.10	260.20		
7.00	S&F socket outlet 2 in 1, 3 pin, 6 and 16amp flush type complete all as specified and as directed.		Each	5.00	224.60	1123.00		
8.00	S&F Ceiling rose, 3 phase/PVC/polycarbonate, isolated body complete all as specified and as directed.		Each	14.00	65.70	919.80		
9.00	S&F 5 stepped type, 80 w, electronic fan regulator flush type on the existing phenolic laminated sheet etc. as reqd complete all as specified and as directed.		Each	2	365.90	731.80		
10.00	Supply and fixing ceiling fans complete with blades, down rods, electronic regulator and accessories, 230 V, 1200 mm sweep. Min air delivery 210 CFM with service value 6.00 BEE. Five star rated with brushless direct current motor (BLDC) with out remote type with accessories like holding down rod, hanging clamp, blades, fan hook including writing MES No. (as per standard practice) and flexible three core copper conductor cable of suitable size for connecting with ceiling rose complete all as specified and as directed.		Each	2.00	2649.47	5298.94		
11.00	Supply and fix exhaust fan, steel body AC single Phase, heavy duty, 230V of size 300m sweep with, blades, and frame and louvers including writing MES No. (as per standard practice) and flexible three core copper conductor cable of suitable size for connecting with ceiling rose complete all as specified and as directed.		Each	1.00	3548.59	3548.59		

SCHEDULE 'A' PART-III
INTERNAL ELECTRIFICATION (CONTD…)
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5.00	6	7	8	9
Note:- Cost of cutting chases in wall and making good for concealing conduit/ terminal boxes etc., for point wiring/sub main wiring/ Telephone cables and cutting opening for fixing exhaust fans are included in the lump sum price of Building under Schedule 'A' Part-I								
12.00	S & F factory made sheet metal enclosure double door designed SPN 8 Way DB with 200 Amps rated copper busbar,IP 42, grouter/ fixed flush in wall with and including suitable flat iron complete with, neutral link, dinrail suitable for mountaning MCB's complete all as specified and as directed. (Note:- MCBs shall be measured and paid separately)		Each	1.00	1912.70	1912.70		
13.00	Supply and fixing amp rating 240V, 10KA 'C' curve miniature circuit breaker (MCB) SPN and current carrying capacity 16-32 Amp in the MCB DB complete with connection testing and commissioning complete all as specified and as directed.		Each	1.00	851.20	851.20		
14.00	Supply and fixing amp rating 240V, 10KA 'C' curve miniature circuit breaker (MCB) of single poles and current carrying capacity 6-32 Amp, in the MCB DB complete with connection testing and commissioning complete all as specified and as directed.		Each	6.00	294.00	1764.00		
	Total carried over to BOQ					**49,384.48**		

Signature of the Contractor
Date :

AAD(Contracts)
for Accepting Officer

SCHEDULE 'A' PART-IV
INTERNAL WATER SUPPLY
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9
(Note:- All GI pipes and fittings laid inside rooms shall be concealled and cost of cutting chases and making good shall be deemed to be included in the unit rates of buildings under Sch "A" Part I)								
1.00	Supplying and laying of CPVC pipes outer dia. 16mm SDR 11 & Plain CPVC SDR 11 fittings in trenches complete all as specified.		RM	10.00	139.10	1391.00		
2.00	All as per item No. 01 here-in-before but 20 mm bore complete all as specified and as directed.		RM	20.00	220.70	4414.00		
3.00	S&F for 15 mm bore bib taps(long body), fancy type, chromium plated, cast copper alloy with crutch or butterfly handle, screwed down, screwed for iron pipe, complete all as specified and as directed.		Each	2.00	607.57	1215.14		
4.00	S & F for 15mm dia Pillar taps, cast copper alloy fancy type, with capstan heads, chromium plated, screwed down high pressure, with or without lettered 'Hot' or 'Cold' with long screwed shanks and fly-nuts, screwed for iron pipe and fixed, complete all as specified and as directed.		Each	1.00	676.44	676.44		
5.00	S&F for PVC connections suitable for 15 mm bore pipe with PTMT nut of length 450 mm complete all as specified and as directed.		Each	2.00	150.00	300.00		
6.00	S&F health faucet (ABS Body) mild steel chromium plated suitable for 15 mm dia pipe with 1.25 metre long flexible tube and ABS wall hook including making connections and fixed in position complete in all respect as specified.		Each	1.00	441.15	441.15		
7.00	S&F for 20 mm bore gun-metal, globe or gate valves, with iron wheel head, screwed both ends for iron pipe and fixed complete all as specified and as directed.		Each	1.00	540.11	540.11		
	Total carried over to BOQ					**8,977.84**		

Signature of the Contractor
Date :

AAD(Contracts)
for Accepting Officer

SCHEDULE 'A' PART-V
ROADS
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9
	(Note: Excavation and earth work shall be measured and paid under Schedule 'A' Part-II)							
1.00	M & L for Rolling and consolidating formation surfaces in cutting with power roller exc 8 tonne but nexc 12 tonne including watering as necessary and consolidating to required level , gradient or camber filling the hollows and depressions with approved earth, complete all as specified and directed.		10 Sqm	356.00	26.00	9256.00		
2.00	M&L for Hardcore of guage exc 63mm but n exc. 100 mm, broken stone or boulders deposited, spread and levelled in layers n exc. 15 cm thick, watered and rolled/rammed to a true surface complete all as specified and directed.		Cum	90.00	2614.90	235341.00		
3.00	M&L for PCC (1:4:8) type D-2 (using 40mm graded stone aggregate) as in foundation filling and mass concrete complete all as specified and as directed.		Cum	45.00	5483.10	246739.50		
4.00	M&L 100 mm thick cement concrete type B2 1:2:4 using 40 mm graded stone aggregate for paving etc. complete all as specified and as directed.		Sqm	600.00	629.00	377400.00		
5.00	M & L for 150mm thick soling (or sub base) with broken boulders or quarried stone, inter stices filled, surface formed and rolled and consolidated to required gradient and camber, complete all as specified and as directed.		10 Sqm	224.00	2633.70	589948.80		
6.00	M&L 150mm (compacted thickness) WBM in two layers, 75 mm thick each with course aggregate of grading 2 (63 mm to 40mm size) using granite/trape or basalt, spread , rolled to required gradient and camber complete all as specified and directed.		10 Sqm	224.00	4782.20	1071212.80		
7.00	S&F Polythene film, black, 50micron laid in floors complete all as specified and as directed.		Sqm	2240.00	32.00	71680.00		
8.00	M&L for providing and laying 150 mm thick quality cement concrete minimum flexural strength 40 kg/sqcm (lower control unit) in field after 28 days in pavement slab using min 400 kg of cement/cum with crushed stone aggregate as per design mix including necessary from work, compacting, curing and brooming without using extra cement complete all as specified and directed.		Cum	336.00	7107.10	2387985.60		
9.00	M&L for forming 20mm thick Expansion joints for 150mm thick slab, filled with pre-formed bituminous filler and top 30mm primed and sealed with sealing compound, Grade 'B' complete all as specified and as directed.		10 RM	40.00	910.00	36400.00		
10.00	M&L forming contraction (Dummy) joints 65mm deep 10mm wide, filled with sealing compound, grade "A' complete all as specified and as directed.		10 RM	75.00	608.70	45652.50		
11.00	M&L for finishing surface of concrete roads, pavements, by belting, complete all as specified and as directed.		10 Sqm	224.00	101.50	22736.00		
12.00	M&L for moorum filling as in road berms 100 mm thick, spread, levelled and ramming/rolling including watering, complete all as specified and directed.		10 Sqm	72.00	799.30	57549.60		
13.00	M&L for preparing surfaces by brushing with wire brushes for removing caked mud etc. sweeping with brooms and finally fanning the cleaned surface with gunny bags to remove all loose dirt etc. on unsurface WBM road surface complete all as specified and directed.		10 Sqm	224.00	360.60	80774.40		
	Total carried over to BOQ					**52,32,676.20**		

Signature of the Contractor
Date :

AAD (Contracts)
for Accepting Officer

SCHEDULE 'A' PART-VI
AREA DRAINAGE
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9
	(Note: Excavation and earth work shall be measured and paid under Schedule 'A' Part-II)							
1.00	M & L for PCC (1:4:8) type D-2 (using 40 mm graded stone aggregate) as in foundation filling and mass concrete complete all as specified and as directed.		Cum	27.50	5483.10	150785.25		
2.00	M&L rendering 15 mm thick in CM 1:6 CM on fair faces of brick or concrete surfaces finished even and smooth without using extra cement complete all as specified and directed.		Sqm	527.50	284.49	150068.48		
3.00	M&L for providing cement concrete type B-1 1:2:4 using 20 mm graded stone aggregate for coping etc. and the like including wheathering slightly rounded of cheamfered angles and throting complete all as specified and directed.		Cum	11.50	9259.40	106483.10		
4.00	M&L for brick work with fly ash modular brick, straight or curved on plan exc 6 M mean radius built in CM 1:6 complete all as specified and directed.		Cum	64.40	5731.64	369117.62		
5.00	M&L for precast reinforced cement concrete (1:2:4), type B1 (using 20 mm graded stone aggregate) as in cover slab and drain slab including necessary form work and set in cement mortar (1:6) complete all as specified and as directed. Note: Reinforcement shall be measured and paid separately.		Cum	4.90	11036.80	54080.32		
6.00	S & F for MS TMT 10mm dia and over, cut to length, bent to shape required, including cranking, bending spirally for hooping for columns, hooking ends and binding with mild steel wire (annealed) not less than 0.9 mm dia., or securing with clips complete all as specified and as directed.		Kg	427.58	82.60	35318.11		
	Total carried over to BOQ					**8,65,852.87**		

Signature of the Contractor
Date :

AAD (Contracts)
for Accepting Officer

SCHEDULE 'A' PART-VII
SEWAGE DISPOSAL
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9
	(Note: Excavation and earth work shall be measured and paid under Schedule 'A' Part-II)							
1.00	S&F Sluice Valves, flanged of class 1.6 PN supplied and installed complete size 150mm dia. complete all as specified and directed.		Each	2.00	14500.00	29000.00		
2.00	M&L for Butterfly valve for pipe 100 mm dia with all connections, complete all as specified and directed		Each	4.00	8012.00	32048.00		
3.00	M&L PCC (1:3:6) type C2 using 40mm graded stone aggregate as in foundation , filling and mass concrete complete all as specified and as directed.		Cum	3.00	5862.00	17586.00		
4.00	M&L for brick work with fly ash modular brick, straight or curved on plan exc 6 M mean radius built in CM 1:6 complete all as specified and directed.		Cum	242.00	5731.64	1387056.88		
5.00	Taking down cement plaster on brick or stone walls etc including raking out joints hacking for key scrubing down with water etc complete all as specifiede and as directed.		Sqm	50.00	94.94	4747.00		
6.00	M&L for rendering 15 mm thick in CM 1:4 on fair faces of brick surface finished even and smooth (without using extra cement) complete all as specifiede and as directed		Sqm	814.00	304.24	247651.36		
7.00	M&L for rendering 15 mm thick in CM 1:4 on other than fair faces of brick surface finished even and smooth (without using extra cement) complete all as specifiede and as directed		Sqm	321.00	204.80	65740.80		
8.00	Demolition of brickwork or stone/ boulder masonry, built in cement mortar including all quoins, arches, pillars, etc, but excluding ashlar facings, dressed stonework and precast concrete articles complete all as specifiede and as directed.		Cum	23.00	1135.56	26117.88		
9.00	M&L for PCC 1:2:4 type B-0 (using 12.5 mm graded aggregate) as coping/ benching, kneelers, apex stone bed plates etc., water troughs and the like including weathering, slightly rounded or chamfered angles and throating, complete all as specifiede and as directed.		Cum	6.00	9259.40	55556.40		
10.00	Formwork to edges of concrete flats, treads, breaks in floors, window cills, concrete railings, openings in concrete walls, floor and roofs n exc 20 cm wide and sides of RCC/ PCC bands in walls, horizontal or sloping and similar work exc 5cm but n exc 10cm wide, for rough finished surface of concrete, complete all as specifiede and as directed.		RM	318.00	51.95	16520.10		
11.00	Formwork to soffits of suspended slabs such as roof slabs, floor slabs, landings and similar work, not exceeding 200mm thick (Horizontal or sloping) on flat and for rough finished surface of concrete, complete all as specifiede and as directed.		Sqm	45.84	495.26	22702.72		
12.00	M&L for RCC 1:2:4 type B1 using 20mm graded stone aggregate as in slabs supported on walls, beams and columns in floors, roofs, landings, balconies, canopies, deck slabs and in shelves and the like complete all as specifiede and as directed.		Cum	12.00	7455.00	89460.00		
13.00	M&L for cement concrete type B1 1:2:4 (using 20mm graded stone aggregate in surface channel and drains finished even and smooth using extra cement, complete all as specifiede and as directed.		Cum	7.00	6611.70	46281.90		
14.00	S&F 2000 Ltr rotational moulded polyethylene water storage tanks (cylindrical vertical with closed top) hoisted and fixed in position all as specified in SSR Part I complete all as specifiede and as directed.		Each	12.00	15978.41	191740.92		
15.00	M&L for 150mm bore (internal dia) Reinforced Concrete pipes, class NP-3 with collars, laid in trenches of any depth upto 7.5m and jointed on concrete bed to required slope jointed and plastered complete all as specified and as directed. Note:- Concrete bed shall be measured and paid separately.		RM	1000.00	400.00	400000.00		
16.00	M&L for 200mm bore (internal dia) Reinforced Concrete pipes, class NP-3 with collars, laid in trenches of any depth upto 7.5m and jointed on concrete bed to required slope jointed and plastered complete all as specified and as directed. Note:- Concrete bed shall be measured and paid separately.		RM	200.00	460.00	92000.00		
17.00	M&L for 250mm bore (internal dia) Reinforced Concrete pipes, class NP-3 with collars, laid in trenches of any depth upto 7.5m and jointed on concrete bed to required slope jointed and plastered complete all as specified and as directed. Note:- Concrete bed shall be measured and paid separately.		RM	300.00	600.00	180000.00		
18.00	M & L for PCC (1:3:6) type C-2, using 40 mm graded stone aggregate as in concrete bed to drain pipe including packing under and haunching against sides of pipes after they are laid and tested for 150 mm bore pipe complete all as specified and directed.		RM	985.00	466.89	459886.65		

SCHEDULE 'A' PART-VII
SEWAGE DISPOSAL(CONTD...)
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9
	(Note: Excavation and earth work shall be measured and paid under Schedule 'A' Part-II)							
19.00	M & L for PCC (1:3:6) type C-2, using 40 mm graded stone aggregate as in concrete bed to drain pipe including packing under and haunching against sides of pipes after they are laid and tested for 200 mm bore pipe complete all as specified and directed.		RM	197.00	647.79	127614.63		
20.00	M & L for PCC (1:3:6) type C-2, using 40 mm graded stone aggregate as in concrete bed to drain pipe including packing under and haunching against sides of pipes after they are laid and tested for 250 mm bore pipe complete all as specified and directed.		RM	296.00	698.62	206791.52		
21.00	M&L for MS round bars as in rungs of manholes including bending to shape required framed work such as grills, gratings, etc.. with ends of bars shouldered and / or riveted, or forged into spikes, framed guard bars, barred iron doors, ladders, framed balusters, walk ways, railings, framework of waler tanks and similar work conforming to Fe-290, Gde-E-165, complete all as specified and as directed.		Kg	1200.00	164.86	197832.00		
22.00	Demolition of RCC of any description and in any position not otherwise specifically provided for complete all as specified and as directed.		Cum	10.50	1883.70	19778.85		
23.00	M&L for Cutting into existing manholes , pipe of dia 150mm for connecting new drains, making good to pipe and rendering, cutting out existing benching for and forming branch channel (1/2 round or 3/4section) in cement concrete 1:2:4 type B-1 smooth finished, and reforming benching complete all as specified and as directed.		Each Pipe	75.00	1120.00	84000.00		
24.00	M&L for removing existing branch drain, pipe of dia 150mm for connecting new drains, making good to pipe making good to walling and rendering, cutting out branch channel, and making good benchings complete all as specifiede and as directed.		Each Pipe	75.00	1080.00	81000.00		
25.00	S & F for MS TMT 10mm dia and over, cut to length, bent to shape required, including cranking, bending spirally for hooping for columns, hooking ends and binding with mild steel wire (annealed) not less than 0.9 mm dia., or securing with clips complete all as specified and as directed.		Kg	1700.00	82.60	140420.00		
26.00	M&L for precast reinforced cement concrete (1:2:4), type B1 (using 20 mm graded stone aggregate) as in cover slab and drain slab including necessary form work and set in cement mortar (1:6) complete all as specified and as directed. Note: Reinforcement shall be measured and paid separately.		Cum	10.50	11036.80	115886.40		
27.00	M & L for PCC (1:4:8) type D2 (using 40 mm graded stone aggregate) as in foundation filling and mass concrete complete all as specified and as directed.		Cum	27.00	5483.10	148043.70		
	Total carried over to BOQ					**44,85,463.71**		

Signature of the Contractor
Date :

AAD (Contracts)
for Accepting Officer

SCHEDULE 'A' PART-VIII

EXTERNAL WATER SUPPLY

(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9
	(Note: Excavation and earth work shall be measured and paid under Schedule 'A' Part-II)							
1.00	Demolition of cement concrete of any description and in any position not otherwise specifically provided for complete all as specified and as directed.		Cum	60.00	1631.30	97878.00		
2.00	S&F 3-layer PP-R Polypropylene random copolymer concealed pipes SDR 7.4 (PN 16) UV stabilized and antimicrobial fusion welded having thermal stability for hot & cold water supply including all PP-R plain & brass treaded polypropylene random fittings laid in trenches of 40mm outer dia. complete all as specified and directed.		RM	500.00	341.60	170800.00		
3.00	S&F 3-layer PP-R Polypropylene random copolymer concealed pipes SDR 7.4 (PN 16) UV stabilized and antimicrobial fusion welded having thermal stability for hot & cold water supply including all PP-R plain & brass treaded polypropylene random fittings laid in trenches of 25mm outer dia. complete all as specified and directed.		RM	150.00	182.86	27429.00		
4.00	S&F 3-layer PP-R Polypropylene random copolymer concealed pipes SDR 7.4 (PN 16) UV stabilized and antimicrobial fusion welded having thermal stability for hot & cold water supply including all PP-R plain & brass treaded polypropylene random fittings laid in trenches of 16mm outer dia. complete all as specified and directed.		RM	50.00	120.60	6030.00		
5.00	M&L for 100mm bore HDPE pipe (PN-10) including all fittings and jointing accessories complete all as specified and directed.		RM	1580.00	756.00	1194480.00		
6.00	M&L for 50mm bore HDPE pipe (PN-10) including all fittings and jointing accessories complete all as specified and directed.		RM	320.00	159.00	50880.00		
7.00	M&L for 20mm bore High density polyethylene pipes (HDPE) of pressure rating 6 kgf/cm2 fixed complete to walls, ceilings or in floors etc.complete all as specified and directed.		RM	200.00	149.40	29880.00		
8.00	S&F PP-R Gate valves (brass insert) of size 40mm outer dia complete all as specified and directed.		Each	7.00	737.27	5160.89		
9.00	S&F PP-R Gate valves (brass insert) of size 25mm outer dia complete all as specified and directed.		Each	8.00	545.30	4362.40		
10.00	Making connection of PP-R distribution branch with PP-R main of for 25mm outer dia pipes sizes by providing and fixing tee including cutting and making fusion joint etc. complete all as specified and directed.		Each	6.00	529.10	3174.60		
11.00	Cutting existing steel tubing and screwing existing steel tubing for alteration and addition for 40 to 50mm bore of pipe, complete all as specified and directed.		Each	2.00	224.06	448.12		
12.00	Making connection of PP-R distribution branch with PP-R main of for 50mm outer dia pipes sizes by providing and fixing tee including cutting and making fusion joint etc. complete all as specified and directed.		Each	5.00	703.88	3519.40		
13.00	Making connection of PP-R distribution branch with PP-R main of for 20mm outer dia pipes sizes by providing and fixing tee including cutting and making fusion joint etc. complete all as specified and directed.		Each	5.00	529.10	2645.50		
14.00	M&L for Butterfly valve for pipe 100 mm dia with all connections, complete all as specified and directed		Each	3.00	8012.00	24036.00		
15.00	M&L for Butterfly valve for pipe 50 mm dia with all connections, complete all as specified and directed		Each	8.00	3120.00	24960.00		
	Total carried over to BOQ					**16,45,683.91**		

Signature of Contractor

AAD (Contracts)
for Accepting Officer

SCHEDULE 'A' PART-IX
EXTERNAL ELECTRIFICATION
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9
	(Note: Excavation and earth work shall be measured and paid under Schedule 'A' Part-II)							
1.00	Supply and spreading dry coarse sand in trenches for protection of underground electric cable, spread in uniform 23 cm thickness and punned down gently to 18 cm complete all as specified and as directed. Note:- Punned thickness will be measured at the time of measurement and payment.		CUM	261.00	1974.30	515292.30		
2.00	Supplying and laying un-reinforced precast concrete cable cover, class LV, Type I, flat, size 250 mm x 150mm x 40mm complete all as specified and as directed.		Each	10400.00	44.20	459680.00		
3.00	Supplying and laying un-reinforced precast concrete cable cover of class HV, Type 1, flat, size 300mmx 180mm x 40mm complete all as specified and as directed.		Each	6944.00	57.40	398585.60		
4.00	M&L for Cement concrete (1:3:6) type C-2 using 40mm graded aggregate as in foundation of poles /stay and filling of mass concrete including providing necessary formwork complete all as specified and as directed.		CUM	27.95	5862.00	163842.90		
5.00	M&L for Cement concrete (1:2:4) type B-2 using 40mm graded aggregate as in foundation of poles /stay and filling of mass concrete including providing necessary formwork complete all as specified and as directed.		CUM	1.00	6477.80	6477.80		
6.00	Supply. laying & testing HT cable XLPE insulated, screened, PVC bedded, galvanized steel strip or wire armoured, electric power cables (heavy duty) with stranded aluminium conductor, 11000 volts grade, of size 95 Sqmm 3 core, complete all as specified and as directed.		RM	1830.00	1067.97	1954385.10		
7.00	Supply and jointing cable jointing kit for 11 KV (Earthed) grade cable for Outdoor end termination cold shrinkable type joint complete with jointing material and accessories suitable for 3 core XLPE, armoured, aluminium conductor cable of size 95 sqmm complete all as specified and as directed.		Each	6.00	18387.10	110322.60		
8.00	Supply and fix cable jointing kit for 11 KV grade cable for straight through cold shrink type joint complete with jointing material and accessories suitable for 3 core XLPE armoured. Al conductor cable of size. 50 Sqmm to 95 sqmm, Indoor/Outdoor type complete all as specified and as directed.		Each	4.00	37615.90	150463.60		
9.00	Supply and laying or fixing or passing through pipe, XLPE insulated 1100 volt grade underground cable with aluminium conductor, armored with steel strip or wire cross sectional area 10 Sqmm 4 core complete all as specified and directed.		RM	700.00	174.68	122276.00		
10.00	Supply and laying or fixing or passing through pipe, XLPE insulated 1100 volt grade underground cable with aluminium conductor, armored with steel strip or wire cross sectional area 35 Sqmm 3.5 core complete all as specified and directed.		RM	700.00	269.75	188825.00		
11.00	Supply and laying or fixing or passing through pipe, XLPE insulated 1100 volt grade underground cable with aluminium conductor, armored with steel strip or wire cross sectional area 95 Sqmm 3.5 core complete all as specified and directed.		RM	100.00	526.72	52672.00		
12.00	Supply and laying or fixing or passing through pipe, XLPE insulated 1100 volt grade underground cable with aluminium conductor, armored with steel strip or wire cross sectional area 120 Sqmm 3.5 core complete all as specified and directed.		RM	200.00	648.15	129630.00		
13.00	M&L for steel tubular swaged poles complete with cast iron base plate, finial taper plug, bolts, nuts & screws as specified type 410 SP-55 (11.0 metres long) complete all as specified and directed.		Each	4.00	18443.78	73775.12		
14.00	Supply and install switches air break , Gang operated Triple pole, mechanically operated mounted on insulator and steel frame, with operating and steel frame, with operating mechanism worked from ground level, for current rating upto 200 Amps, 11000 Volts complete all as specified and directed.		Each	2.00	9435.10	18870.20		
15.00	S&F lightning arresters expulsion type complete with all fittings, transmission class 11 KV, discharge capacity 65 kilo amp complete all as specified and directed		Each Set	2.00	10694.90	21389.80		

SCHEDULE 'A' PART-IX
EXTERNAL ELECTRIFICATION (CONTD....)
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9
	(Note: Excavation and earth work shall be measured and paid under Schedule 'A' Part-II)							
16.00	M&L for cross arms, bracings, supports, clamps and back plates fabricated from structural steel sections including nuts, bolts, washers, welding, bending cold or hot, drilling holes for bolts, if any shape or size as indicated or directed and inclusive of one primer coat, one under coat and one finishing coat of paint complete all as specified and directed.		Qtl	2.50	11931.40	29828.50		
17.00	M&L for anti-climbing device, spikes type fabricated out of M.S. bars (round, square and flat) including nuts, bolts, washers and welding in two halves made to shape to suit the cross section of the pole complete all as specified and directed.		Qtl	0.50	12242.60	6121.30		
18.00	S&F danger notice plate of 1.6mm thick mild steel sheet, vitreous enamelled white, with letters, figures and conventional skull and bones in signal red colour and fixed with M.S. clamps, bolts and nuts of approved size, for HT, 25x20cm size complete all as specified and directed.		Each	2.00	223.80	447.60		
19.00	M&L for copper earth continuity conductor of size 20 x 3 mm strip with insulator fixed to wall or drawn in conduit/ pipe as required complete all as specified and directed. Note: Insulator to be fixed on every 50cm distance and cost of insulator is deemed to be included in unit rate of item.		RM	120.00	426.80	51216.00		
20.00	M&L for 100mm bore HDPE pipe PN10 including all fittings and jointing accessories laid in trenches/Road for cable protection complete all as specified and directed.		RM	500.00	756.00	378000.00		
21.00	M&L for 50mm bore HDPE pipe PN10 including all fittings and jointing accessories laid in trenches/Road for cable protection complete all as specified and directed.		RM	50.00	159.00	7950.00		
22.00	Supply and laying GI pipe light grade of size 50mm outer dia for cable protection on pole/wall/floor or in trenches and drain/culvert etc. complete with necessary clamps, nuts, bolts, washers sockets, complete all as specified and directed.		RM	20.00	402.00	8040.00		
23.00	Supply and laying or fixing or passing through pipe XLPE insulated 1100 volt grade underground cable with aluminium conductor armored with steel strip or wire cross sectional area 16 Sqmm 4 core complete all as specified and directed.		RM	300.00	193.50	58050.00		
24.00	M&L for steel tubular swaged poles complete with cast iron base plate, finial taper plug, bolts, nuts & screws as specified type 410 SP-31 (9.0 metres long) complete all as specified and directed.		Each	20.00	12009.70	240194.00		
25.00	M&L for sub main wiring for 2 single core 2.5 sqmm PVC insulated multistranded copper cable with one single core 2.5 sqmm PVC insulated copper conductor for connection from terminal box MCB to street light fitting in PVC conduit, complete all as specified and directed.		RM	150.00	181.21	27181.50		
	Total carried over to BOQ					**51,73,516.92**		

Signature of Contractor

AAD(Contracts)
for Accepting Officer

SCHEDULE 'A' PART-X
COMPOUND WALL
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9
	(Note: Excavation and earth work shall be measured and paid under Schedule 'A' Part-II)							
1.00	M & L for PCC (1:4:8) type D2 (using 40 mm graded stone aggregate) as in foundation filling and mass concrete complete all as specified and as directed.		Cum	30.29	5483.10	166083.10		
2.00	M&L for brick work with fly ash modular brick, straight or curved on plan exc 6 M mean radius built in CM 1:6 complete all as specified and directed.		Cum	175.00	5731.64	1003037.00		
3.00	M&L rendering 15 mm thick in CM 1:6 CM on fair faces of brick or concrete surfaces finished even and smooth without using extra cement complete all as specified and directed.		Sqm	708.64	284.49	201600.99		
4.00	M&L for providing cement concrete type B0 1:2:4 (12.5mm aggregate) coping, kneelers, apex stones, bed plates complete all as specified and as directed.		Cum	6.06	9259.40	56111.96		
5.00	M&L for broken glass in coping on top of wall in cement mortar 1:3, complete all as specified and as directed.		Sqm	75.00	552.03	41402.25		
6.00	M&L preparation of newly plastred surface of wall and applying two coat of cement based paint, complete all as specified and as directed.		Sqm	708.64	70.50	49959.12		
7.00	Formwork to sides of concrete foundations, footings, bases of columns, and similar work; vertical or to batter for rough finished surface, complete all as specified and as directed.		Sqm	1.60	316.72	506.75		
8.00	Formwork to sides of concrete of pillars, posts, struts , piers, columns for rough finished surface square, rectangular, complete all as specified and as directed.		Sqm	9.39	598.08	5615.97		
9.00	S & F for MS TMT 10mm dia and over, cut to length, bent to shape required, including cranking, bending spirally for hooping for columns, hooking ends and binding with mild steel wire (annealed) not less than 0.9 mm dia., or securing with clips complete all as specified and as directed.		Kg	83.00	82.60	6855.80		
10.00	All as per serial item No 9.00 here-in-before but 5mm dia and over upto and including 10mm dia. as in stirrup, spacers band binders, complete all as specified and as directed.		Kg	26.00	85.18	2214.68		
11.00	M&L Framed work as in doors or gates of angle or other section with gusset plates, rails, braces, etc., complete, drilled for fixing of steel sheeting or other covering. Doors, etc. to be prepared for hanging or sliding with and including either hooks and hinges or steel hanging door fittings (exclusive of steel sheeting or other covering, running rails and guides) and hanging; also fastening and fixing complete. Fe-410 W (Gde-E-250) quality B complete all as specified and as directed.		Kg	265.00	119.16	31577.40		
12.00	M&L for RCC M-30 (design mix) using 20mm graded stone aggregate) as in foundation, bases of columns and mass concrete all as specified and directed.		Cum	0.40	7721.87	3088.75		
13.00	M&L for RCC M-30 (design mix) using 20mm graded stone aggregate) as in columns , pillars, piers posts and struts all as specified and directed.		Cum	1.38	8854.74	12219.54		
14.00	M&L for painting to surface of any description, over 10cm width or girth including preparation of newly surface and one coats of red oxide primer and two coat of synthetic enamel paint complete all as specified.		RM	8.19	141.30	1157.25		
	Total carried over to BOQ					**15,81,430.57**		

Signature of the Contractor
Date :

AAD (Contracts)
for Accepting Officer

SCHEDULE 'A' PART -XI
MISCELLANEOUS ITEMS
(LIST OF WORKS AND PRICES)

Srl No	DESCRIPTION OF ITEM OF WORKS	DRG NO	UNIT	NO. OF UNITS REQD	UNIT RATE	TOTAL AMOUNT Rs Ps	PERIOD OF COMPLETION	REMARKS
1	2	3	4	5	6	7	8	9

Rate To be quoted in BOQ

Signature of Contractor
Date :

AAD (Contracts)
for Accepting Officer

SCHEDULE 'B'
LIST OF MATERIALS TO BE ISSUED TO THE CONTRACTOR
(SEE CONDITION 10 OF IAFW-2249)

Srl. No.	Particulars	Rates at which materials will be issued to the contractor		Place of issue of store	Remarks
		Unit	Rate		
1	2	3	4	5	6

----------NIL----------

SIGNATURE OF CONTRACTOR **AAD (CONTRACTS)**
DATED: _____ **FOR ACCEPTING OFFICER**

SCHEDULE 'C'

LIST OF TOOLS & PLANT (OTHER THAN TRANSPORT)
WHICH WILL BE HIRED TO THE CONTRACTOR
(See Conditions 14,15, 34 and 35 of IAFW-2249)

Srl No	Quantity	Particulars	Details of MES Crew Supplied	Hire Changes per unit per working Day Rs	Stand by changes per unit per off Day Rs.	Place of Issue by Name	Remarks
1	2	3	4	5	6	7	8

----------NIL----------

SIGNATURE OF CONTRACTOR **AAD (CONTRACTS)**
DATED: _____ **FOR ACCEPTING OFFICER**

SCHEDULE 'D'

TRANSPORT TO BE HIRED TO THE CONTRACTOR
(See Conditions 16 and 35 of IAFW-2249)

Srl No	Quantity	Particulars unit per Working Day	Rate per unit per Working Day	Place of Issue by Name	Remarks
1	2	3	4	5	6

----------NIL----------

SIGNATURE OF CONTRACTOR **AAD (CONTRACTS)**
DATED: _____ **FOR ACCEPTING OFFICER**

CONTD….

TENDER

TO,

THE PRESIDENT OF INDIA

Having examined and perused the following documents: -

1. Specifications signed by AAD (CONTRACTS).

2. Schedule 'A', 'B', 'C' & 'D' attached.

3. MES standard Schedule of Rates (Part-I of 2009: Specifications & Part-II of 2020: Rates) (here-in after and in IAFW-2249 referred to as the MES Schedule) together with amendment No **1 to 3** for Part-I (2009) and amendment No. **1 to 122** for Part-II (2020) as applicable to the above said schedule.

4. General conditions of contracts (IAFW-2249 –1989 Print) together with amendment **No. 1 to 49** and errata No. **1 to 20**.

5. General condition 31 of IAFW-2249 (General Conditions of Contracts):

WATER WILL BE SUPPLIED BY MES

6. Should this tender be accepted I/We * agree :-

(a) That the Sum of **Rs. 6,70,000.00 (Rupees Six Lakh Seventy Thousand Only)**forwarded as earnest money shall either be retained as part of Performance Security, or be forwarded by the Govt. on receipt of appropriate amount of the Performance Security all as per condition 22 of IAFW-2249.

(b) To execute all the works referred to in the said documents upon the terms and conditions contained or referred to therein at the item rates contained in the aforesaid Schedule 'A' or at such other rates as may be fixed under the provisions of condition 62 of IAFW-2249 and to carry out such deviations as may be ordered vide condition 7 of IAFW-2249 upto a maximum of **10 (Ten) percent** and further agree to refer all disputes, as required by condition 70 of IAFW-2249 to the 'Sole Arbitrator' from the MOD panel of arbitrator (having qualifications as per condition 70 of IAFW2249) to be appointed by the Engineer-in-Chief or in his absence the officer officiating as Engineer-in-Chief or the Director General of Works if specifically delegated in writing by the Engineer-in-Chief, Army Headquarters, New Delhi, whose decision shall be final, conclusive and binding.

* Delete whichever is not applicable.

CONTD….

TENDER (CONTD)

Rs._____(Rupees_____

_____

_____)

Signature _____ in the capacity of _____ duly authorized to

sign the tender for and on behalf of M/S._____

(IN BLOCK LETTERS)

Witness _____ Postal Address_____

Address_____
_____
_____

Telegraphic Address_____

Telephone No._____

ACCEPTANCE

_____ Alterations have been made in these document and as evidenced that these alterations were made before the execution of the contract agreement, these have been initialed by the contractor and Smt MG Marbaniang, AAD (Contracts) The said officer is hereby authorized to sign and initial on my behalf of the documents forming part of the contract.

The above tender is/was accepted by me on behalf of the President of India for the amendedlumpsum/

Lumpsum of Rs._____ (Rupees_____

_____

_____)

on the _____day of _____2025.

Signature _____ dated this _____day of _____2025.

Accepting officer
Appointment: CHIEF ENGINEER (AIR FORCE) SHILLONG ZONE
(For and on behalf of the President of India)

CONTD….

(In lieu page)

GENERAL CONDITIONS OF CONTRACTS (IAFW: 2249-1989)
FOR LUMP SUM CONTRACTS (IAFW: 2159)
MEASUREMENTCONTRACTS (IAFW: 1779 & 1779A)

1. A copy of the MES GENERAL CONDITIONS OF CONTRACTS (IAFW: 2249 - 1989) with Errata and Amendments has been supplied to me/us and is in my/our possession. I/We have read and understood the provisions contained in the aforesaid GENERAL CONDITIONS OF CONTRACTS before submission of this tender and I/We agree that I/We shall abide by the terms and conditions thereof.

2. It is, hereby, further agreed and declared by me/us, that the MES General Conditions of Contracts (IAFW: 2249 - 1989) including Condition 70 thereof pertaining to settlement of disputes by Arbitration and conciliation Act, 1996 containing 33 pages with errata 1 to 20 and Amendment Nos 1 to 49 and Annexure-I attached hereto shall form part of these tender documents.

AAD (CONTRACTS)
FOR ACCEPTING OFFICER

SIGNATURE OF CONTRACTOR
Dated

CONTD….

Annexure-I
(to GCC (IAFW: 2249-1989))

The existing Condition 70 of GCC (IAFW-2249) shall be substituted by the revised Condition 70 as under :-

"70. Arbitration :

(a) Arbitration Where Applicability of Section 12 (5) of Arbitration & Conciliation Act Has Been waived off

(i) All disputes between the parties to the Contract (other than those for which the decision of the CWE or any other person is by the Contract expressed to be final and binding) shall, after written notice by either party to the Contract to the other of them be referred to the Arbitral Tribunal of a Sole Arbitrator (in case of contract sum less than or equal to 100 Crore) or to Arbitral Tribunal of three Arbitrators (in case of contract sum exceeding Rs 100 Crore) from MoD Panel of Arbitrators. The Officers so considered for appointment as Arbitrator, either as Sole Arbitrator or for Arbitral Tribunal, shall be having degree in Engineering or equivalent or having passed Final/ Direct Final Examination of Sub Division II of Institution of Surveyor (India) or similar other Institutes recognized by the Government of India.

(ii) In case of arbitration by sole Arbitrator, the Arbitrator shall be appointed by the authority mentioned in the contract document from the MoD Panel of Arbitrators within a period of thirty days of having received the notice from any of the parties to contract. In case of Arbitral Tribunal consisting of Panel of three Arbitrators, both the parties shall be asked by the Appointing Authority to suggest at least two names out of MoD Panel of Arbitrators within thirty days. The Appointing Authority will appoint two Arbitrators, one Arbitrator each out of the suggested names. The two Arbitrators so appointed shall select one Arbitrator from the MoD Panel of Arbitrators who will be the 'Presiding Arbitrator'. The Serving Officer (s) so appointed as Arbitrator(s), either as Sole Arbitrator or as one of the three Arbitrators in the Arbitral Tribunal, can continue as Arbitrator even after retirement, provided both the parties to the Contract give written consent to this effect. In such case, however, the Arbitrator shall not be entitled for any fee even after retirement.

(b) Arbitration Where Applicability of Section 12 (5) of Arbitration & Conciliation Act Has Not Been waived off

(i) All disputes between the parties to the Contract (other than those for which the decision of the CWE or any other person is by the Contract expressed to be final and binding) shall, after written notice by either party to the Contract to the other of them be referred to the Arbitral Tribunal of a Sole Arbitrator (in case of contract sum less than or equal to 100 Crore) or to Arbitral Tribunal of three Arbitrators (in case of contract sum exceeding Rs 100 Crore) from MoD Panel of Arbitrators. The Officers so considered for appointment as Arbitrator, either as Sole Arbitrator or for Arbitral Tribunal, shall be having degree in Engineering or equivalent or having passed Final/ Direct Final Examination of Sub-Division II of Institution of Surveyor (India) or similar other Institutes recognized by the Government of India.

(ii) In case of arbitration by sole Arbitrator, the Arbitrator shall be appointed by the authority mentioned in the contract document from the MoD Panel of Arbitrators within a period of thirty days of having received the notice from any of the parties to contract. In case of Arbitral Tribunal consisting of Panel of three Arbitrators, both the parties shall be asked by the Appointing Authority to suggest at least two names out of MoD Panel of Arbitrators within thirty days. The Appointing Authority will appoint two Arbitrators, one Arbitrator each out of the suggested names. The two Arbitrators so appointed shall select one Arbitrator from the MoD Panel of Arbitrators who will be the 'Presiding Arbitrator'.

CONTD….

(c) Common for All Arbitration

(i) Unless both parties agree in writing, such reference shall not take place until after the completion or alleged completion of the works or termination or determination of the contract under Condition 55, 56 and 57 thereof.

(ii) Provided that in the event of abandonment of the works or cancellation of the Contract under Condition No 52, 53 or 54 thereof, such reference shall not take place until alternative arrangements have been finalized by the Government to get the works completed by or through any other Contractor or Contractors or Agency or Agencies.

(iii) Provided always that commencement or continuance of any arbitration proceedings hereunder or otherwise shall not in any manner militate against the Government's right of recovery from the Contractor as provided in Condition 67 hereof.

(iv) If the sole Arbitrator or one or more Arbitrators of the Arbitral Tribunal so appointed resign(s) from his/ her appointment or vacate(s) his/ her office or is unable or unwilling to act due to any reason whatsoever, the Authority appointing him/ her will appoint a substitute arbitrator to act in his/ her place in the manner specified hereinabove. In case the Arbitrator resigning in this manner is the Presiding Arbitrator, the other two Arbitrators of the Arbitral Tribunal shall appoint the substitute Presiding Arbitrator.

(v) The Arbitral Tribunal may proceed with the arbitration, ex parte, if either party, in spite of a notice from the arbitrator fails to take part in the proceedings.

(vi) The Arbitral Tribunal may from time to time with the consent of the parties, enlarge the time for making and publishing the award subject to the limit laid down in the Arbitration & Conciliation Act 1996 as amended upto the date on which arbitration proceedings commence.

(vii) The Arbitral Tribunal shall make the award within the period as provided in the Arbitration & Conciliation Act 1996 (as amended upto the date on which arbitration proceedings commence) from the date of entering on the reference or within the extended period as the case may be on all matters referred to it and shall indicate findings alongwith sums awarded separately on each individual item of dispute. The arbitral Tribunal shall give reason for the award in each and every cases irrespective of the value of claims or counter claims.

(viii) The venue of Arbitration shall be such place or place or places as may be fixed by the Arbitral Tribunal in its sole discretion.

(ix) The Award of the Arbitral Tribunal shall be final and binding on both parties to the Contract.

Signature of Contractor
Dated _____

AAD (Contracts)
for Accepting Officer

CONTD….

SCHEDULE OF MINIMUM FAIR WAGES

1. It is hereby agreed that the Schedule of Minimum Fair Wages (SMFW) as published vide Government of India Notification dated *19 Jan 2017 (Revised up to date) form part of these tender documents.

2. My/Our signature hereunder amounts to my/our having read and understood the provisions contained therein and I/We agree that I/We shall abide by the same and those aforesaid documents from part of this tender.

*NOTES:

(i) Schedule of Minimum Fair Wages is not enclosed along with tender but the contractor is deemed to have full knowledge regarding the minimum wages payable to labours as legally effective on the date of submission of tender and his tendered rates shall be deemed to have been based on this (for the purpose of reimbursement of PV for wage escalation of labour the minimum wages legally effective on the date of receipt of tender shall be the basis). The minimum wages notified in the State/Central Gazette governed by any local regulation whichever is higher.

(ii) “Schedule of Minimum Fair Wages” referred to above is available for reference in the Office of Accepting Officer”.

AAD (CONTRACTS)
FOR ACCEPTING OFFICER

SIGNATURE OF CONTRACTOR
DATED: _____

CONTD….

SPECIAL CONDITIONS

1. **GENERAL:**
The following Special conditions shall be read in conjunction with General Condition of Contracts, IAFW-2249(1989 Print) including errata and amendments thereto and IAFW-2159 (Revised 1947) including all errata and amendments thereto. If any provision in this Special Conditions is at variance with the provisions in the abovementioned documents the former shall take precedence there over.

2. **GENERAL SPECIFICATION:**
The term General Specification used in any of the documents forming part of this contract refers to the specifications contained in MES Schedule.

3. **INSPECTION OF SITE:**
Refer conditions 7 and 10 of Notice of Tender (IAFW-2162) and Condition 4 of IAFW-2249. The tenderer shall be deemed to have satisfied himself before submitting his tender as to the nature of the ground and sub-soil (so far as it is practicable), the climatic conditions, the form and nature of the site, nature of work and the means of access to the site(s), the accommodation he may require and in general tenderer shall be deemed to have himself obtained all necessary information on inspection of site as to risk, contingencies and other circumstances which may influence or effect his tender whether he actually inspect the site(s) or not. No extra payment consequent on any mistake or misunderstanding or otherwise on this account shall be allowed.

4. **AVAILABILITY OF LAND FOR STORAGE OF MATERIALS, ACCOMMODATION FOR LABOUR ETC:**
(a) Delete the following in para 1 of Condition 24 of IAFW-2249(1989 Print) General Conditions of Contracts "In the event of the areas of land allotted to him" and the following shall be read in conjunction with Condition 24 of IAFW-2249:-
"The contractor shall be permitted to store his materials including erecting temporary sheds thereof, temporary workshops and the like at the area of land, marked for this purpose on the site plan shown by the GE free of rent, if it is in class A-1 land, for other types of land he shall be charged license fee of Re 1/- per year or part of year for each and every separate area of land allotted to him. Land shall not be allotted to the contractor for construction of huts for accommodation of labour etc. inside the Air force area."

(b) The contractor will not be allowed to make quarter in MD land.

5. **ROYALTIES:**
Condition 14 of General Conditions of Contracts (IAFW-2249) is not applicable to this contract.

6. **CONDITIONS OF WORKING:**All works lie in **RESTRICTED AREA.**The restrictions for entry to work site and conditions of working in restricted area shall be as under:-

(a) **Entry and Exit**. The contractor/his agents/representatives/workmen etc. and his materials, carts, trucks or other means of transport etc will be allowed to enter through and leave from only such gate or gates and at such times as the GE or authorities in charge of the restricted area may at their sole discretion permit to be used. Contractor's authorised representative is required to be present at the places of entry and exit for the purpose of identifying his carts, trucks etc to the person in charge of the security of the restricted area.

(b) **Identity of Workmen**:-

(i) Every workman shall be in possession of an identity card. The identity card shall be issued after a thorough investigation of antecedents of the labourers by the contractor and attested by Officer-in- charge of the unit concerned in accordance with the standing rules and regulations of the unit. Contractor shall be responsible for the conduct of his workmen, agents or representatives.
(ii) **Identity Card or Passes:** The contractor, his agents and representatives are required individually to be in possession of an identity card or pass which will be examined by the security staff at the time of entry into or exit from the restricted area and at any time or number of times inside a restricted area.

CONTD….

SPECIAL CONDITIONS (CONTD…)

(c) **Search:** Thorough search of all persons and transport shall be carried out at each gate and for as many times the gate is used for entry or exit and may also be carried out any number of times at the site within the restricted area.

(d) **Working Hours:** The units controlling restricted areas usually work during six days in a week and remain closed on the 7th day. The working hours available to the contractor's labour and staff are however appreciably reduced because of the time of entry and exit during working hours. The exact working hours, working days and number of working days observed for these restricted area(s) where works are to be carried out shall be deemed to have been ascertained by the contractor before submitting his tender. The tenderer's attention is invited to the fact that the number of working hours for a unit are prescribed in regulations and that these cannot be increased by the Garrison Engineer or authorities controlling the restricted area. The definition of "working days" as given under Condition 1(t) of IAFW-2249 does not apply in cases where the works are carried out in restricted areas.

(e) **Working on Holidays**: The contractor shall not carry out any work on gazette holidays, weekly holidays and other non working days except when he is specially authorised in writing to do so by the Garrison Engineer. The GE may at his sole discretion declare any day as holiday or non working day without assigning any reasons for such declaration.

(f) **Fire Precautions**. The contractor, his agents, representatives, workmen etc shall strictly observe the order pertaining to fire precautions prevailing within the restricted area. Motor transport vehicles, if allowed by the authorities to enter the restricted area, must be fitted with a fire extinguisher in working condition.

7. **CONTRACTOR'S REPRESENTATIVES AND WORKMEN:**

Refer condition 25 of IAFW-2249. The contractor shall employ only Indian Nationals as his representatives, servants and workmen and verify their antecedents and locality before employing them in the works. He shall ensure that no person of doubtful antecedents and nationality is, in any way, associated with works. If for reasons of technical collaboration or other consideration the employment of any foreign national is unavoidable, the contractor shall furnish full particulars to this effect to the Accepting Officer at the time of submission of his tender.

8. **OFFICIAL SECRET ACT:**

Refer condition 2A of General Conditions of Contracts IAFW-2249. The Contractor's attention is invited to the Indian Official Secret Act 1923 (XIX of 1923), particularly Section 5 thereof. The contractor shall be bound by the provisions of this Act.

9. **SECURITY OF CLASSIFIED DOCUMENTS:**

Contractor's special attention is drawn to Condition 2A and 3 of IAFW-2249 (General Conditions of Contracts). The contractor shall not communicate any classified information regarding the work either to Sub-contractors or others without the prior approval of the Engineer-in-Charge. The contractor shall also not make copies of the design/drawings and other documents forwarded to him in respect of the work and shall return all documents on completion of the work or earlier on termination of the contract. The contractor shall alongwith the final bill attach a receipt of his having returned the classified documents as per Condition 3 of IAFW-2249 (General Conditions of Contracts).

10. **MINIMUM WAGES PAYABLE:**

(a) Refer Condition 58 of IAFW-2249. The contractor shall not pay wages lower than minimum wages for labour as fixed by the Government of India/State Govt/Union territory, whichever is higher.

(b) Contractor's attention is also drawn, amongst other things, to the 'explanations' to the schedule of minimum wages referred to above.

(c) The fair wages referred to in Condition 58 of IAFW-2249 will be deemed to be the same as the minimum wages referred to above as updated from time to time.

(d) Schedule of minimum wages are not enclosed along with tender documents. However, the contractor shall be deemed to have verified the minimum fair wages payable as on the bid submission end date.

(e) The contractor shall have no claim whatsoever, if on account of local factors and/ or regulations, he is required to pay the wages in excess of minimum wages as described above during the execution of work.

CONTD….

SPECIAL CONDITIONS (CONTD…)

11. **CO-OPERATION WITH OTHER AGENCIES AT SITE OF WORK:**
The contractor shall permit free access and generally afford reasonable facilities and usual conveniences to other agencies or department workman to carry out their part of the work, if any, under separate arrangements. The contractor shall not be allowed any extra payments on this account.

12. **WATER:**

(a) Water will be supplied by MES to the contractor at points shown on site plan from piped system and shall be paid by the contractor @ **Rs. 3.75 per Rs 1000**/- worth of work done priced at contract rates in case of unmetered supply (where it is not possible to meter the supply of water viz works of repairs, addition/alteration etc involving number of locations) and at **Rs.21.46 @ per 1000 ltrs** in case of metered supply. The contractor shall arrange at his own expense for storage of water and lifting, pumping, carrying or conveying water to the site of work as required. In case water points are not marked on the site plan, the water shall be supplied at one point as decided by GE.

(b) The supply of water may not be continuous. The contractor shall be deemed to have ascertained the hours of availability of water before submitting his tender. MES does not guarantee continuity of water supply and no compensation shall be allowed for intermittent or inadequate water supply and break down in the system. If the supply is not sufficient the contractor shall make his own arrangement to supplement the water supply at his own cost. For this purpose the contractor shall be allowed to install hand pumps, tubewells at the site of work at places as directed by the Engineer-in-Charge without any charges from the contractor on this account. The contractor shall remove the hand pumps, tubewells as and when asked to do so by Engineer-in-Charge or GE and in any case on completion of work unless GE desires that these hand pumps, tube wells be left in position and the contractor agrees to do so without claiming cost thereof from the Department. No compensation whatsoever shall be admissible to the contractor if the GE requires him to remove the pumps before completion of work. Use of water from such sources shall be permitted only if found potable after testing, and fit for use in the work. The water from such sources shall be got tested by the contractor from a laboratory approved by the GE, who shall, after satisfying himself, permit the contractor to use the water from such sources. Testing charges shall be borne by the contractor.

13. **SUPPLY OF ELECTRICITY:**

(a) If the contractor desires to buy electricity, the electric supply will be made available by the MES and shall be charged for the electric energy consumed for execution of works at the following rates :-
   (i) **At Rs 14.56 per unit for lighting.**
   (ii) **At Rs.14.56 per unit for power.**

(b) Electric supply required for works shall be made available by MES at the point(s) as shown on site plan/near site of work (exact location to be ascertained from the GE before tendering) upto a maximum load of 5 KW 3 phase. The main switch and meters to register the electric energy supplied shall be provided and installed by MES. The contractor shall provide all necessary connections, cables, fittings etc from the main switch in order to ensure a proper and suitable supply of electricity for the execution of work. Contractor's all installation shall conform to and be strictly in accordance with Indian Electricity Act and moreover the layout of cable etc. shall be as per plan approved by the Engineer-in-Charge. Any risk either for materials or personnel will be the contractor's responsibility.

(c) GE or his representative shall be free to inspect the power consuming devices or any electrical line provided by the contractor. Any devices or electrical lines provided by the contractor which is not to the satisfaction of the GE, shall be disconnected from the supply, if so required by him.

(d) MES do not guarantee for continuity of supply and no compensation whatsoever shall be allowed for supply becoming intermittent or for break down in the system.

14. **MATERIALS AND SAMPLES:-**

(a) Refer to Condition of 10 of IAFW-2249.

(b) The materials and articles, which have been specified from certain makers/manufacturers, shall be of makes/manufactures as specified. If the manufacturers specified in tender documents make both ISI marked and conforming to ISI, the materials/articles shall be ISI marked.

CONTD….

SPECIAL CONDITIONS (CONTD…)

(c) The materials and articles, which have not been specified in tender documents by makes/manufacturers, shall be as under :

(i) If ISI marked materials are being manufactured, the same shall be ISI marked. For a list of ISI marked manufacturers refer to the website of BIS ie www.bis.org.in.
(ii) If ISI marked materials are not being manufactured, the same shall be conforming to IS Specifications.

(d) Materials of local origin shall be as specified and conforming to samples kept in GE's office. The tenderer is advised to inspect samples of materials which are displayed in the office of GE before submitting his tender. The tenderer shall be deemed to have inspected the samples and satisfy himself as to the nature and quality of materials he is required to incorporate in the work irrespective of whether he has actually inspected them or not.

(e) The contractor shall not procure materials and articles unless the samples are first got approved by the GE.

15. **RECORD OF MATERIALS:**

(a) The quantity of materials such as WPC, paints, bitumen primer and the like, as directed by the Engineer-in-Charge (The quantity of which cannot be checked after incorporation in the works), shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work.

(b) Materials brought to site shall be stored as directed by the Engineer-inCharge and those already recorded in measurement book shall be suitably marked for identification.

(c) The contractor shall produce to the GE receipted vouchers / purchase vouchers for the items as mentioned below in original in respect of the supplies. Vouchers/invoices so produced and verified shall be stamped by the Engineer-in-Charge indicating contract number, name of work, under his dated signature. The contractor shall ensure that the materials are brought to site, in original sealed containers/packing bearing manufacturer's marking except in the case of the requirement of material being less than smallest packing. The materials for which purchase vouchers are to be submitted invariably are:-

(i) Cement
(ii) Steel
(iii) WHB, EWC etc. and all sanitary fittings,
(iv) Steel tubular pole, XLPE cables any type, All size wirings
(v) Acrylic emulsion paint/ Synthetic Enamel Paint/ OBD/Exterior emulsion paint
(vi) HDPE tank, All types door, window, ventilator
(vii) All types of LED light fittings, CPVC pipe,
(viii) Cable junction box, MCCBs,MCBs, DBs, Cable jointing kit
(ix) LT Panel, All types of fans, All types of electrical accessories or fittings, Starters, Pumps, manhole cover, Transformer, DG Set, APFC Panel
(x) GI pipes of various size and grades, DI pipe, All types of valve
(xi) Admixture, Paver block, tiles, granite, Bitumen, WPC etc.
(xii) APP membrane, structural steel, RCC pipe, PVC (SWR) Pipe

(d) However contractor shall produce vouchers / Invoices of all items for which payment is claimed in RAR as 'Material lying at site'.

16. **RECORD OF CONSUMPTION OF CEMENT:**

(a) The contractor shall maintain a pucca bound register with serially numbered pages with all pages initialed by the Engineer-in-Charge against numbering, showing quantities of cement received, used in work and balance at the end of each day. The form of record shall be as approved by the Engineer-in-Charge. The register shall be signed daily by representatives of MES and the contractor in token of verification of its correctness and will be checked by the Engineer-in-Charge, at least once in a week.

(b) The register shall be kept at site in the safe custody of the contractor's representative during the progress of the work and shall on demand be produced for verification to the inspecting officer(s).

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(c) On completion of the work the contractor shall deposit cement register with the Engineer-in-Charge for record.

17. **VALIDITY OF TENDER:**
The tender shall remain open for acceptance for a period of **90 (ninety) days** from the bid submission end date.

18. **TIME AND PROGRESS CHART:**

(a) The time and progress chart, to be prepared as per Condition 11 of IAFW2249. (General Conditions of Contracts), shall consist of detailed network analysis and a time schedule. The critical path network shall be drawn jointly by the GE and contractor soon after acceptance of the tender. The time scheduling of the activities will be done by the contractors as to finish the work within the stipulated time. The contractor is advised to use the facility of computerization available with any organisation who affords such facility in doing the same. On completion of the time schedule a firm calendar date schedule will be prepared and submitted by the contractor to GE who will approve after due scrutiny. The schedule will be submitted in four copies within two weeks from the date of handing over the site. In case the contractor fails to submit the CPM network diagram, the network prepared by the GE/ AGE(I) will be binding on him.

(b) During the currency of the work, the contractor is expected to adhere to the time schedule and this adherence will be a part of the contractor's performance under the contract. During the execution of the work, contractor is expected to participate in the reviews and up dating of the network under taken by the GE(I) / GE/ AGE(I). These reviews may be under taken at the discretion of the Garrison Engineer, either as a periodical appraisal measure or when the quantum of work ordered on the contractor is substantially changed through deviation orders or amendments. Any revision of the time schedule as a result of review will be submitted within a week, by the contractor to the GE(I) / GE/ AGE(I) who will approve after due scrutiny. The contractor will adhere to the revised time schedule thereafter. In case of contractor not agreeing to revised schedule, the same will be referred to the Accepting Officer whose decision will be final, conclusive and binding. GE(I) / GE's/ AGE(I)'s approval to the revised schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to grant of extension of time. Extension of time shall be considered and decided by the appropriate authority mentioned in Condition 11 of IAFW-2249 and separately regulated.

(c) Contractor is expected to mobilise and employ sufficient resources to achieve the detailed schedule within the broad framework of the accepted methods of working and safety.

(d) No additional payment will be made to the contractor for any multiple shift work or other intensive methods contemplated by him in his work schedule even though the time schedule is approved by the department.

19. **REIMBURSEMENT/REFUND ON VARIATION IN "TAXES DIRECTLY RELATED TO CONTRACT VALUE":**
The rates quoted by the contractor shall be deemed to be inclusive of all taxes (including GST on materials, GST on work contracts, labour welfare cess/tax etc.), Duties, Royalties, octroy & other levies payable under the respective Statues. No re-imbursement/refund for variation in rates of taxes, duties, Royalties & other levies, and/or imposition/abolition of any new/existing taxes, Duties, Royalties & other levies shall be made except as provided in sub Para (b)(i) here-in-after.

(i) The taxes which are levied by Govt at certain percentage rates of Contract Sum/ Amount shall be termed as "taxes directly related to Contract value" such as GST on materials, GST on work contracts, labour welfare cess/tax and like as applicable but excluding Income Tax. The tender rates shall be deemed to be inclusive of all "taxes directly related to Contract value" with existing percentage rates as prevailing on last date of bid submission. Any increase in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on last due date of bid submission shall be reimbursed to the Contractor and any decrease in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on last date of bid submission shall be refunded by the Contractor to the Govt/ deducted by the Govt from any payments due to the Contractor. Similarly Imposition of any new "taxes directly related to Contract value" after the last due date of bid submission shall be reimbursed to the contractor and abolition of any "taxes directly related to Contract value" prevailing on last due date of bid submission shall be refunded by the Contractor to the Govt/ deducted by the Govt from the payments due to the Contractor.

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(ii) The contractor shall within a reasonable time of his becoming aware of variation in percentage rates and/or imposition of any further "taxes directly related to Contract value" give written notice thereof to the GE stating that the same is given pursuant to this Special Condition together with all information relating thereto which he may be in a position to supply.The Contractors shall submit the other documentary proof/information as the GE may require.

(iii) The Contractor shall, for the purpose of this condition keep such books of account and other documents as are necessary and shall allow inspections of the same by a duly authorized representative of Govt, and shall further, at the request of the GE furnish, verified in such a manner as the GE may require, any documents so kept and such other information as the GE may require.

(iv) Reimbursement for increase in percentage rates/imposition of "taxes directly related to Contract value" shall be made only if the contractor necessarily & properly pays additional "taxes directly related to Contract value" to the Govt without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Govt. Authority and submits documentary proof for the same as the GE may require.

20. **DAMAGE TO EXISTING STRUCTURE/BUILDING AND CLEANING DOWN:**
Any damage caused to the existing structure during the execution of work shall be made good by the contractor at his own cost. The contractor shall clean all floors, walls, remove cement/lime/paint marks/drops etc., clean the joinery/glass panes etc., touch up all paint work and carryout all other necessary items of work in connection therewith and leave the site of work clean and tidy on completion. Rectification/making good etc. shall conform to the standard of materials and workmanship originally used in the work and finished work shall match with existing work in all respects to the entire satisfaction of the Engineer-in-Charge. In case of any dispute on this account, the matter shall be referred to the GE/AGE(I) whose decision in writing shall be final, conclusive and binding.

21. **ROCK MET IN EXCAVATION:**
If during excavation, rock is met with, GE/AGE(I) will be informed of this in writing by the contractor. The nature/type of rock met with, will be decided by the GE/AGE(I) whose decision is final and binding. If the nature of rock met with is "Hard Rock", the same will be priced and paid as excavation in hard rock as per Condition 62 of IAFW-2249. Those rock shall become the property of Contractor and a credit of Rs. 1300/- (Rupees One thousand three hundred only) per cubic meter on stack measurement after deduction 50% quantity for void etc. shall be deducted from the payment due to the contractor.

22. **ADVANCE ON ACCOUNT AGAINST MATERIALS (APPLICABLE WHERE CONTRACT SUM IS MORE THAN 50 LAKHS):**

(a) Reference: Condition 64 of the General Conditions of Contracts (IAFW-2249), Advance on account. The Condition 64 of the General Conditions of contracts shall deemed to be modified to the extent mentioned here-in-after. Add the following in continuation of para 8 "Provided further, the contractor may be paid advance on account to the full value of the under mentioned materials only, brought on the site, on his furnishing guarantee bond(s) from the Scheduled Bank for the amount of retention money which should otherwise be recoverable from him under the contract :-

(i) All types of sanitary fittings.
(ii) All types of builders hardware.
(iii) XLPE cables any type
(iv) Ceiling /Exhaust fan,LED light fittings
(v) Cable junction box
(vi) LT Panel, Transformer, DG Set, HDPE Tank
(vii) All electrical fittings/fixtures/cables/wires.
(viii) Steel
(ix) RCC Pipe, DI Pipe

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(b) The Bank Guarantee Bond(s) shall be executed for a period and on a form as directed by the Accepting Officer. The contractor shall further arrange to extend the period of Guarantee Bond(s) if and when necessary, as directed by the Accepting Officer or shall furnish fresh Guarantee Bond(s) of similar value in lieu".

23. **FEMALE SEARCHER:**

If the contractor desires to employ female labour for work to be carried out inside the area of a Factory, Depot, Park etc. and female searcher is not borne on the authorised strength of the Factory, Depot, Park etc. at the time of submission of the tender, he shall be deemed to have allowed in his tender for pay and allowances etc. for a female searcher (Class IV servant/GP `D' servant calculated for the period female labour employed by him inside that area. If more than one contractor employees female labour during any month and female searcher(s) has/have to be employed in addition to the authorised strength of the Factory, Depot, Park etc. the salary and allowances paid to the additional female searcher(s) shall be distributed on an equitable basis between contractors employing female labour taking into consideration the values and periods of completion of their contracts. The GE/AGE(I)'s decision in regard to the amount recoverable on this account from any contractor shall be final and binding.

24. **ACCEPTANCE QUALITY OF WORK AND FINISHES:**

(a) To determine the acceptable standard of materials and workmanship/final finishes and layout of fittings etc, the Contractor shall execute sample workmanship under the close supervision of Engineers-in-Charge for all B/R and E/M items in one quarter in GF as directed by GE. These shall be approved by GE and labelled as “SAMPLE ITEMS".These shall serve as guiding sample for the balance works to be carried out under the Contract.

(b) Approval of the stages and workmanship of sample shall be separately entered and approved in stage passing register giving reference to location for easy identification even at a later date.

25. **QUALIFIED TRADESMEN (APPLICABLE FOR WORKS COSTING RUPEES ONE CRORE OR MORE):**

In compliance with the condition 26 of IAFW-2249 (General Conditions of Contracts), the contractor shall employ skilled/semi skilled tradesmen who are qualified and possessing certificate in particular trade from industrial Training Institute (ITI)/National Institute of Construction Management and Research (NICMAR)/ Similar reputed and recognized-Institutes by State/Central Government, to execute the works of their respective trade. The number of such qualified tradesmen shall not be less than 25% of total skilled/semi skilled tradesmen required in each trade. The contractor shall submit the list of such tradesmen alongwith requisite certificates to GE for verification and approval. Notwithstanding the approval of such tradesmen by GE, if the tradesmen are found to have inadequate skill to execute the work of their trades, leading to unsatisfactory workmanship, the contractor shall remove such tradesmen within a week after written notice to this effect by the GE and shall engage other qualified tradesmen after prior approval of GE/AGE(I). GE’s / AGE(I)’s decision whether a particular tradesmen possesses requisite qualification, skill and expertise commensurate with nature of work, shall be final and binding. No compensation whatsoever on this account shall be admissible.

26. **CONCILIATION**

(a) The scope of conciliation shall be restricted to the following types of disputes with financial limits as indicated therein:-

(i) Disputes relating to levy of compensation for delay in completion actual amount of compensation.

(ii) Disputes relating to technical examination of works.

(iii) Disputes relating to interpretation of the provisions of the of the contract with reference to their application to parties.

(iv) Disputes relating to non-return of Schedule ‘B’ stores over-issued to contractor.

(v) Any other dispute having fair chances of being resolved by conciliation and considered fit to be referred to conciliation by the parties.

(vi) For item (i) to (v) each as stated above, the financial limit shall be Rupees two lakhs or one percent of the contract amount whichever is less.

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(b) **COMMENCEMENT OF CONCILIATION PROCEEDINGS**

(i) The party initiating conciliation shall send to the other party a written invitation to conciliate, briefly identifying the subject of the dispute.

(ii) Conciliation proceedings shall commence when the other party accepts in writing the invitation to conciliate.

(iii) If the other party rejects the invitation, there will be no conciliation proceedings. If the party initiating conciliation does not receive a reply within 30 days from the date on which he sends or within such other periods of time as specified in the invitation, he may elect to treat this as a rejection of the invitation to conciliate and if he so elects, he shall inform in writing the other party accordingly.

(c) Number of Conciliators - There shall be a Sole conciliator.

(d) Appointment of conciliator :- All disputes mentioned above shall be referred to the sole conciliator viz Serving Officer not below the rank of Superintending Engineer/Superintending Engineer(QS & C) having degree in Engineering or equivalent or having passed final/direct final examination of Sub division II of Institution of Surveyors(India) to be appointed by the Engineer-in-Chief, Army Headquarters, New Delhi or in his absence the Officer Officiating as Engineer-in-Chief or Director General of works specifically delegated by the Engineer-in-Chief in writing.

(e) Status of Effect of Settlement Agreement:- The settlement agreement signed by the parties as a result of conciliation proceedings shall have the same status and effect as it is an arbitral award on agreed terms.

27. **TESTING OF MATERIALS:**

(a) A LEVEL TESTS FOR WORKS COSTING 100 LAKHS AND ABOVE: The contractor shall set up site laboratory for testing of materials (except Sch „B" materials) for "A" level tests as listed in Appx „A". The contractor shall arrange all equipment/ machines for the tests specified in Appx „A" as „A" level tests at his own cost with prior approval of GE. This cost shall be included in the lump-sum costs quoted by the contractor. The contractor shall employ a competent technical representative as approved by the GE for the purpose of testing and all such tests shall be carried out in the presence of Eng-in-charge. The successful tests result thereof shall be recorded and signed jointly by the contractor and the Eng-in-charge. The charges for these tests i.e. „A" level tests carried out in site laboratory of the contractor shall not be recovered. In case, the contractor has not set up the site laboratory and the tests are carried out in zonal or any other laboratory set up/approved by GE, the recovery shall be made at rates applicable i.e. as given in Appx "A".

(b) A LEVEL TESTS FOR WORKS COSTING UPTO 100 LAKHS: Thecontractor may set up site laboratory at his option for works costing upto 100 lakhs. The other stipulations will be same as specified in preceding para 27(a). However in case the contractor has not set up the site laboratory and tests are carried out in zonal or any other laboratory approved/set up by GE/AGE(I), the recovery shall be made at rates applicable ie as given in Appx "A".

(c) B & C LEVEL TESTS: For tests of "B" &"C" level as indicated in Appx "A", the contractor shall provide all facilities for testing of materials for "B" level at CTL Jorhat / CTL Shillong or MES (CE/CWE/GE) lab at his own cost. The lump sum/rates quoted by the contractor shall deemed to be inclusive of these tests. The rates of various tests conducted in laboratory of MES are indicated in Appx "A". The Contractor shall bear the actual charges of "C" level tests (to be done in NABL Accredited Labs / IITs) irrespective of rates indicated in Appx "A". Wherever it is convenient to get "B" level test done at NABL Accredited Labs/ IITs, the same can be done at the cost of the contractor and no separate recoveries will be made by the Department for the same. The rates of "C" level tests have been indicated in Appx "A" for the purpose of recovery, if these test are done in MES laboratories, if facilities are available.

(d) The recoveries on account of testing charges wherever applicable shall be effected from the running account payments due to the contractor payable after completion of the respective tests or whenever the test is due whichever is earlier.

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28. **QUALITY CONTROL PLAN:**
The contractor shall prepare a "QUALITY CONTROL PLAN" which will be approved by GE. A format for this plan is enclosed at Appendix 'B'. The contractor shall designate one appropriately qualified person responsible for total quality control. The contractor shall submit alongwith Appendix 'B', his comprehensive plan on how he would ensure that the laid down requirements in Appendix 'B' are implemented effectively.

29. **FEES CHARGABLE BY POLICE AUTHORITIES FOR VERIFICATION OF ANTECEDANTS:**
Fees chargeable by police authorities for verification of antecedents of contractor's representative, labours etc shall be borne by the contractor. The quoted rate shall be deemed to be included this aspect and nothing extra shall be admissible at later stage.

30. **TOOLS & PLANTS, MACHINERY AND TRANSPORT:**

(a) Contractor shall arrange any additional T & P and any other T & P required during execution of work without any extra cost to the Govt so as to ensure timely completion of the work. The contractor shall also make suitable arrangement to ensure uninterrupted execution of work in case of failure of any of above referred T & P. The vintage of the Tools & Plants shall be as specified here-in-before.

(b) For mobilizing T & P, no payment shall be allowed to the contractor. However advance on account for materials procured shall only be allowed in terms of contract conditions, provided the materials are approved for incorporation in the work.

(c) Contractor's vehicles shall be allowed to ply on perimeter / link roads as directed by GE. A proper discipline shall be followed in this respect in consideration with flight safety norms.

(d) The locations(s) of Hot Mix Plant (s), Concrete Batching Plant(s) etc if required **shall be as decided by GE in consultation with the users at Contractor's own cost.** The location for these plants may be changed as directed by GE as per requirements and the Contractor shall have no claims whatsoever on this account.

(e) The minimum **requirement** of T & P, Machinery & Transport:-

Srl No.	Special T&P/ machinery / Transport	Minimum Nos required
1	Fully automatic concrete batching plant	01
2	Vibrators needle type	03
3	Vibrators Plate type	03
4	Transit Mixer	02
5	Road Roller	01
6	Survey Equipments	02
7	Tower / Builder's hoist	01
8	Steel shuttering with spans, props etc (sqm)	2000
9	Concrete Pumps	02
10	Concrete mobile weigh batchers	02
11	Crane	01
12	Bull dozers	01
13	Lodder	01
14	Sand screener capacity 10 cum/hrs	01
15	Tractor with trolley	01
16	Tipper	01
17	Drilling Machine	01
18	Welding machine	4
19	Gen Set 5/10 KVA	01
20	Total station	01
21	Concrete Cube testing machine (Hydraulically operated)	01
22	Excavators(Power shovels/draglines)	01

31. **CONTRACT SUPERVISION (REFER CONDITION 25 OF IAFW-2249):**

(a) The Engineering staff shall be deployed by the contractor for the supervision of the work to be carried out under the subject contract agreement all as stipulated in Condition 25 of General Conditions of Contracts (IAFW-2249).

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(b) Engineering establishment as detailed are to be employed by the Contractor without fail and responsibility in this regard shall solely rest with the contractor. It is to be ensured that they are deployed at site during execution of work, failing which the work will be stalled / suspended. Concerned AGE/JE is responsible for keeping necessary record of the same and non-compliance if any shall immediately be brought out to the notice of GE.

(c) **Engagement of engineers for the subject work shall be as under as stipulated in IAFW 2249 (General Conditions of Contract):-**
Two Degree holder (Civil Engineer)from a Govt recognized Institution with minimum experience of five years in buildings and two diploma holder (OneCivil Engineer and One Electrical engineer diploma holder)from a Govt recognized Institution with minimum experience ofeight years also should have good knowledge of Project Management software like MS Projects / Primavera in all Projects.

32. **OUTPUT OF POWER ROAD ROLLER:**

(a) Reference Condition 15 of IAFW-2249.

(b) Where road roller(s) is/are hired by the department to the contractor, a log book for each road roller shall be maintained by the department for recording hours of working of the road roller. In case, however, when the contractor procures road roller(s) from sources other than the department, a logbook for each road roller shall be maintained by him for recording hours of working of the road roller. Entries in the logbook shall be signed by the contractor or his authorised representative and by the Engineer-in-Charge at the site of work.

(c) To ensure proper consolidation, roller must work for at least number of days assessed on the basis of output given here under. If the roller has not worked for the number of days so assessed, recovery shall be effected from the contractor for the number of days falling short of the days assessed on the basis of output stipulated. The recovery shall be effected as under :-

(i) Where road roller is hired out only by the department to the contractor, at rates given in Sch `C'.

(ii) Where road roller is hired by the contractor only from sources other than the department, Rs.1500/- per working day of 8 hours for 8 to 10 ton power roller.

(iii) Where road roller is hired by the contractor from the department and also from sources other than the department, at higher of the two rates given in Schedule `C' of contract and clause 32(b) above.

(iv) The above provision shall not, however, absolve the contractor of his responsibility of properly consolidating surface as required under the provisions of the contract.

(v) Output per day of 8 hours work of power roller (8 to12 Tonne) :-

(a)	Consolidation of formation surfaces/sub-grade	-	1850 Sqm
(b)	Consolidation of stone soling 20 to 23 cm thick	-	518 Sqm
(c)	-ditto- but for 15 cm thick,	-	800 Sqm
(d)	Consolidation of water bound macadam (Stone metal) 7.5 cm consolidated thickness including spreading and consolidation with binding materials	-	372 Sqm
(e)	Consolidation of water bound macadam (Stone metal) 11 cm consolidated thickness including spreading and consolidation with binding materials.	-	158 Sqm
(f)	Consolidation of one coat surface dressing	-	774 Sqm
(g)	Consolidation of two coat surface dressing	-	558 Sqm
(h)	Consolidation of 2.5 cm thick premixed carpet including seal coat	-	600 Sqm
(i)	Consolidation of 2 cm thick premixed carpet including seal coat	-	744 Sqm
(j)	Consolidation of bituminous mixture, 2 parts of broken stone metal and one part of sand and bitumen, consolidated thickness 4 cms	-	372 Sqm
(k)	Consolidation of 15 cm thick earth	-	1800 Sqm
(l)	Consolidation of broken stone base	-	500 Sqm
(m)	Granular sub base with vibratory roller	-	50 cum/hour
(n)	Wet mix macadam with vibratory roller	-	60 cum/hour
(o)	Dense/ semi-dense asphaltic concrete	-	As decided by BOO ordered by CWE.

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33. **HANDING OVER SITE**

33.1 Site for execution of work will be available as soon as the work is awarded. In case it is not possible to make the entire site available on the award of work the contractor will have to arrange his working programme accordingly. No claim whatsoever, for not giving entire site on award of work and for giving site gradually, will be tenable.

34. **APPROACHES**

34.1 The contractor shall make arrangements for and provide at own cost all temporary approaches, if required to the site(s), after obtaining approval in writing of the GE to the layout of such approaches.

35. Contractor's attention is drawn to 'THE PROHIBITION OF EMPLOYMENT AS MANUAL SCAVENGERS ANS THERE REHABILITATION ACT 2013' and rules made there by the concerned authorities from time to time. Since the work involves repairs to wxisting sewer lines, contractor shall make adequate arrangements for temporary plugging, pumping out the sewage/sludge etc to properly cover tankers using mechanical neans and disposal of the same to the designated places as decided by the GE in consultation with users and local bodies.

36. **WATCH/LIGHTING**

36.1 The contractor shall at his own cost take all possible precautions to ensure safety of life and property by providing necessary, barrier, light, watchman etc., during the progress of work and as directed by the Engineer-in-Charge.

37. **CONDITIONS OF CONTRACT AND EXECUTION OF WORK.**

37.1 The material and workmanship shall satisfy the job specifications contained in latest relevant Indian Standard Codes as applicable as on the date of issue of tender by the Accepting Officer where the job specifications stipulate requirement in addition to those contained in the Indian Standard Codes of practice, these additional requirements shall be satisfied by the contractor.

38. **APPROACHED**

38.1 The contractor shall make arrangements for and provide at own cost all temporary approaches, if required to the site(s), after obtaining approval in writing of the GE to the layout of the such approaches.

39. **LOCATION OF BUILDINGS AND WORKS**

39.1 There may be some changes in location/siting of building shown in site (layout) Plan(s) to suit local conditions and/or department requirements. The contractor shall have no claim what-so-ever consequent to such changes in the location/siting of works.

40. **GUARANTEE FOR ANTI TERMITE TREATMENT:**

40.1 Should the GE at any time during construction or reconstruction or prior to the expiration of period of ten years after the certified date of completion of work as per contract finds that buildings have been infested with termites, the contractor shall on demand in writing from the GE, specifying the building(s) complained of, not with-standing that the same may have been inadvertently passed, certified and paid for, forthwith undertake to carry out such treatment as may be necessary to render the said building(s) free from termite infestation at his own expenses for a period of ten years from the certified date of completion and in the event of his failing to do so within a period to be specified by the GE in his demand aforesaid, the GE shall undertake such treatment at the risk and expenses in all respect of the contractor. The liability of the contractor under this condition shall not extend beyond the period of ten years from the certified date of completion unless the GE has previously given Notice to the contractor. General Condition of Contracts (IAFW-2249) shall deemed to be amended to the extent mentioned above.

40.2 An amount of security deposit equal to 5% on the amount of Anti Termite Treatment at contract subject to minimum of Rs. **50,000.00** (Rupees Fiffty only) as security deposit towards the guarantee for efficiency of Anti Termite Treatment shall be retained from the RAR/Final bill amount whichever is more shall be retained by the Government from the contractor's payment (i.e final bill) as security deposit for the efficiency of the anti-termite treatment carried out and will be refunded to the contractor after expiry of the period of ten years from the certified date of completion by the GE provided there are no termite infestation in the buildings.

CONTD….

SPECIAL CONDITIONS (CONTD…)

41. **GUARANTEE OF WATER PROOFING TREATMENT OF RCC ROOF:**

41.1 The contractor shall furnish guarantee in favour of the GE/AGE (I) for efficiency of water proofing treatment during the guarantee period. The guarantee period shall be 08 (Eight) years from the certified date of completion of the work as per contract. Any leaking in the RCC roof noticed during this period shall be carried out by the contractor forthwith, on demand in writing from the GE specifying the roof of building(s) effected, not withstanding that the water proofing treatment to the roof might have been inadvertently passed, certified and paid for, the contractor shall carry out the rectification of the leakage, if any, in roof at his own expense during the guarantee period and in event of his failure to do within a period to be specified by the GE in his demand aforesaid, the GE shall undertake such rectification work at the risk and expense in all respect of the contractor.

41.2 An amount of security deposit equal to 5% on the amount of Water Proofing Treatment at contract subject to minimum of Rs. **1,00,000** (Rupees One lakhonly) whichever is more shall be retained by the Government from the contractor's payment (i.e final bill) as security deposit for the efficiency of the water proofing treatment carried out and will be refunded to the contractor after expiry of the period of eight years from the certified date of completion by the GE provided there are no leakage in the buildings. The contractor may alternatively furnish a deposit receipt in favour of GE in lieu of sum to be retained as security deposit for water proofing treatment from any approved Scheduled bank.

42. **YARD STICK FOR PAYMENT THROUGH RAR:**

42.1 Contractor shall submit yard stick for items of Schedule 'A' Part I item No. 1.00 of BOQ in duplicate to GE within one month of acceptance of tender, indicating percentage of payment to be made for each stage of the building along with supporting details i.e. detailed estimate. GE shall technically checked the Yard stick and forward to CWE for final approval of Yard stick.

42.2 There may be certain changes in yard stick percentages as submitted by the contractor while approving the yard stick due to market rates of various materials and due to policy of department of withholding sufficient amount for later stages of the building. Contractor shall not have any claim on this account and percentage payment to be made for each stage as approved shall be final and binding on the Contractor.

42.3 Payment against building(s) shall be made as per approved yard stick percentage.

43. **INSPECTION OF THE WORKS:** Refer Condition 46 of IAFW – 2249 (General Conditions of Contracts).

SIGNATURE OF CONTRACTOR **AAD (CONTRACTS)**
DATED: _____ **FOR ACCEPTING OFFICER**

CONTD….

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APPENDIX 'A'

Srl No.	Materials	Test	Method of testing	Frequency of tests	Level of test	Rate per test Rs. Ps.	Remarks
1	2	3	4	5	6	7	8
1.	Bricks	i. Compressive strength	IS-3495 (Part-II)	As per IS-5454 as given under:	A	180/-	Checks for visual and dimensional characteristics shall also be carried out as per IS5454.
		ii. Water absorption	IS-3495 (Part-II)		A	150/-	
		iii. Efflorescence	IS-3495 (Part-I)		A	180/-	
				Lot size / Sample size / Permissible No of defective bricks			
				1001 to 10000 / 5 / 0			
				10000 to 35000 / 10 / 0			
				35001 to 50000 / 15 / 1			
2.	Coarse aggregate	i. Sieve Analysis	IS-2386 (Part-I)	One test for every 15 Cum of aggregates or part thereof brought to site.	A	120/-	**LEGEND:** A – Site Lab B – Zonal Lab /CTL C – National test house/ SEMT wing, Pune/ Engg. College
		ii. Flakiness Index	IS-2386 (Part-I)	-do-	A	190/-	
		iii. Estimation of deleterious material	IS-2386 (Part-I)	One test for every 100 Cum of aggregates or part thereof brought to site.	A	120/-	
		iv. Organic impurities	IS-2386 (Part-I)	One test for source of supply	C	120/-	
		v. Moisture Content	IS-2386 (Part-II)	Regularly as required	A	120/-	
		vi. Specific Gravity	IS-2386 (Part-II)	One test for each source of supply	B	120/-	
3.	Fine aggregate	i. Sieve analysis	IS-2386 (Part-I)	One test for every 15 Cum of aggregates or part thereof brought to site.	A	180/-	
		ii. Test for clay, silt and impurities	IS-2386 (Part-I)	-do-	A	90/-	
		iii. Specific gravity	IS-2386 (Part-II)	One for each source of supply	B	180/-	
		iv. Moisture content	IS-2386 (Part-II)	Regularly as required subject to 2 tests per day when being used.	A	180/-	
		v.Test for organic impurities	IS-2386 (Part-II)	One test for each source of supply.	C	180/-	

SPECIAL CONDITIONS (CONTD…)

1	2	3	4	5	6	7	8
4.	Structural concrete (M-15 Grade and above)	i. Slumptest or compacting factor test or Vee-bee time	IS-119	The minimum frequency of sampling of concrete each grade shall be as under:- Qty of concrete in the work in Cum / No. of samples 1-5 / 1 6-15 / 2 16-30 / 3 31-50 / 4 51 and above / 4+1 for each additional 50 Cum or part thereof.	A	180/-	Random sampling shall be carried to cover all mix units.
		ii. Compressive strength	IS-516		A	120/-	As per IS-456 clause 15.9.2 for frequency of sampling.
5.	(a) PCC Block for walling (Hollow Block)	i. Compressive Strength	IS:2156-1984 (Appx B)	08 blocks out of 14	A	60/-	Sample: 14 blocks from consignment of every 5000 blocks or part thereof.
		ii. Water absorption	IS:2156-1984 (Appx B)	03 blocks out of 14	B	120/-	
		iii. Density	IS:2156-1984 (Appx C)	03 blocks out of 14	B	90/-	
	(b) PCC Solid block for walling	i. Compressive Strength	IS-2185	12 blocks out of 18	A	60/-	Sample: 20 blocks from consignment of every 5000 blocks or part thereof as per IS. These blocks to be checked for dimension and weight.
		ii. Water absorption	-do-	03 blocks out of 18	B	120/-	
		iii. Density	-do-	03 blocks out of 18	B	120/-	
6 .	Cement flooring tiles/ terrazzo tiles	i. Water absorption	IS-1237 (Appx D)	06 tiles out of 18	B	180/-	Sample of 18 tiles from each source of supply selected at random.
		ii. Wet transverse strength	IS-1237 (Appx E)	06 tiles out of 18	B	144/-	
		iii. Resistance to wear	IS-1237 (Appx F)	06 tiles out of 18	C	540/-	

CONTD….

SPECIAL CONDITIONS (CONTD…)

1	2	3	4	5	6	7	8
7.	Burnt clay roofing tiles (machine made as per IS-2690 (Part-I)) Length: 150mmto 250mm Width: 100mm to 200mm Thickness: 35mm to 50mm	i. Water absorption	IS-3495 (Part-II)	06 tiles out of 12	B	216/-	Sample: 12 tiles from each source of supply selected at random
		ii. Compressive strength	IS-3495 (Part-I)	06 tiles out of 12	A	180/-	
8.	Mangalore Pattern roofing tiles	i. Water absorption	IS-654 (Appx A)	06 tiles out of 32	B	180/-	Sample: 32 tiles from each consignment of 3000 tiles or part thereof. These tiles shall be checked for dimensions.
		ii. Breaking load	IS-654 (Appx C)	06 tiles out of 32	B	120/-	
9 .	Timber	i. Specific gravity and weight	IS:1708-1960	Minimum 3 samples from a lot of 4Cum or 250 pieces of seasoned timber.	B	120	
		ii.Moisture content	-do-	-do-	B	120/-	
10 .	Water for construction purposes	a)Test for acidity	IS-456 & 3015	One at the stage of approval of source of water.	B	240/-	Also refer clause 5.4 of IS-456 and its subsequent sub-clauses regarding suitability of water.
		b)Test for alkalinity	-do-		B	240/-	
		iii.Test for solid content	-do-		C	300/-	
11.	Welding for steel work	Visual inspection test	IS:822-1970 Clause 23.1	100% by visual inspection	Work site	360/-	Specialised tests, their method and frequency to be decided on consideration of their importance by the Accepting officer

CONTD….

SPECIAL CONDITIONS (CONTD…)

1	2	3	4	5	6	7	8
12.	Timber paneled & glazed Door/ Window shutters (including factory made shutters)	i.Dimensions, sizes, workmanship and finish	IS:1003-1977 (Part-I)	Frequency of sampling from each lot shall be as under:- Lot size / Sample size 26 to 50 / 5 51 to 100 / 8 101 to 150 / 13 151 to 300 / 20 301 to 500 / 32 501 to 1000 / 50 1001 and above / 80	A	180/-	
		ii.Strength test		From each lot 5% of the factory made shutters shall be manufacturer tested for strength tests.			
		a)Slamming	IS: 1303-1990				
		b)Impact indentation	-do-				
		c)Shock resistance	-do-				
		d)Edge loading	-do-				
13.	Plywood (IS: 303-1989)	Moisture content	IS:1734-1983 (Part-I)	Six test pieces cut from each of the boards selected as per table shall be subjected to tests.	C	240/-	Sampling shall be as per IS:7835-1975 Tables.
14.	Wood particle board (Medium density) (IS:3097-1985)	i.Density	IS:2360 (Part-III)	Three test specimens from each sample (size: 150mm x 75mm)	A	60/-	Sampling shall be as per IS:3487-819.
		ii.Moisture content	IS:2360 (Part-III)	-do-	A & B	60/-	
		iii.Water absorption	IS:2360 (Part-16)	-do- (Size: 300mm x 300mm)	A	60/-	
		iv.Swelling due to surface absorption	IS:2360 (Part-17)	-do- (Size: 125mm x 100mm)	A	60/-	
		v.Swelling in water	IS:2360 (Part-17)	-do- (Size: 200mm x 100mm)	A	60/-	
		vi.Modulus of rupture	IS:2360 (Part-4)	Three test specimens as per IS:2380-77	B	90/-	
		vii.Screw withdrawal strength	IS:2360 (Part-4)	Three test specimens as per IS:2385	C	120/-	

CONTD….

SPECIAL CONDITIONS (CONTD…)

1	2	3		4	5	6	7	8
15.	PAVEMENT QUALITY CONCRETE	(a)	Compresive strength (Cube)		As per IS			Independent test after 28 days
		(b)	Flexural strength (Beam)		As per IS			Independent test after 28 days
16.	DRY LEAN CONCRETE (DLC)				As per IS			Independent test after 28 days
17.	AGGREGATE TEST	(a)	40mm		As per IS			
		(b)	20mm		As per IS			
		(c)	10mm		As per IS			
		(d)	Sand		As per IS			
		(e)	Dust		As per IS			
		(f)	Moorum		As per IS			
18.	CEMENT TEST OPC Cement 43 Grade							
		(a)	Physical test		As per IS			
		(b)	Chemical test		As per IS			
19.	BITUMEN TEST							
		(a)	Modified Bitumen		As per IS			
		(b)	Bitumen		As per IS			
20.	WATER				Initially then quarterly			
21.	GLASS GRID 100t				Min one test per lot			
22.	ADMIXTURE				Min one test per lot			
23.	CURING COMPOUND				Min one test per lot			

CONTD….

SPECIAL CONDITIONS (CONTD…)

1	2	3	4	5	6	7	8	
24.	FILLER BOARD				Min one test per lot	'C'	-do-	
25.	POLYURETHENE SEALENT				Min one test per lot	'C'	-do-	
26.	PRIMER JOINT				Min one test per lot	'C'	-do-	
27.	RECRON				Min one test per lot	'C'	-do-	
28.	PQC CORE TEST				As per Inspecting Team's Requirement	'C'	-do-	
29.	DBM CORE TEST (Each layer)				As per IS	'C'	-do-	
30.	DAC CORE TEST (Each layer)				As per IS	'C'	-do-	
31.	Polythene sheet 400 micron			IS-2508	Only Independent test one test/lot	'C'	-do-	
32.	K VALUE				As per IS	'C'	-do-	

Note: (i) To be collected and dispatched under arrangement of GE to independent testing Lab on progress of work or as advised by visiting officers.

(ii) All above material should be sent for testing if there is any change of source. Change of source will also require revision of Design mix formula .

SIGNATURE OF CONTRACTOR
DATED : _____

AAD (CONTRACTS)
for ACCEPTING OFFICER

CONTD….

SPECIAL CONDITIONS (CONTD…..)
APPENDIX 'B'

FORMAT FOR QUALITY CONTROL PLAN

(This format shall from part of Contract Agreement)

(To be submitted by Contractor with 30 days of commencement of contract)

PART-I

1. Contract Agreement Reference No :
2. CPM network prepared and approved by GE/AGE(I) :
3. Resource scheduling done based on CPM :
4. Site Laboratory (with equipments) set up as per Contract Agreement (CA) :
5. Concrete mix design submitted and approved :
6. Preliminary works completed to standard engineering practice :
7. Arrangements for water made (including testing of water) :
8. Arrangement for electric supply made :
9. Materials

Sl No	Item	Source as per CA	Contractor's plan of sourcing	Refer to testing clause	Agency for testing	Responsibility

10. List of all T&P, make and numbers that the contractor would deploy at site of work :
11. Name of person nominated by contractor for exercising quality control :
12. Qualifications/Experience of person at Serial No 11 above. :
13. Names of supervisors with their qualifications/ experience employed by contractor :
14. Confirmation that contract requirement relating to quality of all materials and quality standards of workmanship and finishes and acceptance criteria are explained and understood by all :
15. Confirmation that requirement of tests to be conducted on materials before approval of sample and during execution, test on workmanship, tests before acceptance including the testing procedure, sampling techniques, frequency and agencies responsible for testing are understood and shall be complied with. :
16. Method to be adopted for maintaining records of test result :
17. Certificate that contractor shall maintain a log of all materials received at site as per the following format :

Cont'd…

SPECIAL CONDITIONS (CONTD…..)

APPENDIX 'B' (Contd…)

Sl No	Date	Material	Quantity received	Source	Whether as per approved sample or not	Tests carried out by supplier	Tests to be carried out before incorporation

18. General Remarks by contractor of his plan of actions to ensure that quality standards :

Date : (Signature of Contractor)

PART – II

(To be completed by GE / AGE(I) before forwarding for approval by CWE)

1. Verification of Serial No 2 to 8 of Part-I :
2. Verification of Serial No 9 to 18 of Part-I :
3. Confirmation that Stage Passing Register laying down the stages and authority responsible for approving the same has been prepared, shown to contractor and kept at site :
4. Confirmation that all sites as require by contract had been handed over to contractor on the date fixed in Work Order No 1 :
5. Confirmation that arrangements for Govt liability in supply of water and electricity has been made and no hold up on this account is expected. :

Dated : {Signature of GE/AGE(I)}

Approved by CWE

PARTICULAR SPECIFICATIONS

A. **GENERAL: APPLICABLE TO ALL SCHEDULE 'A' PARTS**

1. **GENERAL**

1.1 General Rules, Special Conditions and Preamble to various Parts given in the MES Standard Schedule of Rates Part I, 2009 and Part – II, 2020 shall deemed to be applicable to the work under this contract. Materials and workmanship required to be incorporated in these works and services shall be as given under specification and workmanship in the various trade section of the MES Standard Schedule of Rates mentioned here-in-before duly modified by the Particular Specifications and notes on the drawings.

1.2 Particular specifications given here-in-after are only to particularise, amend or to emphasize the aforesaid specifications in the MES Schedule which is not repeated here.

1.3 Particulars specifications given hereinafter shall be read in conjunction with the provision in the MES Schedule and the relevant Pt IS referred to therein. In case of any discrepancy, the provision in these particular specifications shall take precedence. The contractor shall not take cognizance of notes appearing on the drawings regarding the bearing capacity of soil as the same are for departmental purpose only while designing the foundations etc.

1.4 Lump sum quoted by the tenderer shall be deemed to include for any details/items of work and/or construction which are obviously and fairly intended and which may not have been included in these documents but which are essential for the execution and entire completion of the work.

1.5 Decision of the Accepting officer as to whether any minor detail of work and/or construction is obviously and fairly intended to be included in the contract shall be final, conclusive and binding.

1.6 However, some of the items which shall be deemed to be essential for execution and entire completion of work are detailed as under: -

(a) Reinforcement for any RCC member not indicated in the drawings (minimum reinforcement as per relevant IS shall be provided).
(b) Dwarf wall in situation like verandah, passage, ramp, etc. not indicated.
(c) Lintels over doors, windows and opening not indicated.
(d) Fittings to doors, windows, ward robes and such other built-in-furniture items not shown.

1.7 In all the above and similar cases, the details indicated elsewhere in the drawings which are similar or near to the missed items of work shall be followed. In the absence of any other similar or near details, minimum essential requirement to complete the work from structural and utility point of view shall deemed to be included in lump sum quoted. In the event of dispute, decision of the Accepting officer thereon shall be final, conclusive and binding.

2. **SCOPE OF WORK**

2.1 This contract covers all the works as described in Schedule `A', Particular Specifications and as shown on drawings and other documents forming part of this tender.

2.2 In Particular Specifications, approved list of manufacturers for panelled doors, pressed steel door frames and other items have been given. Contractor shall obtain the aforementioned items from one of the approved manufacturers only subject to prior approval of sample by the GE. However, it will not absolve the contractor from ensuring that the products supplied are conforming to the relevant IS in all respects before procuring the sample for approval by the GE and before procuring the product for incorporation in the work.

3. **DIMENSION**

3.1 In laying out building and other structure, the dimensions mentioned in drawing or deduced there from shall be strictly followed.

3.2 Where specifications for any item of work are not given in MES schedule or in these Particular Specifications, specifications as given in relevant Indian Standard or Code of practice shall be followed.

PARTICULAR SPECIFICATIONS (CONTD…)

3.3 Any drawing which is mentioned in the drawings forming part of the tender but not specifically mentioned in the list of drawings shall deemed to be forming part of the tender. The tenderer shall see such drawings/details in the office of Accepting Officer/ GE.

3.4 In all above and other similar cases, the details indicated elsewhere in the drawings which are similar or near similar to the missed out items of work shall be followed. In the absenceof any other similar or near similar details, minimum essential requirement for completion of the work from structural and utility point of view and in sound Engineering practice shall deemed to be included in the lump sum quoted. In the event of any dispute, decision of the Accepting Officer shall be final and binding.

4. **MATERIAL**

4.1 All materials to be supplied by the contractor for incorporation in the work shall conform to relevant specifications. In case specification of any materials needed for incorporation in the work is not contained in any of the contract documents, the specifications of such materials proposed to be incorporated in work shall be got approved in writing from the GE before their incorporation in the work, whose decision with regard to quality shall be final and binding.

4.2 All manufactured materials/articles, other than those manufactured in contractor's workshop and those of local origin shall bear IS certification mark. In case any articles does not bear any IS certification mark being not manufactured with ISI mark, but conforming to or superior to the relevant IS specifications and is proposed to be incorporated in the work, samples of the same shall be got approved in writing from GE before their incorporation in the work. The contractor shall submit sufficient evidence to the GE to show that such articles conform to the relevant specifications. However no price adjustment shall be made for such materials conforming to IS specifications but not bearing IS certification mark. Contractor shall use materials/articles conforming to IS mentioned in the MES Schedule. The contractor will however be permitted to use materials/articles conforming to IS of subsequently revised edition up to the date of submission of tender for which no price adjustment is to be made either in the lump sum quoted by the contractor or while pricing deviation.

4.3 Manufactured materials/articles shall be brought at site in original sealed covered containers/packing bearing manufacturer's marking unless the quantity required is a fraction of a smallest packing.

4.4 Materials of proprietary nature such as paints, water proofing compound, chemicals for anti-termite treatment and the like quantity of which cannot be checked after incorporation in the work shall be measured and recorded in the measurement book as soon as those are brought at site. These measurements shall be signed both by the Engineer-in-charge and the authorised representative of the contractor. Original stamp receipted bills alongwith the relevant invoices from the manufacturers or their authorised dealers (if any) in support of having brought the full quantity required for incorporation in the work, shall be produced to Engineer-in-charge.

4.5 In case of non-availability of test certificate for a particular item from National Test Laboratories and such test certificate is considered necessary by the GE, he may ask the contractor to get the test carried out of the item from a laboratory as approved in writing by the GE.

4.6 The contractor shall carry out such instructions that are given to him in writing by the Engineer -in- Charge to ensure that full quantity of such materials goes in to work.

4.7 Integral water proofing compound (WPC) shall conform to IS -2645 and the brand shall be got approved in writing from the GE. Preparation of mix and method of mixing shall be as per manufacturer's instructions. While pricing deviation for omission the quantity shall be considered as 3% by weight of cement.

4.8 Lime shall conform to IS-712. Classes of lime for different items of works shall be as specified in MES Schedule.

4.9 Letters conveying approval of samples/materials by GE/Engineer-in-charge will interalia, mention source of supply, name of manufacturers, trade name/brand (if applicable) and reference to clause of tender documents containing specifications of particular materials.

PARTICULAR SPECIFICATIONS (CONTD…)

4.10 **MATERIALS:** The materials such as bricks, sand, stone aggregate, metal for road etc shall be the best quality and shall comply with the specifications given here-in-after.

4.11 **SOURCE OF MATERIAL :** Unless otherwise specified here-in-after, source of fine aggregate, course aggregate and bricks shall be as under:-

Srl No	Material	Source
(i)	Coarse aggregates	Chandil / Pakur / Silta / Saltora conforming to IS 383
(ii)	Fine & coarse sand	Kasai River/ Fecoghat/Kaliaghat/Bhasraghat
(iii)	Broken stone boulder for soling / sub-base	Chandil / Pakur / Silta conforming to IS 383
(iv)	Bricks	Locally available best quality

B. **FOR SCHEDULE 'A' PART-I : BUILDING WORKS**

1 EARTH WORK

(a) Excavation and earthwork shall be deemed to be in any type of soil which shall include soft/loose soil, hard/ dense soil, soft/ disintegrated rock. However, during excavation if any of the above type of soil is met with the excavation in such earth shall be deemed to have been included in the quoted lump sum of the building in Schedule 'A' Part I. In case rock is met with the same shall be ordered as deviation.

(b) If during excavation, rock is met with, GE/AGE(I) will be informed of this in writing by the contractor. The nature/type of rock met with, will be decided by the GE/AGE(I) whose decision is final and binding. If the nature of rock met with is "Hard Rock", the same will be priced and paid as excavation in hard rock as per Condition 62 of IAFW-2249. Those rock shall become the property of Contractor and a credit of **Rs. 1300/- (Rupees One thousand three hundred only)** per cubic meter on stack measurement after deduction 50% quantity for void etc. shall be deducted from the payment due to the contractor. The quantity of hard rock shall be equivalent in hard rock excavated (cum)

(c) In no circumstances blasting will be permitted for excavating hard rock.

(d) Lump sum quoted by the contractor shall not include use of timbering to uphold sides of excavation. If timbering is ordered by the GE in writing, it will be paid extra as a deviation to the extent required/ordered in writing.

(e) Lump sum quoted by the contractor shall be deemed to include surface excavation not exceeding 30 cm deep but averaging 15 cm deep over the area occupied by the buildings including plinth protections, ramps, platforms, steps, court-yard, shaft etc, and up to a width of 3 metres beyond the edge of the plinth protection around the building and the like shall be carried out. If buildings are sited in areas after site development no further surface excavation need be carried out unless ordered by the GE in writing. In such cases, surface excavation as mentioned above shall be omitted through a deviation order.

(f) The ground levels shown in the drawings are ground levels after surface excavation is carried out, there after excavation in trenches or over areas will be carried out as shown on drawings. Excavation made deeper than what is shown in drawings shall be made good by the contractor with PCC (1:5:10) type E2 without any extra cost to the Government.

(g) Lump sum quoted by the contractor shall be deemed to include for bailing pumping and de-watering foundation trenches of buildings, if water is met with or accumulated from any source or cause. No additional payment will be admissible in this account. In the event of deviation no price adjustment shall be made for cost of bailing, pumping etc, and de-watering as specified herein before whether these are actually required and done at site or not.

(h) Deviations for earthwork in excavation shall be priced as excavation by mechanical means (Hydraulic excavation)at the rates in MES schedules Part-II subject to addition of contractor's quoted percentage.

PARTICULAR SPECIFICATIONS (CONTD…)

1.2 FILLING IN TRENCHES, PLINTH AND UNDER FLOORS:

1.2.1 Filling of excavated earth in foundation trenches, in plinth and under floor, etc. shall be carried out as per Clause No. 3.19.1, 3.19.2 and 3.19.3.1 of MES Schedule Part-I.

1.2.2 Tenderers shall quote their Lump Sum taking into consideration that entire quantity of earth obtained from excavation shall be utilized for filling purpose like returning, filling in trenches, etc., to achieve plinth height as shown on drawing including removal of surplus soil and bringing in approved earth to make up shortage, if any. Additional Earth, if required for Filling, as mentioned above, shall be obtained from Outside the MD Land without any Extra Cost to the Department. Earth, obtained thus shall be got approved from GE in writing prior to incorporation in the Work. No Royalties and Transportation Charges shall be Paid Extra by the Govt. for Earth obtained from Outside MD Land.

1.2.3 All filling of earth will be spread in layers not exceeding 25cm thick (each layer) watered and well rammed. Thickness of filling shown in drawing is after consolidation.

1.3 REMOVAL OF SOIL:

1.3.1 Soil obtained from excavation and not approved for filling and surplus soil obtained from excavation and soil obtained from surface excavation and surface dressing shall be disposed off to a distance exc. 50 metre but n.exc. 100 metre, deposited and spread as directed by the Engineer-in-Charge.

1.4 HARD CORE:

1.4.1 Hard core shall consist of broken stone aggregate of gauge not exceeding 63 mm and shall be well graded so that after consolidation it provides a dense and compact sub grade. Hard core shall be watered well rammed and consolidated. The thickness of hard core shown on drawings shall be thickness after consolidation.

1.5 ANTITERMITE PRECONSTRUCTION SOIL TREATMENT:

1.5.1 Pre-constructional antitermite chemical treatment to bottom and sides of foundation trenches/pits, filling in trenches/pits, and below floors, junctions of walls and floors, external perimeter of buildings/plinth protection and surroundings of conduits/pipes of all buildings, shall be carried out all as specified in MES Schedule part-I.

1.5.2. Antitermite treatment shall be got done through approved specialist agency which is member of Indian Pest Control Association holding valid licence as par clause 13 of Insecticides CT-1968, Persons employed to do the antitermite treatment shall be qualified as per rule 10 of the rules framed under the inspection rules 1971. The specialist agency may be any one from the list circulated by E-in-C's Branch which are located and active in the area with prior approval of GE(I)/GE/AGE(I). However, if the contractor himself is enlisted for antitermite treatment also, he may carry out antitermite work himself.

1.5.3 Antitermite treatment shall be carried out with chemicals 'chloropyriphos' 20% concentration conforming to IS as specified in the MES Schedule, purchased directly from the manufacturer or his authorised agents. Contractor will not be allowed to use chemicals banned by the Government of India and contractor shall have no claim on this account. Chemical brought to site in sealed containers and bearing IS mark shall only be used in the work. Purchase vouchers shall be produced before the GE(I)/GE/AGE(I) for verification and defacement against this contract.

1.5.4 The defects liability period of antitermite treatment shall be 10 years and the contractor shall be responsible to keep the entire buildings free from termite infestation for a period of 10 years after the date of taking over the completed buildings from the contractor. The contractor may obtain a similar guarantee from the specialist firm engaged by him for the purpose and the same shall be submitted on demand.

1.5.5 Should the GE(I)/GE/AGE(I) at any time during the construction or prior to expire of said guarantee period of 10 years find that the buildings have been infested with termite, the contractor, on demand in writing from the GE(I)/GE/AGE(I), will forthwith undertake to carry out such treatment which may be necessary to render the buildings free from termite infestation at his own expense till expiry of the guarantee period of 10 (TEN) years. The guarantee shall also be applicable to wood work and joinery as specified hereinafter.

PARTICULAR SPECIFICATIONS (CONTD…)

1.5.6 In the event of contractor's failure to comply with the GE(I)/GE/AGE(I)'s direction within the stipulated period, the work shall be got carried out at the risk and the cost of the contractor. The liability of the contractor under this condition shall not be however extended beyond the period of 10 (TEN) years from the certified date of completion unless the notice was served on the contractor previously to rectify such defects.

1.5.7 The contractor shall provide a plaster plate of requisite size in situation as decided by the Engineer-in-charge on the wall of each of the building. The plate shall be 10 mm thick in cement sand mortar (1:4) to indicate the CA NO., Name of the contractor, the date of completion of the work and the probable date of expire of 10 years guarantee for Antitermite treatment by engraving and painting (black). The cost of plaster plates is deemed to be included in the unit rates of the buildings.

2. **CONCRETE WORK**

2.1 **MATERIALS**

2.1.1 **CEMENT**

2.1.1.1 **TYPE OF CEMENT**:

2.1.1.1.1 Cement shall be Ordinary Portland Cement Grade 43 conforming to IS 8112-1989 (Latest revision) or Portland Pozzollana Cement, conforming to IS 1489-1991.

2.1.1.1.2 The following checks and procedures shall be followed before the cement supplied by the contractor is accepted and is approved for incorporation in the works.

2.1.1.2 **PROCUREMENT:**

2.1.1.2.1 Cement (OPC & PPC) supplied by the contractor shall be procured from any of the E-in-C's Branch approved cement manufacturers as listed in List of Makes. Further, Cement can be procured even from manufacturers approved by E-in-C's Branch after date of receipt of tenders also.

2.1.1.2.2 The following conditions shall have to be satisfied before PPC can be allowed to be used in the work:-

(a) PPC should meet the strength criteria as per IS-8112-1989.

(b) The minimum period before striking off formwork while using PPC shall be as given here-in-after.

(c) In cold climate regions where temperature is lower than 15 0C only OPC shall be used.

(d) Mixing of OPC & PPC shall not be allowed in a work. However with the strict control of ground executives different buildings can have different types of cement.

(e) While procuring PPC, the following requirements are to be ensured and certificate to the effect is to be obtained by the contractor for each batch from the manufacturer and the same shall be submitted to the GE(I) / GE / AGE(I) for approval to procure PPC.

  (i) The quality of fly ash is strictly as per IS-1489 (Part-I) – 2002.

  (ii) Fly ash is inter-ground with clinker not mixed with clinker.

  (iii) Dry fly ash is transported in closed containers and stored in silos. Only pneumatic pumping should be used.

  (iv) The fly ash received from thermal power plants using high temperature combustion above 10000C should be used.

2.1.1.2.3 Cement (OPC & PPC) supplied by the contractor shall be procured from any of the manufacturers approved by HQ E-in-C's Branch as per approved list of manufacturer given in tender.

2.1.1.3 **TESTING OF CEMENT:**

2.1.1.3.1 The manufacturer/ main producer shall carry out testing of cement in accordance with the relevant BIS provisions. The Contractor shall submit the Manufacturer's Test Certificate in original along with the Test Sheet giving the result of each physical test as applicable and the chemical composition of the cement or authenticated copy thereof, duly signed by the manufacturer with each consignment. The Engineer-in-Charge shall record these details in the **Cement Supply and Acceptance Register**, after due verification. Identification marks on cement bags by the manufacturer shall be as per relevant BIS codes.

2.1.1.4 **INDEPENDENT TESTING:**

2.1.14.1 The GE shall organise independent testing of random samples of cement drawn from each lot as per laid down frequency specified here-in-after from the **IIT/NIT/ NABL ACCREDITED LABORATORIES** as per IS: 3535-1986 (Method of sampling hydraulic cement), IS: 4031 (Method of physical test for hydraulic cements) and IS:4032-1985 (Method of chemical analysis of hydraulic cement). In order to undertake departmental testing, requisite facilities along with materials, conveyance etc. shall be organised by the contractor without any extra cost to the Govt. The cost of all tests including cost of conveyance shall be borne by the contractor. The record of such tests would be maintained in the **Cement Test Register**.

2.1.1.4.2 The contractor at his own cost shall provide all facilities required for the tests including cost of tests samples, conveyance etc and the same shall be deemed to be included in the quoted rate.

2.1.1.4.3 The entire lot of cement brought by contractor to the site shall be removed by contractor under his own arrangements and at his own cost, whenever the test results are not found within the prescribed acceptable limits. Decision of GE in this respect shall be final and binding.

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2.1.1.5. **STORAGE** : Cement shall be stored over dry platform at least 20 cm high in such a manner as to prevent deterioration due to moisture or intrusion of foreign matter. In case of store rooms, the stack shall be at least 20cm away from floors and walls. Inspection of storage shall be carried out by Engineer-in-Charge or J.E(Civil) specifically detailed by him in writing, once a day. It shall be ensured that tested and untested cement are segregated and stored separately with distinct identification.

2.1.1.6 **DOCUMENTATION** : The Contractor shall submit original vouchers from the main producer/manufacturers for the total quantity of cement supplied under each consignment to be incorporated in the work. All consignments received at the work site shall be inspected by the GE alongwith the relevant documents before acceptance. The original vouchers and the Test Certificates shall be defaced by the Engineer-in-Charge and kept on record in the Office of GE duly authenticated and with cross reference to the Control Number recorded in the cement Supply & Acceptance Register. The Cement Supply & Acceptance Register will be signed by J.E(Civil), Engineer-in-Charge, GE and the Contractor. The Accepting Officer may order Board of Officers for random check of cement and verification of connected documents. The entire quantity of all cement shall also be suitably recorded in the Measurement Book for record purposes before incorporation in the work and shall be signed by the Engineer-in-Charge and the Contractor.

2.1.1.6.1 Identification of cement bags by the manufacturer shall be as per relevant BIS codes.

2.1.1.6.2 The entire lot of cement brought by contractor to the site shall be removed by contractor under his own arrangements and at his own cost, whenever the test results are not found within the prescribed acceptable limits. Decision of GE in this respect shall be final and binding.

2.1.1.6.3 The cost of all tests including cost of transportation shall be borne by contractor irrespective of test results.

2.1.1.6.4 The proper accommodation for storage, preservation and maintenance of cement shall be provided by the contractor at his own cost till consumption of cement in the work at the place(s) decided by GE. All arrangement for proper storage and safety of cement as indicated here-in-before shall be made by contractor at his own cost. The cement storage accommodation shall be provided with two locks at each door. The key of one lock at each door shall remain with the Engineer-in-Charge or his representative and that of other lock shall remain with the contractor's authorised agent at site of works so that cement is removed from the godown/storage accommodation according to daily requirement with the knowledge of both the parties.

2.1.1.6.5 Schedule of supply of cement shall be worked out by contractor in consultation with GE so as to match with the progress contemplated in CPM Chart prepared for this Project.

2.1.1.7 **WEIGHING AND PAYMENT FOR CEMENT**

2.1.1.7.1 Random samples of 5 bags per 100 bags shall be got weighed by Engineer-in-Charge in presence of representative of contractor from each consignment to arrive at average weight per bag for calculating total quantity of cement brought at site for incorporation in the work. All arrangement for weighing of cement bags including labour etc shall be provided by the contractor at his own cost.

2.1.1.7.2 The quantity of cement arrived at as aforesaid shall be considered for payment as applicable in accordance with IAFW-2249, General Conditions of Contracts.

2.1.1.7.3 Payment for cement shall only be allowed after production of original purchase vouchers, manufacturer's test certificates for each consignment of cement brought by contractor and after the results of random testing carried out by GE are found satisfactory as specified herein-in-before.

2.1.2 **AGGREGATES**

2.1.2.1 **COARSE AGGREGATE (STONE)**: Aggregate for all concrete work shall be crushed stone aggregate conforming to IS-383 and para 4.4 of MES Schedule Part I, 2009. The aggregate also shall be in conformity with the samples kept in GE's office.

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2.1.2.2 **FINE AGGREGATE (COARSE SAND)** : Fine aggregate shall be sand conforming to the requirements to IS-383 and para 4.4 of MES Schedule Part I Sand shall be washed in clean water if the deleterious materials exceed the permissible limit. Use of sand confirming to grading Zone-IV of IS-383 shall not be allowed in RCC/PCC work.

2.1.2.3 Grading of coarse aggregate unless otherwise specified shall be as follows :-

SL. NO	DESCRIPTION		GRADING
(a)	For Reinforced/Structural Element of depth/thickness 50mm or under	:	12.5mm graded
(b)	For Reinforced/Structural Element of depth/thickness over 50mm	:	20mm graded
(c)	For Plain Cement Concrete thickness exc. 25mm but not exc. 75mm	:	20mm graded
(d)	For Plain Cement Concrete thickness 75mm and over	:	40mm graded
(e)	Lime Concrete	:	40mm graded

2.1.2.3.1 Grading of coarse aggregate for plain and reinforced concrete i.e. stone aggregate shall be as per para 4.4.7 (I) of MES Schedule part-I and grading of fine aggregate sand shall be as per para 4.4.7(2) of MES Schedule Part-I

2.1.2.4 **AGGREGATES FROM NATURAL SOURCES**: Aggregates obtained from natural sources shall not be contained any type of harmful materials and shall be cleaned in all respect.

2.1.3. **WATER**: Refer para 4.9 & 4.9.2 of MES Schedule Part I. Water shall conform to clause 5.4 IS-456-2000 in case water is drawn by the contractor from his own sources as mentioned in Special Condition Clause No 12 of these tender documents.

2.1.4 **PLASTICISER**

2.1.4.1 Plasticiser shall be used in Design Mix Concrete as per the Manufacturer's instruction without any Extra Cost to achieve the desired Results Strength and Workability. However Plasticiser should confirm to IS-9103.

2.1.5. **MINI WEIGH BATCHING PLANT**

2.1.5.1 The concrete shall be mixed in mini concrete batching plant and nothing extra shall be payable on this account. The plant will have output capacity of 10 to 12 cum/Hr (approx). The batching plant shall have digital weighing accuracy of ±1% (one percent) with automatic water meter, drum capacity 800 liters, batch capacity 600 liters, three bin hopper, heavy duty balanced chassis 600x16pneumatic tyre 4 Nos & 3000 OPUSB(Offline USB Drive). The plant shall have computer printout facility for concreting. Where requirement of concrete is less than 1 cum in a day, the same can be mixed by using normal weigh batcher and hopper type mixer.

2.2. **WORKMANSHIP**

2.2.1. **MIX OF CEMENT CONCRETE**

2.2.1.1. Unless otherwise specified, the mix of cement concrete required in various situation shall be of the following specified grade/proportions. Plain cement concrete and the RCC shall be provided in accordance with IS: 456-2000: -

Sl No	Situation	Mix of cement concrete	Type with nominal size of graded aggregates
1	2	3	4
(a)	Sub-base of floor	1:3:6	Type C-2 (40mm)
(b)	Lean concrete below RCC column footing	1:3:6	Type C-2 (40mm)
(c)	All RCC work.	-	M-25 / M-30 (Design Mix) as per drawing, M-25 (design mix) where not mentioned in the drawing.
(d)	**Lean concrete for wall** foundation and dwarf wall.	1:3:6	Type C-2 (40mm)

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1	2	3	4
(e)	PCC in foundation for steps in GF and lean concrete to RCC post of staircase.	1:3:6	Type C-2 (40mm)
(f)	PCC cills, PCC drain, PCC plinth protection and hopper of gully trap and PCC blocks.	1: 3: 6	Type C-1(20mm)
(g)	All PCC bed block and pre-cast Jalli.	1:2:4	Type B-0(12.5mm)
(h)	All plain cement concrete not described in (a) to (g) above	1:2:4	Type B-1(20mm)
(j)	Plain cement concrete below plinth beam where shown on drawings	1: 3: 6	Type C-2 (40mm)

2.2.1.2. All concrete except lime concrete shall be machine mixed. Where small quantities of concrete work is involved, hand mixing may be permitted by the Engineer-in-charge at his own discretion subject to addition of 10% extra cement.

2.2.1.3 Compaction of RCC slabs, beams, foundations and columns shall be done with approved mechanical vibrator using pin and surface type vibrators for all concrete works including for flooring concrete. In other locations compaction may be achieved manually as directed by Engineer-in-charge.

2.2.1.4 For batching, workability, mixing, form work, removal of form work, transportation, placing, compaction, construction joint, curing, protection and inspection of concrete refer relevant paras of MES Schedule Part-I.

2.2.1.5.Sampling and testing of concrete of grade M-30 / M-25 shall be done as per paras 4.11.17 of MES Schedule part I. Testing of concrete as mentioned in IS-456 shall be carried out by the contractor without any extra cost to the department i.e. cost of samples, transportation and testing will be borne by the contractor. Sampling and strength testing of concrete shall be done in accordance with clause 15 of IS-456.

2.2.2.**SPECIFICATIONS FOR DESIGN MIX CONCRETE : M-30/M-25 GRADE**

2.2.2.1 CEMENT : Cement shall be all as specified here-in-before.

2.2.2.2 **AGGREGATE**

2.2.2.2.1 **AGGREGATE**: Aggregate shall conform to IS-383 and shall be type basalt/trap granite stone. Aggregate shall be non porous, hard, strong, durable, clear and free from various impurities and adherent coating and shall not contain deleterious materials exceeding the limits specified in the above referred IS. When required by the Engineer-in-Charge, the contractor shall, at his own expenses, carry out any test laid down in IS-383 and IS-2386 to verify that the aggregate complies with the requirements of the IS. The test will be carried out in NABL Accredited Labs / IITs.

2.2.2.2.2 **COARSE AGGREGATE** : Coarse aggregate shall consist of crushed graded stones aggregates. The maximum size of the aggregate shall be 20mm graded as per IS-383.

2.2.2.2.3 **FINE AGGREGATE** : Fine aggregate shall consist of naturally occurring coarse sand, fine aggregate shall conform to Zone II/III of Table IV of IS-383.

2.2.2.2.4 **SAMPLES** : Samples of the aggregate proposed to be used shall be approved by GE, prior to bulk delivery of the same at site of work. Field tests for determining the contents of silt, clay etc., for fine aggregate shall be carried out by the Engineer-in- Charge, from time to time to ensure that material brought to site are in conformity with the samples, approved by the GE.

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2.2.2.2.5 **GRADING OF AGGREGATES** : The grading of coarse and fine aggregate shall be as per MES Schedule. Fine aggregate shall have grading as per grading Zone II/III of IS 383-1970. The grading of coarse and fine aggregate shall be checked as frequently as possible. The frequency for the aggregate testing shall be determined by Engineer-in-Charge to ensure that the specified grading is being maintained.

2.2.2.2.6 **TESTING OF AGGREGATE** : Aggregate shall be tested for following properties in accordance with IS mentioned against them:-

(i)	Aggregate particle size & shape	:	IS-2386 Part I
(ii)	Presence of deleterious materials	:	IS-2386 Pt II 1969
(iii)	Aggregate specific gravity, density, voids absorption and bulk density	:	IS-2386 Pt III
(iv)	Aggregate crushing value aggregate impact value and aggregate abrasion value	:	IS-2386 Pt IV
(v)	Aggregate soundness	:	IS-2386 Pt V

2.2.2.3 MIX PROPORTION AND STENGTH

(i)	Type of cement	:	Ordinary port-land cement, Grade 43 as per IS : 8112-1989 or Portland pozzolana cement (PPC) as per IS 1489
(ii)	Aggregate/cement ratio by weight	:	As per mix design based on IS-10262 &IS-456
(iii)	Workability	:	As per clause 7 of IS-456 – 2000
(iv)	Water cement ratio	:	As per mix design based on IS-10262 & IS-456.
(v)	Degree of quality control	:	Good (Refer IS-10262)
(vi)	Minimum cement content	:	As per table 5 of IS-456, 2000 and as specified here-in-after.
(vii)	Maximum size of aggregate	:	20 mm graded
(viii)	Maximum water cement ratio	:	As per table 5 of IS-456, 2000
(ix)	Exposure Condition	:	Moderate for M-25 & Severe for M-30.

2.2.2.3.1 As soon as possible, after receiving the order to commence work, the Contractor at his own expense shall get concrete mix designed from any Govt. approved laboratories/Govt. Engineering Colleges to satisfy the requirements of Clause 2.2.2.3 of Particulars Specifications here-in-above.

2.2.2.3.2 The actual proportion of concrete mix shall be arrived at by means of number of trial mixes. First trial mix shall be prepared with the mix proportions arrived at as per mix design and other trial mixes with suitable adjustment in mix proportion.

2.2.2.3.3 Six number of cubes of size 15x15x15 cm shall be made from each trial mix. The concrete cubes shall be tested as per IS-516-1959. Out of six cubes, three will be tested after seven days and remaining three after 28 days from the date of casting.

2.2.2.3.4 The test after seven days is intended to give an early indication of the strength likely to be achieved at 28 days.

2.2.2.3.5 The concrete shall be deemed to comply with the strength requirements of clause 15 and 16 of IS-456-2000 is met with.

2.2.2.3.6 Whenever, there is any change in the type of grading of materials, the mix should be re-designed and modified suitably to give the desired compressive strength.

2.2.2.3.7 On the basis of above test results, the final design mix will be approved by the CWE. The approval of the CWE shall not relieve the contractor of his responsibility for obtaining the required minimum strength of quality concrete in the work.

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2.2.2.3.8 Minimum cement content for the M-30/M-25 (Design Mix) concrete shall not be less than as in Table 5 of IS-456, 2000. In case cement content as per actual Design Mix works out lower than minimum cement content specified, the contractor shall provide minimum cement content as per Table 5 of IS-456, 2000 and nothing extra shall be payable on this account.In case cement content as per actual Design Mix works out more than minimum cement content specified, the contractor shall provide cement content as per approved design mix and nothing extra shall be payable on this account.

2.2.2.3.9 **BATCHING AND MIXING OF CONCRETE**

2.2.2.3.9.1 (a) The mixing of cement and aggregate for cement concrete for all grades and type of RCC work shall be done by weigh batching. Calibration certificate of weigh batcher including that of water measuring device shall be submitted by the contractor. Contractor to arrange for periodic calibration through approved agencies as directed by GE(I)/GE /AGE(I).

(b) The batching shall be as per clause 10.2 of IS 456: 2000 and clause 4.11.3 (Excluding provision vide sub clause 4.11.3.1) of MES SSR Part – I.

(c) In case the quantity of design mix concrete in the entire work is less than 1000 cum, mixing of concrete shall be done in a mechanical mixer using weigh batching and as per clause 10.3 of IS : 456- 2000. Weigh batching shall be done preferably with layer type weigh batcher. The concrete mixer shall confirm to IS : 1791 and fitted with water measuring (metering) devices. In case, the quantity of design mix concrete in the entire work is more than 1000 cum, the contractor shall install fully automatic Computerized Batching Plant of capacity minimum 10 to 12 Cum per hour at site, complete quality assurance and control shall be maintained by the contractor in accordance with the latest IS codes and best modern Engineering practice. The record of quality assurance shall be submitted to the GE and shall be kept on record.

2.2.2.3.10 **TRIAL MIX.** The actual trial mix proportion will be arrived at by means of number of trial mixes by changing the water cement ratio, proportions of fine and coarse aggregate, fineness modules of aggregate by changing their grading and proportions etc Attempts shall be made to make the mix design as economical as possible.

2.2.2.4. **CAST-IN-SITU/PRECAST ARTICLES**

2.2.2.4.1 PCC cills, RCC shelves, isolated lintels up to 1.5 metre length (i.e not in continuation with lintel band) may be cast in situ or pre-cast at the discretion of the contractor. In the event of deviation with regards to these items pricing shall be done at appropriate rate in SSR for cast-in-situ articles. Top of such pre-cast articles should be distinctly marked.

2.2.2.4.2.Unless otherwise specified all pre-cast concrete articles shall be set and jointed in CM (1:3).

2.2.2.4.3 For Assembly and cover to reinforcement refer paras 4.11.7 to 4.11.8.4 of MES Schedule Part-I.

2.2.3 **LINTEL/TIE**

2.2.3.1.Lintel beam and SLT shall be provided as per detailed section shown in structural drawings irrespective of whether the sections are shown or not on architectural drawings. Where lintel band is provided, SLT and lintel with chajja, wherever occurring shall be cast monolithic with the lintel band and the reinforcement higher of the two shall be provided.

2.2.4. **FORMWORK (STEEL)**

2.2.4.1 Refer Para 4.11.6 and 7.15 of MES Schedule Part I.

2.2.4.2 Formwork to exposed surfaces of concrete (other than mentioned here-in-after) shall be steel formwork having rough and smooth surfaces in contact with concrete. However, in case of any deviation involving formwork to surface exposed to view, the pricing shall be done at the rate of timber formwork for rough finish and in case of unexposed concrete surface, pricing shall be done at the rate of timber work, clean sawn subject to contractor's percentage. The contractor shall have no extra claim on account of this.

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2.2.4.3. Form work to exposed surfaces of RCC column, beams, lintels etc which are continuous with the plaster surface of wall and/or to the surface of concrete which are buried in contact with masonry or otherwise shall also be steel formwork.

2.2.4.4 Exposed surfaces of concrete such as soffits/sides of RCC chajjas, beams, staircase & shelves etc which are not in continuous with the adjoining plastered surface shall be finished as specified in Para 4.11.16.2(a) of MES SSR Part-I. However, the soffit of RCC roofs/floor slab shall be plastered with 5mm thick cement mortar (1:3).

2.2.4.5 Application of non-adhering oils such as used engine oils or bitumen based paints to inside of forms is not permitted.

2.2.4.6 Provide steel props/spans/bracings of sufficient strength such as ACROW props, ACROW spans from ACROW Indian Ltd, Bombay or similar as approved by the GE. Use of conventional wooden ballies (posts), struts that is other than steel member are not permitted.

2.2.5. **CURING AND MINIMUM PERIOD FOR STRIKING OF FORM WORK WITH POZZOLANA CEMENT USED**

2.2.5.1. Where pozzolana cement is used for RCC works in various locations, the minimum period for curing and striking off form work shall be as under in lieu of that specified in clause 4.11.6.3 of MES Schedule Part-I :-

(a) Curing :- Structural RCC work, PCC work, Plastering etc. : 14 days
(b) Striking off Form Work Walls, columns and vertical sides of beams-3 to 4 days
(c) Slabs (Props left under) : 8 days
(d) Beams soffits (Props left under) : 9 days
(e) Removal of props to slab/beam
  (i) Spanning upto 6 M span : 16 days
  (ii) Spanning beyond 6 M span : 22 days
(f) For cantilever portion of slab form work with support shall be retained until the completion of the casting of the entire frame work of the building.

2.2.5.2 In case of bad weather, periods mentioned above may be revised at the discretion of the Engineer-in-charge. The contractor shall be deemed to have considered the above provision before tendering and quote lump sum accordingly. No claim will be entertained if longer periods required for striking off form work and curing and all such effected matters is looked into upon the use of pozzolana cement as a result being issued to the contractor in full or in part or for bad weather.

2.2.5.3. Curing shall be carried out using pump of suitable capacity at no extra cost to the Govt. all as directed by Engineer-in-Charge.

2.2.6 **FINISHES**

2.2.6.1 For finishing to exposed surfaces of concrete, refer at paras 4.11.16.1, 4.11.16.2(b),(c),(d),(e) and 4.11.16.3 of MES Schedule Part-I. If thickness of plaster in cement mortar (1:3) as specified in para 4.11.16.2(b) of MES Schedule Part-I is required to be increased in excess of 5mm to achieve fair and even surface, it shall be provided by the contractor without any extra cost to the Govt.

2.2.6.2 All projections of chajja, RCC roof, cills and the like shall be provided with proper drip course and/or throating in plastering only and not while casting RCC/PCC.

2.2.6.3. All flat surface exposed to weather shall be weathered with imperceptible slope for smooth run of rainwater.

2.2.7. **LIME CONCRETE**

Surrounds of Squat Pattern WC's shall be lime concrete as specified in Para 4.10.1, 4.10.2 and 4.10.4 of MES Schedule Part-I. The contractor may use quick lime or factory made dry hydrated lime powder at his option without any price adjustment. Mixing of lime concrete shall be done by hand mixing.

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2.2.8. **PCC DRAIN**

2.2.8.1. PCC drain shall be provided wherever shown and as per detail shown on drawing. PCC for drain shall be of concrete 1:3:6 type C-1, finished even and smooth with using extra cement.

3. **BRICK WORK**

3.1. **MATERIALS**

3.1.1. CEMENT: - Cement shall be as specified for concrete work.

3.1.2. SAND :- Sand for mortar shall be as specified in para 5.4, 5.4.1 of MES Schedule Part I.

3.1.3 **BRICK** - Refer to paras 5.6.1 to 5.6.9 of MES Schedule part I. The brick shall have minimum compressive strength of 75 kg/sq cm and shall be of sub class `B' bricks as mentioned in para 5.6.5 of MES Schedule Part I. Testing of bricks shall be carried out by the contractor without any extra cost to the Govt including the cost of transportation. The dimensions of bricks shall not vary more than ± 8%. The size of bricks shall be 230 x 115 x 75 mm/ 250 x125 x75mm whichever is available.

3.1.4. WATER – Refer to para 5.17 of MES Schedule Part I.

3.2. **WORKMANSHIP**

3.2.1 **Brick Work** – Mix of mortar for brick shall be as follows:-
Brick work in various situations irrespective of whatever shown on drawing shall be in cement mortar (1:6) for full brick work and in cement mortar (1:4) for half brick/less than half brick work.

3.2.2. **HALF BRICK WALLS** – Refer para 5.36 of MES Schedule Part I. Brick work in half brick walls exceeding 1200 mm in height from floor level shall be reinforced with 2 Nos. 8 mm dia deformed bars at every fourth course, starting from top of the first course from floor level.

3.2.3. The foundation for half brick wall shall be provided as shown in foundation plan. Where foundations for half brick wall are not indicated the same shall be constructed off the sub base level. Half brick thick walls in first floor shall be constructed of the RCC floor slab.

3.2.4 The drawings for this work are prepared taking size of half brick thick as 11. 5 cm and full brick thick as 23 cm, deviation if any shall also be priced on this basis irrespective of actual size of bricks used. Centerline dimensions of buildings shown in drawings shall however, be maintained. No adjustment for extra cost due to increase in width of foundation concrete, walls, lintels, bands and blocks shall be admissible to the contractor on this account.

3.2.5 BLANK.

3.2.6. Any hole left in the brick work for the purpose of centering/scaffolding shall be filled up with PCC (1:3:6) type C1 mixed with WPC as per manufacturers instructions.

3.2.7 Flat iron 40 x 6 mm of size 300 mm bent to shape shall be provided at junction of column & brick work for building work. Spacing of FI shall be 300 mm c/c.

3.2.8 Dwarf wall as shown in structural drawing shall be provided below plinth beam along the external perimeter of building .

3.3. **BEARING PLASTER** – Refer to clause No 5.34.14 of MES Schedule Part I, tops of walls bearing the edges of RCC floors, roofs, slabs, beams shall be finished with a layer of cement mortar 1: 4, 15 mm thick and the plastered surface shall be white washed. Two layers of chisel craft paper shall be provided above the plastered surface. Where the bottom of slab does not coincide with the level of brick course after cement plaster, the level shall be make up either by increasing the thickness of slab at bearing or where feasible by using brick tiles so that the bearing is directly on the plaster.

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3.4 **DAMP PROOF COURSE (DPC)**

3.4.1 DPC shall be 40 mm thick in PCC 1:2:4 type B0 (using 12.50 mm graded stone aggregate) with water proofing compound as per manufacturer's instruction. Layer of 200 micron polythene film shall be laid over a coat of hot blown bitumen, grade 85/25 @ 1.50 Kg /Sqm over the cement concrete.

3.4.2 DPC shall be laid to the full width of wall. Level of the DPC at the openings shall be lowered equal to the thickness of the floor finish. The continuity of the DPC shall be maintained. The workmanship shall be all as specified in clause 5.42 of SSR Part-I, 2009. DPC shall not be provided on dwarf wall, RCC wall and RCC columns and at the location where RCC / PCC plinth band or RCC plinth beams are provided above the ground level.

4. **STEEL AND IRON WORK**

4.1 Unless otherwise specified quality of steel to be incorporated in the work shall be as follows:-

(a)	High strength deformed steel bars produced by Thermo Mechanical treatment process (In short called as TMT bars) for reinforcement.	Of Grade Fe-500 and meeting all requirement of IS 1786-1985 (Reaffirmed) 1990 (except water retaining structure).
(b)	Mild steel for miscellaneous works	Shall conform to IS-432(Part-I). Grade wherever not shown/ indicated otherwise shall be of Grade I.
(c)	Structural steel	Standard quality - Conforming to IS- 2062. Ordinary quality – Confirming to IS -1977.
(d)	Galvanised steel sheets (Plain & corrugated)	Conforming to IS-277.
(e)	Fabric reinforcement for concrete	Conforming to IS-1566

4.2 Deviations on items of steel work shall be priced at applicable rates in MES Schedule subject to contractor's percentage for respective parts of Schedule 'A'.

4.3. WELDING

4.3.1 Welding of steel and ironwork shall be done in an approved manner with electric arc welding.

4.3.2 Electric welding shall be metal arc welding as specified in Para 10.15 of MES SSR Part-I.

4.3.3 Welding electrodes shall be of quality suitable for welding of structural steel and shall comply with requirement of IS-814 for covered electrodes for metal arc welding of mild steel.

4.3.4 Unless otherwise specified/shown on drawings, the fillet welds shall be of 6mm size.

4.3.5 All butt welds shall be full penetration welds.

4.4 **REINFORCEMENT**

4.4.1 Reinforcement shall be bent to shape and provided as indicated on the drawings.

4.4.2 All laps and crossings shall be tied with mild steel wire (annealed) of size not less than 0.9 mm dia.

4.4.3 Unless otherwise shown on drawing, bends to a radius of 4 times the dia of bar shall be provided in respect of tor steel bars. In case of deviations involving reinforcement bars, the length of each bar for the purpose of calculating laps shall be taken as 10 (Ten) metres for omit purpose.

4.4.4 Unless otherwise shown on drawings, spacing of chairs shall be 75 cm centre to centre below the cranked up bars to keep the bars in position.

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4.5 **REINFORCEMENT STEEL AND STRUCTURAL/NON-STRUCTURAL STEEL**

4.5.1 **TYPE OF STEEL**

4.5.2 **REINFORCEMENT STEEL**

(a) Unless otherwise specified, mild steel TMT bars shall be of grade Fe 500 meeting requirements of IS-1786-2008 shall be used of approved primary manufacturers / main producers as given in list of makes.

4.5.2.1 BLANK

4.5.2.2 **Galvanised Steel sheet & Fabric Reinforcement for concrete**: These shall be ISI Marked and shall be procured directly from Main manufacturers.

4.5.2.3 However reinforcement steel, structural steel, Galvanised steel sheets and Fabric Reinforcement shall be permitted to be procured from authorised dealers of main producers in case of contracts of value less than 7.5 lakhs.

4.5.2.4 Steel sections for railing, gates, fencing, guard bars, grills, steel chowkhat, holdfasts etc., which do not constitute structural members, can be procured from main producers/secondary producers/BIS marked manufacturers or their authorised dealers at the option of Contractor without any price adjustment. Tests for such steel sections shall not be insisted by GE. However steel section for large size gates of ammunition sheds shall be treated as structural members.

4.5.2.5 The contractor should place their demand/requisition of steel with the producers with adequate lead time.

4.5.2.6 Steel, when procured from main producers/approved secondary producers/ authorized conversion agents of main producers (stipulated in above paras) shall have to be procured directly from the storage depots of the main producers/approved secondary producers/authorized conversion agents and not from their authorized agents/dealers.

4.5.2.7 The particulars of the manufacturer/supplier of steel shall be submitted by the contractor to the GE(I)/GE/AGE(I) prior to procurement for every lot of steel separately. The original documents in support of purchase of steel shall be verified by the site staff and GE(I)/GE/AGE(I) and certified true copy of the result shall be retained in GE/AGE(I)'s office. The form given here in after will be used for this purpose.

PARTICULAR SPECIFICATIONS (CONTD…)

STEEL SUPPLY & ACCEPTANCE REGISTER

1. CA No & Name of Work :
2. Contract No :
3. Name of Manufacturer's TC No :
4. Manufacturer :
5. Random Test Details (a) Physical test report from _____ (Name of NABL approval lab/Engg College) vide their letter No _____
   (b) Chemical test report from _____ (Name of NABL approval lab/Engg College) vide their letter No _____
6. Type of Steel, Dia & Qty(a)type : TMT/CRS (b) Dia___ mm (c) Actual Wt ___ MT (d) Conversion Wt ____ MT

	Chemical Test							***Mechanical Test***						
	Carbon %	Sulphur %	Phosph Orous %	Sulphur + phosphorus %	Manganese %	Silicon %	Corrosion Resistant element	*Wt per meter*	*Yield Stress (N/mm^2) 0.2% proof*	*Tensile Strength (N/mm^2)*	*Elongation % (As per IS)*	*Bend test*	*Rebend test*	*Remarks*
1	2	3	4	5	6	7	8	*9*	10	*11*	12	*13*	14	*15*
As per IS 1786-2008														
As per manufacture's test certificate														
As per independent test														

Remarks with Signature Accepted/Rejected

Contractor Junior Engineer Engineer-in-Charge GE

Remarks of BOO/Inspecting Officer/CWE

Cont'd…

PARTICULAR SPECIFICATIONS (CONTD…)

4.6 **TESTING OF STEEL**

4.6.1 The manufacturer is to carry out inspection and testing of steel in accordance with the relevant BIS provisions. The contractor shall submit the manufacturer's test certificate in original along with the Test Sheet giving the result of each mechanical test as applicable and the chemical composition of the steel or authenticated copy thereof duly signed by the manufacturer with each consignment. The Engineer-in-Charge shall record these details in Steel supply & Acceptance Register (as per proforma given herein-before) after due verification and send a certified true copy of test sheet to GE(I)/GE/AGE(I) for his records. The GE/ CWE shall also organize independent testing of random samples of steel drawn from various lots from National Test House, SEMT Wing CME, Regional Research Labs and NABL approved Labs etc as per the recommended minimum frequency shown in table as per given here-in-after. Samples from each lot should be tested for quality and elongation. The elongation shall be as admissible as per IS. Cost of samples, transportation and testing shall be borne by the contractor. The record of such checks would be maintained in the steel test register.

4.6.2 Testing shall not be insisted upon for steel given in Para 4.5.2.7 above.

4.6.3 The GE/CWE shall also organize independent testing of random samples of steel drawn from various lots from National Test House, SEMT Wing CME Pune, Regional Research Laboratories, Govt approved laboratories, Zonal laboratories as per the recommended minimum frequency as in Para 4.6.4 .In addition samples from each lot shall be tested for quality and elongation as per the parameters given in STEEL SUPPLY & ACCEPTANCE FORM. The elongation shall be as admissible as per IS. Cost of samples, transportation and testing shall be borne by the contractor.

4.6.3.1 Nominal mass of any size of finishes bar/section of steel shall be checked as specified in relevant IS code. The nominal mass so determined shall be recorded in steel testing register giving crossb reference to consignment number. Nominal mass of any size of finished bar/section of steel if found to be beyond the tolerance limits on minus side as specified in relevant IS code, the same shall be rejected and the contractor shall remove the same at his own cost without any extra cost to the Government. However, if the weight of steel section is beyond the tolerance limit on higher side the same can be provided with approval of the GE but without any extra cost to the Government.

4.6.3.2 If any Test Result of any particular size of Bar/Section of steel of any consignment is not found satisfactory as specified in relevant IS, the contractor shall remove the same at his own cost and no claim of contractor shall be entertained on this account.

Cont'd…

PARTICULAR SPECIFICATIONS (CONTD…)

4.6.4 **FREQUENCY FOR TESTS ON STEEL**

4.6.4.1 **REINFORCEMENT STEEL**

Nominal mass test, Tensile test, Bend and Rebend test shall be carried out as per minimum frequency given below: -

Nominal Size of Bars		Frequency
(a)	Bar size less than 10 mm	One sample (3 specimens) for each test for every 25 tonnes or part thereof.
(b)	Bar size 10 mm to 16 mm	One sample (3 specimens) for each test for every 35 tonnes or part thereof.
(c)	Bar size over 16 mm	One sample (3 specimens) for each test for every 45 tonnes or part thereof.

4.6.4.2 **STRUCTURAL STEEL:** Thetests shall be carried out as per the frequency mentioned below: -

Type of Test		**Frequency**
(a)	Tensile Test	One test for every 25 tonnes of steel or part thereof.
(b)	Bend Test	One test for every 25 tonnes of steel or part thereof.

NOTE:

1. **For various tests, Acceptance criteria, tolerance etc refer to Steel supply & Acceptance Form given here in before and relevant BIS Codes.**

2. **For steel items, for which, independent test is mandatory, as specified above, payment shall be released against these items only after satisfactory test report. Any item of steel, failing in test, shall be rejected and the particular consignment removed from the site by the contractor at his own cost.**

4.6.5 **DOCUMENTATION**:The contractor shall submit original purchase vouchers from the manufacturer for the total quantity of steel supplied under each consignment to be incorporated in the work. The GE along with the relevant documents before acceptance shall inspect all consignments received at the work site. The original Vouchers and the testy certificates shall be defaced by the Engineer-in-charge and kept on record in the office of the GE duly authenticated and with cross reference to the control number recorded in the Steel Acceptance Register. The Steel Acceptance Register will be signed by JE, Engineer-in-charge, GE and contractor. The entire quantity of all steel items shall be suitably recorded in the Measurement Book as not to be abstracted, before incorporation in the work and shall be signed by the Engineer-in-charge and the contractor.

4.7 **STORAGE ACCOUNTING, PRESERVATION AND MAINTENANCE OF STEEL**

4.7.1 The storage, accounting, preservation and maintenance of steel supplied by the contractor shall be done as per standard engineering practice till the same is incorporated in the work and the cost of the same shall be deemed to be included in the unit rate/amount quoted by the tenderer. The GE shall inspect at regular interval to verify that steel lying at site is stored, accounted, preserved and maintained as per the norms. The steel shall be stored so as to differentiate each consignment separately. If the GE is not satisfied with the storage / preservation of any size of bar/section of steel, he may order for any test(s) of steel as applicable for that size of bar/section of steel and as specified in tender documents /relevant IS code to recheck the acceptability criteria for the same. The contractor shall bear the cost of necessary testing(s) in this regard and no claim whatsoever shall be entertained.

4.7.2 Movement of steel shall be recorded in In/Out steel register as per following Performa. Each entry in the register shall be signed by the contractor and Engineer-in-Charge.

PARTICULAR SPECIFICATIONS (CONTD…)

IN/OUT STEEL REGISTER

Sl No	Date	Steel IN			Steel OUT			Qty Balance
		Qty (Tons)	Section	Control No	Qty (Tons)	Section	Reasons*	
1	2	3	4	5	6	7	8	9

*** Note**:

(i) The following reasons may be mentioned for taking out steel from storage: -
- (a) For testing purpose
- (b) For use in work
- (c) Rejected steel taken out of site

(ii) All the transaction in the register shall be signed by Contractor / his representative and Engineer-in-Charge/JE.

4.7.3 SCHEDULING AND SUPPLY: Schedule of supply of steel will be finalised by GE in consultation with contractor and the same will be incorporated in CPM chart so that supply of steel is monitored in a way to avoid any delay in completion of the work. The schedule of supply of steel will be vetted by Accepting Officer or Officer nominated by the Accepting Officer from time to time

4.8 **MEASUREMENTS AND PAYMENT OF STEEL**

4.8.1 The entire quantity of all steel items shall also be suitably recorded in the Measurement Book for record purposes as 'Not to be abstracted' before incorporation in the work and shall be signed by the Engineer-in-Charge and the contractor.

4.8.2 The nominal mass conversion factors for various steel Sections/ size of finished bars as given in relevant IS codes shall be considered standard for measurement.

4.8.3 The contractor shall plan to procure steel considering the time required for approval and testing of steel well in advance. No extension of time on grounds of delay in procurement of steel or testing/retesting of steel shall be admissible.

4.8.4 The payment of steel shall only be allowed after production of original purchase vouchers, test certificates by the contractor for each consignment of steel and results of testing carried out by the department are found satisfactory after testing as specified herein before.

4.8.5 **FABRICATION OF STEEL STRUCTURES:** The structural steel work shall be carried out all as specified in MES Schedule Part-10 and in drawings. The quality of fabrication and erection at site shall be got certified to the satisfaction of the department by any of the certifying bodies as approved by Accepting Officer.

4.9 **MILD STEEL HOLDFAST**

4.9.1 Mild Steel holdfast shall be of size as mentioned in the drawings. This shall be treated with two coats of coal tar and sanded before fixing in position, and shall be embedded in PCC Block (1:3:6) type C-1 of size 250 mm x 150 mm x thickness of wall unless specially shown otherwise in drawings.

4.10 **MILD STEEL GUARD BARS/OTHERS STEEL ITEMS**

4.10.1 Mild Steel guard bars shall be of 10mm x10mm square bars of Fe-310-0 or Fe-410-0 quality. Guard bars shall be provided to all windows & vents as shown on drawings. The guard bars shall be painted with 2 coats of synthetic enamel paints over a coat of red oxide zinc chrome primer.

4.10.2 Other steel items like steel door with PGI sheet covering, railing in balconies and stair case, water meter and switch board boxes etc shall be fabricated to the dimensions as shown on drawings as directed by Engineer-in-Charge. Welding, bolting, drilling holes etc shall be as specified in MES Schedule.

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PARTICULAR SPECIFICATIONS (CONTD…)

7. ROOF

7.1. RCC ROOF SLAB

(a) RCC roof slab shall be laid to a slope as shown on drawings.

(b) RCC roof slab shall be projected all as shown on drawings.

(c) Full bearing of RCC roof slab shall be provided on external walls.

(d) Exposed surfaces of soffit of RCC slab shall be plastered in CM (1:3) 5 mm thick and finished even and fair.

7.2 WATER PROOFING TREATMENT TO RCC ROOF

7.2.1 Unless otherwise specified, water proofing treatment to RCC roof shall be with non-woven polyester reinforcement and with mineral finish on top to be laid by torch application method. Technical characteristics (minimum) of the membrane shall be as under: -

Characteristic	Minimum requirement
(i) Weight	4 Kg/Sqm with mineral, minimum 3.5 Kg/Sqm of membrane weight.
(ii) Thickness	Minimum 3mm without considering mineral.
(iii) Softening point	Min 150 ^{0} C.
(iv) Cold flexibility	(-) 5 ^{0} C.
(v) Reinforcement	Non-woven polyester 160 gsm.
(vi) Tensile strength Length wise Cross wise	650N / Sqcm 350N/ Sqcm
(vii) Elongation Length wise Cross wise	> 30% > 30%

7.2.2 FOR NON ACCESSIBLE ROOF

7.2.2.1 Roof treatment shall be done all as specified here-in-below and as directed by Engineer-in-Charge: -

(a) 15 mm thick rendering in CM (1:4) shall be carried out to prepare the surface when the RCC of slab is green.

(b) The roof surface shall be thoroughly wire brushed, washed, dried up and cleaned of all foreign matter and dust prior to the application of primer.

(c) On the dried surface one coat of approved bituminous primer @ 0.40 litre per Sqm shall be applied.

(d) APP Membrane to be laid on primed surface by torch application method. The overlaps shall be of minimum 75 mm on longitudinal direction and 100 mm on the transverse direction.

(e) All joints and changes in direction should be curved with cement mortar (1:3) mixed with SBR latex @ 200 ml/ bag of cement and cured well before laying membrane by torching.

(f) All angles and abutments should be sealed well with extra care to ensure full bondage.

(g) Water proofing treatment shall be done by the approved authorized applicators of firm specified here-in-after and are certified by firms.

(h) The top surface shall be finished by a coat of aluminium paint @ 100 gms/sqm.

PARTICULAR SPECIFICATIONS (CONTD…)

7.2.3 **FOR ACCESSIBLE ROOF**

7.2.3.1 Roof treatment shall be done all as specified here-in-below and as directed by Engineer-in-Charge: -

(a) 15 mm thick rendering in CM (1:4) shall be carried out to prepare the surface when the RCC of slab is green.

(b) The roof surface shall be thoroughly wire brushed, washed, dried up and cleaned of all foreign matter and dust prior to the application of primer.

(c) On the dried surface one coat of approved bituminous primer @ 100 gms per Sqm shall be applied.

(d) Membrane to be laid on primed surface by torch application method. The overlaps shall be of minimum 75 mm on longitudinal direction and 100 mm on the transverse direction.

(e) All joints and changes in direction should be curved with cement mortar (1:3) mixed with SBR latex @ 200 ml/ bag of cement and cured well before laying membrane by torching.

(f) All angles and abutments should be sealed well with extra care to ensure full bondage.

(g) Membrane shall be covered with PCC tile of size 250x250x22 mm laid over 10 mm screed bed in CM (1:6) with WPC 3% by weight of cement all as per SSR clause No.13.16 and 13.29. Joint of tile shall be filled with cement slurry mixed with water proofing compound.

(h) Water proofing treatment shall be done by the approved authorized applicators of firm specified here-in-after and are certified by firms.

7.2.4 Contractor shall obtain guarantee certificate from the company immediately after completion of the work. This guarantee will be in addition to guarantee of water proofing treatment to be furnished by the contractor to the department.

7.2.5 Contractor shall submit original invoice and manufacturer's test certificate of membrane directly from company to the GE(I)/GE/AGE(I) before membrane is accepted. GE/AGE(I) shall compare the result with the tender specification and membrane shall be accepted if the results meet the technical characteristics specified here-in before.

7.2.6 The contractor shall furnish guarantee in favour of the GE/AGE(I) for efficiency of water proofing treatment during the guarantee period. The guarantee period shall be 8 (Eight) years from the certified date of completion of the work as per contract. Any leaking in the RCC roof noticed during this period shall be carried out by the contractor forthwith, on demand in writing from the GE specifying the roof of building(s) effected, not withstanding that the water proofing treatment to the roof might have been inadvertently passed, certified and paid for, the contractor shall carry out the rectification of the leakage, if any, in roof at his own expense during the guarantee period and in event of his failure to do within a period to be specified by the GE in his demand aforesaid, the GE shall undertake such rectification work at the risk and expense in all respect of the contractor.

7.2.7 An amount of security deposit equal to the individual security deposit calculated, based on the amount of water proofing treatment at contract rates for the buildings for which treatment to be carried out shall be retained out of the final bill amount towards security deposit for water proofing treatment and will be refunded to the contractor after expiry of the period of eight years from the certified date of completion by the GE. The contractor may alternatively furnish a fixed deposit receipt in favour of GE in lieu of sum to be retained as security deposit for water proofing treatment from any approved Scheduled bank.

PARTICULAR SPECIFICATIONS (CONTD…)

8. **FLOORING & FLOOR FINISHES**

8.1 **GENERAL**

8.1.1 Refer Clause No. 13.2, 13.4, 13.5, 13.8, 13.14, 13.21, 13.23, 13.24, 13.25, 13.27, 13.32, 13.38, 13.39, 13.40, 13.46 and 13.47 of MES Schedule Part-I are to be adopted for Materials and Workmanship of laying Floor/ Skirting/Dado.

8.1.2 Floors shall be laid to levels or to falls as shown on drawings and as directed by Engineer-in-Charge. Floor finish shall be extended over dwarf wall, doors and other opening.

8.1.3 Steps and connected ramps shall have the same type of floor as adjoining floors of building unless otherwise specifically mentioned and shall have chequered finish in addition.

8.1.4 Where two different types of floors finish are provided on either side of the opening the superior floor finish shall be extended over the opening.

8.1.5 Finishing layer of floor shall be carried out through all opening and over dwarf wall.

8.1.6 PCC floor shall be finished even and smooth using extra cement unless otherwise specifically mentioned in drawings of schedule of finishes or here-in-after.

8.1.7 Polythene film shall conform to IS-2508, 150 micron thick of 01.0 Grade. If full width of polythene film is not available, the same shall be laid with an overlap of 150mm pasted with the resin solution.

8.1.8 Sub floor may not be laid in panels. Under layer and toping layer of casting PCC/terrazzo cast-in-situ floor shall be laid in square or rectangular panel as directed by Engineer-in-Charge. Length of panel shall not exceed 2.0 Metre. Length width ratio shall not exceed 1.5 times in rectangular panels. However, area of each panel shall not exceed 2.0 Square Metre as specified. Cement concrete in sub base or sub floor and top wearing coat shall be laid separately and not monolithically.

8.1.9 Polished Kota stone shall be of machine cut polished and thickness shall be all as mentioned in the schedule of finishes. The stone slab shall be hard, even, sound, durable, tough, and regular in shape of shade as approved.

8.2. **BLANK**

8.3. **NON-SKID COLOURED CERAMIC TILE FLOORING**

8.3.1 Provide non skid ceramic tiles as per the specification indicated in schedule of finish. In case specification not mentioned in schedule of finish the area of each tile 400x400 mm laid over 20mm thick screed in cement mortar 1:4 over 40mm thick PCC 1:2:4 type B-0 over 75mm thick sub-base with PCC 1:3:6 type C-2 over rammed earth. Tiles shall be set in neat cement slurry and shall be pointed and jointed in white cement mixed with pigment/colour cement matching with the shade of tiles. Tiling shall be done as specified in Clause 13.40 of SSR Part-I. Floor tiles shall be Grade B-IIa as per IS 13712.

8.3.2 In first and second floor tile shall be laid over 20mm thick screed in cement mortar 1:4 over RCC slab.

8.4. **GLAZED COLOURED CERAMIC TILE IN SKIRTING/DADO**.

8.4.1 Glazed Ceramic tiles in skirting/dado shall be provided as per specification mentioned in schedule of finishes. In case specification not mentioned in schedule of finish the same shall be of 6 to 7 mm thick glossy finish with minimum size 450x350mm. Tiles shall be laid over 10mm thick screed in cement mortar 1:3. Tiles shall be set in neat cement slurry and shall be pointed and jointed in white cement mixed with pigment/colour cement matching with the shade of tiles. Dado shall be provided upto the lintel height. Height of skirting shall be 10cm. Tiling shall be done as specified in Clause 13.40 of SSR Part-I. Colour of tiles as approved by the GE. Wall tiles shall be Grade BIII as per IS 13712.

PARTICULAR SPECIFICATIONS (CONTD…)

8.5 **KOTA STONE FLOORING AND SKIRTING**

8.5.1 Kota stone in skirting/dado shall be provided as per specification mentioned in schedule of finishes. In case specification not mentioned in schedule of finish than the Kota stone of 20-22mm thick minimum shall be provided over 20mm thick screed bed in Cement Mortar 1:6 at the top surface of steps as shown on drawings with bull nose finished. The kotastone shall be provided in single piece with round edge nosing and shall be pointed and jointed in white cement mixed with pigment/colour cement matching with the shade of stone. Single piece kotastone shall be used for treads & risers of steps. Width of kotastone for tread/riser shall be as indicated in drawings.

8.6 **PCC FLOORING :-** 50mm thick PCC 1:2:4 laid in staggered panels of linear dimension exc. 1.0 Sqm divided by 4mm thick glass strips equal to thickness of floor finished even and fair without using extra cement over 100mm hardcore over well compacted earth.

8.7 BLANK

8.8 BLANK

8.9 BLANK

8.10& 8.11 **BLANK**

8.12 **PCC PLINTH PROTECTION WITH DRAIN**

8.12.1. Plinth protection with drain shall be provided as shown on drawings. If not shown on drawings, PCC plinth protection shall be provided with 75 mm thick PCC 1 : 3 : 6 type C-1 over 100 mm thick (consolidated thickness) hard core of broken stone aggregate over rammed earth. The surface of plinth protection shall be finished even and smooth without using extra cement. The concrete in plinth protection shall be laid in alternate bays of length n.exc 3 m and shall be laid in slope 1 in 12. The joints shall be 12 mm thick filled with bitumen mastic.

PARTICULAR SPECIFICATIONS (CONTD…)

9. **PLASTERING AND POINTING**

9.1. **CEMENT PLASTER**

9.1.1. **GENERAL**

9.1.1.1 For sand to be used in cement plaster, refer paras 14.5, 14.5.1 and 14.5.2 of MES Schedule Part-I

9.1.1.2 The thickness of plaster mentioned here-in-after is finished thickness exclusive of dubbing out. Plastering in one coat/first coat as applicable and dubbing out shall however, be done in one operation.

9.1.1.3 The external and internal finishes (Plastering) shall be returned in the jambs of the openings upto frames and taken around cupboards and similar projections.

9.2.1 **INTERNAL PLASTER**:-Internal plastering shall be provided as per specification mentioned in schedule of finishes. In case specification not mentioned in schedule of finish than the Internal surfaces of walls including internal surfaces shall be plastered with 10 mm thick in cement mortar (1:6) finished the surface even and fair.

9.2.2. **EXTERNAL PLASTER**:-External plastering shall be provided as per specification mentioned in schedule of finishes. In case specification not mentioned in schedule of finish than theexternal surfaces of walls shown shall be plastered with 15 mm thick CM in two layers of 10 mm thick rendering coat and 5 mm thick setting coat in cement mortar (1:4). The base layer to be finished rough with scouring mark to receive the second layer which will be applied after not less than a gap of two days over the first one with water proofing compound as per manufacturer's instruction. For the purpose of pricing deviation, the percentage of water proofing compound shall be taken as 3% by weight of cement. The plaster shall be continued to the exposed surfaces of RCC column., beams etc. which are continuous with plastered surfaces of walls. All external plaster shall be taken up to 15 cm below GL. The finishing of External plaster shall be of even and smooth without using extra cement. Provide plaster groove as per the details given in drawing.

9.2.3 **CEILING**:-Exposed surfaces of Ceiling shall be plastered with cement mortar (1:3), 5 mm thick finished the surface even and smooth without using extra cement.

9.2.4 **BLANK**

10. **FINISHES**

10.1 **BLANK**

10.2 **WHITE WASHING**

10.2.1 Three coats of white wash shall be provided at all the locations as indicated in drawings. White wash shall be provided all as specified in clause 15.12 of SSR Part-I.

10.3 **OIL BOUND DISTEMPER :-**

10.3.2 Two coats of OBD over a coat of primer shall be provided at all the internal locations as indicated on drawings. OBD, Distempering shall be done all as specified in clause 15.14 of SSR Part-I.

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10.2 **EXTERNAL ACRYLIC BASED ANTI FUNGAL WEATHER PROOF PAINT:-**

10.2.1 GENERAL: For paints and allied materials refer para 17.2 of MES Schedule part-I.

10.2.2 Water based acrylic weather proof exterior emulsion paint and primer for the same shall be of premium quality and the make is approved by GE and shade of colour shall also be approved by GE.

10.2.3 The paint shall be 100% acrylic and semi acrylic paint shall not be used in the work.

10.2.4 **PREPARATION OF SURFACES**

10.2.4.1 The surface to be applied with acrylic weather proof emulsion paint shall be cleaned to remove loose dirt, lichen, algae, fungi or any organic growth by use of stiff brush. Then the surface is washed well and allow water to dry .All cracks, voids and minor damages shall be patched/repaired prior to application of paint over primer and wall care putty.

PARTICULAR SPECIFICATIONS (CONTD…)

10.2.5 **Primer coat**- A coat of acrylic primer as per manufacturer's instruction. This shall be allowed to dry for 4 to 6 hours before application of final paint.

10.2.6 **Finishing Coat**- Acrylic paint of premium quality 100% acrylic shall be applied in at least two coats as per manufacturer's instruction. The acrylic paint shall be prepared as per manufacturer's instruction by adding water in proportions as suggested. No other thinner shall be used. Paint shall be applied with brushes or roller and shall be allowed to dry for 3 to 4 hours between subsequent coats.

10.2.7 Before application of primer for plastered surfaces, the surfaces shall be cleaned, made free of loose particles, dust, grease, chalk, fungus and mould. One coat of acrylic emulsion weather coat paint with water (1:1 by volume) as a self priming coat shall be applied as per manufacturer's instructions and as directed by Engineer-in-Charge.

10.2.8 The acrylic emulsion weather coat paint shall be applied by brush or roller. No strainer or colorants shall be used. The paint shall be stirred well before use. The primer coat shall not be left without application of top coats for a long period of time.

10.2.9 The paint shall be as per manufacture's original colour as available no mix of tint shall be made into original shades. The primer and paint proposed to be used at site shall be from the same manufacturers. The mixing of different brand of primer and paint will not be allowed at site.

10.2.10 The work shall be executed through closed supervision of the respective manufacturer's supervisor of the paint proposed to be used at site for better quality control. However, the contractor does not absolve from his responsibility in respect of this specialized work as per contract conditions. The Contractor shall be responsible to ensure the proper quality control of the work and entire completion of work to the full satisfaction of GE.

10.2.11 **GUARANTEE**

10.2.11.1 The Contractor shall ensure that all the buildings painted with acrylic paint free from fungal infections, light patches, discolouring, etc. during the guarantee period of 07years and he shall be responsible to upkeep the entire surface for a period of 07 years from the certified date of completion of work. An amount of security deposit equal to 5% on the amount of treatment (acrylic painting) at contract subject to minimum of Rs. **50,000** (Rupees fifty thousand only) towards the guarantee for efficiency of treatment (acrylic painting) shall be retained from the RAR/Final bill amount. This amount shall be released to the contractor after satisfactory expiry of Seven years guarantee period. The GE may accept BGB from schedule bank or a fixed deposit receipt (pledged in GE's favour) from any Nationalized Bank for the said guarantee amount for a period of Seven years, in which event no further amount will be recovered from the final bill/RAR on this account. The defects liability period under condition 46 of General conditions of contract (IAFW-2249) shall be deemed to be amended to the extent mentioned above for acrylic painting works. If any defects observed during the guarantee period, the same shall be rectified by the contractor without any extra cost to the Dept.

10.2.11.2 Necessary guarantee shall be obtained from the manufacturer applicator on non judicial stamp paper giving seven years guarantee from the certified date of completion of contract to keep the building free from fading, peeling off paint during guarantee period. The guarantee does not however, absolve the Contractor from his responsibility in respect of this specialized work as per contract conditions. The Contractor shall be responsible to ensure that the buildings are kept free from fading or peeling off the acrylic paint during guarantee period.

10.2.12 **SCAFFOLDING**

10.2.12.1 The exterior painting works shall be carried out by using scaffolding. No jhoola is permitted for the work under any circumstances. Suitable protection shall be provided for workmen. The contractor shall be solely responsible for all the labours deployed for the work for identity/security. In case of any accident/injury fatal or partial disability, the contractor shall be solely responsible for settling all claims, compensation. Department shall have all rights to recover any sum indicated/claimed by labour commissioner/court directives. Engineer-in-charge will have to ensure that contractor has provided all adequate and required means for the workmen and contractor may get his personnel insured as per labour law.

PARTICULAR SPECIFICATIONS (CONTD…)

10.2.12.2 Scaffolding or staging more than 3.5m above the ground or floor, swung or suspended from any overhead support or erected with stationery support shall have a guard rail properly attached, braced and otherwise secured at least 1m high above the floor or platform of such scaffolding or staging and extending along the entire length of the outside and ends thereof with only such opening as may be necessary for the delivery of materials. Such scaffolding or staging shall be so fastened as to prevent it from swaying from the building or structure.

10.2.12.3 Working platform, gangways and stairways should be so constructed that they should not sag unduly or unequally, and if the height of the platform or the Gangway of the stairway is more than 3.5m above ground level they should be closely boarded, should have adequate width and should be suitably fenced, as described herein before.

10.2.12.4 Every opening in the floor of a building or in a working platform be provided with suitable means to prevent the fall of persons or materials by providing suitable fencing or railing whose minimum height shall be 1.0m.

10.2.12.5 Safe means of access shall be provided to all working platforms and other working places.

10.2.12.6 The rates quoted are deemed to included the above provision of scaffolding and no extra will be payable to contractor on this account. The scaffolding shall be removed only after obtaining clearance of Engineer-in-Charge/Garrison Engineer after considering the quality of the work undertaken.

10.2.13 **STAGEWISE APPLICATION FOR ACRYLIC EMULSION WEATHER COAT PAINT**

10.2.13.1 Work for acrylic emulsion paint shall be carried out as per stage given below. Before starting the work for next stage prior approval of Engineer-in-charge shall be obtained.

Stage I : Preparation of surface.
Stage II : Applying priming coat
Stage III : Applying first coat of weather coat acrylic emulsion paint
Stage IV :Applying second coat of weather coat acrylic Emulsion paint

10. **PAINTING (ETC)**

10.1 **GENERAL**:

(a) For paints and allied materials refer para 17.2 of MES Schedule part-I.

(b) Synthetic enamel paint shall be of 1st grade quality and of any of the makes as approved by GE(I)/GE/ AGE(I).

10.2 **WORKMANSHIP**:

(a) For painting works generally refer para 17.3 of MES Schedule Part-I.

(b) In case under coat is not done within 6 months of application of primer, the primer coat shall be redone without any extra cost. Priming coat shall be done to doors and windows surfaces etc, irrespective of the fact that these surfaces are already having shop coat of painting.

10.3 **PAINTING TO STEEL AND IRON WORK**:

(a) GENERAL: Refer para 17.8 of MES schedule part-I.

(b) All steel work exposed to view, other than galvanised surfaces, shall be treated with one finishing coat and one under coat of readymade synthetic enamel paint over a coat of red oxide primer as per clause 17.8 of MES Schedule Part-I.

(c) Surfaces of structural steel viz trusses, purlins, rafters and bracings etc which are hidden under ceiling shall be given two coats of red oxide primer, one coat before fixing and other coat after fixing. Hold fasts shall be tarred in two coats and sanded.

(d) Reinforcement shall be left untreated.

(e) Timber and steel (other than reinforcement) surfaces embedded in earth, concrete or wall/plaster shall be applied with two coats of coal tar. Tarred surfaces shall be sanded before use.

(f) Hidden wood work in ceiling shall be treated with two coats of creosote oil.

PARTICULAR SPECIFICATIONS (CONTD…)

(g) Unless otherwise specified here-in-after, wood work and joinery including particle board/plywood surfaces exposed to view shall be treated with one under coat and one finishing coat of synthetic enamel paint over one coat of pink primer as per clause 17.6 of MES Schedule Part-I.

(h) Priming coat shall be applied to surfaces of steel/wood work before fixing in position and under coat and finishing coat after fixing in position. Surface inaccessible for applying further coats shall be painted before fixing.

(i) Fly proof wire mesh shall be 0.48mm nominal dia and average width of aperture shall be 1.40mm.

(j) Fly proofing of GI wire mesh shall be treated with a coat of synthetic aluminium paint including preparation of surfaces.
NOTE :- Surface of steel and wood work except that of particle board shall be prepared before painting/polishing.

11. **SUNDRIES ITEMS**

11.1 MS RUNGS: Mild Steel rungs shall be provided as per details shown on drawings. Portion of rungs embedded in masonry wall shall be hot tarred. MS rungs 16 mm dia tor/round bar shall be provided as directed by Engineer-in-Charge at spacing as shown on drawings for climbing up to roof top for maintenance of roof, water supply lines, overhead service water tank etc. The rate for rungs shall deemed to be included in the lump sum amount quoted against buildings.

11.2 **MILD STEEL RAILING**:

Mild steel railings shall be provided all as shown on drawings. The size and type of sections shall be all as shown on drawings. MS railing shall be painted with two coats of synthetic enamel paint over a coat of red oxide primer.

11.3 **STAINLESS STEEL RAILING** :-

Stainless steel railing shall be provided at locations and as per the details shown on drawings. Stainless steel hand rail, verticals and horizontal sections shall be of medium grade stainless steel (SS -304 grade) and of size as indicated in drawing.

11.4 **PEG SET**:

Provide peg set with number of pegs as and where indicated in drawings. The pegs shall be as shown on drawing and of aluminium.

11.5 **CURTAIN ROD**:-

Curtain rod shall be provided at locations all as shown in drawing. Curtain rod shall be aluminium and dia. shall be as per drawing.

11.6 **CUP BOARD :-**

Cup board shall be provided at locations all as shown in drawing.

11.7 **PLAIN MILD STEEL DRYING WIRE/ MOSQUITO NET WIRE :-**

Plain mild steel drying wire/mosquito net wire shall be provided at locations all as shown in drawing.The diameter of plain mild steel wire shall be 4 mm dia., class 3 and shall be fixed using standered straining clamp complete all as shown in drawing.

PARTICULAR SPECIFICATIONS (CONTD…)

11.8 **FIRST MANHOLE** (Size 90cm x 90cm x 90cm): The manhole shall be of the following specifications: -

Excavation and earthwork	Soft/loose soil. Surplus soil shall be removed to a distance n.exc 50m and deposited as directed by Engg-in-Charge.
PCC foundation	PCC (1:3:6) type C2.
Walling	Brick work in CM (1:4).
Internal Plastering	15mm thick in CM (1:3) mixed with integral water-proofing compound as per manufacturer's instruction and surface finished even and smooth without using extra cement
External Plastering	15mm thick in CM (1:6) from top upto 150mm below GL.
Benching	PCC (1:2:4) type B1.
RCC cover slab	-M-25 (design mix) and TMT bars as reinforcement.
Rest	all as per drawings.

11.9 **HDPE WATER TANK**: Rotational moulded water storage tank double layered wall construction ISI marked seamless perfection made of stabilized food grade polyethylene to ensure resistant to weather, temperature, corrosion and leakage including ball valve and shall be fixed on roof slab over PCC platform as shown on drawings and as directed. The water tank shall be provided all as per manufacturer's instructions. The overflow pipe shall have perforated polythene cap screwed to one end. Washout pipe shall be provided with GI plug. Ball valve shall be of brass 25 mm bore high pressure, with polythene float and fly nut. The inlet, outlet, overflow and wash out pipe shall be measured under respective item of Schedule 'A'. Capacity of water tank shall be as indicated in the drawings.

11.10 **WPC**: Integral water proofing compound (WPC) shall conform to IS -2645 and the brand shall be got approved in writing from the GE(I) / GE /AGE(I). Preparation of mix and method of mixing shall be as per manufacturer's instructions. While pricing deviation for omission the quantity shall be considered as 3% by weight of cement.

11.11 **CRUMPLE JOINT**: Crumple joints shall be provided all as shown in drawings.

11.12 **SPOUT**: Provide spout all as shown on drawings.

PARTICULAR SPECIFICATIONS (CONTD…)

12. **SANITARY APPLIANCES**

12.1 All vitreous china sanitary appliances, shall be of white colour as approved by GE unless otherwise is mentioned elsewhere.

12.2 **WASH HAND BASIN (WHB)**: Refer Para 18.32 and 18.89 of SSR Part-I 2009. Wash hand basin shall consist of the following:-

(a) Where WHB is shown shall be of size 550 x400mm. Make shall be as per list of approved make.

(b) Rubber plug and chromium plated chain.

(c) 32mm chromium plated waste fittings with PVC waste pipe.

(d) Bottle trap shall be chromium plated conforming to IS – 5219 and also refer to Para 18.22.1 of MES Sch Part-I. Whether shown on drawings or not each WHB shall be provided with one number bottle trap.

(e) A pair of cast iron spout brackets.

(f) Polythene connecting pipe of 300 mm long including brass union with nuts and bolts both sides.

12.3 **COUNTER TYPE WHB**: Provide vitreous china coloured oval shape wash hand basin (560 x 450mm) with granite slab, counter top duly made to shape for seating the basin, 18 mm to 20 mm thick with top polished and edges bull nosed smooth rounded. Colour of Granite slab and WHB shall be as approved by GE. The accessories for WHB shall be as specified here-in-before. Fixing of WHB and granite slab shall be as directed. Full length mirror to a height of 600 mm shall be provided to WHB. The granite slab shall be placed over brick partition at suitable intervals duly plastered and finished matching adjoining surfaces.

12.4 **MIRROR**: Looking mirror shall be of selected quality 5.5 mm thick and it shall be mounted on 6 mm thick, AC plain building board with all round teak wood bead frame of size 25 x 12mm with polished with varnishing and shall be embedded in wall. Each WHB shall be provided with one mirror and size of the mirror shall be 450mm x 600mm.

12.5 **INDIAN TYPE WC** : Irrespective of whatever indicated on drawings all Indian type WCs shall be first quality, vitreous china, coloured, earthenware pan of size 580 x 440mm and shall be provided with one dual mode flushing cistern complete with all fittings and flushing pipe etc, 100 mm dia ‘P’ or ‘S’ trap and a pair of vitreous china foot rest (ISI marked). Fixing of pan, flushing cistern and foot rest shall be all as specified in clause No 18.86 of SSR Part-I.

12.6 **EUROPEAN TYPE WC (EWC)**:EWC shall be first quality, vitreous china, coloured, Earthen ware pan with integral ‘P’ trap, seat cover etc and shall be of Make as per list of approved makes, dual mode flushing cistern shall also be of Make as per list of approved makes and fittings shall be of as per list of approved makes. EWC along with seat cover and dual mode flushing cistern shall be complete and shall be fixed as specified in clause 18.87 of SSR Part-I.

12.7 **TOILET PAPER HOLDERS**: Toilet paper holder shall be as per clause No 18.92 of SSR-Part-I and it shall be Cat No. BA-88 of Prayag or equivalent from the approved make list. Each EWC shall be provided with the toilet paper holder. Each EWC & IWC shall be provided with the toilet paper holder.

12.8 **HEALTH FAUCET** : Provide health faucet of Cat No. BEA-41 of Prayag or equivalent from the approved make list. Each EWC & IWC shall be provided with the Heath Faucet.

PARTICULAR SPECIFICATIONS (CONTD…)

12.9 **SOAP NICHE/SOAP TRAY**: Irrespective of whether shown on drawings or not one soap tray shall be provided along with each WHB. Soap tray shall be of Cat No. BA-25 of Prayag or equivalent make given in the list of makes.

12.10 **TOWEL RAIL**: Towel rail shall be as per clause 9.22 of of MES Sch Part-I. Towel rail shall be of Cat No. BA-20 of Prayag or equivalent make given in the list of makes.

12.11 **LOW LEVEL FLUSHING CISTERN**: Each WC/EWC/IWC shall be provided with PVC low level dual mode flushing cistern of 10 lit capacity complete along with 32 mm/40mm flushing PVC pipe and PVC supply pipe 15mm bore. It shall be of Cat No. 502 (DUO FLUSH) of Prayag or equivalent make given in the list of makes.

12.12 **URINALS**: Urinal shall be of size 635 x 395 x 420 mm or equivalent. Each urinal shall be provided with chromium plated flushing pipe and brass CP urinal spreader etc complete. It shall be of Make as per list of approved makes.

12.13 **PARTITION WALL IN URINAL**: Irrespective whatever shown on drawing, Granite slab (Pre-polished) partition shall be provided all as shown in drawing.

12.14 **ANGLO INDIAN WC**: Anglo Indian WC shall be first quality, vitreous china, coloured, earthenware pan with integral 'P'. trap, seat cover etc and flushing cistern fittings shall be of Make as per list of approved makes. WC along with seat cover and flushing cistern shall be complete and shall be fixed as specified in clause 18.87 of SSR Part-I.

12.15 **BLANK**

12.16 **SOIL, WASTE, VENT PIPES AND FITTINGS**: These shall be of ISI marked as approved by GE and shall be provided as follows:-

(a) Soil waste, vent pipes and fittings shall be 100mm bore.

(b) All pipes and fittings shall be cast iron conforming to IS: 3989 and bearing ISI marked in following locations:-

(i) Inside buildings and through walls.

(ii) Below plinth level including 30 cm below ground level including duck foot bend and 3.0 m beyond outer face of external walls or upto first man hole. However, in case of soil/waste pipes running in shaft, these pipes shall be connected to man hole inside the shaft. From the man hole, 100 mm dia SWG pipe line shall be provided instead of running each soil pipe beyond outer face of external wall. Waste pipe shall be of cast iron upto gully trap.

(iii) In other locations as mentioned here-in-after.

(c) Waste pipes from sanitary fittings/ground sink upto nahani trap shall be 40 mm bore galvanised steel tube medium grade. Waste pipes from nahani/floor trap upto gully trap including necessary fittings etc shall be of cast iron.

(d) Cast iron vent pipes shall be 1.00 m above top of roof with necessary slotted vent cowls.

(e) Cast iron soil and waste pipes and fittings shall be centrifugally cast conforming to IS: 3989 and bearing ISI marked. All the soil and vent pipes along the walls shall be without ears and shall be fixed to wall with MS flat iron clamps of suitable size as directed by Engineer-in-Charge at suitable spacing.

(f) Joints between cast iron to cast iron pipes and fittings shall be with run lead caulked joints.

(g) Whether indicated on drawings or not, soil pipes to take salvage from water closets and waste pipes to take water from nahani traps shall be provided all as specified here-in-before.

PARTICULAR SPECIFICATIONS (CONTD…)

(h) Accessories such as bends branch pieces (single/ double) etc shall be provided as required. All exposed accessories above ground level shall have oval and access doors (Standard Pattern) to enable access to each straight section of the joints.

(i) Each and every joints of soil/ waste pipe getting embedded in the floor/walls shall be tested with smoke test as laid down in MES Schedule Part-I, in presence of Engineer-in-Charge and he will certify that the joints are satisfactory and no leakage is noticed. A copy of this certificate shall be forwarded to the GE for record.

(j) Lump sum tendered by the contractor for the buildings shall be deemed to include the cost of soil, waste and vent pipe and fittings as mentioned here-in-before.

12.17 **NAHANI TRAP**: Refer para 18.25 of MES Sch Part-I. Nahani trap shall have internal dia of minimum 75mm, long body conforming to IS: 153989 and bearing ISI marked.

12.18 **FLOOR TRAP**: Refer para 18.26 of MES Sch Part-I. Floor trap shall have internal dia of minimum 75mm, long body conforming to IS: 153989 and bearing ISI marked.

12.19 **GULLY TRAP** : Salt glazed stone ware gully trap shall be with rebated top square body type P with 15 cm x 10 cm cast iron grating and 100 mm bore outlet. Trap shall be set in cement concrete foundation of size 25.5 cm x 53.5 cm and 30 cm x 30 cm concrete hopper with 75 mm thick rebated concrete curb with 32.5 cm x 32.5 cm cast iron grating jointing of gully trap with drain pipe shall be in CM (1:3).

12.20 **RAIN WATER PIPE(RWP)**: RWP shall be of HDPE and shall be of 160 mm dia(irrespective of whatsoever show on drawing) and shall be fixed to the locations as indicated in drawing and as directed by the Engr-in-Charge.

13. **BUILDER'S HARDWARE**

13.1 **GENERAL**: Irrespective of what is shown on drawings all articles of builder's hardware except butt hinges shall be of aluminium anodised matt finish. Size shall be as given on drawings/ Schedule of fittings. All aluminium builder's hardware shall be fixed with cadmium coated steel screws.

13.2 **TOWER BOLT**: Refer para 9.3 of MES Schedule part-I. The diameter of bolts shall be 10 mm and the bolts shall be barrel bolts.

13.3 **BUTT HINGES**: Refer para 9.7.2 of MES Schedule part- I. Butt hinges shall be of pressed steel/cold rolled mild steel medium weight. Butt hinges for PVC door shall be stainless steel and shall be fixed with cadmium coated steel screws.

13.4 Ball catch shall be of cast brass all as specified in para 9.14 of SSR part-I.

13.5 All fittings shown on schedule of fitting/drawings shall be provided whether the same have been specified or not.

13.6 All builder's hardware shall be ISI marked except bow handles which shall conform to relevant IS.

13.7 **HOLD FAST**:

(a) Flat iron hold fasts/lugs shall be provided by welding as and where shown on drawings.

(b) Irrespective of what is shown on drawings, flat iron hold fasts shall be embedded in PCC (1:3:6) type C-1, block of size 15 cm (height of block) x 30 cm (length of block) x width of masonry wall.

(c) Hold fasts shall be hot tarred and sanded before fixing.

13.8 **FAN HOOKS**:

(a) Cast iron box with hook shown in drawings shall be provided in ground floor where RCC roof is provided. Fixing / hanging arrangement of fan in ground floor / first floor, where CGI / Galvalume sheet roof is provided, shall be with GI pipe all as shown in drawings, wherever ceiling fan/fan hook is indicated in drawings.

PARTICULAR SPECIFICATIONS (CONTD…)

(b) The contractor may at his discretion provide mild steel box fabricated out of 1.40 mm thick mild steel sheet in lieu of cast-iron box without any price adjustment.

(c) Laminated white circular sheets shall be fixed to cover metallic boxes provided for fans & pendent lights in the ceiling so that no portion of metallic boxes is visible. The cost of fan hook with CI box as specified above in RCC slab and fixing / hanging arrangement with GI pipe in case of CGI sheet roof shall be deemed to be included in the cost tendered by the contractor for buildings under Schedule "A' Part–I.

14 **PVC DOOR FRAME AND PVC DOOR SHUTTER**

14.1 Provide PVC door as per the details given in drawings. In case details are not given in drawings all toilet/WC doors shall be of PVC door, the door and frame shall be provided as per the specifications mentioned here-in-after. Size of the door shall be 2100mm height and 800mm width).

14.2 **DOOR FRAME**: These shall be all as specified in Item No. 08095 of SSR Part-II-2020.

14.3 **DOOR SHUTTER**: These shall be all as specified in Item No. 08104 of SSR Part-II-2020 and shall be with particle board core.

15. **ALUMINIUM DOORS/WINDOWS/VENTILATORS**

15.1 Provide powder coated aluminium doors/windows/ventilators/partitions all as shown on drawings and as specified in clause No.10.37 of SSR part I and and shall be fixed as per manufacturer's instructions. Aluminium doors, windows, ventilators and partitions shall be of powder coated . The doors, windows, ventilators shall be aluminium sections as shown in drawings and sketches mentioned above including necessary jointing cleats, glazing clips, rubber packing, aluminium snap beading, screws, anodized aluminium fittings of suitable sizes such as peg stays, hinges, handles etc as approved by GE. In the event of non availability of any section given in drawings, alternative equivalent or higher sections thickness not less than the thickness indicated shall be provided by the contractor with prior approval of the GE without any extra cost to Govt. on this account.

15.2 The shutters shall be made out of specially extruded tubular sections with provision for weather stripping made in the vertical jambs. The cleats for mechanical horizontal/vertical joints of the fixed frame and shutters shall be specially extruded aluminium sections so as to avoid any play between jointed members.

15.3 The active leaf shall have unity lock. The active leaf shall also have one concealed sheet bolt which can be operated from inside.

15.4 For fixing glazing, the glass shall be encased in EPDM gaskets so as to avoid direct contact between metal and glass.

15.5 The aluminium sections shall be coated with epoxy polyester powder coating. Thickness of coating shall be 50 microns Colour of coating shall be as approved by GE. Chemical and mechanical properties of sections shall comply with requirements given in IS 733-1983 and as specified in Clause No. 10.37(a) to 10.37 (g) of MES Schedule Part–I.

15.6 Glazing shall be provided all as shown in drawings, wherever thickness of glazing not indicated shall be 5mm thick.

15.7 All aluminium windows and ventilators shall be provided with steel grills all as shown on drawings. Steel grills shall be painted with two coats of synthetic enamel paint over a coat of zinc chrome primer. Colour/tint of paint shall be as approved by GE.

15.8 **PEG SET**: Provide peg set with number of pegs as and where indicated in drawings. The pegs shall be as shown on drawing and of aluminium.

PARTICULAR SPECIFICATIONS (CONTD…)

16 to18. **BLANK**

19 WOOD WORK (CARPENTER'S AND JOINERY)

19.1. **TIMBER** :- Timber to be incorporated for the work shall be well seasoned, free from defects, moisture content not exceeding the limits laid down in IS 287-1973 all as mentioned in paras 7.3 to 7.7 of MES Schedule Part-I before timber is given preservative treatment.

19.1.1. Unless otherwise indicated here-in-after, specifications for various timber/wood work shall be as under:-

19.1.1.1 Timber for all wood work and joinery (except particle board) shall be of second class hard wood of any of the following species:-

(a) HOLLOCK (b) KOKKO (c) CHAMPAK (d) GAMARI
(e) CHAPLASH (f) KALASIRISH (g) BONSUM

19.1.1.2 Strength factor for timber shall be 70 and above. Durability shall be class I or II with any treatability class.

19.1.2. **TOLERANCE**

19.1.2.1. Tolerance in the dimensions shown on drawings for the wrought faces of carpenter's work, joinery etc shall be permissible as specified in MES Standard Schedule Part-I.

19.1.3. **SEASONING**

19.1.3.1. All timbers shall be kiln season by suitable process conforming to IS-1141-1973 (Code of practice seasoning of timber) to maintain the moisture content as permissible tabulated in clause 7.7 of SSR Part-I (2009) and in IS-287-1973 (recommendations of maximum permissible moisture content for timber used for different purpose). Seasoning shall be carried out in two stage i.e. to remove partial moisture content before treatment and then to maintain desired percentage of moisture content after treatment.

19.2 **PARTICLE BOARD**

19.2.1 **UNVENEERED PARTICLE BOARDS** :-This shall be flat pressed three layers type conforming to IS-3087 for general purpose. Adhesive used for bonding purpose shall be BWP type (phenol bonded) exterior grade.

19.2.2. **VENEERED PARTICLE BOARDS**:- Veneered particle board used for paneling of door shutters shall be solid core Exterior grade conforming to IS-3097-1980 manufactured from natural wood and further veneered on both sides with strong veneers commercial or decorative as the case may be for the surface, bonded with water proof high quality phenol formaldehyde synthetic resin adhesive (BWP type) conforming to IS-848-1979 and hot pressed at high pressure and temperature conforming to IS-3097-1980 (specification for veneered particle boards). Veneered particle board shall be three layered flat pressed and type I (commercial) or type 2 (decorative) as indicated in drawings.

NOTE :- 19 mm thick particle board shall mean 18/19/20mm thick and no price adjustment shall be made if particle boards of 18/19/20 mm are used.

PARTICULAR SPECIFICATIONS (CONTD…)

19.3 **GLUE**

19.3.1 Adhesive and glues for putting together the joints of the door shutters shall be as per IS-1003-1991 (Part I) and may be synthetic adhesive (BWP type) conforming to IS-848-1974 or IS-852-1969, as required.

19.4. **WORKMANSHIP**

19.4.1 **CARPENTER'S WORK/JOINERY ETC GENERALLY**

19.4.1.1 Dimensions of various parts of joinery viz. Panelled joinery or gauged or glazed joinery shown on drawings shall supersede those stipulated in MES Schedule part-II. However, rates of joinery as given in MES Schedule shall apply to the above types of joinery as shown on drawing in the event of deviation.

19.4.2. **SURFACE FINISH**

19.4.2.1 Unless otherwise specified here-in-after, all timber work shall be wrought except surfaces of timber in contact with or buried in PCC block/concrete/plaster and all wood work not exposed to view shall be left clean sawn.

19.4.2.2. Counter sunk for bolts and the like on wrought surfaces shall be plugged with tightly fitting plugs and plane flush with surrounding surfaces.

19.4.2.3. All door timber frames and shutters shall be finished smooth with well planed faces. Panels of shutters shall be flat and well sanded to a smooth and level surface. Defective knots where permitted on surfaces exposed to view shall be completely bored or cut out and tightly plugged with the same timber species and properly glued in. The grains of the plug shall be run in direction of the piece.

19.4.2.4. All the surface of door fame and shutter which are required to be painted ultimately shall be covered with a suitable priming coat as specified in IS-3536. 'Ready mixed paint wood primer' In case of doors intended to be polished priming coat of suitable polish be applied before delivery.

19.4.3. **FIXING**

19.4.3.1. Use of nails in joinery and other timber in building items unless otherwise specified or shown on drawings is prohibited. Wood screws of appropriate sizes shall be used.

19.4.4 Chowkhats for door shall be fixed as the work proceeds. However, if the contractor chooses to fix the chowkhats after the masonry work is over, he may do so by providing PCC blocks of size 300 mm x 150 mm x thickness of wall for embedding hold fast for which no price adjustment to be made.

19.4.5. **PANELED JOINERY**

19.4.5.1 Paneled door shutters shown on drawings shall be of approved factory made, kiln seasoned, chemically treated 2nd class hard wood vacuum pressure treated with centrifugal and anti termite preservatives. Fully paneled shutters shall be provided with panels of 12 mm thick veneered particle board commercial facing on both sides bonded with phenol formaldehyde synthetic resin as per IS 3097. Shutter shall conform to IS-1003(Part-I). Thickness of panelled shutters shall be 35 mm thick. Timber paneled shutter shall be constructed in the form of timber frame work of styles and rails.

19.4.5.2. Styles and rails including lock rail of shutters shall be one piece only and shall be jointed to each other by mortise and tenon joints at right angles. Rails more than 150 mm width shall have two tenon. The bottom, lock and top rails shall be inserted 25 mm short of the width of the styles.

19.4.5.3 The panels shall be fixed by providing groove and beading as detailed in drawings. The edges of the panel shall fit into groove to the full depth leaving an air gap of 1.5 mm. Edges of the panel insert shall be sealed by application of glue.

PARTICULAR SPECIFICATIONS (CONTD…)

19.4.5.4 All members of the shutter shall be straight without any warp or bow. They shall have smooth well planned surfaces at right angle to each other. The right angles of the shutter shall be checked by measuring the true diagonals from one extreme corner to the opposite one and the difference between the two diagonals shall be within tolerance limit of ± 3 mm. The contact surfaces of mortise and tenon joints shall be glued before putting together. All joints shall be pinned with bamboo pins two per tenon in staggered position.

19.4.5.5. Factory made penelled door shutters shall be procured by the contractor from approved factory specialised in manufacturing such door shutters after getting samples approved by the GE in writing.

19.4.6 **TIMBER TREATMENT**

19.4.6.1. All timber for wood work and joinery and built in furniture to be manufactured at site and to be incorporated in the work shall be treated as per IS-401 by modified hot and cold treatment as given below, with copper chrome arsenic/ Acid cupric chromate composition/chromate zinc chloride/copper chrome boron/Zinc metal Arsenate. The minimum quantity of chemical impregnated per cubic metre of timber shall be as specified in table 2 group 5 of the said IS.

19.4.6.2. Water shall be kept boiling in the tank and in other tank (which should not be porous but may be of cement concrete) preservative solution shall be maintained at room temperature of 16 degree C to 25 degree C. Timber shall be immersed in boiling water for 15 to 25 minutes and instantly transferred to the preservative solution and kept there for another 15 to 40 minutes. Surfaces of timber to be fabricated after above treatment i.e. to be cut/ bored holes and planned shall be given two coats of surfaces applications of above preservative, first coat having dried completely before application of second coat. It should be ensured that temperature of boiling water is never less than 90 degree C.

19.4.6.3 Rate in MES Schedule for item for wood work and joinery shall be deemed to include above preservative treatment.

19.4.6.4. Other treatment to timber surfaces such as tarring, Painting and oiling etc, as specified here-in-after shall however be carried out in addition to the above preservative treatment.

NOTES :-

1. The above treatment is not applicable to particle board and plywood and teak wood members which shall be treated as specified in relevant IS.
2. The composition and concentration of chemical shall be as per clauses 3.4.2 and 3.4.3 of IS-401.
3. The absorption of the preservative shall be as per table 2 group 5.
4. The contractor may, however, use factory made pressure treated timber with the prior approval of the GE in writing without any price adjustment. Contractor to produce certificate from the factory of such treatment.

19.4.7. **TESTING**

19.4.7.1. METHODS OF TESTS: -

Besides carrying out field testing of moisture content by means of moisture meter or oven and determining penetration of preservative through chemical test as specified in IS 401-1982 to check treatment of timber, following type of tests for door shutters shall also be carried out.

(a) Slamming test (b) Impact indentation test (c) Shock resistance test
(d) Edge loading test.

19.4.7.2 **Slamming test**: - The door shutter shall be placed in a horizontal position suitably hinged at three equidistant places on one of the long edges. The other edge is lifted up so as to form an angle of 30 degree at the hinged edge and allow to be dropped under its own weight, several times on the rail. General Condition shall be noted after every ten drops. There shall be no visible damage caused in any part of the door shutter by the first fifty drops.

PARTICULAR SPECIFICATIONS (CONTD…)

19.4.7.3. **Impact Indentation Test (Dynamic Punching Test):-** The shutter shall be placed horizontally resting equally on all the four edges. The steel ball shall be allowed to fall freely from a height of 750 mm at any randomly selected 10 different points on the surface of the shutter avoiding nearness of edges. If any indentation is noticed the depth shall be measured by the dial gauge and shall be recorded. The general behavior of the shutter and de-lamination if any shall also be noticed.

19.4.7.4. **Shock Resistance Test:-** The shutter shall be supported in a vertical plane in flush with a fixed frame. The rope with the leather ball shall be tied on the central vertical line just above the shutter. The length of the rope shall be such that the ball may strike the shutter at a point 200 mm above the bottom end on the vertical central line of the door. The ball shall be pulled away from the shutter to a height of 1.25 m from the plane of striking place and shall be allowed to strike the shutter several times in quick succession. The general condition will be noted periodically after every five blows. The test shall be repeated with the shutter turned up side down in the vertical plane. There shall be no visible damage in any part of the shutter after the fifteen blows on each end.

19.4.7.5 **Edge Loading Test:-** The shutter shall be hung vertically pivot supported on one longitudinal edge and the load shall be applied on the top of the other longitudinal edge parallel to the plane of the shutter. The loading shall be done in steps of not more than 20 kg at a time until a load of 100 kg is reached and kept constant at the load for a period of half-an-hour. The deflections shall also be measured immediately on removal of the load.

NOTE :- The test may also be done on suitable testing machine if racking action may be simulated. The deflection of the edge at the maximum load shall not be more than 0.5 cm. On removal of the loads there shall not be any residual deflection of more than 0.5 mm, failing which the test may be repeated on the other edges in the reverse direction.

19.4.7.6 Testing of panelled shutters shall be got done from National Test House or any other approved laboratory. Cost of testing including transportation charges shall deemed to be included in lump sum quoted by the contractor for buildings.

20. **FLUSH DOOR SHUTTER**.

20.1 Flush door shutter shall be 35mm thick, solid core, non-decorative, factory made shutter made as per IS: 2202 (Part I- 1999) and shall be ISI marked. Styles & rails shall be without any joint and be made of non coniferous timber hard wood (HOLLOCK / MIRANTEE/BONSUM/ BEN TEAK) with moisture contents not more than 12% and dimension as given in IS code. The face panel comprising of plywood or cross band and face veneers shall be glued by hot pressed process. Over all finished thickness of face panel shall not be less than 3mm while the thickness of face veneer shall not be less than 0.6mm block board core shall fully confirm to the requirement specified in the IS code. All timber used shall be well seasoned and chemically treated. Adhesive shall be phenol formaldehyde synthetic resin BWP type specified in IS: 848- 2006. All dimensions shall be finished dimension & manufactures test certificate for test specified in IS: 2202 (Part-I) shall be rendered.

20.2 The Contractor shall submit machine numbered paid vouchers from the Authorized Dealer/ Manufacturer for the total quantity of the flush door shutters supplied under each consignment. At the time of delivery of flush doors at site, delivery challan along with invoices shall be submitted by the Contractor to the GE. Each consignment received at the site shall be inspected by the GE. The original invoice shall be defaced by the GE and kept on record at the site office. The Exicse duty gate pass shall be submitted by the contractor to the GE for each consignment of flush door shutter dispatched by the factory holding valid BIS certification and brought at site for incorporation in the work. On receipt of the shutters at site, the samples of door shutters shall be tested in any approved laboratory as instructed by the GE. From each lot of approximately 700 shutters, one shutter shall be selected at random by the GE. The cost of replacement of the door shutters selected as samples, their transportation to the laboratory and the cost of testing by the laboratory shall be borne by the contractor and shall be deemed to be included in the lump sum rates quoted in Schedule "A" Section-I.

20.3 Flush doors to be provided with teakwood beading as per drawings and painted to match the door. Bottom of the flush door be painted with two coats of synthetic enameled paint before installation.

PARTICULAR SPECIFICATIONS (CONTD…)

C. FOR SCHEDULE 'A' PART- II :SITE CLEARENCE, SITE DEVELOPMENT AND EARTHWORK

1. **GENERAL**: Refer relevant parts of particular specification here-in-before. The work shall be executed as specified in Schedule 'A' and all as specified in SSR Part-I.

D. FOR SCHEDULE 'A' PART-III: INTERNAL ELECTRIFICATION

1. GENERAL

1.1 All work of internal electrification shall be executed by fully qualified licensed electricians all as per the latest Engineering practice and latest IS codes/IS Rules. Certificate of qualification shall be produced on demand by Engineer-in-Charge.

1.2 The position of electrical fittings and fixtures shown on the drawing are tentative. The exact position of these shall be as directed by Engineer-in-Charge at the time of execution without any price adjustment.

1.3 The runs of wires and locations of fittings shall be marked on walls and soffits of roof/floor slabs for the wiring in a sample building for the approval of Engineer-in-Charge. The contractor may have to realign the wiring and or relocate the fittings required for final approval at no extra cost to Government. The entire work shall be completed on the basis of sample building. Layout wiring diagram in duplicate shall be submitted by the contractor and got approved from GE before execution of work.

1.4 Looping -in-system of wiring shall invariably be used through the installation.

1.5 All equipment, fittings and accessories, materials etc selected for the wiring installation shall conform to the relevant India Standards wherever these exist.

1.6 Contractor shall produce samples of all the fittings accessories materials etc for approval by the GE, before incorporation in work.

1.7 Not withstanding, the fact that equipment has passed the inspection carried out during the stage of manufacture, the contractor is not relieved from his obligations to conform to the quality, workmanship, guarantee of performance etc.

1.8 Any defective material, equipment or workmanship which may come to the notice of the GE or his representative after installation shall be liable for rejection and the contractor shall have to replace such materials, equipment etc or rectify the defects at his own cost.

2. **NON METALLIC (PVC) CONDUIT AND CONDUIT ACCESSORIES**:-

2.1 Refer clause 19.29 of MES Schedule Part-I, 2009. Conduit and conduit accessories shall conform to relevant IS i.e., IS-9537 (Part-III) 1983 and IS-3419-1976. Conduit shall be non-metallic PVC conduit and shall show no appreciable un-evenness. Interior and ends of conduit and conduit accessories shall be free from bushes, ducts fins and the like which may cause damage to the cables. Each length of PVC conduit shall be marked with the following :-

(a) Manufacturer's name or trade mark.
(b) Country of manufacture.
(c) Nominal size of the conduit.

2.2 Each length of conduit shall also be marked with the ISI certificate mark.

2.3 PVC flexible conduit of size as directed by the Engineer-in-charge shall be provided as required without any extra cost.

PARTICULAR SPECIFICATIONS (CONTD…)

2.4 **CONDUIT ACCESSORIES**

2.4.1 The conduit fittings shall be made of non metallic PVC and shall be fabricated or cast to shape. They shall be so designed and constructed that they match with the corresponding conduit sizes without any adjustment and ensure reliable mechanical protection to the cable. The inside edges, through which the cables are intended to pass, shall be smoothly routed in order to prevent abrasion of cables. Entries for conduits shall be formed so as to prevent the conduit being screwed or pushed right through to the interior of the fittings. Inspection fittings shall be so constructed that the screws fixing the cover do not deform the conduits or damage the insulation of the cable enclosed. The screwed couplers shall be screwed inside throughout their entire length.

2.4.2 The axis of any elbow shall be a quadrant of a circle plus a straight portion at each end. Normal bend shall be large sweep being giving a diversion of 90 degree in the turn of the conduit half normal bends shall be similar to normal bends both as regards of curvature and length of straight portions at the ends but shall give a diversion in the run of the conduit of 45 degree small circular boxes shall be provided with lugs of thickness not less than 2.8 mm for fixing the covers. Large circular boxes shall have been lugs of thickness not less than 4mm for fixing the cover fittings shall be adequately protected against rust and corrosion both outside and inside with medium protective coating.

2.4.3 Fitting shall comply with the dimensions given in appropriate tables of IS-2667-1976 subject to a tolerance of + 5 percent.

2.4.4 Each fittings shall be marked clearly and with the following information :-

(a) Name or trade mark of manufacturers.
(b) Country of manufacture.

2.4.5 The fittings may also be marked with the ISI certification mark.

2.4.6 Conduit pipes shall be fixed by heavy gauge saddles, secured to suitable wood plugs or other plug with screws in an approved manner at an interval of not more than one metre, but on either side of couplers at bends or similar fittings, saddles shall be fixed at a distance of 30 cm form the center of such fittings. All necessary bends in the system, including diversion shall be done by bending pipes or by inserting suitable solid or inspection by normal bends, elbows or similar fittings, or fixing inspection boxes whichever is more suitable. Conduits fittings shall be avoided as per as possible on conduit system exposed to weather, where necessary solid type fittings shall be used. All the screws shall be cadmium coated.

2.4.7 The conduit of each circuit or section shall be completed before conductors are drawn in the entire system of conduit after erection shall be tested for mechanical and electrical continuity through out and permanently connected to earth wire by means of suitable earthing clamp efficiently fastened to conduit pipe in a workman like manner for a perfect continuity. Gas or water pipes shall not be used as earth medium.

2.4.8 Inspection type conduit fittings such as inspection boxes, drawn boxes, bends, elbows and tees shall be so installed that they can remain accessible for such purposes as to withdrawal of existing cables or the installing of additional cables.

2.4.9 Separate conduit shall be used for normal light, fan, exhaust fan.

2.4.10 In case of concealed wiring, conduits shall be laid before casting in the slab so as to conceal the entire run of conduits and ceiling outlet boxes. Vertical drops shall be buried in columns or walls.

2.5 All wiring from FFL upto a height of 100 cm shall be drawn through non-metallic (PVC) conduit, concealed in walls. Contractors rates for the respective item of point wiring shall included for this provision.

2.6 All the fan boxes shall be covered with 3 mm thick hylem sheet and contractors rate for the point wiring for fan shall include this provision.

PARTICULAR SPECIFICATIONS (CONTD…)

2.7 **LAYING OF CONDUITS**

2.7.1 Conduits shall be laid either recessed in walls and ceilings or on surface on walls and ceilings or partly recessed partly on surface, as required.

2.7.2 Same rate shall apply for recessed and surface conduiting in this contract.

2.7.3 Stranded copper conductor insulated wire of size as per schedule of quantities shall be provided in entire conduiting for loop earthing.

2.7.4 GI wire of suitable size to serve as a fish wire shall be left in all conduit runs to facilitate drawing of wires after completion of conduiting.

3. **PVC INSULATED 1100 VOLTS GRADE CABLE (PVC SHEATHED/UN- SHEATHED)**

3.1 The conductor shall be of stranded copper wires. The insulation shall consist of compounded polyvinyl chloride. The thickness of insulation shall not be less than nominal value mentioned in appropriate table of IS-694-1964. The composition of the compound of sheath shall be polyvinyl chloride. The compound shall be designed to offer adequate mechanical protection. The value of thickness of sheath shall be in accordance with those specified in appropriate tables. The insulation on and sheath shall be applied by the extrusion process and each shall form a compact homogeneous body.

4 **Switch Boxes.**

4.1 Switch boxes suitable to house switches of required ratings, and fan regulators as required shall be provided. In case the number of switches in one box is not tallying with that available in standard manufacturer, the box accommodating the next higher of switches shall be provided without any extra cost. In case fan regulator/regulators is/are to be provided at a later dated, suitable provision for accommodating such regulators shall be made in the switch boxes and blank of covers shall be provided without any extra cost.

4.2 Switch boxes shall be so designed that accessories are mounted on a grid plate with tapped holes for brass machine screws leaving ample space at the back and on the side for accommodating conductors, check-nuts and screwed bushes at conduit entries etc.. The grid plates and MS boxes shall be fitted with a brass earth terminal. Boxes shall be attached to conduits by means of check-nuts on either of their walls. Moulded front covers made from high impact resistant and ultra violet stabilized engineering plastics shall be fixed by means of counter sunk chromium plated brass machine screw. No timber shall be used for any supports. Switch boxes shall be located with bottom at 1200mm above floor level unless otherwise indicated.

4.3 **Boxes for Socket / Telephone / Call Bell Outlets**

4.3.1 Outlet boxes shall be suitable for housing switches socket outlets / telephone outlets / buzzers and any other outlet as required. These shall be so designed that accessories are mounted on a grid plate with tapped holes for brass machine screws leaving ample space at the back and on the sides for accommodating conductors, check nuts and screwed bushes at conduit entries etc.. The grid plates and MS boxes shall be fitted with a brass earth terminal. These shall be attached to conduits by means of check nuts on either sides of their walls. Moulded front covers made from high impact resistant, flame retardant and ultra violet stabilized engineering plastics shall be used to mount the outlets and shall be fixed to the outlet MS boxes by means of counter sunk chromium plated brass machine screws. No timber supports shall be used. Boxes shall be located at skirting level or bottom at 1200mm from floor or inside raceways on laboratory work table., as indicated in drawings and / or as directed.

4.4 **WIRES**

4.4.1 Wiring shall be carried out with PVC insulated 660/1100 volt grade FRLS unsheathed single core wires with electrolytic annealed stranded copper (unless otherwise stated) conductors and conforming to IS: 694/1990,IS: 1554. All wire rolls shall be ISI marked. All wires shall bear manufacturers label and shall be brought to site in new and original packages. Manufacturer's certificate certifying that wires brought of site are of their manufacture shall be furnished as required.

4.4.2 The smallest size of conductor for lighting circuits shall have a nominal cross-sectional area of not less than 1.5 sq.mm, while minimum size of power wiring shall be 4 sq.mm.

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4.5 **CEILING ROSES**

4.5.1 Ceiling rose shall be of bakelite and shall comply with IS:371 specification for ceiling roses having two or three terminal plates of outside diameter not less than 63.5 mm (flexible cord shall be fixed to ceiling rose in such a way that no weight is carried by the clamping screws). The flexible cord shall be of 23/0.0076 twin core with copper conductor.

5. **MCB & DB**

5.1 MCBs shall have quick make and break non-welding silver alloy contacts both on the manual and automatic operation. MCBs shall be of thermal magnetic type with inverse time delay over current tripping having a short circuit rupturing capacity as specified in Schedule 'A'. In case of multiple breakers, the tripping must be on all the poles and operating handle shall be common. Pressure clamp terminals for stranded solid conductor insertion are acceptable upto 4 Sqmm aluminium and for higher ratings, the terminals shall be suitably extended for bolted lugs connections. All terminals shall be suitably shrouded.

5.2 MCBs shall be housed in distribution board fabricated out of sheet steel and shall have IP-52 degree of protection to prevent entries of dust, ample clearance between the conductors and sheet steel body shall be maintained in order to obviate any chance of short circuit. Movable conduit entry plates shall be provided at top and bottom to facilitate drilling holes to suit individual requirement. The MCBs shall be mounted on a high grade rigid insulating support and connected by electrolytic copper bus bars phase separation barriers made out of are resistant materials shall be provided between the phase. Bus bars shall be colour coded for phase identification and shall be PVC shrouded.

5.3 Distribution board of IP-52 degree of protection shall be recessed in wall niche/mounted on the surface of the wall with necessary clamps, bolts as required. Distribution boards shall be provided with proper identification name plate, danger mark etc. DB shall be provided with a circuit list giving details of each circuit. All the outgoing circuit wiring shall be provided with identification ferrule giving the circuit number and phase. Identification of all out going circuit shall also be written on the expose portion of DB.

5.4 Suitable earth terminal shall be provided on the distribution board for bending to earth.

5.5 Distribution Boards and MCBs shall of the same make.

6. **SWITCHES/SWITCH SOCKET OUTLET/SOCKET OUTLET**

6.1 The base shall be made of vitrified ceramic material or tough non ignitable molded insulating material. The covers, cover plates and actuate members shall be of tough, non ignitable insulating material. Current carrying parts shall be made of brass, copper phosphor bronze, aluminium alloy or any other suitable material, springs shall be of corrosion resistant metal. Attachment fitting screw and other non current carrying Paris shall be of mild steel aluminium alloy or insulating material. All socket outlets and switch socket combination shall have ISI certification mark.

7. **UTILITY FITTINGS**

7.1 The detailed dimension/indicated in the drawing/MES Schedule Part-I shall be followed. These fittings shall be pre-wired with twin core copper flexible cable of size 23/.0076. About 15 cm long extra flexible cable shall be provided with fittings for making connections with the connector.

7.2 Four Nos brass machine screw with washer, 4 mm dia 15 mm length, shall hold the shade.

7.3 Acrylic sheet shall be 3 mm thick and conform to IS-7569-1981.

7.4 Provision of earth connection is to be made in the fittings for earthing.

7.5 Brass lamp holders shall conform to relevant IS.

7.6 All fittings shall conform to relevant IS.

7.7 Fittings shall be fitted on PVC block of suitable size as directed. Cost for this provision shall deemed to be included under respective item of fittings.

8 **MOUNTING HEIGHT**

8.1 Mounting height of various fittings shall be as under :-

(a)	Bracket light fittings	-	215 cm from FFL.
(b)	Ceiling light fittings	-	These shall be fixed centrally in the soffit of slab or as directed.
(c)	Switch	-	100 cm from FFL.
(d)	Socket outlet	-	-ditto -

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8.2 Position of fittings not indicated above shall be as directed by the Garrison Engineer.

9. **EARTHING**

9.1 All the non-current carrying metal parts of electrical installation shall be earthed properly. All metal conduits, cable sheathe, switch gear, distribution board, light fittings. MS boxes and parts made of metal shall be bounded together and connected by means of specified earthing conductors to an efficient earthing system. All earthing shall be in conformity with IE Rules.

9.2 Neutral conductor pipe conveying water, structural steel work, metallic enclosures, metallic conduits and lighting protection system conductors shall not be used as means of earthing an installation of even as a link in an earthing system. The electrical resistance measured between earth connection at the main distribution board and any other point on the completed installation shall be low enough to permit the passage of current necessary to operate circuit breakers and shall not exceed phase. All metal clad switches and other equipment carrying single phase current, shall be connected to earth by a single connection. All metal clad switches, carrying medium voltage shall be connected with earth by two separate and distinct connections.

9.3 No earth pit shall be made within 2 metre of a wall or foundation. Effort shall be made to locate them in grass lawns or near flower beds or water taps. The distance between two earthing pits shall be at least 2 metres.

9.4 On testing after completion, if desired earth resistance is not obtained the earthing will be redone at no extra coat.

10 **TESTING**

10.1 After completion of wiring, contractor shall submit three copies of wiring diagram.

10.2 The installation after completion of work shall be tested for insulation, wiring continuity, earth continuity, earth resistance polarity test etc, as specified in IS/IE Rules. The test shall be carried out in the presence of Engineer-in-Charge and shall be signed by Engineer-in-Charge and the contractor. Three copies of test sheet shall be submitted by contractor. In case the earth result does not fall within the specified limit, as mentioned in IS/IE Rules, the contractor shall dig earth pit beyond 2.5 m deep to obtain the desired earth results.

10.3 All testing instruments labour, materials and incidentals necessary to conduct the test shall be arranged by the contractor at his own cost.

11 **RECORD DRAWING**

11.1 Record drawings for the complete work of wiring on tracing cloth with two prints as directed by the Engineer-in-Charge shall be submitted by the contractor at his own expense.

12. **ROUTINE AND COMPLETION TESTS**

12.1 **Installation Completion Tests:**

12.1.1 At the completion of the work, the entire installation shall be subject to the following tests:

(a) Wiring continuity test
(b) Insulation resistance test
(c) Earth continuity test
(d) Earth resistivity test

12.1.2 Besides the above, any other test specified by the local authority shall also be carried out. All tested and calibrated instruments for testing, labour, materials and incidentals necessary to conduct the above tests shall be provided by the contractor at his own cost.

12.2 **Wiring Continuity Test:**

12.2.1 All wiring systems shall be tested for continuity of circuits short circuits and earthing after wiring is completed and before installation is energized.

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12.3 **Insulation Resistance Test**

12.3.1 The insulation resistance shall be measured between earth and the whole system conductors, or any section thereof with all fuses in place and all switches closed and except in concentric wiring all lamps in position of both poles of the installation otherwise electrically connected together, a direct current pressure of not less than twice the working pressure provided that it does not exceed 1100 volts for medium voltage circuits. Where the supply is derived from AC three phase systems, the neutral pole of which is connected to earth, either direct or through added resistance, pressure shall be deemed to be that which is maintained between the phase conductor and the neutral. The insulation resistances lower than one mega ohm. The insulation resistance shall also be measured between all conductors connected to one phase conductor of the supply and shall be carried out after removing all metallic connections between the two poles of the installation and in those circumstances the insulation shall not be less than that specified above. The insulation resistance between the frame work of housing of power appliances and all live parts of each appliance shall not be less than that specified in the relevant standard specification or where there is no such specification, shall not be less than half a mega ohm or when PVC insulated cables are used for wiring 12.5 mega ohms divided by the number of outlets. Where a whole installation is being tested a lower value than that given by the above formula subject to a minimum of 1 Mega ohms is acceptable.

12.4 **Testing of Earth Continuity Path**

12.4.1 The earth continuity conductor including metal conduits and metallic envelopes of cable in all cases shall be tested for electric continuity and the electrical resistance of the same along with the earthing lead but excluding any added resistance of earth leakage circuit breaker measured from the connection with the earth electrode to any point in the earth continuity conductor in the completed installation shall not exceed one ohm.

12.5 **Testing of Polarity of Non-Linked Single Pole Switches**

12.5.1 In a two wire installation a test shall be made to verify that all non-linked single pole switches have been connected to the same conductor throughout, and such conductor shall be labeled or marked for connection to an outer or phase conductor or to the non-earthed conductor of the supply. In the three of four wire installation, a test shall be made to verify that every non-linked single pole switch is fitted to one of the outer or phase conductor of the supply. The entire electrical installation shall be subject to the final acceptance by GE.

12.6 **Earth Resistivity Test**

12.6.1 Earth resistivity test shall be carried out in accordance with IS Code of Practice for earthing IS 3043.

12.7 **Performance**

12.7.1 Should the above tests not comply with the limits and requirements as above the contractor shall rectify the faults until the required results are obtained. The contractor shall be responsible for providing the necessary instruments and subsidiary earths for carrying out the tests. The above tests are to be carried out by the contractor without any extra charge.

12.8 **Tests and Test Reports**

12.8.1 The contractor shall furnish test reports and preliminary drawings for the equipment to the GE for approval before commencing supply of the equipment. The contractor should intimate with the tender the equipment intended to be supplied with its technical particulars. Any test certificates etc. required by the local Inspectors or any other Authorities would be supplied by the Contractor without any extra charge.

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E. FOR SCHEDULE 'A' PART-IV: INTERNAL WATER SUPPLY.

11. GENERAL:- Refer relevant parts of particular specification here-in-before. The scope of work shall be executed as specified in Schedule 'A' and all as specified in SSR Part-I (2009).

F. FOR SCHEDULE 'A' PART-V : ROADS

1. GENERAL : Refer relevant parts of particular specification here-in-before. The work of shall be executed as specified in Schedule 'A'.

2. **ROAD ROLLER:**

2.1 Road roller required for execution of work will not be issued under Schedule 'C'. The contractor shall arrange the same at his own cost. Rolling should be done by 8 to 10 ton power roller and as par 20.A.21.4 of MES Schedule Part -I

3. **WATER BOUND MACADAM**

3.1 WBM work shall be carried out all as specified in Sch 'A' and as per para 20.A.21.1 of MES Schedule Part-I.

3.2 **PREPARATION OF SURFACES**:-
Preparing surfaces of unsurfaced water bound macadam/concrete should be done by brushing with wire brushes and as specified in Sch 'A'. For all other requirement refer Para 20.A.21.2 of MES Schedule Part-I.

3.3 **SPREADING COARSE AGGREGATE**
WBM should be prepared by spreading coarse aggregate of grading 2 (63 to 40mm size) of crushed stone of 100mm compacted thickness. For spreading method & requirement refer Para 20.A.21.3 of MES Schedule Part-I

3.4 **APPLICATION OF SCREENING**
Refer Para 20.A.21.5 of MES Schedule Part-I.

3.5 **SPRINKLING AND GROUTING**
Refer Para 20.A.21.6 of MES Schedule Part-I.

3.6 **APPLICATION OF BINDING MATERIALS**
Refer Para 20.A.21.7 of MES Schedule Part-I.

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3.7 **SETTING AND DRYING**
Refer Para 20.A.21.8 of MES Schedule Part-I.

3.8 **PLYING OF CONSTRUCTION TRAFFIC**
Refer Para 20.A.21.9 of MES Schedule Part-I.

3.9 **SURFACE EVENNESS OF WBM COURSE**
Refer Para 20.A.21.10 of MES Schedule Part-I.

3.10 **RECTIFICATION OF DFECTIVE CONSTRUCTION**
Refer Para 20.A.21.11 of MES Schedule Part-I.

3.11 **PATCHING OF POT HOLES ALONG WITH REMOVAL OF RUTS AND DEPRESSIONS**
Refer Para 20.A.21.12 of MES Schedule Part-I.

3.12 **QUANTITIES OF MATERIAL FOR WBM**
Refer Para 20.A.21.13 of MES Schedule Part-I.

3.13 **BROKEN STONE SOLING**
3.13.1 The soling should be done by broken boulder or quarried stone of 150mm thick (spread thickness) as per Sch 'A' by 8 to10 ton power roller and interstices filled to their full depth and as per Sch 'A' and per Para 20.A.20.1 of MES Schedule Part-I.

3.14 **TACK COAT/ PRIME COAT**
3.14.1 Tack/Prime coat of paving bitumen of grade and at the rate shall be provided as given in the Schedule 'A' and all as specified in para 20.A.24 including sub para 20.A.24.1, 20.A.24.2, 20.A.24.3 and 20.A.27.2 of MES Schedule Part-I (2009).

3.14.2 The binder shall be applied uniformly with the help of either self propelled or towed pressure prayer with self heating arrangement and spraying nozzles, a system capable of spraying bitumen at specified rate and temperature so as to provide a uniform and even spread of bitumen. The tack coat shall be applied just ahead of the oncoming bituminous construction. Temperature of application of binder shall be between 165^{0}C and 180^{0}C.

4.00To 9.15 BLANK

9.16 SEMI DENSE CONCRETE
9.16.1 Semi dense concrete work shall be carried out as specified in Sch 'A' and as per para 20.B.4 of MES Schedule Part-I.

9.16.2 **BINDER PAVING BITUMEN GRADE VG-30/ VG-10**

9.16.2.1 The Bitumen to be used in the work shall be straight run of VG-30 confirming to IS:73; "Specification for Paving Bitumen". Paving bitumen shall meet the following requirements:-

Characteristics	Requirements for Paving Bitumen(vg-30)	Requirements for Paving Bitumen(Vg-10)	Test Method
Flash Point, Cleveland Open Cop, ^{0}C, Min	220	220	IS: 1202
Softening Point, ^{0}C, Min, R&B	47	40	IS: 1205
Penetration at 25 ^{0}C, 100gm, 5s, 1/10mm	50-70	80-10	IS: 1203
Viscosity Ratio at 60 degree cen max	4	4	-
Ductility at 25^{0}C, cm, Min	40	75	IS: 1208
Matter soluble in trichloroethylene, percent by mass, Min	99	99	IS: 1216
Absolute Viscosity at 60^{0}C, Poise	2400	800	IS: 1206
Kinematic Viscosity at 135^{0}C, cSt, Min	350	250	IS: 1206

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9.16.2.2 The Bitumen shall be procured from the any of the main producer as specified here-in-after. Bitumen shall be supplied in "Bulk", however, to ensure that no adulteration with original material takes place in transit, the contractor shall deploy proper watch and ward during transit of material. The contractor at his option may supply "Packed" bitumen without any extra cost to the Government.

9.16.2.3 The contractor shall submit original purchase vouchers alongwith manufacturers test certificate for the total quantity of bitumen supplied under each consignment to be incorporated in the work. All consignments received at the work site shall be inspected by the Garrison Engineer along with the relevant document. The original purchase vouchers and the test certificate shall be verified for subject contract and defaced by the Engineer-in-Charge and kept on record in the office of the Garrison Engineer duly authenticated and with cross reference to the consignment. The Accepting Officer may order a Board of Officers for random check of bitumen and verification of connected documents during the currency of contract. If the Garrison Engineer desires separate test of bitumen, the test shall be got done through a recognised and approved laboratory/institution as approved by Garrison Engineer at no extra cost to Government.

9.16.2.4 The bitumen brought at site shall be recorded in Measurement Book "Not to be abstracted" and shall be signed by both the contractor and Engineer-in-Charge.

9.16.2.5 **TESTING OF PAVING BITUMEN**:-
The contactor shall submit test certificates for each consignment brought at site. The material shall also be tested at site in the field laboratory at the stipulated frequency as specified. Any test which is required as per MES Schedule/IS Code/as specified in this documents, but can not be performed at the field lab established by the contractor, shall be carried out at a laboratory as specified for Job Mix Formula, expenditure of such tests shall be borne by the contractor. In addition, GE may also get it tested independently from approved laboratory mentioned here-in-before. Contractor shall provide all facilities like material, transportation of materials to lab, etc. for testing by the Deptt. the test charges for such independent tests only shall be borne by the Deptt.

9.16.3 **COARSE/FINE AGGREGATE/FILLER**

9.16.19.1 The combined grading of fine and coarse aggregates shall conform to the following:-
Nominal adequate size shall be 19.5mm as per grading as under:-

	Grading		
(i)	Nominal agg size	:	19.5mm(wearing course)
(ii)	Thickness of layer	:	30-55mm
	IS sieve(mm)		**Percent passing**
(i)	26.5	:	-
(ii)	19	:	100
(iii)	119.2	:	79-100
(iv)	9.5	:	70-88
(v)	4.75	:	53-71
(vi)	9.36	:	42-58
(vii)	19.18	:	34-48
(viii)	0.6	:	26-38
(ix)	0.3	:	18-28
(x)	0.15	:	12-20
(xi)	0.075	:	4-7

Note:- The combined grading shall not vary from the low limit on one sieve to the high limit on the adjacent sieve.

9.16.4 **COARSE AGGREGATE**

9.16.4.1 Coarse aggregate shall be crushed material retained on IS 9.80mm sieve and shall be crushed stone from approved quarry and shall consist of angular fragments clear, tough and durable rock, free from disintegrated pieces and organic or other deleterious matters and adherent coatings. The aggregates shall not contain more than 6 percent by weight of flat/longed pieces (Flat piece is one having ratio of "width/thickness" of more than 4, elongated piece is where the ratio "length/width" is more than 4). The aggregates shall preferably be hydrophobic and of low porosity. If hydrophanous aggregates are to be used, which in normal circumstances shall be avoided, bitumen shall be treated with anti-stripping agents of appropriate quality in suitable doses.

PARTICULAR SPECIFICATIONS (CONTD…)

9.16.4.2 The aggregates shall satisfy the following physical requirements:-

Property	Test	Specification
Cleanliness (dust)	Grain size analysis (IS-2386 Part-I)	Max 5% passing 0.075mm sieve
Particle shape	Flakiness Index (IS:2386 Part-I)	Max 25%
Strength*	Los Angeles Abrasion Value (IS:2386 Part-IV)	Max 35%
Impact Value	*Aggregate Impact Value (IS:2386 Part-IV)	Max 30%
Durability	Soundness-(IS:2386 Part-V)	
	Sodium Sulphate	Max 12%
	Magnesium Sulphate	Max 18%
Water Absorption	Water absorption (IS:2386 Part-III)	Max 1%
Stripping Value	Coating and Stripping of Bitumen Aggregate Mixtures (IS:6241)	Minimum Retained Coating 95%
Water Senstivity**	Retained Tensile Strength (AASHTO T 283)	Min 80%
Polishing	Polished Stone Value (BS : 812 Part 114)	Min 55%

Note:- 19. *Aggregate may satisfy requirements of either of these two tests.

9. **Water sensitivity test is only required if the minimum retained coating in the striping test is less than 95%.

9.16.5 **FINE AGGREGATES**

9.16.5.1 Fine aggregates shall be the fraction passing IS 2.80 mm sieve and retained on 90 micron sieve, consisting of crusher-run screening, natural sand or a mixture of both. It shall be clean, hard, durable, dry and free from injurious, soft or flaky pieces and organic or other deleterious substances. It shall be free from sulphates, chlorides and other materials (including products of decomposition), which may be liable to break down during drying or subsequently when exposed to weather.

9.16.6 **FILLER**

9.16.14 Filler shall be Ordinary Portland Cement as specified in clause here-in-before.

9.16.7 **DESIGN CRITERIA**

9.16.23.1 Semi dense concrete mix should be properly designed so as to satisfy certain criteria needed to assure adequate stability and durability. The mix as designed and laid should satisfy the requirements as given under:-

CRITERIA SPECIFIED VALUE

(a)	Number of compaction blows on each end of Marshall specimen	75
(b)	Marshall stability in, kN at 60 degree C, min	14
(c)	Flow value (mm)	2-3.5
(d)	Percentage void in mix (to prevent bleeding)	3-5
(e)	Percentage void in mineral aggregates filled with bitumen.	60-75
(f)	Binder content % by weight of total mix.	5.5%
(g)	Percent voids in the mineral aggregate (VMA)	As per Table-500-12 of MORTH
(h)	Retained tensile strength % (AASHTO T 283)	Minimum 80%

Note:- Tensile strength ratio(TSR) shall be determined in accordance with AASHTO T 283.If TSR of mix conditioned with one freeze and thaw cycle is less than 80%, 2% hydrated lime or liquid anti-striping agent shall be used in the mix to satisfy this criteria, cost of which shall deemed to be included in quoted rate.

PARTICULAR SPECIFICATIONS (CONTD…)

9.16.8 **IMMERSION COMPRESSION TEST**

9.16.24.1 Compression test to measure the loss of Marshall Stability due to effect of water on the mix shall be conducted and if the index of retained stability is less than 75, the mix shall be rejected or the aggregates shall be processed to increase the index to minimum 75. The immersion test shall be conducted on Marshall test specimens of design job mix after storing in 1% sodium chloride solution v/v distilled water for 24 hours at 60^0C and then tested for Marshall Stability values. The percentage loss or retention of stability values after immersion in water with reference to the values obtained before immersion is expressed in terms of percentage in the original values.

9.16.9 **JOB MIX FORMULA**

9.16.9.1 The blending of various types and sizes of aggregate shall be done on the basis of the requirements as indicated and optimum bitumen content within the specified limits.

9.16.9.2 The tenderer shall submit their job-mix formula to the Garrison Engineer for approval of design mix from CWE office before 30 days of start of Semi dense concrete work duly tested from CRRI, New Delhi/IITs/NITs/Highway Research Station, Chennai/SEMT Wing, CME, Pune/ National Test House or any other Govt lab.

9.16.9.3 The approved Job-mix formula shall remain effective unless and until the same is modified and re-approved by the CWE office. More than one Job-mix formula is permissible based upon the properties of actual materials available during construction of the project and also due to change in sources of material, if any.

9.16.9.4 For any change in source of supply of aggregate, fresh mix design shall be got done and testing charges shall be borne by the contractor.

9.16.9.5 Semi dense concrete shall be designed to satisfy the criteria as specified in clause here-in –before.

9.16.10 **PERMISSIBLE TOLERANCES**

9.16.27.1 The permissible variations of various ingredients shall be within the following limits:-

Description		Permissible variation	
		Base/binder course	Wearing course
(a) Aggregate passing 19mm sieve or larger	:	+8%	+7%
(b) Aggregate passing 119.2mm, 9.5mm	:	+7%	±6%
(c) Aggregate passing 4.75mm	:	±6%	± 5%
(d) Aggregate passing 9.36mm, 19.18mm, 0.6mm	:	±5%	± 4%
(d) Aggregate passing 0.3mm, 0.15mm	:	±4%	± 3%
(e) Aggregate passing 0.075	:	±2%	± 19.5%
(f) Binder content	:	±0.3%	± 0.3%
(g) Mixing temperature	:	± 10^0C	±10^0C

9.16.11 **MIXING AND TRANSPORATION OF MIX**

9.16.119.1 The temperature of binder at the time of mixing shall be in the range of 150 degrees Celsius to 165 degrees Celsius and of the aggregates in the range of 150 degrees Celsius to 170 degrees Celsius, provided that at no time, the difference in temperature between the aggregates and binder shall exceed 14 degrees Celsius. The mixing shall be thorough to ensure that a homogeneous mixture is obtained in which all particles of the mineral aggregates are coated uniformly and temperature of mix shall not exceed 165 degrees Celsius.

PARTICULAR SPECIFICATIONS (CONTD…)

9.16.119.2 Hot Mix Plant shall be used for mixing of aggregate and the binder. The binder shall be heated to the specified temperature. The aggregate shall be suitably warmed or heated before loading into hot mix plant. The correct quantity of each size of aggregate shall be fed into mixer with specified quantity of binder.

9.16.119.3 The mix shall be transported from the mixing plant to the point of use in suitable tipper vehicles specified here-in-after. The vehicles employed for transport shall be clean and be covered using suitable covers in transit to ensure that the temperature of mix does not fall below 125 degrees Celsius at the time of laying**.**

9.16.12 **SPREADING OF THE MIX**

9.16.19.1 Spreading of Semi dense concrete shall be done by means of self-propelled mechanical paver. SDAC shall not be laid during rainy weather or when the surface is damp or wet. The transverse joints or construction joints shall be placed in the vertical plane after cutting back to the original thickness of previously laid mix. The vertical cut faces shall be painted with bitumen (VG-10 Grade) prior laying of fresh mix.

9.16.13 **ROLLING AND COMPACTION**

9.16.13.1 The rolling and compaction process for asphaltic concrete (SDAC) shall be in four stages as described here in after. The rolling shall be done with 8 to 10 tone Roller used on static mode.

9.16.13.2 Rolling operations shall be conducted when the mix is neither too hot nor too cold, so that shoving or hair cracks may be eliminated. Rolling shall be continued till the density is achieved at least 98% of that of the Laboratory Marshall Specimen and all roller marks are eliminated.

9.16.13.3 Rolling operations shall be completed in all respects before the temperature of the mix falls below 100 degrees Celsius.

9.16.13.4 The wearing surface shall be cold rolled with pneumatic tyred roller for a minimum of 100 passes in a period of 3 month or opening to traffic whichever is earlier and as directed.

9.16.14 **PAVEMENTS EDGES**

9.16.14.1 Pavement shall be laid to correct width and alignment. To achieve straight and vertical edges, the contractor shall cut back the edges to correct width and alignment by removing extra mix spread. Nothing extra shall be paid on this account.

9.16.15 **BINDER CONTENT**

9.16.15.1 The binder content of the mix shall be tested periodically and as directed by the GE using centrifugal extraction method or such method approved by the GE. The test shall generally be carried out in accordance with established practices and as directed by the GE. Whenever, there is a deviation in the resultant binder content from approved job-mix-formula values, it shall be reported to the GE immediately and necessary correction made at the job-mix plant as directed.

9.16.16 **RATE**

9.16.16.1 Unit rate quoted for SDAC shall include for cost of all material, plant/tools/machineries and labour required in all operations described here-in-before. Rate shall include cost of all materials, plant, machinery and labour required in all the operations described above including all cartages and lifts involved.

9.16.17 **QUALITY CONTROL**

For quality control of SDAC refer Para 20.B.4.9 of MES Schedule Part-I 2009.

9.16.18 **CONTROL ON LAID THICKNESS**

For control on laid thickness of SDAC refer Para 20.B.4.10 of MES Schedule Part-I 2009.

9.16.19 **PLANTS/MACHINES TO BE USED FOR SEMI DENSE ASPHALTIC CONCRETE/ ASPHALTIC CONCRETE & BITUMINOUS CONCRETE :-**

For plants/Machines to be Used for Semi Dense Asphaltic Concrete/Asphaltic Concrete & Bituminous Concrete refer Para 20.B.4.11 of MES Schedule Part-I 2009.

9.17 **INSPECTION AT HOT MIX PLANT**

9.17.1 Before commencement of work, contractor shall submit the details of hot mix plant Proposed to be used by them for subject work for approval of GE. The plant will be inspected by GE for its effective functioning and approved if found satisfactory.

9.17.2 It shall be ensured that while preparation of asphaltic concrete at mixing plant, the representative of the contractor and Engineer-in-Charge or his representative detailed for the purpose shall be present to ensure that specified quantity of bitumen is incorporated in the work. The detailed record of bitumen so mixed shall be entered daily in the MB as Not to be Abstracted duly signed by Engineer-in-Charge and contractor.

9.18 **BITUMINOUS MACADAM**

9.18.1 For execution of bituminous macadam stages i.e preparation of surface, application of tack coat, preparation of mix, temperature, transportation of mixed materials, laying, joints in bituminous macadam, rolling, pavement edges, quality control, bitumen content & bituminous grading, acceptance criteria of field density, control on laid thickness & rectification refer clause 20.B.3 of section 20B (Runway Pavement & Hard Standing Flexible & Rigid Pavement) of MES Schedule 2009 Part-I Specifications. Anti stripping agent shall be added to the aggregate to reduce the stripping time of aggregate all as per manufacturer's instructions without any extra cost to the Government.

9.18.2 **RATE**

9.18.2.1 For the purpose of tendering, the contractor may base his rate on the assumption that the bitumen content of the mixture shall be 4% for bituminous macadam by weight of total mix compacted to 100% density in Marshall mould.

9.18.2.2 If the actual quantity of bitumen required to be used as a result of the laboratory test is found to be less than the quantity of bitumen i.e. assumed above, necessary minus adjustment in the cost of less bitumen used shall be made at the rate contained in Standard Schedule of Rates, Part-II 2020 for paving bitumen, adjusted with the percentage quoted for the particular Schedule. However if the bitumen content required to be used as a result of the laboratory test is found to be more than the specified content, the same shall be provided without any extra cost to the Govt. Except adjustment in cost due to variation in bitumen consumption as stated above, no other factor whatsoever shall be considered for adjustment in quoted rates.

9.18.2.3 Laboratory charges for the job mix formula design, cost of materials, its cartage and other incidental expenses shall be borne by the contractor and nothing extra shall be paid on this account. The tenderer shall submit their job-mix formula to the Garrison Engineer for approval of design mix from CWE office before 30 days of start of Bituminous macadam work duly tested from CRRI, New Delhi/IITs/NITs/Highway Research Station, Chennai/SEMT Wing, CME, Pune/ National Test House or any other Govt lab.

9.18.2.4 The approved Job-mix formula shall remain effective unless and until the same is modified and re-approved by the CWE office. More than on Job-mix formula is permissible based upon the properties of actual materials available during construction of the project and also due to change in sources of material, if any.

9.18.2.5 In the event of change in materials available at site due to unforeseen agencies, a fresh job mix formula shall be arrived at within the specified limits in the laboratory as approved by CWE. Nothing extra is payable on this account.

PARTICULAR SPECIFICATIONS (CONTD…)

9.19 **MOORUM / RED BAJRI**

9.19.1 Moorum/ Red Bajri shall be obtained from approved pits and quarries of disintegrated rocks containing siliceous material and natural mixture of clay of calcareous origin. These shall not contain any admixtures of ordinary earth. Red Bajri shall be dark red in colour consisting of coarse grains, free from mica and other foreign matter. Size of moorum/Bajri shall vary from dust to 40 mm gauge.

9.19.2 Ground where stacks are proposed to be made shall be dressed to a uniform slope and all lumps, depressions etc., shall be removed. Sample of moorum shall be got approved from the Engineer-in-Charge, before the material in bulk is brought to site. Moorum shall be stacked in convenient units of one cubic meter in between aggregate stacks in each length of 100 M as per requirement. The stacks shall be made with wooden boxes open at both ends and of 2x2x0.25 M dimensions. These shall always be kept at site for stacking and check measurements. The stacks shall be uniformly distributed along the road. Supply of moorum shall be completed for the entire work or for a complete length of one Km or as directed by Engineer-in-charge. The soaked CBR value of moorum shall be not less than 20.

9.20. **HOT APPLIED THERMOPLASTIC ROAD MARKING**

9.20.1 **GENERAL**

9.20.1.1 The work under this section consists of marking traffic strips using thermoplastic compound meeting the requirement specified herein. The thermoplastic compound shall be screeded /extruded on to the pavement surface in molten state by suitable machine capable of controlled preparation & laying with surface application of glass beads at a specific rate. Upon cooling to ambient pavement temperature, it shall produce an adherent pavement marking of specified thickness and width and capable of resisting deformation by traffic. The colour of the compound shall be white or yellow (IS:Colour No 356) as directed by the Engineer in charge.

9.20.2 **THERMOPLASTIC MATERIAL**

9.20.2.1 **GENERAL**: The thermoplastic material shall be homogeneously composted of aggregate, pigment, resins & glass beads reflectorizing.

9.20.2.2 **REQUIREMENT**: The pigment glass beads and aggregate shall be uniformly dispersed in the resin. The material shall be free from all skins, dirt & foreign objects & shall comply with requirement indicate in Table 800.3 of MOST specification.

9.20.2.3 **APPLICATION :** Before application, the surface should be thoroughly dry free from dust, moisture, grease, oil, loose particles or earliest coating. All road surface repairs should be carried out before applying the thermoplastic materials. The compound is preheated to around 180 degree (+10degree C) & transferred into machines. It reaches the surface by gravity and glass beads can be super imposed through a dispenser or manual spray for reflectivity. Alternatively, the compound can be screeded/extruded out from larger motorized machine as well. Normal application thickness is 2.5mm as per M.O.S.T (INDIA) specification, the use of suitable primer is recommended when Road marking compound is to be applied on concrete surfaces.

9.20.2.4 **APPLICATION PRECAUTION:** Before application the surface should be free from dust, oil spillage ,grease, loose particles, moisture cracks/unfilled space, excessive bleeding of bitumen. Previous cold paint marking must be removed by wire brushes before application. At the time of application road should be closed for at least 30min, for traffic passing. For better result use pre heater to melt the compound instead of directly heating the material in a trolley machine. Frequently check that the Glass beads dispenser sprinkles required quantity uniformly. Thermoplastic can be applied over both types of surfaces like asphalt bituminous and concrete surface. For concrete surface recommended primer should be used.

PARTICULAR SPECIFICATIONS (CONTD…)

G. FOR SCHEDULE 'A' PART VI : (AREA DRAINAGE)

1. GENERAL:- Refer relevant parts of particular specification here-in-before. The scope of work shall be executed as specified in Schedule 'A' and all as specified in SSR Part-I (2009).

H. FOR SCHEDULE 'A' PART VII : (SEWAGE DISPOSAL)

1. GENERAL:- Refer relevant parts of particular specification here-in-before. The work of sewage disposal shall be executed as specified in Schedule 'A'.

2. **LAYING OF PIPES**

2.1 The pipes shall be laid all as specified in para 18.69 of MES Schedule Part-I.

2.2 **JOINTING OF PIPES**

2.2.1 Jointing of pipes shall be carried out all as specified in paras 18.70 and 18.70.1 of MES schedule part-I.

2.3 **PIPES**

2.3.1 **RCC PIPES**. RCC pipe shall conforming to IS and shall be of approved make and jointing shall be done in cement mortar (1:1) as specified in MES Schedule Part-I.

2.3.2 **SWG PIPES.** Salt glazed stone ware pipes and fittings shall be of grade-A conforming to IS-651 and shall be of approved make and jointing shall be done in cement mortar (1:1) as specified in paras 18.70 and 18.70.1 of MES Schedule Part-I.

2.4 **PCC BEDDING, HAUNCHING AND COMPLETELY SURROUNDING**

2.4.1 RCC/SWG pipes shall be bedded in PCC (1:3:6) type C-2 and hunched against the sides of pipes after they are laid and tested as specified in paras 18.79.3 and 18.79.4 of MES schedule part-I.

2.5 **MANHOLES**

2.5.1 Manholes shall be built all as specified in paras 18.78 to 18.78.10 of SSR Part I and Schedule 'A' and as shown on drawing.

J. FOR SCHEDULE 'A' PART-VIII: EXTERNAL WATER SUPPLY

1. GENERAL

1.1 The scope of work in this tender covers external water supply to various buildings included in the tender and include all works described in various items of Schedule `A', particular specifications and as shown on drawings.

1.2 The layout of various water supply equipments, pipe lines, valves, fittings/accessories etc shown on various drawings attached with this tender are tentative. The exact layout shall be as directed by Engineer-in-Charge and approved by GE in writing to suit the site requirements. Contractor to ensure written approval before taking work in hand. No price adjustment, whatsoever, shall be admissible to the contractor on this account.

1.3 During the currency of work and also on completion of work, testing shall be carried out for all items and installation as a whole as per MES Schedule Part-I/IS/Manual on water supply and treatment of Min of Wks & Housing to the entire satisfaction of GE and a record shall be maintained duly signed by the Engineer-in-Charge and the contractor. For testing of pipe lines para 5.4.3 and Appx 10 of Manual on water supply and treatment of Min of Wks & Housing shall strictly be followed IS-3114 and IS-2065 shall specifically be road and adopted strictly.

1.4 Before laying the pipes, the detailed map of the area showing pipe alignment, sluice valves scour valves, air valves and fire hydrants alongwith the existing intercepting sewers, telephone and electric cables and gas pipes will have to be studied. Care should be taken to avoid the existing sewer, telephone and electric cable and gas pipes. The pipe line may be laid on the side of the street where the population is dense.

PARTICULAR SPECIFICATIONS (CONTD…)

1.5 Suitable flange joints in 80 mm bore GI pipe line as one joint per 100 Mtr to be provided by the contractor and cost of the same deemed to be included in item rate of the pipe.

2 PROTECTION AGAINST POLLUTION NEAR SEWERS AND DRAINS

2.1 Para 9.11.1 of the Manual of Water Supply and Treatment of the Min of Wks & Housing shall be referred to and adopted.

3. **TESTING**

3.1 The contractor will be responsible for the arrangement of testing. All the equipment, testing material etc will be arranged by the contractor at no extra cost.

3.2 The entire installation shall run continuously for 72 hours under normal operating conditions and contractor's engineer, fully familiar with the system and equipment, shall be available at the site during the period for proper rectifications of any defects.

3.3 Tendered rates shall be deemed to include all connected works required for supplying, erecting, testing and commissioning of the entire equipments (s) fittings and items like foundation for equipment, supplying laying and connecting of water main, various valves, fittings, hydrants etc. Minor connected items of work for proper execution and functioning of equipments though not specifically mentioned in the tender are deemed to be included in the rates quoted.

3.4 Unless otherwise specified civil engineering works required for installation of the equipments such as grouting of panels and equipments, trench cutting and back filling in ground and in road crossing etc shall be deemed as included.

3.5 Wherever the pipe line crosses the existing path/road, the contractor shall dig only half the width of path/road at a time. If required, necessary diversion shall be displayed and watchmen posted to prevent accident at crossings and diversions. After the pipes are laid, tested and trenches refilled, the road/path disturbed shall be made good to conform to the original specifications. No extra payment shall be made to the contractor for the aforesaid incidentals. The contractor shall be deemed to have visited site of works and made himself familiar.

3.6 The water supply work shall be carried out by properly skilled and licensed fitters/plumbers under the supervision of qualified supervisions/engineers. The contractor shall on demand by Engineer-in-Charge produce such evidence of qualifications of his workmen/supervisors engineers at the time of commencement of work and at any time thereafter/during the currency of work.

3.7 The items covered in this tender and incorporated in work shall be capable of delivering satisfactory service/operation without detriment to its life under the climatic conditions of site.

3.8 Before taking up any items of work under this tender in an area not belonging to Min of Def. the contractor shall give adequate notice to the local authorities through MES, He shall be responsible for making adequate arrangements for lighting, watching and guarding excavations which might be a source of danger to the public.

4 **ANCHORAGE/THRUST BLOCKS**

4.1 Thrust blocks of the shape, size and design as directed by GE shall be provided at abrupt changed in direction/gradient. Location of thrust blocks shall be as ordered by GE in writing.

4.2 Anchorages for valves fittings shall be provided as per design/ size approved/ordered by GE.

5. **MATERIALS**

5.1 **MILD STEEL GALVANISED TUBES (PIPES) AND FITTINGS**,:- Refer para 18.4 of MES Schedule Part-I. The pipe and fittings shall be ISI marked.

6 **WORKMANSHIP**

6.1 GENERAL REQUIREMENT - Refer paras 18.41 of MES Schedule Part-I. Generally layout of water supply work is shown on drawings. The exact position of water supply line and fittings shall as directed by the Engineer-in-Charge. The mounting height of various fittings shall be as directed by Engineer-in-Charge.

PARTICULAR SPECIFICATIONS (CONTD…)

6.2 LAYING OF GI PIPE - Refer para 18.46 of MES Schedule Part-I. Union shall be provided at suitable places as directed by the Engineer-in-Charge so that in case of repairs etc long length of pipes is not disturbed.

6.3 The contractor shall use proper bends/elbows/tees at turning/corners. Bending of pipes is not permitted.

6.4 The contractor shall provide screwed plug to all open bends of fittings at the end or days work and at time when the work is stopped.

6.6 FIXING OF WATER SUPPLY FITTINGS - Fixing shall be done as per manufacturer's instruction and as directed by Engineer-in-Charge.

6.7 TESTING - The water supply installation shall be tested hydraulically at a pressure of 2.5 kg/sq cm in the presence of Engineer-in-Charge for at least 24 hours continuously to ensure leak proof connections. All arrangement for testing shall be made by contractor at his own cost. Any leakage/defects noticed shall be rectified to entire satisfaction of Engineer-in-Charge.

6.8 Necessary cuttings/leaving/forming holes, chases, etc., in wall floors and ceiling and making good involved shall be deemed to have been included in the unit rate quoted by the contractor for respective buildings under schedule 'A' part I. No adjustment shall however be made on this account for pricing any deviation in quantities indicated as provisional items under Schedule 'A'.

7. **SLUICE VALVE** :- For sluice valve for water purpose refer IS code 14846 – 2000 Sluice Valve for water works purposes (50 to 1200 mm size). And for fixing of sluice valve refer clause 18.57 of MES Schedule SSR part I 2009.

8. CUTTING OF PIPE

The cutting of pipe for inserting valves or fittings shall be done in a neat and workman like manner without damage to the pipe so as to leave a smooth end at right angle to the axis of the pipe.

GUN METAL GATE VALVE

9.1 These shall be of gun metal fitted with wheel head (iron or gun metal), screwed/ flanged both ends for iron pipe opening full way and of the size as specified. These shall conform to IS 778. The weight of Gun Metal Gate valve shall be as under:-

Size in	Flanged ends mm	Screwed ends mm
15mm	1.021	0.567
20mm	1.503	.68
25mm	2.495	1.077
32mm	3.232	1.559
40mm	4.082	2.268
50mm	6.691	3.232
65mm	10.149	6.804
80mm	13.381	8.545

9.2 **FIXING GUN METAL GATE VALVE**

The Gun Metal Gate Valve shall be carefully examined and cleared of all foreign matter before being fixed. The Gun Metal Gate Valve shall be fitted in the pipe lines in a workman like manner. The joints between Gun Metal Gate Valve and pipe shall be made leak-proof when tested to a pressure of 17.5 Kg/sq cm.

9. RUN LEAD JOINTS

The spigot shall be centered in the adjoining socket by tightly caulking in sufficient turns of tarred gasket or hemp yarn to leave unfilled half the depth of socket for lead. When gasket or hemp yarn has been caulked tightly home, a jointing ring shall be placed round the barrel and against the faces of the socket. Molten pig lead shall then be poured in to fill the remainder of the socket, the lead shall then be solidly caulked with suitable tools and hammers of not less than 3 Kg weight, right round the joint to make up for the shrinkage of the molten metal on cooling and shall be preferably finished 3 mm behind the socket face. The pipes shall be perfectly dry before run lead joint are made, otherwise blow holes may occur in the lead.

PARTICULAR SPECIFICATIONS (CONTD…)

10.1 The lead shall be heated to proper temperature so that when stirred it will show a rapid change of colour. Before pouring, all scum shall be removed. Each joint shall be made in one continuous pouring. Care shall be taken that no grass enters the joint. Spongy or imperfectly filled joints shall be burnt out and re-poured.

10.2 The joint runner shall fit snugly against the face of the socket and the outside of the pipe shall be dammed with clay to form a pouring lip to provide for filling the joint flush with the face and to the top of the socket.

10.3 Any deviation either in plan or elevation less than 11-1/40 shall be effected by lying the straight pipes round a flat curve of such radius that the minimum thickness of lead at the face of socket shall not be reduced below 6 mm or the opening between the spigot and socket increased beyond 12 mm at any joint. A deviation about 2-1/40 can be effected at each joint this way.

H. FOR SCHEDULE 'A' PART-IX : EXTERNAL ELECTRIFICATION

1. GENERAL

(a) The scope of work covers external electrification to various buildings in this tender and includes all works described in various items of Schedule 'A' particular specification and as shown on drawing.

(b) The layout of various electrical equipments, fittings /accessories cables etc shown on various drawings attached with this tender are tentative. The exact layout shall be as directed by Engineer-in-Charge and approved by GE in writing to suit the site requirements. Contractor to ensure written approval before taking work in hand. No price adjustment what so ever, shall be admissible to the contractor on this account.

(c) Specifications of materials and workmanship in these external electrification services shall be as specified in MES Schedule part-I except where specifically mentioned in these documents. General rules, preamble to various section rates, special Conditions, methods of measurements) etc. given in SSR shall apply to this contract unless otherwise specified in these documents.

(d) Particular specifications given here-in-after are brief and are only to particularise, amend or emphasize, the aforesaid specifications which are not repeated here. In case of any discrepancy between the two, specification mentioned herein shall take precedence over MES Schedule Part-I.

(e) The tenderer shall be responsible for supplying, installing, laying, connecting, testing and commissioning of the item covered in the tender all as specified and directed by Engineer-in-Charge. The work shall be of high standard and executed as per sound engineering practice as per MES Schedule part-I/IE Rules/IS/National Electrical code.

(f) During the currency of work and also on completion of work, testing shall be carried out for all items/the installation, as a whole as per SSR/IE Rules/IS to the entire satisfaction of GE and a record shall be maintained duly signed by the Engineer-in-Charge and the contractor.

(g) Pre-commissioning test by EI/SEI will be carried out and record maintained. The contractor will be responsible for the arrangement of testing and all the equipment testing material will be arranged by the contractor at no extra cost.

(h) The entire installation shall run continuously for 72 hours under normal operating conditions and contractor's Engineer, fully familiar with the system and equipment, shall be available at the site during this period for proper setting of relays instruments etc, and rectifications of any defect.

PARTICULAR SPECIFICATIONS (CONTD…)

(j) The equipments for testing and arrangements for testing shall be made by the contractor without any extra cost.

(k) The tenderer shall include all requisite components, accessories like cable glands, lables, CTs selector switches shrouds, indication lamps, Auxiliary relays, contractors, toggle switches, handles, cable indicating dice, brackets and clamps of all kinds, instrument wiring, instrument fuses interlocks and all such other requirements for completion of the entire work though not specifically brought out in schedule 'A' and in these specifications. The tendered rates shall be deemed to include all connected works required for supplying, erecting, testing and commissioning of the entire equipments (s) fittings and items like foundation for equipments, supplying and connecting of cables in ducts etc.

(l) Unless otherwise specified civil engineering works required for installation of the equipments such as grouting of poles and equipments, trench cutting and back filling in ground and in road crossing etc shall be deemed as included.

(m) Wherever the cable crosses the cable existing path/road, the contractor shall dig only half the width of path/road at a time. If required, necessary diversion shall be provided. Proper warning signs and red light shall be provided. Proper warning signs and red light shall be displayed and watchman posted to prevent accident at crossings and diversions. After the cables are laid, tested and trenches refill the road/path disturbed shall be made good to conform to the original specifications. No extra payment shall be made to the contractor for the aforesaid incidentals. The contractor shall be deemed to have visited site of works made himself familiar.

(n) The electrical works shall be carried out by properly skilled and licensed electricians under the supervision of qualified electrical supervisor/engineers as per IE Rules. The contractor shall, on demand by Engineer-in-Charge produce such evidence of qualifications of his workman/supervisors/engineers at the time of commencement of work and at any time thereafter/during the currency of work.

(o) The items covered in this tender and incorporated in work shall be capable of delivering satisfactory services/operation without detriment to its life under the climatic conditions of the proposed site.

(p) Before taking up any item of work of this tender in an area not belonging to Min of Defence, the contractor shall give adequate notice to the local authorities through MES. He shall be responsible for making adequate arrangements for lighting, watching and guarding excavation which might be a source of danger to the public.

2. **TESTING OF TRANSFORMER/ LT PANELS**: Transformer/ LT panels shall be factory tested at manufacturer's /OEM premises before dispatch by representative of accepting officer. Intimation to this effect should reach Accepting Officer at least one month in advance enabling Testing Officer to plan his visit. Presence of contractor or his auth representative is required during factory testing.

3. **TIMELY PLANNING/PROCUREMENT ACTION BY CONTRACTORS**: E/M machines/equipments such as Transformer/ LT panels shall be brought at site by the contractor only after 80% progress of Bldg portion to avail manufacturer's guarantee/warrantee in full. However, necessary approval of make, drgs & technical details be submitted by the contractor to GE for his approval well in advance.

4. **SAMPLES AND MATERIALS**:

(a) All materials (equipments, fittings, cables, accessories etc) to be incorporated in the work shall invariably be ISI marked and of the make specified herein and approved by GE. In case any ISI marked items is not available in the country, this shall be conforming to BIS and of the make approved by GE.

(b) Amendments to IS issued till the date of receipt of tender are deemed to be included.

(c) In case IS does not exist for any item, it shall conform to current BS/DIM/ASTM, specifications in which case a copy of relevant standard shall be submitted by tenderer.

PARTICULAR SPECIFICATIONS (CONTD…)

(d) The contractor after obtaining written approval of makes and samples shall obtain the materials and equipments from an approved firm or through their authorised agents and shall ensure that the materials are brought to site in the manufacturer's original packing with seal intact.

(e) The contractor is required to submit the original purchase vouchers/cash memos duly machine numbered and dated in respect of purchase of the articles showing all taxes, makes and numbers as required. The purchase vouchers shall be produced to the Engineer-in-Charge for his verification of materials/equipments and obtaining the approval/permission of GE for release of payments/incorporation in works. On approval, the Engineer-in-Charge shall deface the purchase vouchers etc and the defaced original with two Xerox copies thereof shall be submitted by the contractor to the Engineer-in-Charge for reference and record. The original purchase vouchers shall be released to contractor after technical check and payment of final bill.

(f) No foreign exchange shall be made available by the Department.

(g) Manufacturer's test certificate as per IS test procedures for items to be incorporated in works shall be submitted by the contractor.

(h) The tenderer shall submit detailed catalogues/pamphlets/drawings for all electrical equipment(s) like UG & LT cables etc. to the GE for approval within one month of placement of work order. The GE would convey the decision in two weeks and thereafter procurement action be organised accordingly by the contractor.

5. **RECORD**:

(a) On completion of work, the contractor shall submit one tracing (on cloth) and three copies of detailed layout plans showing :-

(i) Detailed schematic diagram and S/S layouts, with equipment details.

(ii) Schematic circuit and control wiring diagram for LT panels.

(b) The following record for all buried cables :-

(i) Size and type of cable and manufacturer's number.

(ii) Location of the cables in relation to buildings, roads, etc., with depth.

(iii) Cross section showing where cables are laid in pipes or ducts giving their sizes, type and depths together with an indication of any spare ways available.

(iv) Position and type of all joints.

(v) Location of other cables which run alongwith or across the cable route.

(vi) Position and depth of all pipes, ducts etc which are met as obstructions to the cable route.

(vii) Accurate lengths from joint to joint.

(viii) Record of service lines as per MES Schedule Part-I para 19.96.

6. **SAFETY PROCEDURE AND PRACTICES**: In all electrical installations such as sub-stations industrial establishments, transmission and distribution lines and cable networks, safety procedures & instructions for working on low, medium and high voltage mains and apparatus and safety practices listed in IS-5216 (Guide for safety procedures and practices in electrical works) shall be followed to the extent applicable. The contractor shall provide workmen with safety devices and appliances.

7. **FIRE SAFETY**: All electrical equipment shall satisfy the requirements laid down in IS-1646 Code of practices for fire safety of buildings (general) electrical installation and IS-3034. Code of practice for fire safety of industrial buildings, electrical generating and distributing stations to the extent applicable.

8. **CABLES**:

(a) LT cables shall be XLPE insulated and PVC sheathed, heavy duty armoured all as specified in the Schedule 'A' conforming to IS-1554.

(b) CABLE HT 11 KV GRADE : The HT underground XLPE cables shall be 11KV grade, 3 core of size as specified in Schedule ‘A’ with Aluminium conductor suitable for earthed system.

(c) All cables shall bear ISI certification marks, These shall be brought at site wound on cable drum, ends sealed and with manufacturer's test certificates.

(d) All jointing with termination lugs/sleeve shall be done with hydraulic/ mechanical crimping tools.

PARTICULAR SPECIFICATIONS (CONTD…)

(e) All LT electric power cable connected in switches, bus bar, LT feeder pillar box etc will be connected with suitable size of aluminum lugs and entry of cable only through proper size of steel/brass chromium plated cable gland all as directed by Engr-in-Charge.

(f) **METHOD OF LAYING**: Before laying the cable, the trench shall be provided with a layer of sand of 8 cm depth for the purpose of cushioning. After the cable has been uncoiled and laid into the trenches over the rollers, the cable shall be lifted slightly over the rollers beginning from one end by helpers standing about 10 metres apart and drawn straight. The cables then be taken of the rollers by additional helpers lifting the cable and then laid straight into the route. When the cable has been properly straightened, it shall be covered with sand to a depth of 15 cm. This is then gently punned down to a depth of 10 cm above top of the upper most cable, thus providing a good bedding for the protective cable cover on warning covers which are placed centrally over the cable i.e. by PCC slab. The punning shall be done by hand, mechanical punners shall not be used.

(a) Depth of trench :- Depth from the upper surface of the street pavement or ground to the cable axis shall be not less than the following :-

(i) Upto 1.1 KV Working pressure - 45 cm plus radius of complete cable.

(ii) 3.3 KV to 11 KV Working pressure - 75 cm plus radius of complete cable.

(iii) 22 KV to 33 KV Working pressure - 1 metre plus radius of complete cable.

(g) **SPACING BETWEEN CABLES**: When more or one multicore or trefoil group of single core cables are laid in the same trench a horizontal inter axial spacing of 25 to 40 cm shall be provided to reduce the effect of mutual heating and to ensure that a fault occurring in one cable will not damage the adjacent cable. When more than one tier of cable is necessary, riddle soil is filled in the trench, and rammed down to from a solid bed to a height of 25 to 40 cm above that of the first tier. The mesh of the screen through which the soil is riddled should not exceed 12.5 mm IS sieve. The width of the trenching shall in no case be less than 45 cm so as to facilitate working on the cables. Laying of cables in tier formation shall be avoided as far as possible as loading of cables is very much reduced.

(h) **ROAD CROSSING**: When crossing roadways, it is preferable to cross on the askew to reduced the angle of the bend as the cable enters leaves the road crossing. The cable shall pass through pipe as per schedule ‘A’.

(i) **PULLING OUT THE CABLE**: On no account the cable shall be allowed to twist or kink as it is likely to spring the Armour and fracture the paper insulation and outer serving of the cable. When pulling cable around the bends, rollers should be placed sideways so as to keep the cable on a safe radius. Cable drums should be rolled in the direction of arrow.

(j) **MINIMUM BENDING RADIUS**: Cables shall always be bend (or straightened) slowly they should never be bend to a small radius. The minimum safe bending radius for impregnated paper insulated cables may be taken as given below, but wherever possible larger radius should be used.

(k)

Voltage	Minimum Bending Radius		
	Single core cables	Multicore cables	
		Unarmoured	Armoured
Upto 11 KV	20 D	15D	12 D
Upto 33 KV	30 D	25 D	20 D

Note: ‘D’ is the overall diameter of the cable. At joints and terminations, the individual of Multiple cable should never be bent so that the radius of the bend is less than15 times the diameter over the insulation.

PARTICULAR SPECIFICATIONS (CONTD…)

(l) **CABLE LAID IN DUCTS OR FLOORS**: Cables ducts should be of such dimensions that the cables laid in it do not one another. If necessary, cables may be fixed with clamps on angle iron support on the walls of the ducts. The duct should be covered with removable slabs or chequered plates and filled with sand after the cables have been laid without any cost to the Government.

(m) **CABLE LAID ALONG THE SURFACE OF THE WALLS**: Cables may be fixed by means of suitable MS clamp or bracket or on cable racks, as directed. If necessary the cable may also be laid in a chase cut in the wall and fixed with clamps. The chase may be covered with removable MS Sheet.

(n) **CABLE ENDS**:All PVC cables upto 1100 volts grade shall be terminated into the equipment by means of compression type cable glands having screwed nipples and check nuts. The cable conductors shall be provided with crimp type solder less cable socket or connections to suit the termination arrangement of the equipment. The crimping shall be done by hand/hydraulically operated crimping tool after applying conducting jelly. The insulation of the cable shall be removed just immediately before the crimping is done.

(o) The cable route indicator shall be made out of 50x50x6 mm angle iron, 60 cm long embedded in PCC (1 : 4: 8) type D-2 of size 30x30x30 cm with the letter 'HT CABLE' OR "LT CABLE" suitably inscribed on CI circular plate 8 mm thick 100 mm dia. The cable route indicator shall be provided at the ends, for loops, at bends and at interval of 25 m in straight runs and cost for the same shall be included in the quoted rate of respective cable items & no extra cost will be admissible on this ground.

(p) **TESTING OF CABLES**: All low voltage and high voltage cables shall be tested by Megger before laying and after carrying out the joints. The cables shall be tested for: -
- (i) Continuity.
- (ii) Absence of cross phasing.
- (iii) Insulation resistance to earth.
- (iv) Insulation resistance between conductors.

(q) **TESTING AFTER LAYING AND JOINTING**: - The following tests shall also be carried out after the initial laying and jointing work.
- (i) Insulation resistance test : Sectional and overall
- (ii) Continuity Test : Sectional and overall

(r) Insulation resistance test shall be carried out between conductor and earth by Meggar of the following voltages:-
- (i) LT Cables - 0.5/1 KV
- (ii) HV Cables - 2.5/5 KV

9. **CABLE JOINTS**:

(a) Jointing work shall be carried out only by a licensed/experienced cable jointer. Sufficient surplus cable approximate 3 meter shall be left on each end of the cable and on each side of underground joints at the time of original installation. A caution board indicating "CAUTION CABLE JOINTING WORK IN PROGRESS" shall be displayed to public and traffic where necessary.

(b) Cable Terminations and Straight Through Joint Kits for 3 Core HT XLPE Cables

(c) The cable termination kits and straight through joint kits shall be suitable for HT. XLPE cables shall be of Cold shrinkable or push type on with sound and proven technology. These terminations shall be environmental friendly and flame retardant.

(d) Each kit shall contain all necessary components accessories, including cable glands wherever required cable lugs (climping type) jointing materials and consumables to make a complete termination in such a way that no live parts of the terminal and connecting lugs are exposed.

(e) The basic raw material used in construction of Joints and termination shall be Electrical grade EPDM (Ethylene Propylene Diene Monomer Rubber component) with high dielectric strength. The termination shall control the voltage distribution of electrical field effectively and shall minimize the surface stress by uniformly redistributing the electrical field over the entire surface of the insulator. Insulators shall be made of silicon rubber with long service life.

PARTICULAR SPECIFICATIONS (CONTD…)

(f) The straight through joint kits shall be suitable for underground buried installation with uncontrolled back fill and chances of flooding by water and suitably designed to be protected against rodent and termite attack. Further the joint kit shall be in capsulated in cast resin compound to provide excellent mechanical & moisture protection in sub soil conditions. While making termination, earthing connection & outer protection arrangement shall also be ensured.

(g) The test certificates from the manufacturer for termination kits and straight through joints kits shall mention results of the following tests :
- (i) AC Voltage withstand test (as per IEC-68)
- (ii) Partial discharge test
- (iii) Impulse withstand test (as per IEC-68)
- (iv) Load cycling test (as per VDE-2078)
- (v) Thermal short circuit test.
- (vi) DC Voltage withstand test.
- (vii) Humidity test (as per IEC-166)
- (viii) Dynamic short circuit test (as per VDE-2078)
- (ix) Salt Fog (outdoor termination only) test.
- (x) Impact test (for joints only)

(h) Cable End Termination and Straight Through Joint for 1100V Grade XLPE Cables Cable end termination Indoor/outdoor type and straight through joints for LT cables shall be based on cast resin system. The casting resin shall be polyurethane based having compatibility with the cable components. The termination kit shall be designed for easy installation and have a crutch sealing arrangement for environmental protection on the core trifurcation. The joints shall be suitable to withstand mechanical impact, heat shock tests, load cycle test and water, penetration test (under pressure) without damage to the outer sheath.

(i) **CABLE TERMINATIONS**: All cables shall be terminated at the end in such a manner as to prevent ingress of moisture into cable. The type and method of termination of cable shall depend on the voltage of cable and type of termination whether indoor or outdoor. All types of cable and termination joints shall be indicated as prescribed by the cable manufacturers. However, the following types of joints and termination systems are common and popular.
- (i) Tapex system
- (ii) Heat shrinkable sleeve type
- (iii) Push on or slip on type technique

10. **EARTHING**:

(a) All the non-current carrying metal parts of electrical installation shall be earthed properly. All metal conduits, cable sheathe, switch gear, distribution board, light fittings. MS boxes and parts made of metal shall be bounded together and connected by means of specified earthing conductors to an efficient earthing system. All earthing shall be in conformity with IE Rules.

(b) Neutral conductor pipe conveying water, structural steel work, metallic enclosures, metallic conduits and lighting protection system conductors shall not be used as means of earthing an installation of even as a link in an earthing system. The electrical resistance measured between earth connection at the main distribution board and any other point on the completed installation shall be low enough to permit the passage of current necessary to operate circuit breakers and shall not exceed phase. All metal clad switches and other equipment carrying single phase current, shall be connected to earth by a single connection. All metal clad switches, carrying medium voltage shall be connected with earth by two separate and distinct connections.

(c) No earth pit shall be made within 2 metre of a wall or foundation. Effort shall be made to locate them in grass lawns or near flower beds or water taps. The distance between two earthing pits shall be at least 2 metres.

PARTICULAR SPECIFICATIONS (CONTD…)

(d) **TESTING**:

(i) After completion of wiring, contractor shall submit three copies of wiring diagram.

(ii) The installation after completion of work shall be tested for insulation, wiring continuity, earth continuity, earth resistance polarity test etc, as specified in IS/IE Rules. The test shall be carried out in the presence of Engineer-in-Charge and shall be signed by Engineer-in-Charge and the contractor. Three copies of test sheet shall be submitted by contractor. In case the earth result does not fall within the specified limit, as mentioned in IS/IE Rules, the contractor shall dig earth pit beyond 2.5 m deep to obtain the desired earth results.

(iii) All testing instruments labour, materials and incidentals necessary to conduct the test shall be arranged by the contractor at his own cost.

(iv) On testing after completion, if desired earth resistance is not obtained the earthing will be redone at no extra coat.

12. **MOULDED CASE CIRCUIT BREAKERS**

12.1 Moulded case circuit breakers (MCCB) or fuse free breakers, incorporated in switchboards wherever required, shall conform to IS 13947: 1993 in all respects. MCCBs shall be suitable either for single phase 240 Volts or 3 Phase 415 Volts AC 50 Hz supply. MCCB cover and case shall be made of high strength heat resisting and flame retardant thermosetting insulating material. Operating handle shall be quick make/break, trip-free type. Operating handle shall have suitable ON, OFF and TRIPPED indicators. Three phase MCCBs shall have a common handle for simultaneous operation and tripping of all the three phases. Suitable are extinguishing device shall be provided for each contact. Tripping unit shall be of thermal/ magnetic type provided on each pole and connected by a common tripe bar such that tripping of any one pole causes three poles to open simultaneously. Thermal/magnetic tripping device shall have IDMT characteristic for sustained over loads and short circuits.

12.2 Contact trips shall be made of suitable are resistant sintered alloy. Terminals shall be of liberal design with adequate clearances.

12.3 MCCBs shall be provided with following accessories, if specified in drawings/schedule of quantities:

(a) Under voltage trip
(b) Shunt trip
(c) Alarm switch
(d) Auxiliary switch

12.4 MCCBs shall be provided with following interlocking devices for interlocking the door a switch board.
(a)Handle interlock to prevent unnecessary manipulations of the breaker
(b) Door interlock to prevent door being opened when the breaker is in ON position.
(c) De-interlocking device to open the door even if the breaker is in ON position.

12.5 MCCBs shall have rupturing capacity as specified in drawings/schedule of quantities All MCCB shall be provided with adapter terminal for facilitates higher sizes of cable/links

13. **Metering, Instrumentation and Protection.**

13.1 Ratings, type and quantity of meters, instruments and protective devices shall be as per drawings and schedule of quantities.

14. **Ammeters**

14.1 Ammeters shall be of moving iron type. Moving part assembly shall be with jewel bearings. Jewel bearings shall be mounted on a spring to prevent damage to pivot due to vibrations and shocks.

14.2 Ammeters shall be manufacture and calibrated as per IS 1248

14.3 Ammeters shall normally be suitable for 5 A secondary of current transformers.

14.4 Ammeters shall be capable of carrying substantial over loads during fault conditions.

PARTICULAR SPECIFICATIONS (CONTD…)

15. **Voltmeters**

15.1 Voltmeters shall be of 3 phase electro dynamic type and shall be provided with a maximum demand indicator if required.

16. **Watt meter**

16.1 Wattmeter shall be of 3 phase elector dynamic type and shall be provided with a maximum demand indicator if required.

17. **Power factor meters**

17.1 3 phase power factor meters shall be of electro dynamic type with current and potential coils suitable for operation with current and potential transformers provided in the panel. Scale shall be calibrated for 50% lag- 100% - 50% readings. Phase angle accuracy shall be +40.

18. **Energy and reactive power meters**

18.1 Trivector meters shall be two element, integrating type, KWH, KVA . KVA hour reactive meters. Meters shall confirm to IEC 170 in all respects. Energy meters, KVA and KVARH meters shall be provided with integrating registers. The registers shall be able to record energy conception of 500 hours corresponding to maximum current at rated voltage and unity power factor. Meters shall be suitable for operation with current and potential transformers available in the panel.

19 **Steel tubular Swaged poles &Struts.**

19.1. Steel tubular swaged poles & struts shall comply with the requirements of IS-2713-1980 (Part-I to III) specification. The steel tubular swaged poles shall be seamless well finished, clean and free from harmful defects. The poles shall be straight, smooth, cylindrical and the end of the pole shall be cut square. The base plate shall be of cast iron and base plate, finial taper plug, screws and bolts shall be conform to the dimensions given in Appx 'C' of IS 2713 – 1980 (Part-I to III) specifications. The poles shall be painted with two coats of black bituminous paint in all interval surfaces of poles with base plate & taper plug and external surfaces upto the portion embedded in concrete with base plate, the remaining external surfaces shall be painted with two coats of Aluminium paint over a coat of red oxide primer.

J. FOR SCHEDULE 'A' PART X : (COMPOUND WALL)

1. GENERAL:- Refer relevant parts of particular specification here-in-before. The scope of work shall be executed as specified in Schedule 'A' and all as specified in SSR Part-I (2009).

K. FOR SCHEDULE 'A' PART XI:- (MISCELLANEOUS ITEMS)

1 to 2 **BLANK**

3. **PVC INSULATED 1100 VOLTS GRADE CABLE (PVC SHEATHED/UN- SHEATHED)**

3.1 The conductor shall be of stranded copper wires. The insulation shall consist of compounded polyvinyl chloride. The thickness of insulation shall not be less than nominal value mentioned in appropriate table of IS-694-1964. The composition of the compound of sheath shall be polyvinyl chloride. The compound shall be designed to offer adequate mechanical protection. The value of thickness of sheath shall be in accordance with those specified in appropriate tables. The insulation on and sheath shall be applied by the extrusion process and each shall form a compact homogeneous body.

4 BLANK

5. **EARTHING**

5.1 All the non-current carrying metal parts of electrical installation shall be earthed properly. All metal conduits, cable sheathe, switch gear, distribution board, light fittings. MS boxes and parts made of metal shall be bounded together and connected by means of specified earthing conductors to an efficient earthing system. All earthing shall be in conformity with IE Rules.

5.2 Neutral conductor pipe conveying water, structural steel work, metallic enclosures, metallic conduits and lighting protection system conductors shall not be used as means of earthing an installation of even as a link in an earthing system. The electrical resistance measured between earth connection at the main distribution board and any other point on the completed installation shall be low enough to permit the passage of current necessary to operate circuit breakers and shall not exceed phase. All metal clad switches and other equipment carrying single phase current, shall be connected to earth by a single connection. All metal clad switches, carrying medium voltage shall be connected with earth by two separate and distinct connections.

PARTICULAR SPECIFICATIONS (CONTD…)

5.3 No earth pit shall be made within 2 metre of a wall or foundation. Effort shall be made to locate them in grass lawns or near flower beds or water taps. The distance between two earthing pits shall be at least 2 metres.

6 **TESTING**

6.1 After completion of wiring, contractor shall submit three copies of wiring diagram.

6.2 The installation after completion of work shall be tested for insulation, wiring continuity, earth continuity, earth resistance polarity test etc, as specified in IS/IE Rules. The test shall be carried out in the presence of Engineer-in-Charge and shall be signed by Engineer-in-Charge and the contractor. Three copies of test sheet shall be submitted by contractor. In case the earth result does not fall within the specified limit, as mentioned in IS/IE Rules, the contractor shall dig earth pit beyond 2.5 m deep to obtain the desired earth results.

6.3 All testing instruments labour, materials and incidentals necessary to conduct the test shall be arranged by the contractor at his own cost.

7 **LT PANEL**

7.1 PANEL CONSTRUCTION

7.2 The panel shall be fabricated out of 2mm thick for indoor and 3.15 mm thick for outdoor CRCA sheet steel using CNC machine. The enclosure shall have welded frame. Construction with CRCA frame sections with minimum 5mm thickness.

7.3 The door, partitions and covers shall be CRCA sheet steel of 2mm thickness. The whole enclosure shall be pretreated as per eight tank pre treatment process and finished with powder coating of shade RAL 7032.

7.4 The panel shall be compartmentalized (Modular) design. The front portion of the enclosure shall be for mounting switch gear and the rear potion for mounting bus bar, cables etc. Suitable ventilation in the form of louvers shall be provided for better heat dissipation. All live parts shall be shrouded with FRP material of fire retardant grade.

7.5 The whole panel shall be fabricated and assembled in a CPRI (Central Power Research Institute Bangalore) certified factory and certificate in original to the effect for IP 52, IP-53 (As per IS-13947 (Part-I) 1993 and IP-5x, IP x 2 as per IS 13947 (Part-I) 1993, Temperature rise (as per IS-8623 (Part-I) 1993, Short circuit withstand strength in accordance with Sub clause 8.2.3 of IS 8623 (Part-I) 1993/IEC Pub 439-1 (1985) shall be produced by the contractor for verification of the Engr-in-Charge and GE.

7.6 **BUS BAR**

7.6.1 The aluminium bus bar shall be suitable sleeved with heat shrinkable PVC sleeves. The bus bar system shall have a short circuit with stand capacity of 50 KA for 1 second. The bus bar shall be supported on SMC type, single grip type support. The enclosure shall be provided with an earth bus bar of minimum 50 x 6 mm aluminium.

7.7 **INTERNAL WIRING**

7.7.1 The power wiring of the panel shall be done using suitable size copper cables ISI marked and shall have superior mechanical, electrical and thermal properties.

7.8 **DOCUMENTATION**

7 .8.1 The contractor shall provide the complete wiring diagram of the panel duly certified by the manufacturer for record.

7.9 **BATTERY BANK:-** shall be provided as described in Schedule ‘A’.

7.10 **CABLE ROUTE INDICATOR:** The cable route indicator shall be made out of 50x50x6 mm angle iron, 60 cm long embedded in PCC (1 : 4: 8) type D-2 of size 30x30x30 cm with the letter ‘HT CABLE’ OR “LT CABLE” suitably inscribed on CI circular plate 8 mm thick 100 mm dia. The cable route indicator shall be provided at the ends, for loops, at bends and at interval of 25 m in straight runs.

PARTICULAR SPECIFICATIONS (CONTD…)

8 **LIGHT FITTINGS**:

8.1 **Industrial Type Fittings**

(a) These fittings shall be suitable for use with incandescent/ fluorescent / mercury vapor / sodium / Metal halide lamps as per requirements and generally as described below.

(b) The angle or cut-off for fittings with filament lamp shall not exceed 70° C and that for fittings with fluorescent tubes shall not exceed °.

(c) The luminous output of each reflector shall not be less than:

(i) 60 % in case of vitreous enameled reflectors with incandescent and mercury vapour lamps.

(ii) 80 % in case of anodized aluminium reflectors with lamps as in (a) above

(iii) 75 % in case of fluorescent lamps irrespective of the type of reflector used

(d) The distribution of light shall be such that at least 80% of the total luminous flux from the fitting shall be in the lower hemi-sphere.

(e) Fluorescent lamps to be mounted in the false ceiling shall be installed so as to form a continuous luminous ceiling, shall have an inside reflector of non fluorescent power occupying 2/3 of inner surface of the tube so that the lamp radiates light mainly in the direction determined by the position of the reflective material.

8.2 **Bulkhead and Well glass Type**

(a) Bulkhead, weather-proof type fittings for use in corridors, staircase landings covered porches and low ceiling factories, shall be suitable for direct fixing to wall ceiling and to accommodate upto 150 watts incandescent lamp. The fitting shall be provided with housing of cast aluminium alloy with stove enameled finish, prismatic heat resistant glass cover hinged on to the housing and fixed by screws, Neoprene gaskets shall be provided to make the fitting weather-proof. For mechanic protection to glass cover, a protective wire-guard of 3 mm galvanized wire with stove enameled finish paint shall be provided.

(b) Well glass, weather-proof type fittings for use in building exteriors and common walls, shall be suitable for conduit mounting and to accommodate upto 200 wail incandescent lamps or 125 W HPMV lamp. The fitting shall be provided with stove enameled cast aluminium housing with top conduit entry, heat resistant clear glass cover unit hinged to the housing and fixed by screws. The fittings shall be provided with neoprene gaskets and wire guards as started in above item.

8.3 **High Bay / Low Bay Type**

The fittings shall be suitable for mounting to the roof structure and for this purpose suitable hook for suspension shall be provided. The fitting shall be suitable for up 400 watts mercury vapor/metal halide lamp. If mercury vapor lamp or metal halide is provided, the fitting shall be provided with an integral housing for the control gear complete with ballast, capacitor, fuse-gear and internal wing. The integral housing shall be designed for efficient cooling of the unit. The canopy and housing shall be of cast aluminium, stove-enameled and the reflector shall anodized aluminium sheet, mirror polished.

8.4 **ACCESSORIES FOR LIGHT FITTINGS**: Fluorescent tube lamps, fitting and accessories .shall comply with following standard.

IS 1596 :1976 Capacitor for use in tubular fluorescent , high pressure mercury and low pressure sodium vapour discharge lamp circuit.

IS 1777 :1978 Industrial luminaries with metal reflectors

IS 2215 :1983 Starters for fluorescent lamps

IS 2418 (Part 1 to 4) :1977 Tubular fluorescent lamps for general lighting services

IS 3323 :1980 Bi-Pin lamps holders for tubular fluorescent lamps

IS 3324 :1982 Holders for starters for tubular fluorescent lamps

IS 3287 :1965 Industrial light fittings with Plastic reflector

PARTICULAR SPECIFICATIONS (CONTD…)

8.5 **Reflectors**

(a) The reflectors shall be made of CRCA sheet steel/ aluminum / silvered glass/chromium plated sheet copper as indicated for the above-mentioned fittings unless otherwise specified.

(b) The thickness of steel / aluminum, shall comply with relevant Standards specified. Reflectors made of steel, shall have stove-enameled/Vitreous-enameled/Epoxy-coating finish. Aluminum used for reflectors, shall be anodized / epoxy stove-enameled/mirror polished. The finish for the reflector shall be indicated above mentioned fitting.

(c) Aluminium paint on the reflectors of flameproof fittings is prohibited.

(d) Reflectors shall be free from scratches or blisters and shall have a smooth and glossy surface having an optimum light reflecting co-efficient such as to ensure the overall light output specified by the MANUFACTURER.

(e) Reflectors shall be readily removable from the housing for cleaning and maintenance without disturbing the lamps and without the use of tools, they shall be securely fixed to the housing by means of positive fastening device of captive type.

8.6 **Lamp/Starter Holders**

(a) Lamp Holders shall comply with relevant Standards. They shall have low contact resistance, shall be resistant to wear and shall be suitable for operation at the specified temperature without deterioration in insulation value. They shall hold the lamps in position under normal condition of shock and vibration met within normal installation and use.

(b) Lamp Holder for the fluorescent lamps shall be of the spring loaded bi-pin rotor type. Live parts of the lamp holder shall not be exposed during insertion or removal of the lamp or after lamp has been taken out. The lamp holder contacts shall provide adequate pressure on the lamp cap pings when in working position.

(c) Starter holders for fluorescent lamps shall conform to the relevant Standards. All material used in the construction of the holder shall be suitable for tropical use.

(d) The Starter Holder shall be so designed that they are mechanically robust and free from any operational difficulties. They shall be capable of withstanding the shocks met within normal transit, installation and use.

8.7 **Ballasts**

(a) The ballasts shall be designed, manufactured and supplied in accordance with the relevant Standards. The ballasts shall be designed to have a long service life and low power loss.

(b) The electronic ballasts shall be fully enclosed in an aluminum housing shall include a divided wiring compartment to separate the power leads from the control leads. All leads to be color coded.

(c) The electronic ballast shall be multi voltage capable and operate from a voltage range of 180 – 305V at 50Hz.

(d) Ballasts shall be mounted using self locking, anti-vibration fixings and shall to remove without demounting the fittings. They shall be in dusting combustible enclosures.

(e) The ballasts shall be of the inductive, hea1y duty type, filled with then insulating, moisture repellent polyester compound filled under pressure or Ballasts shall be provided with tapings to set the voltage within specified. End connections and taps shall be brought out in a suitable block, rigidly fixed to the ballast enclosure. The ballast wiring shall bed wire. They shall be free from hum. Ballasts which produce humming be. replaced free of cost by the VENDOR.

(f) Separate ballast for each lamp shall be provided in case of multi lamp fittings.

8.8 **Lamps**

(a) The fluorescent lamps shall be "Cool Day Light" type unless otherwise specified and shall be provided with features to avoid blackening of lamp ends. The fluorescent lamps shall have a high lumen output of 3250 lumens. The lamp shall have triple coil electrode with an anode ring and a tri-band Phosphor coating.

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(b) The lamps shall be capable of withstanding small vibrations and the connections at lead in wires and filaments/electrodes, shall not break under such circumstances.

(c) Lamps/Tubes shall conform to relevant Standards and shall be suitable for supply voltage and frequency specified.

8.9 **Starter**

(a) Starters shall have bimetal electrodes and high mechanical strength. starters shall be replaceable without disturbing the reflector or lamps and without the use of any tool. Starters shall have brass contacts and radio interference capacitor.

(b) The starters shall generally conform to the relevant Standards.

8.10 **Capacitors**

(a) The capacitors shall have a constant value of capacitance and shall be connected. across the supply of individual lamp circuits

(b) The capacitors shall be suitable for operation at supply voltage and shall have a value of capacitance so as to correct the power factor of its corresponding lamp circuit to the extent of 0.95 lag or better.

(c) The capacitors shall be hermetically seated preferably in a metal enclosure to prevent seepage of impregnate and ingress of moisture.

8.11 **Spare Parts**

(a) Unit prices of the items shall be quoted together with catalogue numbers.

(b) The unit prices shall not however be limited to the above items. The VENDOR may recommend additional spare items and quote the unit prices of the respective items.

8.12 **Tests and Test Reports**

(a) Type tests, acceptance tests and routine tests for the lighting fittings and covered by this specification shall be carried out as per the relevant standards for the respective fittings and their accessories.

(b) The MANUFACTURER's type and routine test certificates shall be submitted for tests conducted as per relevant standards for the fittings and accessories. The BIDDER shall submit with his proposal copies of available test certificates of the fittings offered.

8.13. **JUNCTION BOX**

8.13.1 Junction box shall be provided as described in Sch 'A'/ BOQ.

9. **BLANK**

10. **Moulded Case Circuit Breakers**

10.1 Moulded case circuit breakers (MCCB) or fuse free breakers, incorporated in switchboards wherever required, shall conform to IS 13947: 1993 in all respects. MCCBs shall be suitable either for single phase 240 Volts or 3 Phase 415 Volts AC 50 Hz supply. MCCB cover and case shall be made of high strength heat resisting and flame retardant thermosetting insulating material. Operating handle shall be quick make/break, trip-free type. Operating handle shall have suitable ON, OFF and TRIPPED indicators. Three phase MCCBs shall have a common handle for simultaneous operation and tripping of all the three phases. Suitable are extinguishing device shall be provided for each contact. Tripping unit shall be of thermal/ magnetic type provided on each pole and connected by a common tripe bar such that tripping of any one pole causes three poles to open simultaneously. Thermal/magnetic tripping device shall have IDMT characteristic for sustained over loads and short circuits.

10.2 Contact trips shall be made of suitable are resistant sintered alloy. Terminals shall be of liberal design with adequate clearances.

10.3 MCCBs shall be provided with following accessories, if specified in drawings/schedule of quantities:

(a) Under voltage trip
(b) Shunt trip
(c) Alarm switch
(d) Auxiliary switch

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10.4 MCCBs shall be provided with following interlocking devices for interlocking the door a switch board.
(a)Handle interlock to prevent unnecessary manipulations of the breaker
(b) Door interlock to prevent door being opened when the breaker is in ON position.
(c) De-interlocking device to open the door even if the breaker is in ON position.

10.5 MCCBs shall have rupturing capacity as specified in drawings/schedule of quantities All MCCB shall be provided with adapter terminal for facilitates higher sizes of cable/links.

11. **MASONRY VALVE PIT :-**

Valve pit shall be constructed all as mentioned in Sch 'A' and as directed the specifications for valve pit shall be as under:-

Sr	Item	Specifications
1	2	3
a	Excavation	In soft/ loose soil and surplus be removed to as distance not exc 50 m & deposited as directed by Engr-in-charge
b	Foundation	PCC (1:4:8) Type D-2
c	Wall	Burnt brick work with subclass 'B' bricks with CM (1:4)
d	Benching/ Copping	PCC (1:3:6) Type C-1
e	Bed	PCC (1:3:6) Type C-1
f	Plastering	10 mm thick in CM (1:4) internally and externally upto 15 CM below ground level. Top of PCC bed shall be plastered in CM (1:4) and finished even and smooth using extra cement. Top portion of valve pit as shown on drawing shall be plastered 10 mm thick cement mortar (1:4)
g	Butt hinges	Shall be purpose made of heavy pattern suitable for fixing to cover plate as approved by GE.
h	Hasp & staple	75 mm long mild steel safety type.
j	MS Sheet	MS Sheet shall be 3 mm thick as approved by GE.
k	Angle Iron	Angle iron shall be of 40 x 40 x 6 mm thick as approved by GE.
l	Painting	(i) All steel members except covered under ser No (ii) below shall be painted with two coats of synthetic enamel paint over a coat of red oxide primer. (ii) Steel surface which are embedded in concrete/ brick masonry shall be treated with two coats of hot tar and sanded before fixing.

12. **Rubber Gasket for Jointing :-** Refer to Para 18.11 of MES SSR Schedule, Part-I 2009 and jointed as per relevant IS/manufacturers instruction.

13. Ductile Iron Pipe and Fitting:-:- Refer to Para 18.94 of MES SSR Schedule, Part-I 2009 and IS code 8329- 2000 Specification for centrifugally cast (spun) ductile iron pressure pipes for water, gas and sewage (Third revision) and IS code 9523- 2000 Specification for ductile iron fittings for pressure pipes for water, gasand sewage (First revision).

14. **TRANSFORMER:**

14.1 Transformer shall be subjected to following routine and supplementary tests at the manufacturers works as per IS 1180 and Departmental representative will witness the tests on the transformer. Successful tenderer shall make all arrangements for witnessing the test by the representatives of the Accepting Officer.

[a] Measurement of winding resistance.
[b] Ratio, polarity and phase relationship.
[c] Impedance voltage. Load losses.
[d] No load losses and No load currents.
[e] Insulation resistance test.
[f] Induced over voltage withstand test.
[g] Separate source voltage withstand test.
[h] Zero phase sequence impedance measurement.
[j] Power frequency high voltage test.
[k] Dielectric test on oil.

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Test Certificates shall be given for all type tests, actual measurements in tabular farm and also for derived values of losses, impedance voltage, and regulation for stated loading condition. Copies of all test results, inspection and guarantee certificates shall be furnished in duplicate to the Department for approval.
The insulation resistance of the HV &LV windings shall be measured by a megger of specified voltage and duly calibrated. The value of insulation resistance shall be in accordance with the specifications and IS 1180.

14.2 **DRAWINGS:**
The manufacturer shall supply all technical literature and drawings considered necessary for the installations operation and maintenance of the transformer and its fittings. These shall essentially include:

[a] Detailed foundation drawings.
[b] Drawings showing overall dimensions of the transformer.
[c] Drawings showing position of fittings. Winding diagram and terminal markings.
[d] List of parts with reference number.
[e] Instruction for the installation and commissioning of transformer.
[f] Any other relevant data, which would be assistance for efficient operation, maintenance and repair of fittings and accessories.
[g] Manual of instructions for the operation &maintenance of fittings. In hard and soft copies.

14.3 **TECHNICAL AND GUARANTEED PARTICULARS**:
The tenderer shall furnish all guaranteed technical particulars as called for in schedule particulars, which are subject to guarantee, shall be clearly marked. Tenders lacking information in this respect may not be considered.

14.4 **PROTECTION**:
The following protection shall be provided for the transformers Detecting or actuating devices required, which form part of the transformer construction shall be incorporated.
[a] Buchholz alarm and trip including control cabling upto VCB (for transformer 500 KVA and above).
[b] Thermal alarm and trip (winding temp. &oil temp.)
[c] Low oil level alarms for the conservators.
In addition to above differential protection scheme and restricted earth fault protection scheme shall be provided and interconnected with breaker for tripping and annunciation.

14.5 **INSTRUCTION MANUALS:**
Three (3) copies of operation, maintenance and erection shall be supplied one month prior to the dispatch of the transformer. The manuals shall be bound volumes and shall contain all the drawings as per section 21.0 and information required for erection, operation and maintenance of the transformer. The manual shall include the following particulars amongst others.
[a] Marked erection paints identifying the component parts of the transformers as dispatched, with the assembly drawings.
[b] Detailed dimensions, assembly and description of all the components and catalogues for operation of the transformer and accessories.
[c] List of spare parts.

14.6 **DEVIATIONS FROM THE SPECIFICATION:**
All deviations from this specification shall be separately listed in schedule in the absence of which, it will be presumed that the provisions of the specification are complied with the purchaser.

14.7 Pre-commissioning test by EI/SEI will be carried out and record maintained. The contractor will be responsible for the arrangement of testing and all the equipment testing material will be arranged by the contractor at no extra cost.

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14.8 **ERECTION &COMMISSIONING:**

Immediately on arrival, the transformers shall be carefully examined for any damage, during transit, paying particular attention to bushing and cooling tubes, etc; The presence of rust, condensation anywhere within the tank the pink color of silica gel breathers provided with transformers filled with dry air or a considerable drop in the pressure of nitrogen or other inert gas filled in transformers suggests absorption of moisture by the transformer windings. In such cases, the transformer shall be dried out before being commissioned. The transformer shall be filled with oil as soon as possible after satisfactory inspection and set up in their permanent locations even though not expected to be energized for some time. Before moving the transformer, the bi-directional rollers shall be fixed in the required direction. The transformer may then be moved to its permanent position. Care shall be taken when using any levers etc; that they do not touch the tank or cooling tubes and only the skids or rollers shall be used for this purpose. Transformers shall be lifted by lugs provided for the purpose and simultaneously use shall be made of all lugs to avoid any imbalance and it shall be ensured that all cover bolts are tightened. Where it is necessary to use jacks for lifting, the projections / lugs provided for the purpose of jacking shall be used. Jacks shall never be placed under valve or cooling tubes. In certain circumstances, jacks may be placed under stiffening curbs on the tank base. Care shall be taken that spanners or other tools used in connection with work are not dropped into or left inside the tank. The transformer shall be placed on suitable roles or tracks on level concrete plinth with bearing plates of sufficient size and strength where the transformer is in the final position; the wheels shall be locked to prevent accidental movement of the transformer. After positioning of transformer all electrical power and control connections shall be done in approved manner.

The transformer should be installed on suitable pedestal and foundation. The scope of this civil work is included in this item of the work. Contractor should provide all the hardware that is necessary for erection. The drawings, layout, etc; should be got approved by Department. The power transformer shall be installed, tested, commissioned and integrated the same with the substation network in all respects.

After the transformer has been installed at the place of operation and beforeconnecting it to supply the transformer tank shall be earthed by two separate and distinct connections through two separate lines and the two earthing terminals provided on the, transformer. Gas or water pipes shall not be used for such connections.

All connections of the earthing system shall be visible for periodical inspection. Copper strip as specified shall be used for earthing to the earth pit. The transformer neutral point shall be brought out and solidly earthed by copper strip of suitable size. (GI, copper strips and required earth stations are to be provided by the tenderer).

For connection of transformer neutral point with LV neutral bus bar the current carrying capacity of conductor shall be same as the conductors used for solid earthing of neutral. Following tests shall be conducted on the transformer prior to its commissioning.

1. Insulation resistance of the winding between phases and between individual phases and earth on both HV and LV sides and also between HV and LV windings.
2. Winding resistance.
3. Dielectric strength of transformer oil.
4. Voltage ratio test at all tap positions.
5. Polarity test.
6. Operation of Buchholz relay.
7. Proper functioning of all protective relays.

After commissioning of the transformer the LV side voltage shall be measured at different tap positions. The transformer shall be kept on no load for 24 hours, and shall be ascertained that there is no undue temperature rise before loading the transformer.

14.9 **GENERAL:**

The work shall be carried out according to general specification mentioned and the detailed specifications and also according to any other instructions issued by the Department. No claim on account of the ignorance of specifications shall be entertained. In their own interest, the tenderers are advised to submit their *offers* after a thorough study of the requirements of the specifications and site conditions.

14.10 **GUARANTEE PERIOD.**

Transformer, its fittings and accessories shall be guaranteed for trouble free operation for a' period of Twelve Months from the date of completion. Any problem related to transformer within guarantee period shall be attended immediately at free of cost. The supplier has to furnish performance bank guarantee for 10 % of the quoted amount.

PARTICULAR SPECIFICATIONS (CONTD…)

15. SILENT DG SET.

15.1 The work shall be executed as specified in Schedule 'A' and all as specified in SSR Part-I.

15.2 **TESTING OF GEN SET**

15.2.1 The DG Set shall be tested in the presence of GE or his accredited Rep before issue of completion certificate. The Period for Testing of DG set shall be as under :-

(a)	(i) ½ Hours	-	No load
	(ii) 1½ Hours	-	25% on load
(b)	2 Hours	-	50% on load
(c)	3 Hours	-	75% on load
(d)	4 Hours	-	100% on load
(e)	01 Hour	-	110% on load

(f) Consumable stores including, Lub, oil and load for testing generating sets during all tests is to be included in the unit rate quoted by the contractor. Water required will also be arranged by the contractor. However quoted rate inclusive of of diesel (DHPP) required for testing of D G Set.

(g) The test results will be compared with the standard performance parameters given by the manufacturer for the equipment. Contractor is bound to retest and satisfy the GE in case of variation in results.

(h) Six copies of test reports shall be prepared by the contractor & Engineer-in-Charge and given to GE for approval before issue of completion certificate.

15.2.2 **RUNNING TEST**

(a) The performance test in respect of Gen Set shall be conducted to check standard performance parameters given by the manufacturer. All tests shall be done by an officer nominated by the Accepting Officer.

(b). All DG's shall run individually on NO LOAD & FULL LOAD successfully without any problem both in manual & PLC mode separately.

(c) Adjust droops for speed & voltage. All testing and maintenance charts shall be displayed in DG House along with operating instructions to operators.

15.3 **Maintenance Period**- Contractor shall be responsible for the safe and successful running of the Gen sets for a period of one year or 500 running hours whichever is early and any repair/replacement of any parts, during above period shall be carried out by the Contractor without extra cost to Government.

ARBORICULTURE

1 All the Trees/Lawn/Hedges shall be true to their names and species. Plants/ trees with robust root growth and free from disease/ decay shall be planted. Red earth to be filled in the pits shall be of loamy earth suitable for developing of the plants. The type of pesticides, manures, fertilizers shall be appropriate to the type of the plants and shall be as directed by the Engineer-in-charge.

2 **PEST CONTROL**

2.1 The soil and pits shall be treated with prophylactic measures against pest by spraying insecticides and fungicides either by systematic type or contract type as directed by the Engineer-in-charge.

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3 **LAWN**

3.1 Lawn site shall be cleared of all unwanted growth, debris, wastes, etc. and then levelled along contours as directed by the Engineer-in-charge. The under layer shall be filled with good quality red earth and the top layer shall be filled with a mixture of garden manure, river sand and red earth in the ratio as specified. The lawn site shall be thoroughly watered and weeds which sprout through the manure shall be removed periodically once in 15 days. The lawn area shall then be treated with a dressing of 5% BHC or other approved pesticides according to the site conditions. Plantation of grass shall be according to the type of grass and as directed by the Engineer-in-charge. Adequate watering to the extent as required and as ordered shall be supplied to the whole lawn area depending upon the weather conditions.

4 **PLANTS/ TREES**

4.1 Plants/ trees shall be planted in the trenches/ pits to a depth as directed. Filling up the trenches/ pits shall be with mixture of red earth, river sand and manures in proportions as specified with 5% BHC. The plants shall be planted at the intervals as specified and directed.

5 **MAINTENANCE OF ARBORICULTURE WORKS**

5.1 Trees/Lawn/Hedges provided shall be maintained for the period as indicated in Schedule 'A'. Plants provided shall be concurrently maintained by the Contractor up to the completion date,

6 **DE-WEEDING**

6.1 De-weeding shall be carried out in lawns and areas surrounding the plants at 15 days intervals. Lawns shall be kept free of both monocot and dicot weeds. The dicot weeds are to be eliminated using de-weedicides and the monocot weeds are to be de-weeded manually. The surrounding areas are to be kept free of seed producing weeds.

7 **MOWING**

7.1 Mowing of the lawns shall be carried out periodically according to the growth of the grass and type of grass developed. The height of the grass to be cut shall be according to the type of grass and as directed by the Engineer-in-charge.

8 **FORKING, MANURING AND TOPPING**

8.1 Periodical forking, manuring and topping shall be carried out for the requisite growth of the plants/ grass and depending upon the growth of the weeds. The moisture level of the lawn shall be brought to the minimum. The lawn area shall be given a general forking to loosen the top 25mm layer of soil. Periodical topping layer shall be provided all over the lawn areas during the maintenance period with a mixture of manure, sand, loamy soil and red earth in the proportion of 1:2:2:1. The lawn shall then be given a Drenching for settlement of the topping layer.

9. **FERTILIZERS**

9.1 Fertilizers like Urea, Ammonium Sulphate and Complex fertilizers shall be applied periodically and depending on the need due to change in climatic conditions. Urea is to be applied once in 6 months period as granules/ Foliar Spray depending upon the colour and growth desired. Complex fertilizers shall be applied in powdered form once in a year to replenish major N.P.K. nutrient. Before application of fertilizers the moisture content of the lawn areas shall be reduced to lowest level and after application of fertilizers the lawn areas shall be flooded for a period of 2 weeks. Plants/ trees shall be treated with combination of garden manure and fertilizers once in 3 months with due regards to the growing period and additional quantity of fresh red earth shall be applied once in a year for the area covered by the plants.

10 **PESTICIDES**

10.1 General pests like white ants, cut worms, ants, etc. found in lawn shall be eradicated using dust or solution of Aldrin. Downy mildew which occurs in patches shall be treated with solution of Blitox.

11 **WATERING**

11.1 Adequate watering shall be given at intervals as required depending upon the climatic conditions and growth of the plants/ grass. Sufficient water shall be provided to moist the top 50cm to 75cm layer of soil.

PARTICULAR SPECIFICATIONS (CONTD…)

12 **TRIMMING AND PRUNNING**

12.1 Trimming and pruning shall be carried out periodically depending upon the growth and shape desired. One major pruning shall be carried out once in year before spring season.

13. **RECORDS OF PLANTS/ TREES**

13.1 Proper records shall be maintained in respect of the plants/ trees provided and area covered in a bound register and shall be produced to the GE at the time of claiming RAR/ FB and at the time of handing/ taking over.

14. **DEPLOYMENT OF PERSONNEL**

14.1 Contractor shall deploy such personnel who are well conversant with the gardening works and maintenance of the same to achieve desired growth. The Contractor shall submit the list of workmen deployed by him for maintenance of Arboriculture works to Engineer-in-charge and shall be kept on record.

15 **PENALTY**

15.1 Contractor shall deploy Gardener daily for watering and maintenance of plants/ lawn during the maintenance period. If Gardener is not deployed/ found absent an any day, penalty at double the rate quoted per day for maintenance will be levied for each day's absence.

16-27 **BLANK**

28. **SEWAGE TREATMENT PLANT**

28.1 SCOPE OF WORK FOR STP & UNDERGROUND SUMP : The Scope of work to be provided by the bidder is as follows Erection of equipment auxiliaries likes piping, electrical, instrumentation, etc.

28.2 **CIVIL WORK FOR STP**

28.2.1 The construction work of STP units shall be carried out all as specified in Srl item No 1.00 to 1.11 of Schedule 'A' of BOQ, as shown in Drawing and as specified here-in-after.

28.2.2 All RCC foundation, wall, beam and slab shall be M-30(Design mix).

28.2.3 All MS work shall be as per drawings.

28.2.3 All steel reinforcement bar shall be of grades Fe 500(D) and shall be CRS (Corrosion Resistant Steel).

28.2.4 The internal surfaces shall be plastered as per drawings

28.2.4 The external exposed surfaces shall be plastered as per drawings.

28.2.5 Internal plastered surfaces of walls/bottom & top slab shall be applied with two coats of air drying, algae, fungal growth resistant & anti corrosion epoxy paint for holding drinking water. The paint should be free from any bad taste and odour for potable water.

28.2.6 Lean Concrete in Foundation Bed shall be provided as shown in drawings.

28.2.7 All Other Details shall be provided as shown in drawings.

28.2.8 PVC water bar/water stop seal shall be provided at all construction joint of tank. The PVC water bar shall be 3.5 mm minimum thickness and 200 mm minimum width.

28.2.9 Cast iron steps pattern-1 shall be provided on RCC wall and spacing of steps shall be as per standard practice or as per IS.

29.2.10 For STP works Sulphate Resistant cement conforming to IS 12330:1998 shall be used.

28.3 **ELECTRICAL WORK OF STP**

28.3.1 All items shall be executed all as specified in various schedules of work as applicable.

28.4 **PERFORMANCE & GUARANTEE :**

28.4.1 The bidder shall guarantee the sewage treatment plant to perform as per the stated capacity and treated waste water quality characteristics requirement specified for a period of 24 months from the date of commissioning and handing over.

28.4.2 In case any short fall in the plant performance the contractor shall undertake to rectify and make the plant proper functional at this own cost, to achieve the specified performance conditions.

28.4.3 The plant shall be guaranteed against performance in either achieving the quantity output or in achieving the treated water quality.

PARTICULAR SPECIFICATIONS (CONTD…)

28.4.4 The Contractor shall also guarantee the materials, quality, workmanship and the satisfactory performance of all the equipments, accessories, auxiliaries, piping, electrics, instruments, installations, constructions, buildings, facilities, infrastructure etc. under his scope of work for a period of 24 (Twenty Four) months from the date of preliminary acceptance. Any defects noticed during the guarantee period shall be replaced/ rectified immediately without any extra cost to the Department.

28.5 **BEFORE EXPIRY OF DEFECT LIABILITY** :- Before expiry of 60 days from the date of defect liability period, a Board of Officers shall be **ordered by CWE** to ensure that the performance of the plant is satisfactorily and otherwise issues notice to contractor prior of expire date of defect liability period as per condition 46 of IAFW-2249 and with with intimation to Accepting Officer.

28.6 **O & M MANUAL**

28.6.1 The contractor shall deliver to the Engineer-in-charge in duplicate copies of draft operational and maintenance manuals for the plant after setting necessary standards basing on the tests conducted for a period as indicated. The Manual shall fully and clearly set out the OEM/contractor's own recommendations, and instructions for the satisfactory operation and maintenance of the plant or equipment. The manual shall also accompany drawings showing the electrical wiring, piping, handling and erection instructions. Draft manual shall, during the testing and commissioning of the works, be carefully checked by the contractor and updated and modified to ensure that it is fully descriptive and applicable to the final process plant as installed and as found to behave under operational conditions.

28.6.2 The Manual shall also highlight handling of emergency situations during chlorine leaks, electric short-circuiting, personnel safety and hygiene.

28.6.3 The various aspects of the manual shall be discussed under highlighted captioned heading. There shall be a table of contents as usual. The draft manual shall be in English and local language and may include manufacturer's standard literature, but the contractor shall fully supplement the literature by his own descriptive text explanations and drawings. After satisfactory trial run & performance guarantee test the draft manual shall be approved by the Department and one copy shall be communicated to the contractor.

28.6.4 The manual shall also include a check list for the mechanical and electrical plant procedures for their smooth operation and maintenance broadly as follows:-

28.6.5 Calibration, adjustment, checking and lubrication of each component.

28.6.6 Attention at weekly, monthly or other regular intervals to ensure reliable trouble free operation.

28.6.7 Complete overhaul, dismantling, reassembling and re-commissioning of various components.

28.6.8 Identification and selection of suitable lubricants of various Indian Manufacturer's suitable equivalents.

28.6.9 Fault findings.

28.6.10 Maintenance of protective coatings.

28.6.11 Monitoring of water quality throughout the works including sampling techniques, laboratory and reporting procedure.

28.6.12 The work shall not be considered as complete unless this requirement has been met and the cost for the same shall be deemed to be included for the unit quoted rate.

28.7 **TRAINING OF MES PERSONNEL**

22.7.1 The contractor should arrange training program / certificate course for minimum 05 person of MES regarding operation and maintenance of Sewage Treatment Plant (STP) by the OEM representative to get fully conversant with plant without any extra cost to the department.

28.8 **TESTING :**

28.8.1 The SEWAGE TREATMENT PLANT (STP) shall be tested in presence of GE and OEM /Contractor's personnel and demonstrate the overall suitability of the plant for the functionintended together with proof of specified performance for 72 hours duration before issue of completion and the cost will deemed to be inclusive of quoted rate.

28.8.2 The treated water should be get tested by the bidder from NABL certified laboratory and thereport confirming that the parameters are within the limit, as mentioned in the documents, should be submitted before seeking completion of the plant. The cost of the testing shall be borne by the bidder.

PARTICULAR SPECIFICATIONS (CONTD…)

28.9 **COMPLETION DRAWINGS**:

28.9.1 The contractor shall submit six copies completion drawing with Auto CAD-2008 with soft copy of the following drawings duly signed by him and the Engineer-in-Charge before completion.

28.9.2 Overall layout plan for Sewage Treatment Plant (STP) showing entire details of the installation complete with actual layout of various element / equipment, various pipe lines (indicating clearly different sizes in different colours), various valves and all other information necessary for record, maintenance and operation, all as directed without any extra cost.

28.9.3 The contractor shall obtain necessary statutory clearance from the concerned State Pollution Control Board before the installation and operation of STP and the cost for the same shall be borne by the contractor and nothing extra will be admissible.

28.10 -Blank-

28.11 Treated sewage parameters are as given in BOQ Srl Item No. 1.00 to 1.11.

29.0 **AMC,MANNING AND OPERATION OF SEWAGE TREATMENT PLANT DURING DEFECTS LIABILITY PERIOD AND AFTERWARDS.**

29.01 The scope of work (Phase-III) includes AMC &manning, operation, maintenance, repair and servicingof STP Plant including commissioned by the contractor upto 5 Yrsfrom the date of commencement of work of Phase-III. Work catered in phase-III shall commence from next day to certified date of completion of work catered in Phase-I. Itwill be the responsibility of the contractor to ensure safe, smooth and un-interrupted functioning of the system in sound condition during the period mentioned above.

29.02 GE's decision as to what constitutes major/minor repairs shall be final and binding. Major repairs to STP Plant are outside the scope of work in Phase-III. BOQ rates shall be deemed to include for minor repairs, repairs and rectification necessitated due to negligence or wrong operation of the STP Plant on the part of the operator/helper. The decision of the GE whether the repair/rectification is necessitated due to negligence on the part of operator/helper or not shall be final and binding. However all repairs & rectification whichever arise during defect liability period of Phase-I has to be carried out as intimated by GE.

29.03 The contractor will submit the following details to the Engineer-in-Charge regarding each person employed by him under this contract:-

(a) Name of person along with photograph.
(b) Father’s Name.
(c) Date of Birth.
(d) Qualification & Experience certificates

29.04 Operator & Helper shall have minimum essential qualification as following:-

(i)	Operator for STP Plant	Industrial training Institute pass certificate from a recognized institute in attendant operator (chemical plant)/fitter/ general mechanic with 2 years experience in relevant field.
(ii)	Helper for STP Plant	Class-X Passed

29.05 The contractor shall provide the necessary attendance record of both employee of STP Plant before enter STP Plant & before leave from STP Plant of every shift with digital system i.e. biometric attendance and print out duly signature of AGE & contractor before any payment and also record proper log book attendance register for reflecting and measurements parameters of the plant up to the entire satisfaction of Engineer-in-Charge/GE. The cost of biometric machine shall be borne by contractor.

PARTICULAR SPECIFICATIONS (CONTD…)

29.06 Before commencement of work and taking over any installation for operation, contractor is responsible for submitting original qualification and/or experience certificates of persons to be employed, in original and two photocopies to the Engineer-in-Charge for approval. During currency of the contract, the department reserves the right to ask for such certificates fo verification. The decision of the GE regarding the suitability of an individual to a particular trade shall be final and binding.

29.07 -Blank-

29.08 The persons working in installations are expected to exhibit a high sense of discipline, punctuality and neat turnover. They should not leave the installation unattended without prior permission of Engineer- in-Charge. The persons employed by the contractor shall be amenable to the discipline and other security and standing order issued by the defence authorities for installation from time to time. The persons employed should not consume alcohol / liquor, drugs or smoke while on duty. They shall work in cohesion with other defense employees and shall be capable of working as a team. They shall take orders / instructions from Engineer-in- Charge of the installation and shall implement the same with great care and interest. Accepting officer has the right for removal of those persons whose performance and integrity is not satisfactory. The contractor shall replace such persons with one week notice. The decision of the Accepting Officer in this regard shall be final and binding and the contractor shall have no claim of compensation on this account.

29.09 The contractor shall take all the necessary safety measures pertaining to plant and operator room of STP Plant in addition to over all safety and security to plant and plant room.

29.10 The operating staff shall have thorough knowledge of safety precautions during emergency cases. They must be able to operate and use all safety equipment. Necessary notice boards containing instructions for precautions to be followed in case of emergency/risks/accidents shall be displayed at locations as required and as directed by Engr-in-Charge.

29.11 The particulars for qualification and experience of staff/operators shall be submitted to GE for approval before engaging them for maintenance & operation purpose.

29.12 The maintenance technicians/trade-men employed by the contractors shall be fully responsiblefor maintaining the STP Plant in consultation with Engineer-in-Charge.

29.13 The persons employed by the contractor shall have the basic knowledge of using firefighting equipment such as fire extinguisher etc., maintained in the installation. In case of fire, they shall ring the fire gang and telephone the fire fighting units nearby and Engineer-in-Charge andGE for further help / assistance. They will not risk their lives containing the fire / damage tothe installations. No compensation shall be paid by the department to the persons employed for any injuries/death caused in such acts.

29.14 Safety procedures and instructions for working in low, medium and high voltage mains and apparatus and safety practices listed in IS: 5216-1982 guide for safety procedures and practicesin electrical works shall be followed to the extent applicable.

29.15 The contractor will provide and maintain first-aid/medical box and first aid chart in operator room and quoted rate shall be deemed to include cost of the same.

29.16 The operating personnel shall be provided with protective clothing, hand gloves (shock-proof), safety belt, rubber shoes, ladder, torch etc. all mandatory safety measures as per IE rules and dress with logo & shoes shall have to be issued by the contractor to the operator & the operator shall be well equipped with hand tools mobile phones without camera to carryout necessary repair & to make contact with concerned JE in any emergency situation and quoted rate shall deem to include it.

29.17 The contractor shall be solely responsible for settling all claims/ compensation whatsoever arising due to accident/fatal injury or partial disability/loss of life or any reason for the workmen employed by him. Department will have the right to recover any sum indicated /claimed by labour commission /court directive. GE however will ensure that the contractor has provided all adequate and required means to the workmen. The contractor may get his personnel insured at his own discretion. However nothing extra shall be paid to contractor on this account.

PARTICULAR SPECIFICATIONS (CONTD…)

29.18 In the event of any loss or damage caused to the installation or any part thereof except for the accepted risks as per condition 48 of IAFW -2249 shall be made good by the contractor under his own arrangement as per the specifications given in the contract without any extra cost to the Government. If the contractor fails to do so, the government reserves the right to make recoveries from the contractor for the loss or damage at double the prevailing market rates as decided by the Garrison Engineer whose decision shall be final and binding.

29.19 The contractor shall be responsible for any claim whatsoever arising due to any reason for the workmen employed by him.

29.20 The contractor shall comply with all the provisions laid down in Contract Labour Act such as payment of minimum fair wages

29.21 The contractor shall make payment to the persons employed directly in their bank account and quarterly employee payment statement i.e. bank account statement of contractor shall be submitted to department for proof of the same.

29.22 Contractor will ensure that during the operation of the STP Plant, a constant vigilance is kept on the running of STP Plant by the operator and in case of any break down or any abnormality whatsoever will be reported to Engr-in-Charge and The contractor will be responsible to keep the STP Plant and surroundings the fenced area or area of distance of 6 meters neat and cleanevery day.

29.23 All the stationeries required during the currency of the contract like registers, log books etc are covered under the scope of work and quoted rate shall be deemed to include cost of the same.

29.24 The contractors shall provide necessary spares, tools/tool box at operation room to attend any minor repair and to measure and record the performance of plant required at site to the entire satisfaction of the Engineer-in-Charge.

29.25 The contractor is required deployment in such a manner that no person/individual should work for a period of more than 11 (Eleven) months.

29.26 No individual deployed will work more than 48 hours in a week.

29.27 Routine maintenance schedule shall be carried out by the contractor and the same shall be recorded in proper preformed duly countersigned by the GE/Engineer-in-Charge.

29.28 The contractor shall provide printed log sheets to the operators & helper on duty for STP Plant separately for each day as directed by the Engineer-in-charge and entries as per the running ofthe STP Plant will be recorded by the operator which shall be produced for verification on demand.

29.29 The contractor shall ensure that daily, weekly, monthly, quarterly yearly, half yearly and yearly schedules of maintenance of Plant etc as per OEM's instructions are carried out properly and same shall be recorded in proper format duly countersigned by GE/Engineer-in- Charge. The quoted rates shall deem to include it.

29.30 All records of testing and consumables shall be recorded on the register maintained at plant on daily basis by the contractor and signed by Engineer-in-Charge and the same shall be randomly checked by GE/ his representative at any time.

29.31 Any material / repairs not covered in this work and required for efficient operation shall be brought to the notice of Engineer-in-Charge for immediate supply of spares / repairs through other agencies.

29.32 On completion of each month, a completion certificate is to be issued by the Engineer-in-Charge for purpose of interim payment to contractor supported by attendance register duly signed by contractor and Engineer-in-Charge.

29.33 In case the operators & helper are not employed on any shift or operators & helper are found absent from the installation and recovery shall be made at double the minimum fare wages as applicable at that time.

29.34 In case the Plant remains un-operational i.e. in case the Contractor fails to run the plant for a period of more than 2 hours, the GE shall take over the plant immediately after 24 hours and the cost of running/maintaining the plant shall be recoverable from the contractor in addition to a penalty of Rs 1,500/- per Hour per plant for entire period of non operation.

PARTICULAR SPECIFICATIONS (CONTD…)

29.35 All records of testing and consumable shall be recorded on the register maintained at plant on daily basis by the contractor and signed by AGE and the same shall be randomly checked by AGE/ GE/ his representative at any time.

29.36 In case the any plant un-operational more than 2 hours, the cost of consumable material shall be recovered @ Rs 500/- per Hour or present market rate of consumable material whichever higher.

29.37 Two months consumables material and required spares shall always be kept at site and the costfor the same shall be borne by the contractor.

29.38 Water & Electricity shall be supplied by the MES to contractor free of cost for Phase-II.

29.39 Maintaining necessary records such as log sheet, manpower and attendance register etc shall beprovide by the contractor and signed by AGE and the same shall be randomly checked by GE/his representative at any time.

29.40 The contractor shall ensure that daily, weekly, monthly, quarterly yearly, half yearly and yearly schedules of maintenance of Plant etc as per manufacturer's instructions are carried out properly and records maintained. The quoted rates shall deem to include it.

29.41 The installation shall be operated on all days irrespective of holidays and Sundays. Contractor shall be required to employ reserve operators if performance of contract is discontinued consequent on labour regulation statute on working of personnel on national holidays etc and also on any day when operator(s) is/are absent from duty. However nothing extra shall be paidto contractor on this account.

29.42 After completion of 22 months of manning & operation period, the contractor shall arrange 10 days training for operation of STP Plant to the DEL/ person deployed by the department, without any extra cost before handing over the plant.

30 **SITE SURVEY OF SEWER LINE**

30.1 Before excavation for sewer line work, contractor in presence of Engr-in-Charge shall examine the entire excavation route for any hidden service/utility lines (power/communication cable, water pipeline etc.) buried in the ground with the help of suitable portable tracer/locater device to avoid any disruption to E/M services in the station, Cost of portable tracer/locator device shall be deemed to be included in quoted rate.

30.2 Damage to utility line if any (underground cable /pipeline) during execution of work will be made good by the contractor at his own cost within 24 hours of occurrence of such event and nothing extra shall be paid to the contractor.

31.0 DEMOLITION/ DISMANTLING:

31.1 The contractor shall ensure that after repairs all demolished / Un-serviceable material shall be deposited in MES store yard & leaving the plant room area neat & clean & tidy where such repair shall be carried out all as directed by the Engineer – in –Charge.

31.2 Demolition statement shall be enclosed with each RAR & final bill. Details of stores incorporated in work shall be indicated with RAR & Final bill.

31.3 Precautionary measure prior to demolition/dismantling as described in SSR Part-I shall be adopted and work shall be executed as per Schedule 'A'.

PARTICULAR SPECIFICATIONS (CONTD…)

32.0 **PRECAUTIONARYMEASURE**: Precautionary measure prior to demolition/dismantling as described in SSR Part-I shall be adopted and work shall be executed as per Schedule 'A'.

33.0 **REMOVAL/SITE CLEARANCE**: Old demolished materials considered under Schedule of credit shall be the property of the contractor and will remove by the contractor at his custody. Debris and other materials if any shall be disposed off and sites to be cleared to the entire satisfaction of the GE.

SIGNATURE OF CONTRACTOR
DATED: _____

AAD (CONTRACTS)
FOR ACCEPTING OFFICER

PARTICULAR SPECIFICATIONS (CONTD…)

APPROVED NAMES OF MANUFACTURERS

APPROVED CEMENT MANUFACTURERS FOR MES WKS ON REGIONAL LEVEL

1	The Associated Cement Companies Ltd Brand: "ACC"	414-421 Splendor Forum (4th Floor) 3 District Centre – Jasola New Delhi – 110 044 Ph 011-46583600	All grades
2	M/s Ultra Tech Cement Ltd Brand: "ULTRATECH"	'B' Wing, 2nd Floor,Mahakali Caves Road Andheri (East), Mumbai – 400 093 Ph 022-66917800	All grades
3	The India Cement	Dhun Building, 827 Anna Salai Chennai-600002	All grades
4	M/s Dalmia Cement (Bharat) Ltd Brand: "DALMIA" INFRA PRO"	Dalmiapuram Dist : Tiruchirappalli Tamilnadu- 621 651	All grades
5	Century Cements Brand: "CENTURY	Industry House, 159 Churchgate Reclamation, Mumbai – 400 020 Ph 022-22023936	All grades
6	Saurashtra Cement Brand: "SAURASHTRA"	Gala No A-1, Ground loor,UdhyogSadan No 3 MIDC, Central Road,Andheri (East) Mumbai-400093 ,Ph 022-32955557/67 Mob : 9320290081	All grades
7	The Ramco cement ltd (formerly Madras Cements Brand: "RAMCO"	Auras Corporate Centre 98-A DrRadhakrishnanSalai Mylapore : Chennai-600 004 Ph 044-28478666	All grades
8	Mangalam Cement Ltd Brand: "MANGALAM"	PO Adityanagar, Morak Dist – Kota, Rajasthan-326 520 Ph 9351468076	All grades
9	Birla Corporation Ltd Brand: "BIRLA"	Birla Building (3rd& 4th Floor) 9/1 RN Mukherjee Road, Kolkata – 700001, Ph 033-30573700	All grades
10	Orient Cement Brand: "BIRLA- A1"	5-9-22/57/D, 2nd& 3rd Floor GP Birla Centre, Adarsh Nagar Hyderabad – 500063 Ph 044-23688600	All grades
11	Nuvoco Vistas Corporation Ltd (Formerly lafarage Cement) Brand: "NUVOCO"	Equinox Business Park Tower – 3, East Wing, 4th floor LBS Marg, Kurla (WEST) Kurla Mumbai Maharashra 400 070	All grades
12	Shree Cement Brand: "SHREE"	Bangur Nagar Beawar-Dist Ajmer Rajasthan – 305 901 Ph 01462-228101-06	All grades
13	JK Cement Brand: "J K"	Kamla Tower Kanpur 208 001	All grades
14	JK Lakshmi Cement Ltd Brand: "J K LAKSHMI"	Jaykaypuram Dist – Sirohi Rajasthan Ph 02971-244409/10	All grades
15	JaypeeRewa Cement Brand: "JAYPEE"	Jaypee Nagar PO Jayapee Nagar Rewa 486 450 MP	All grades
16	Ambuja Cement Ltd Brand: "AMBUJA"	Kodinar ,PO Ambujanagar Taluka – Kodinar, Dist – Junagadh Gujarat – 362715 Ph 02795-237000	All grades

PARTICULAR SPECIFICATIONS (CONTD…)

17	M/s JSW Cement Ltd, AP Brand: "JSW PSC"(Portland Slag Cement)	JSW Centre Opp NIMRDA Ground Bandra Kurla Complex Bandra (EAST) Mumbai 400 051 Tele 022 4286 5047 Fax 022 2650 2001		PSC
18	M/s Kesoram Industries Ltd Brand: "BIRLA SHAKTI"	M/s Kesoram Industries Ltd Unit 12th 1201 of T 19 Towers Indira Nagar Ginwala Compound, M.G Road, Secunderabad, Hyderabad, Telangana – 500003, Tele : +91-40-4334-4555 Fax : +91-40-4334-4534		(a) OPC 43 Gr (b) OPC 53 Gr (c) PPC
19	M/s Sagar Cements Limited, Hydrabad Brand: "SAGAR"	M/s Sagar Cements Limited Plot No. – 111, Road No. – 10, Jubilee Hills, Hyderabad- 500033 Tele : +91-40-2335-1571, 2335672 Fax : +91-402335673		(a) OPC 43 Gr (b) OPC 53 Gr (c) PPC (d) PSC
APPROVED CEMENT MANUFACTURERS FOR MES WKS ON REGIONAL LEVEL				
1	M/S Star Cement Meghalaya	Lumshnong, PO : Khaliehriat, Dist- Jantia Hills, Meghalaya Phone 03655278215/16/18	(a) OPC (b) PPC	Only for CE Shillong Zone and CE(AF) Shillong area
2	M/S Meghalaya Cements Ltd, Brand: “TOP CEM”	Lohia House, MG Road Factory Bazar, Guwahati-781001 Phone 0361260367/671	(a) OPC (b) PPC	Only for Eastern region
3	M/S Hill Cement Company Ltd, Brand: “TAJ”	M/S Hill Cement Company Ltd Shivam Complex, Bharalumukh, A.T Road, Guwahati, Assam – 781009, Mob – 7086085118, Ph – (0361) 2735527, 2735552	(a)OPC-43 (b)OPC-53 (c)PPC	Only for Eastern region
4	M/s Green Valliey Industries Ltd., Brand: “MAX Cement”	M/s Green Valliey Industries Ltd. 4th Floor, LB Plaza, GS Road, Bhangagarh Guwahati, Assam – 781005, Tele : +91-361-2465482/84 Fax : +91-361-2465481	(a)OPC-43 (b)OPC-53 (c)PPC	Only for Eastern region
NOTES:				
Any other manufacturer(s) approved by HQ E-in-C’s Branch, New Delhi but left out in the above list shall also be deemed to be included in the list provided the name(s) of such manufacturer(s) has not been deleted/ removed by HQ E-in-C’s Branch.				
Any other manufacturer(s) approved by HQ E-in-C’s Branch, New Delhi during the pre-tender and post tender stage/after acceptance/during execution shall also be deemed to be included in the list.				
If any of the above manufacturer is banned/ removed by HQ E-in-C’s Branch, New Delhi during the pre-tender and post tender stage/after acceptance/during execution, the same shall be deemed to be deleted from the list.				
Any other manufacturer(s) approved by HQ E-in-C’s Branch, New Delhi but left out in the above list shall also be deemed to be included in the list provided the name(s) of such manufacturer(s) has not been deleted/ removed by HQ E-in-C’s Branch.				
Any other manufacturer(s) approved by HQ E-in-C’s Branch, New Delhi during the pre-tender and post tender stage/after acceptance/during execution shall also be deemed to be included in the list.				
If any of the above manufacturer is banned/ removed by HQ E-in-C’s Branch, New Delhi during the pre-tender and post tender stage/after acceptance/during execution, the same shall be deemed to be deleted from the list.				

PARTICULAR SPECIFICATIONS (CONTD…)

3.15.3.2 The particulars of the manufacturer/main producer of cement along with the date of manufacture shall be submitted by the contractor for every lot of cement separately. The documents in support of the purchases of cement shall be verified by the site staff and GE / AGE(I).

APPROVED TMT MANUFACTRUERS :-

Unless otherwise specified, mild steel TMT bars shall be of grade Fe500D meeting requirements of IS-1786-2008 shall be used of following approved primary manufacturers / main producers :-

Srl No	Company Name	Brand	Type of Steel
1	2	3	4
(i)	RashtriyaIspat Nigam Limited (RINL), Visakapatnam steel plant Visakhpatnam- 530 031, India. Tel (91 891) 518226, 518376 Fax (91 891) 518316 Email: cmdvsp@itpvis.ap.nic.in	RINL	All
(ii)	TATA Iron and Steel Company (TISCO or TATA Steel) Bombay House 2, 4 Homi Modi Street Mumbai - 400001, India. Tel: (9122) 2049131, Fax: (9122) 2049522, 2870840, Email: corpcomm@jsr.tatasteel.com (Br office for North: Jeevan Tara Bldg, patel Chowk, New Delhi)	TATA	All
(iii)	Steel Authority of India Ltd (SAIL), Central Marketing Organization, Northern Region 17th Floor, scope Minar, Laxmi Nagar, Distt Centre, Delhi-110092	SAIL	All
(iv)	M/S Kamachi Industries Ltd, ABC Trade Center 3rd Floor (Inside Devi theatre Complex), Old No. 50, New No 39, Anna Salai, Chennai – 6000002 , India. Tel 91 04442961100, Fax 91 04442961122, Email Sales@Kamachitmt.com	KAMACHI	TMT Bars of Gde Fe 500, Fe 500D, Fe 550 & Fe 550D, HCRM (Size 8-40mm)
(v)	M/S Shyam Metalics & energy Ltd, Viswakarma, Ist floor 86C, Topsia road ,Kolkata 700046, Ph 033-40113000 www.shyammetalics.com	SEL	TMT Bars of Gde Fe 500 & Fe 500D (Size 8-32mm)
(iv)	M/s JSW Steel Ltd, JSW Center , Bandra Kurla Complex Centre Bndra (East) Mumbai-400051, Maharashtra Ph: 022-42861000 Fax 022-42863000	NEOSTEEL	TMT Bars of Gde Fe 500, Fe 500D CRS(Size 8-40mm)
(vii)	M/s Electrotherm(India) Ltd, Survey No 72, Palodia, Via-Thaltej, Ahmedabad, Gujarat-382115, Tel -91 2717234553/660550, website: www. electrotherm.com	ET TMT	TMT Bars of Gde Fe 500, Fe 500D, & CRS, (Size 8-36mm)
(viii)	M/s Tulsyan NEC Limited, APEX Plaza, Ist floor, Old No 3, New No 77, Nungambakkam High Road, Chennai, -600034 (TN) Tel -04461991060/61991045, Mob No 9840354010/9677088334 website: www. tulsyannec.in	TULSYAN TMT	TMT Bars of Gde Fe 500, Fe 500D, & Fe 550, (Size 8-32mm)
(ix)	M/s Incredible Industries Limited (Formerly M/s Adhunik Industries Limited), Lansdowne Towers, 2/1A, Sarat Bose Road, 6th Floor, Kolkota-700020, Tele 033-66384700, Fax 91-33-22890285	ADHUNIK Fe 500 SD	TMT Bars of Gde. Fe 550D (Size 8-32mm)
(x)	M/S Gallantt Metal Ltd, Ward 10 BC . Plot No 123, Ground Foor , Gandhi Dham Kutch, Gujarat – 37020 Tel 91 2836228164, Fax 91 2836235787 Email :gml@gallantt.Com, website www.gallantt.com	GALLANTT TMX	TMT Bars of Gde Fe 500, Fe 500D, CRS
(xi)	M/s Rashmi Metaliks Limited, W.B Premlata Building, 39, Shakespeare Sarani 6th Floor, Kolkata-700017, Tele- 033-22894255/56 Fax- 033-22894254, Email : mkt.domesticdip@rashmigroup.com Website : www.rashmigroup.com	RASHMI TMT	TMT Bars of Gde Fe 500 (Size 8-32mm), Fe 500D& 550D (Size 8-32mm), Fe 500D CRS(Size 8-20mm) Fe-550D CRS (25mm & 32mm dia.)

PARTICULAR SPECIFICATIONS (CONTD…)

(xii)	M/s Bajrang Power & Ispat Ltd., Raipur Vill- Borjhara Uria Industrial Area, Raipur- 493221 (CG) Tele- 0771-4288019/29/39 Fax- 0771-232601/602, 4288123 Email – infobir@giekgriyo.co.in	GOEL TMT	TMT Bars of Gde Fe 500 & Fe-500D (Size 8-32mm) Fe 500D CRS(Size 10-25mm dia.)
(xiii)	M/s Real Ispat & Power Ltd, Vrindavan Near IDBI Bank Civil Lines, Raipur -492001, CG Tel : 91 7714224000, Fax 91 7714224010 Email real@realispat.Com, Website www.realispat.com	G K TMT	TMT Bars of Gde Fe 500, Fe 500D (Size 8-36mm)
(xiv)	M/S Steel Exchange India Ltd. My Home Laxminivas Apartments, Ameerpet, Hyderabad-500016, AP. Tel - 040-23403725, Fax - 040-23413267, Email : info@seil.co.in	SIMHADRI TMT	TMT Bars of Gde Fe 500, Fe 500D HSCRM
(xv)	M/s Super Smelters Ltd, Kolkata, Premlata, 39, Shakespearwe Sarani, 3rd Floor Kolkata -700017 Tel & Fax: 91 33 2289-2734/36 Email: info@supershakti.in, Website: www.supershakti.in	SUPER SHAKTI	TMT Bars of Gde Fe 500, Fe 500D, Fe 550, (Size 8-32mm)
(xvi)	M/s Shyam Industries Ltd.(WB) Shyam Towers, EN – 32, sector –V, Salt lake, Kolkata-700001, TEL-033-40074007, Fax-033-40074010, Email :-markting@shyamsteel.com, web- www.shyamsteel.com	SHYAM	TMT Bars of Gde Fe 500D & CRS
(xvii)	M/s Jai Balaji Industries Ltd , 5 Bentek street, Kolkata – 700001 , Delhi Office 510, Block-B, Navraung House 21 Kasturba Gandhi Marg, New Delhi – 110001 011 -43620219 , 43620220, Mob 7838272772/ 99558936103, Email Info@jaibalajigroup.com	BALAJI SHAKIT	TMT Bars of Gde Fe 500 & Fe 500D
(xviii)	M/s MSP Steel & Power Limited, Chhattisgarh 16/S, Block-A, New Alipore, Kolkata- 700053 Tele- 033 4005 7777/2398 2239, Mob- 7381028976 Email :- contactus@mspsteel.com Web- www.mspsteel.com	MSP TMT 500D	TMT Bars of Gde Fe-500D (Size 8-32mm)

STRUCTURAL STEEL

(a) Definition of structural steel as given in clause 10.4 of SSR Part-I shall be applicable. Standard quality steel of grade E-250(Fe-410W quality A) conforming to IS-2062 for all types of steel structures including those subject to dynamic loading shall be used.
(b) Ordinary quality structural steel wherever mentioned shall be conforming to IS-1977 of grade E-165(Fe-290) and shall be used for steel doors, collapsible gate, window guard bars or grills, steel gates, hand railings, fencing posts etc.
(c) Unless otherwise mentioned, galvanized steel sheets (plain or corrugated) where applicable shall be conforming to IS-277 . Grade of zinc coating to be used shall be the one which is given in Clause 10.29.1 of SSR Part-I.

4.5.2 SOURCE OF PROCUREMENT

4.5.2.1 STRUCTURAL STEEL:- Structural steel shall be procured by the contractor from approved primary manufacturers / main producers of steel like RashtriyaIspat Nigam Limited (RINL), Visakhapatnam / TATA Iron and Steel Company (TISCO or TATA Steel), Mumbai / Steel authority of India Ltd (SAIL), Delhi/ M/S Jindal Steel & Power Ltd, Gurgaon.

Notes :-
1. Any other manufacturer(s) approved by E-in-C's branch, New Delhi but left out in the above list shall also be deemed to be included in the list provided the name(s) of such manufacturer(s) has/have not been deleted/approved by E-in-C's Branch.
2. Any other manufacturer(s) approved by E-in-C's branch, New Delhi during the pre-tender and post tender stage/after acceptance/during execution shall also be deemed to be included in the list.
3. If any of the above manufacturer is banned/ removed by E-in-C's branch, New Delhi during the pre-tender and post tender stage/after acceptance/during execution, the same shall be deemed to be deleted from the list.

PARTICULAR SPECIFICATIONS (CONTD…)

O. **APPROVED NAMES OF MANUFACTURERS**

1.1 Approved names of manufacturers of various items/equipment(s) to be incorporated in this work unless otherwise specified elsewhere in this documents shall be as under :-

SL NO	DESCRIPTION OF ITEMS	MAKE/BRAND
1	**2**	**3**
1	Chloropyriphos ATT Chemicals	
1(a)	Chloropyriphos ATT Chemicals	Dursban /Trishu/ De-Nocil
2	PVC/UPVCDOOR,WINDOW&CHOWKHAT	
2(a)	PVC/WPC/UPVC Door, Window and Chowkhat/ PVC wall panelling/PVC False ceiling/WPC Jalli/SPC flooring	Kumar Arch Tech Pvt Ltd/Sintex India Ltd/Rajshri Plastiwood Ltd/Fenesta/Accura Polytech Pvt Ltd (Accucel)/Duroplast
2(b)	Solid Panel Foam Doors (Solid/ Glazed Panel)	Fenesta/Kesar/LG Hausys
2(c)	FRP Door Shutters	Fibrewats Technology/KraftoDoor SyntexIndia Ltd/Jain Wood Industries (Jayna)
2(d)	Natural FibreThermo Composite Door / Window Shutter and Frames	Duro-sam(AB Composite Pvt Ltd)/ ELCONS
3	WOODEN DOORS & WINDOW	
3(a)	Wooden Panel / Flush Doors &Windows	MP Wood Products Indore/Kitply Kampur/ /Ambika Timber Works/ Premier Wood Craft (P) Ltd Kolkata/ Jain Doors Pvt Ltd, Sonipat
3(b)	Steel Roller Door/ Shutter	Gandhi Automation Pvt Ltd /Metal King/ M/s Swastik Rolling
4	DOOR & WINDOW FITTINGS	
4(a)	Aluminium door &Window fittings	Bharat/Classic/Alans/Argent/Hettich/Halco
4(b)	MS Stove Enamelled	Mowjee/Crown/ Everlite/Harrison/Godrej
4(c)	Floor Springs/Door Closers/ Accessories	Hardwin/Godrej/Dorset/Everest/Universal
5	FALSE CEILING & PARTITION PANELING	
5(a)	PVC False Ceiling	Sintex/Armstrong /Saint Gobain
5(b)	Gypsum Silicate Tiles in False Ceiling/ Wall lining	Armstrong/India Gypsum/Lafarge/Saint Gobain/Everest
5(c)	Mineral Fibre False Ceiling	Saint-Gobain India Pvt. Ltd-Gyproc/ /Armstrong/ Lloyed insulation
5(d)	Metal False Ceiling	Armstrong / Unifloors /Nittobo
5(e)	Cement FibreBoard Ceiling.	Everest/Lafarge /Armstrong
5(f)	Anodized AluminiumPartition System with Solid/Glazed panels.	Jindal/Hindalco /Indalco
6	CUPBOARDS,CABINETS&WARDROBE	
6(a)	PVC kitchen(Cupboards, Cabinets &Wardrobes)	Kumar Arch Tech Pvt Ltd/ Rajshri Plastiwood /Sintex/Commander/ Accura
7	PVC,GALVANIZEDSHEETS, PANELLING, PARTITIONS, HUTS&BATHROOMS	
7(a)	RigidfoamPVC sheets from 0.50mm thickness to 40mm thickness	Kumar Arch Tech Pvt Ltd/Rajshri Plastiwood/ Sintex/ Marino
7(b)	AluminiumComposite Panel	Aluiminium Tech industries /Aluco Bond /AlstoneInternational
7(c)	Polycarbonate Sheets	Pre-star Infrastructure Pvt Ltd/GE /FibrewaysTechnology
8	SANITARYWARES	
8(a)	CP Bath Fittings	Jaquar& Company Pvt Ltd /Marc/Kohler/Grohe
8(b)	Wash Basins/WCs/ Urinals (Vitreous China sanitary Ware)	Jaquar& Company Pvt Ltd/ Kohler/ Toto
8(c)	SS Sink / Basins	Diamond /Nirali / Kraus / Grohe / Kohler
8(d)	Stainless Steel Kitchen Sink	PrayagPolymer Pvt Ltd/Jain Brothers Sanitation Pvt Ltd/ Cera/ Kohler
8(e)	PTMT Float valves / Ball cocks, Cockroach Traps, Glass shelf, Bath Fittings.	Prayag Polymer Pvt Ltd /Polytuff/Cera
8(f)	SS Plate Rack	Suyog/ Prayag/ Nirali

PARTICULAR SPECIFICATIONS (CONTD…)

1	2	3
8(g)	Kitchen Cabinets and Trolleys.	Godrej Interio /Zuari/ Kitchencrafts/ Kitchen concepts/Dream kitchens
8(h)	Concealed Cisterns	Kohler/Grohe/ Hindware/Jaquar
8(j)	Normal or Dual-Flow PVC Flushing Cistern.	Hindware/Cera/Parryware/Jaquar& Company PvtLtd/Johnson
8(k)	Glass Mirror.	Jaquar& Company Pvt Ltd/ Cera/ Saint Gobain /Modiguard
8(l)	PVC/Acrylic Toilet Mirror Cabinet.	Parryware/ Prayag/ Commander/ Kohler
8(m)	Stainless Steel Towel Rail / Towel Rack / Towel Ring / Soap Dish / TPH,etc	Kohler/Grohe/Jaquar
8(n)	Toilet Seat Cover.	Parryware/Cera/Hindware/Jaquar
8(o)	Shower Panels.	Grohe/Kohler/Jaquar/Lauret/Hindware
8(p)	Bath Suites.	Kohler/Grohe/Jaguar
8(q)	Glass WHB.	Zircon/Sparkle glass/Seabird
8(r)	Glass Frameless Shower Enclosures /Cubicles	Cera/Hindware/Lauret/Jaguar
9	PAINT/DISTEMPER/POLISH/VARNISHES	
9(a)	Acrylic Washable Distemper / Oil bound Distemper (Interior)	Berger/Nerrolac/ British paint/ Jotun India Pvt ltd/ Jenson and Nicholson
9(b)	Interior Acrylic / Plastic Emulsion Paint.	Berger/Dulex/Asian paint
9(c)	Synthetic Enamel Paint.	Berger/ Jenson and Nicholson /Asian paint
9(d)	Cement Base Paint.	SnowcemPlus/Berger Paints/ Jenson &Nicholson /Asian paints(Gutucem)/Durocem super cement paint
9(e)	Exterior Acrylic Emulsion Weather Coat Paint	Berger/Dulex/Asian paint
9(f)	French Polish	Berger /J&N/Asian
9(g)	Thermo-plastic Road Marking Paint/Retro-Reflective Paint.	Asian Paint Ltd (Apcomark) /STP/Shalimar Paints Ltd/Berger Paints Ltd/Jenson & Nicholson Paints Ltd
9(h)	Water Based Road Marking Paint.	Jenson & Nicholson Paints Ltd /Berger Paints Ltd/ Asian Paint Ltd (Apcomark)/Shalimar Paints Ltd/Jotun India Pvt Ltd
9(j)	Acid Resistant Paint	Asian/ Berger/Jotun India Pvt Ltd
9(k)	Epoxy Paint/Polyurethane Paint	Fosroc/STP/Pidilite/Asian/Berger / Johnson & Nicholson/Jotun India Pvt Ltd
9(l)	Black Bituminous Paint	Asian/ Berger/ STP
9(m)	White Cement/Wall Putty	Birla white /Berger/J&N/Shalimar paints/ Asian paints/JK white
10	PLYBOARD/MDF&HDF BOARDS	
10(a)	Plywood/Pre-laminate Plywood/Marine Plywood/ Veneered	Greenply Industires Ltd (Ply-Board Division)/ Kitply/Century /National Plywood IndsPvt Ltd / Novapan/Jain wood Industries Ltd.
10(b)	Veneered Plain Particle Board/Pre-laminated Particle Board.	Greenply industries ltd (Ply and Board Division)/ Kitply/ Centurilam/ Novapan
10(c)	Gypsum Board.	Lafarge GysumIndia Ltd /Saint-Gobain India Pvt. Ltd /Gyproc Business(Brand : Gyproc)
10(d)	MDF/HDFBoards	Kitply/National / GreenplyIndustiresLtd (Engineered Panel Division) / Balaji Action Buildwell /Novapan
11	TILES	
11(a)	Vitrified Floor Tiles/ Ceramic Glazed Wall Tiles/Ceramic Glazed Floor Tiles/Wall Tiles/Floor and Wall Tiles	Kajaria Ceramics Ltd/ Asian Granito India Ltd Varmora Granito Pvt Ltd / M/s H&R Johnson India/ Somany Ceramics Ltd
11(b)	Precast Concrete Interlocking Paving Blocks.	Ultra Tiles Pvt Ltd/Mehtab/Terra Firma/Ecco Scope /Nitco/A-One tiles
11(c)	Precast Plain Cement tiles/Chequered Cement Tiles for Flooring.	Ultra /Bansal/ Nitco/ Anjali/ Multiwyn/ A-One tiles
11(d)	Precast Terrazzo Tiles for Flooring.	Johnson/Ultra Tiles Pvt .Ltd /Nitco
11(e)	Wooden Laminated Floor Tiles/ Parquet Tiles	Vista/Pergo/Haro/Armstrong & Bruce

PARTICULAR SPECIFICATIONS (CONTD…)

1	2	3
11(f)	Glazed Mosaic Tiles.	Hindustan tiles/Johnson/ UltratilePvt. Ltd / M/s National tiles &Industries/Coral /Ceco
11(g)	Heavy duty Exterior vitrified tiles	Johnson (Endura)/Multiwyn /Century/ Cristal / Somany (Durastone) / Orient Bell Ltd
11(h)	Glazed Porcelain elevation wall tiles	Century Porcelain / Porselato Porcelain /Crystal Porcelain
11(j)	Acid resistant Vitrified Tiles	Johnson /Somany /Kajaria
11(k)	Metallic Floor hardener for wear –proof topping	Ironite / Stillonite/Hardonate / Maruti Bitumen Pvt Ltd.
11(l)	Non-Metallic Floor hardener	Fosroc /Fibrex/Sika/BASF/Maruti Bitumen Pvt Ltd. / STP
11(m)	Epoxy resin / Polyurethane based floor coating	Silk Fosroc /BASF /Pidilite Industries/STP
12	Water Proofing Chemicals / Materials	M/s Shivam Tar Products/ M/s Sika India Pvt Ltd./ M/s IWL India Ltd. / M/s STP Limited / M/s Fosroc Chemicals / M/s Hindcon Chemicals Ltd.
13	Tile Adhesive	PidiliteIndustries Ltd / SomanyCeramics Ltd/ Kajaria/ Latecreat STP Ltd
14	FENCING	
14(a)	Barbed Wire, Chain link Fence, Purched Tape Concertina Coils, Gabions	Global Technocrats Pvt Ltd / M/s A-1 Fencing Products Company Pvt Ltd /Shiva Engineering Company
15	STEELANDALUMINIUMJOINERY(DOOR/WINDOWS/ GRILLS/MESH/CUP BOARD/ SHUTTERS/SHELVES ETC.	
15(a)	Pressed Steel Frames for Doors/Windows/Ventilators.	M/s BA Enterprises/M/s MultiwynIndustrial Corporation / M/s DoorwynIndustries / M/s Modern Fabricator/ M/s Madhu Industries / M/s Ajanta Ispat/ Alusys
15(b)	Anodized/Powder-Coated Aluminium Frames.	Jinda/Hindalco/Ajit India/ Indal
15(c)	AluminiumDoors / Windows (Solid/Glazed Panel).	Jindal/ Hindalco/ Balco/ Modern Fabricator / Aluminite Pvt Ltd/ Dhiraj / Ajit India Pvt Ltd / Aluminium Door & Window Systems (Brand : ALUPURE)
15(d)	Frameless Glass Doors	Kesar/Super Aluminium/Haresh Aluminium
15(e)	Metal Rolling Shutters and Rolling Grills/Collapsible Gates.	M/s BA Enterprises / M/s MultiwynIndustrial Corporation / M/s DoorwynIndustries / M/s Modern fabricator
16	WINDOWGLAZING, GLASSSHELVING, GLASS URINALPARTITIONS.	
16(a)	Plain Sheet Glass/Float Glass/Frosted Glass.	Saint-Gobain glass/Modi-guard float glass / Asahi India /Hindustan Pilkington glass works
16(b)	Toughened Glass/Laminated Glass.	Safex/Saint Gobain glass /Modiguard float glass /Kesar / Asahi India /Atul Tuf/ Hindustan Pilkington glass works/ Triveni float glass
16(c)	Solar Control and Thermal Insulations Glass.	Evolite/ Envision/SGG EVO
16(d)	Wired Glass.	Saint Gobainglass / Modiguardfloat glass / Triveni/ Safex/ AtulTuf/Hindustan Pilkington glass works / Kesar
16(e)	Black Painted Glass Tiles.	Kesar/ Marvel /Dream walls Color glass Paladio/ Italia /Mridul/ Mehtabtiles
16(f)	Tinted/ ColouredGlass	Modiguard / Saint Gobian / Atul Tuf
16(g)	Glass Tops Shelves	Kesar/Ozone/Sparkle Glass
17	Components of PLC based Control Panel	L&T/ Siemens / Schneider / CIMCON
18	HDPE/LDPE WATER TANK	
18(a)	HDPE/LDPE Water Tank / LLDPE Septic Tank	Sintex/ Polycon/Jindal/Astral/ Finolex/ Supreme
19	SEWAGEDISPOSAL	
19(a)	UPVC/PPR/HDPE Pipes for Sewerage	M/s Jain Irrigation/Kisan/Prince/Supreme / Oriplast /Finolex
19(b)	RCC Pipes(NP2/NP3)	Any ISI marked and having BIS license makes.

PARTICULAR SPECIFICATIONS (CONTD…)

1	2	3
20	MISCELLANEOUSITEMS.	
20(a)	CI Centrifugally Cast (Spun) Pipes &Fittings	NecoCentri/ Jindal /Tata Metalic/ Sri kalahasti Pipes Ltd/SKF
20(b)	PVC Soil/Waste/Rain Water (SWR) Pipes and Fittings.	Jain irrigation/Kisan/Supreme/Prine/
20(c)	CPVC/UPVC Pipes and Fittings (Water &Sewerage)	Kisan /Supreme/Prince/Jain Irrigation /Oriplast /Finolex/ Fusion
20(d)	HDPE Pipes and Fittings(Water)	Kisan/Jain Irrigation/Supreme/Prince
20(e)	Brass Ball (Float) Valves.	Leader/Kirloskar/Zoloto
20(f)	CI manhole Covers.	AIC/PIC/BIC
20(g)	Construction chemicals like WPC, non metallichardner, epoxy and polymer based	Fosroc /Pidilite /Bauchemic India /Cico / Roff / Maruti Bitumen Pvt Ltd./STP
20(h)	APP Modified waterproofing membranes	Sika India Pvt Ltd /Fosroc Chemicals / STP Ltd /IWL India Ltd/Tiki Tar Danosa (India) Pvt. Ltd/AR Industrie /Torchtar Membrane & Bitumen Pvt Ltd/ Bengal Bitumen /S N Infratech Pvt Ltd
20(j)	Joint sealing compound grade 'A'/'B'	STP /Chowksy Chemical /Tiki Tar Maruti Bitumen Pvt Ltd /SIKA
20(k)	Polyurethane/Poly-sulphide sealant	Chowksay/ Fosroc / Rallithene/ Bostik/ Maruti Bitumen Pvt Ltd /STP/ SIKA
20(l)	CGI & PGI Sheets	Tata/Jindal /Sail
20(m)	AC Sheets	Charminar / Everest
20(n)	Galvalume/ Zincalume(Pre-painted color coated) sheet for roof & wall	Tata blue scope/JSW steel / Dyna
	E/M RELATED PRODUCTS	
21	LT WIRES AND CABLES	
21(a)	PVC insulated cable upto 1100 volts copper/ Aluminium conductor sheathed/unsheathed as per IS 694:1990	Havells India Ltd /Finolex/Nicco/KEI Industries Ltd / Polycab / RR Kabel Ltd
21(b)	PVC LT XLPE insulated PVC sheathed heavy armoured/ un-armoured cables upto 1100 volts, Aluminium/Copper conductos, solid/stranded as per IS 1554 Pt1: 1998 & IS 7098 Pt1: 1998	Havells India Ltd / Gloster Cables Ltd / Asian Cable /Finolex / Universal/ KEI Industries Ltd
22	HT XLPE CABLES	
22(a)	HT XLPE Cables	Havells India Ltd / Gloster Cables Ltd / KEI Industries Ltd/ Asian (RPG) / CCI /Finolex / Universal
23	OTHER CAT OF CABLES/ABC CABLE	
23(a)	Other Cat of Cables/ABC Cable	Ravin Cables Ltd / Dynamic Cables Ltd. / KEI Industries Ltd
24	PIPES CI, DI & GI PIPES	
24a)	CI pipes & Fittings	Electrosteel/ Kesoram/Tisco
24(b)	DI Pipes and fittings	Jindal Saw ltd /Electrosteel Castings Ltd /Tata Metaliks
24(c)	ERW MS Pipes/GI Pipes & Fittings	Tata / Jindal Steel / Prakash
25	PVC, UPVC & HDPE PIPES & FITTINGS	
25(a)	PVC Pipes & fittings	KisanMouldings Limited/ Ori-plast/Supreme / Prince/Finolex.
25(b)	UPVC/CPVC/RWP	Supreme/Prince /Finolex/Ori-plast.
26	SWR Pipes	
26(a)	PVC (SWR) Pipes	Supreme / Prince /KisanMouldings Limited
27	PPR Pipes & Fittings	
27(a)	PPR Pipes & Fittings	Savoir-Faire Manufacturing Co Ltd /Prince /Finolex/ Fusion/ KPT
28	Other Category of Pipes	
28(a)	Composite Pipes/ASTM Plumbing Pipes /Submersible Pipes/Micro Irrigation Pipes Casing Pipes/ Underground Drainage Pipes Hot &Cold Water Distribution System	KisanMouldings Limited /Jain Irrigation Systems Ltd /Savoir-Faire Manufaturing Co Pvt Ltd / Vectus Industries Ltd.
29	PVC CONDUIT /STEEL CONDUIT	
29(a)	Conduit Steel ERW	Tata /Jindal / Kalinga/ BEC

PARTICULAR SPECIFICATIONS (CONTD…)

1	2	3
29(b)	PVC Conduit	Presto-Plast/ Finolex/ Indo American Electrical / Kalinga Gold/ Richa Cables Pvt Ltd
30	PVC casing caping	
30(a)	PVC casing caping	Presto Plast/ Finolex/ Kalinga/ Payal.
31	Light Fittings/Lamps	
31(a)	Flame Proof fittings/ Fans/ Exhaust Fan/ Switches/ Sockets/LED lights	Bajaj /Crompton Greaves/ Sudhir/Orient Electric
31(b)	LED Light fittings External/Internal	Philips/ Wipro/ GE/ Havells india ltd/ Orient Electric
31(c)	High Mast Light	M/s Wipro/ Crompton Greaves /Bajaj / Philips
31(d)	Street light, security light, halogen, flood light and accessories	Philips/Wipro/Havells/GE/Orient Electric
32	Switch Fuse/Changeover switch	
32(a)	Switch Fuse/Changeover switch	Lauritz Knudsen Electrical & Automation (L & T)/ Legrand/ ABB /Siemens/ GE/ Schneider/ C&S Electric
33	MCB, RCCB, RCBO & DB	
33(a)	MCB, RCCB, RCBO & DB FOR MCBS, MCCB, Isolators	Legrand /Hager /Siemens /ABB /Lauritz Knudsen Electrical & Automation (L & T) /Schneider/ GE /C&S Electric
34	MCCB & Air CB	
34(a)	MCCB	Schneider /GE / Legrand/ Lauritz Knudsen Electrical & Automation (L & T)/Siemens /ABB / Indo Asian/ C&S Electric
34(b)	Air CB	Legrand /Hager/ Siemens/ ABB/ Lauritz Knudsen Electrical & Automation (L & T)/ Schneider/GE / C&S Electric
35	Switches/Buzzer/Socket/Bell/Ceiling Rose/Regulator (Piano / Modular)	
35(a)	Switches/Buzzer/Socket/Bell/Ceiling Rose/Regulator (Piano Type)	Havells/ Legrand/ Schneider /Hager / Lauritz Knudsen Electrical & Automation (L & T)
35(b)	Switches/Sockets /Regulator (Modular type)	Havells (Crabtree)/Anchor (Woods) /Legrand / Lauritz Knudsen Electrical & Automation (L & T)/Schneider/ Hager
36	SUB STATION / TRANSFORMER	
36(a)	Power Transformer 33/11 KV Copper wound	BHEL /ABB /Siemens/ Andrew Yule /Alstom / Schneider /Bharat Bijlee/Voltamp /GE/ Crompton Greaves/Indian transformer & Electrical Gurgaon
36(b)	Transformers 11 KV, Distribution type Step Down/Up above 500KVA	Voltamp transformer ltd /Siemens /Crompton Greaves / Alstom / Schneider /ABB / Bharat Bijlee/ Andrew Yule/ Indian transformer & Electrical Gurgaon
36(c)	Transformers 11 KV, Distribution type Step Down/Up above 200 but below 500KVA	BHEL/ABB/Siemens/Schneider/Crompton Greaves / Bharat Bijlee/ Voltamp transformer ltd/ Indian transformer & Electrical Gurgaon
36(d)	Transformers 11 KV, Distribution type Step Down/Up below 200	Any make approved by BEE and bearing 5 Star rating
36(e)	Packaged/Unified Sub Station	BHEL /ABB /Siemens /Andrew Yule /Schneider / Crompton Greaves /Voltamp transformer ltd / Lauritz Knudsen Electrical & Automation (L & T)/ Indian transformer & Electrical Gurgaon
36(f)	HT Switch gear, Vacuum CB & SF6 (11 KV & 33 KV) & Ring Main Units	BHEL /ABB /Siemens/Andrew Yule /Schneider / Lauritz Knudsen Electrical & Automation (L & T)/ Indian transformer & Electrical Gurgaon/ Voltamp
36(g)	Current & Potential Transformers (LT & HT)	AE/English Electric/Schnider / Lauritz Knudsen Electrical & Automation (L & T)
36(h)	Electronic Energy Meter (Temper Proof with Optical Port)	Siemens/L&T/ Havells India Ltd/ Secure Meters / Schneider
36(j)	Electrical Instrumentation/ measuring Instruments Digital/Analog/Energy Meters/Voltmeter/Ammeter	Havells India Ltd / Lauritz Knudsen Electrical & Automation (L & T)/AE/BCH/Secure Meters Schneider
36(k)	Isolation Transformer	Vintek (Brand name :Volina)/ Live Line Electronics/ Instruments & Equipment Co/ APLAB/ Voltamp/ Indian transformer & Electrical Gurgaon

PARTICULAR SPECIFICATIONS (CONTD…)

1	2	3
37	WATER HEATER / ENERGY SAVER / VARIOUS TYPE FAN	
37(a)	Water Heater (Geyser)	Jaquar& Company Pvt Ltd/ Usha/ Ao-Smith/ V-Guard
37(b)	Exhaust/Ceiling Fans/Air Circulators/Wall Mounting	Crompton Greaves/ Havells/ Almonard/ Orient Electric / Usha/GE.
37(c)	Solar Water Heater	Tata Power/ Supreme Solar
38	CAPACITOR BANK / APFC PANEL	
38(a)	Capacitor Bank/APFC Panel	Seimens/ABB / Lauritz Knudsen Electrical & Automation (L & T)/ BCH / Schneider/Crompton greaves/universal switch gear control/ Powerware /Havells/Indian transformer and electricals/Vintec electronics (Volina)/ C&S Electric
39	Thermoplastic Street Light Junction Box/DB	
39(a)	Thermoplastic Street Light Junction Box/DB	Hensel/ Sintex Industries Ltd / C & S Electric Ltd
40	Nature Switch/Street Light Timer	
40(a)	Nature Switch/Street Light Timer	Legrand / Lauritz Knudsen Electrical & Automation (L & T)/ Schneider /GE
41	LT Switch board panel (Indoor wall mounted Prewired)	
41(a)	LT Switch board panel (Indoor wall mounted Prewired)	Havells/Schnieder / Lauritz Knudsen Electrical & Automation (L & T) / ABB / Schneider /Neptune Systems Pvt Ltd./BCH Electric Ltd
42	Lt Control Panel/Feeder Pillar Box	
42(a)	Lt Control Panel/Feeder Pillar Box	Lauritz Knudsen Electrical & Automation (L & T)/ Milestone/ Legrand/ Seimens/ C & S Electric Ltd/ Schneider/Shalabh (India) Ltd/ M/s Power System & Controls/Indian Transformers Gurgoan
43	Steel Tubular Poles	
43(a)	Steel Tubular Poles	Jindal Steel /National Tubing Co /Bombay Tube & Store / Calcutta Poles & Tubes
44	Cable Joint HT hot and cold shrinkable/Push on type	
44(a)	Cable Joint HT hot and cold shrinkable/Push on type	M-Seal/Cab-Seal/Denson/Raychem/Birla-3M/Vishwa Electricals
45	Air Break Switches/HT Isolators/Gang /God Switches	
45(a)	Air Break Switches/HT Isolators/Gang/God Switches	Pactil /BHEL /Jaypuria Brothers
46	DSC/Pin type Porcelain Insulators	
46(a)	DSC/Pin type Porcelain Insulators	BHEL/Jaypuria Brothers/Pactil /Southern Insulators
47	DG SET	
47(a)	Diesel Generating Assembler	Kirloskar green /Jackson/Sudhir/ Greaves Cotton /Mahindra & Mahindra
47 (a)	Diesel set engine above 630 KVA	Kirloskar oil green /Cummins/Ashok Leyland / Eicher/Mahindra & Mahindra/Caterpillar/Ruston
47 (b)	Diesel set engine below 630 KVA	Kirloskar oil green /Cummins/Ashok Leyland / Crompton Greaves/ Caterpillar/Ruston
48	ALTERNATORS	
48(a)	Diesel set Alternators above 630 KVA	Bharat Bijlee/Siemens/Stamford/Mahindra / Alstom/ Kirloskar Electric
48 (b)	Diesel set Alternators below 630 KVA	Cromton Greaves/ Kirloskar Electric / Siemens/Bharat Bijlee/ Stamford/ Mahindra
49	LIGHTNING ARRESTOR / ACSR CONDUCTOR / PREVNTER / RELAY	
49(a)	Lightning Arrestors (station Class)	Oblum/Elpro/BHEL/Jaypuria Brothers
49(b)	ACSR Conductor	Indian Aluminium Ltd /Alind/Bharat Conductors / Dynamic Cables Ltd./Gupta Power Infrastructure Ltd

PARTICULAR SPECIFICATIONS (CONTD…)

1	2	3
49(c)	Single Phase Preventor/ Contractors/ Thermal Overload Relay	ABB / Lauritz Knudsen Electrical & Automation (L & T)/Siemens/ Schnieder
49(d)	Numerical Relay and Relay Control Panel	CG Power and Industrial Solution Ltd/Neptune Systems Pvt Ltd./BCH Electric Ltd
50	BATTERY & BATTERY CHARGER	
50(a)	Battery Maintenance Free	Exide/Amco/ Amaron
50(b)	Battery Chargers	Escorp /Su-Kam/ Microtech/ Exide
50(c)	UPS	APLAB/Luminous/SUKAM/TATA-LIEBERT
50(d)	Inverter	Live Line Electronics/ Microtech / Su-Kam
51	STABILIZER /INDICATOR LAMP / SELECTOR SWITCH	
51(a)	Servo Control LT Voltage Stabilizer	Andrew Yule/ AE / Vintek (Brand name Volina)/V – Guard/Aplab/ Indian transformer & Electrical Gurgaon
51(b)	HT Voltage Stabilizer (11 KV)	Andrew Yule/ Makpower/Recon/ Kirloskar
51(c)	Indicator Lamp/Selector Switch	Lauritz Knudsen Electrical & Automation (L & T)/ Siemens/ GE/ RR Micro / BCH /Schneider
52	VARIOUS TYPE VALVES	
52(a)	Sluice / Reflux / Air release / Foot / Non return valve / Gate Valve / Butterfly valve	Kirloskar/ Leader/ Zoloto/ Audco (L & T) /Venus/ Upadhyaya
52 (b)	Balancing Valves	Leader/Audco/ Advance
52(c)	Gun Metal Valve (Full way, check & globe valve)	Kirloskar/ Leader/ Zoloto/ Audco (L & T) / Upadhyaya
53	VARIOUS TYPE PUMPS / MOTORS / COOLING TOWER	
53(a)	Dol/Star-Delta Starter	Siemens/ Lauritz Knudsen Electrical & Automation (L & T)/ ABB / Schneider
53(b)	Submersible Cable	Finolex/ RR Kabel/ Polycab
53(c)	Centrifugal Pumps	Crompton Greaves/ Kirloskar /KSB
53(d)	Submersible Pumps	Crompton Greaves /Calama/ KSB/ Kirloskar
53(e)	Electric Motors	Lauritz Knudsen Electrical & Automation (L & T)/Siemens/ABB/Kirlosker/Crompton
53(f)	Monoblock Pumps	Crompton Greaves /Kirloskar/KSB
53(g)	Chiller Units/Compressors Scroll/Screw Type	Blue Star/Voltas/York/ Kirloskar/Carrier /Daikin/ Trane
53(h)	Condenser & Chiller Pumps	Crompton Greaves/ Kirloskar/ ABB/ KSB
53(j)	Cooling Towers	Paharpur / Mihir/ Delta
53(k)	Cooling Coil/AHU	Blue Star/Voltas/Zeco/ Lloyd/ Carrier
53(l)	Filters	Blue Star /Airtech/Thermodyne
54	CHLORINATOR ITEMS	
54(a)	Bleaching Dozer	Ion Exchange / RMCO /Avon /Aqua Pura/Maic-India
54(b)	Chloronome Plant/Chlorinator	Aqua/Pearl filters/Advance – 2000/ Jesco
54(c)	PVC Over Head Tank (ISI Mark)	Sintex/ Polywell/ Polycon/ Vectus Industries Ltd
	MISCELLEANIOUS ITEMS	
55	Refrigerant Controls	Sporlan/ALCO /Honeywell/Danfoss/Jhonson Control
56	Strainer	Rapid Control/ Emberland/ Sant
57	Insulation Crosslink Polythene with Adhesive	Torcellen/ Paramount/ Thermoflex
58	Thermometers/Pressure Gauges	Hguru/Taylor / Fiebig
59	Thermostat, Humidistat	Honeywell/ Jhonson Control/ Danfoss/ Siemens
60	Actuators	Siemens/Rapid Control /Honeywell
61	Heaters	Daspass/ Recold/ Bajaj
62	Thermostatic Expansion Valve	Siemens/ Honeywell/ Danfoss
63	Grills/Diffusers/Fire Dampers	Caryaire/Mapro/ Conaire
64	Split type AC/Package Type AC/Window type AC	O- General /Hitachi /Daikin /Carriers/Blue star/Panasonic
65	Fire Pump& Panel	Crompton Greaves/Kirloskar / KSB
66	Fire Hose Reel	Minimax/ Firex/ Safex
67	Stand Post type Hydrant	Minimax/ Firex/ Safex
68	RRL Hose Pipe	Dunlop/Cosmos/ Jayshree
69	Sprinkler Head	Minimax/Tyco/Ceasefire/Safex
70	Pressure Switch	Siemens /Honeywell /Danfoss/Schneider /Rapid Control
71	Gunmetal Valves(Full way, check & globe valve)	Audco(L&T)/ Zoloto/ Kirloskar/ Leader/ Upadhyaya

PARTICULAR SPECIFICATIONS (CONTD…)

1	2	3
72	Fire Extinguishers (All Types)	Minimax/Firex/Newage/Cease Fire
73	Single Head Landing Valves three/four way brigade inlet hose reel drum and shut off nozzle, aluminium branch pipe	Minimax/Safex/Superex
74	20 mm dia Rubber pipe	Jyoti/ Dunlop /Minimax
75	Hooter	Safex/Honeywell / Minimax
76	Balancing Valves	Leader/ Audco/ Advance
77	High Mast Pole	Bajaj Electricals /CG Power & Industrial Solutions Ltd /Jindal Power.
78	BLANK	
79	EOT crane	Safex Energy Pvt Ltd/Pioneer Cranes & Elevators Pvt Ltd/ Dinesh Enterprises/ Microtech Engineers & Co/ Electromech Material System India (P) Ltd/ Hertz Controls (India) Pvt Ltd /Reva Industries Ltd/ NRS TIIB & CO/ Rockwell Hoisto Cranes Pvt Ltd
80	Programmable RTU alongwith Battery	Honeywell/ Safetek/ Motorola/ Siemens/ Cimcon/ Powertek/ Moorley
81	Wireless Modem Suitable For GSM Based Communication	Cimcon/Siemens/Motorola/Honeywell/ Moorley
82	Fire Alarm Control Panel and Related Accessories for Fire Alarm and Fire Detection System	Honeywell (Esser) /Siemens/Tyco/Bosch / Godraj Security System/ Cooper/ Cimcon / Safetek/ Moorley/GST /Ravel/ Agni
83	Indoor/ Smoke Detector	GST/ Tyco /Bosch/ Godraj Security System / Ravel/ Moorley/ Honeywell/ Siemens/ Cooper
84	Addressable Manual Call Point	GST/Tyco/Bosch/Godraj Security System /Ravel / Moorley/ Honeywell/ Siemens
85	Remote Response Indicators	GST/Tyco/Bosch/Godraj Security System/Ravel / Moorley/Honeywell/Siemens
86	Addressable Sounder	GST/Ravel/ Moorley/ Honeywell/Siemens/Godrej Security System
87	Fire Alarm Sounder/ Sirens Electronic Type	GST/Godrej Security System/ Ahuja
88	Mimic Type Panel	BCH /Rittal/ Safetek /Powertek/ Cimcon
89	Amplifier/CD Speaker	Philips/Ahuja/ Piyano
90	Loud Speaker	Ahuja / Piyano/ Philips
91	Chemical Dehumidifier	Aqua Aura/Advance International/ Refair
92	Hot water generator	Rapid Cool/Emerald / Ross
93	Ultrasonic Humididifier	Rapid Cool/Emerald / Stulz
94	VFD	ABB/Schneider/Danfoss
95	Motorized Valve with actuator and thermostat	Danfoss / Belimo / Honeywell
96	Water Flow Switch	Honeywell /Danfoss / Dwyer
97	Flexible pipe connector	Resistoflex / Easyflex / KELD
98	On-line scale preventer	Scale-Off / Delta / Scale Guard
99	Motorized Fire cum Smoke Damper with actuator	Conaire /Jobshan/Controls / Belimo
100	Nitrile Rubber Thermal Insulation	Arma Flex / K-flex/Aerofoam
101	Octagonal poles	Wipro/ Bajaj/CG
102	Sewage Feed Pumps	Grundfos/Aqua/KSB/Kirloskar
103	Filter Feed Pumps	Matherplat/KSB/Kirloskar
104	Air Blowers	Usha/Everest/Welcome
105	Indicating Meter	AE/Rishabh(T&T)/Siemens/BCH
106	Level Switches	Aster/Linear/Pune Techtrol
107	Filter Pressure	Welcome/Pharmatech/Dinshaw(Purav Industries)
108	Bar Screen	As per OEM
109	Activated Carbon Filter	As per OEM
110	Multigrade Filter	As per OEM
111	Remote Terminal Unit (RTU)	Cimcon / Siemens/ABB
112	MS Powder coated CRCA outdoor panel for RTU	CIMCON / BCH/Rittal/Safetek/Powertek
113	Smart digital energy meter for the HVAC	Cimcon/ Siemen/Scheneider or equivalent.
114	Wirless modem suitable for GSM/GPRS based communication syste	Siemens/Motorla/Cimcon

PARTICULAR SPECIFICATIONS (CONTD…)

1	2	3
115	Resistant Temperature Detector (RTD) Sensor with Temperature (Signal Converter)	Baumer / Radix / Masibus / Danfoss
116	Process indicator	Multispan / Radix / Masibus/ ACE
117	Supervisory Control and Data Accusition (SCADA) software	GE FANUC / ROCKWELL AUTOMATION / Lauritz Knudsen Electrical & Automation (L & T)/ SIEMENS / SCHNEIDER
118	Bar Screen	As per OEM
119	Activated Carbon Filter	As per OEM
120	Multigrade Filter	As per OEM
121	Water cooler/Refrigerator/Air Conditioner	Voltas/Blue Star/Whirpool
122	Polymer Modified Bitumen (PMB)	M/s Hindustan Coals Ltd./ M/s Ooms Polymer Modified Bitumen Pvt Ltd/ M/s Tiki Tar and Shell India Pvt Ltd
123	Bitumen Emulsion	M/s Hindustan Coals Ltd./ M/s Ooms Polymer Modified Bitumen Pvt Ltd/ M/s Tiki Tar and Shell India Pvt Ltd/ M/s STP Ltd. / M/s Verma Industries/ M/s Giridhar Techfab Pvt Ltd.
124	HDPE polythene film	Any make having valid BIS license of IS 10889-2004
125	Filler Board/Marking Tape/Backup Rod/Bond Breaker Tape/Armour tape/Curing Compound	M/s STP Ltd M/s Tikitar Industries M/s IWL India Ltd M/s ITS Coatings Pvt Ltd.
126	Glass Grid/ Fiber Glass	Any make having valid BIS license
127	Iron Removal Plant	Aquajio/Pearl Filters Kolkata/RMCO

SIGNATURE OF CONTRACTOR
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PARTICULAR SPECIFICATIONS (CONTD…)

D. LIST OF DRAWINGS

1. The following drawings are included in the tender documents : -

S No	Description of drawings	Drawing No	Sheet No	Date of Drawing	Last revision
1	2	3	4	5	6
1	LIST OF DRAWINGS	APEX/CE AF SHL/STP/AR/02/18	AR-1/1	06-02-18	29-01-25
2	SITE PLAN	APEX/CE AF SHL/STP/AR/02/18	1R/9	10-07-24	-
3	GROUND FLOOR PLAN	APEX/CE AF SHL/STP/AR/02/18	2R/9	10-07-24	20-11-24
4	ROOF PLAN	APEX/CE AF SHL/STP/AR/02/18	3/9	26-02-18	--
5	ELEVATIONS	APEX/CE AF SHL/STP/AR/02/18	4/9	26-02-18	--
6	SECTION AA, BB	APEX/CE AF SHL/STP/AR/02/18	5R/9	10-07-24	-
7	SECTION CC, DD, EE, FF AND PART SECTION KK	APEX/CE AF SHL/STP/AR/02/18	6R/9	10-07-24	-
8	PRESSED STEEL DOOR, WINDOW AND MS RAILING DETAIL	APEX/CE AF SHL/STP/AR/02/18	7/9	26-02-18	10-07-24
9	ROLLING SHUTTER AND MS RUNGS DETAIL	APEX/CE AF SHL/STP/AR/02/18	8/9	26-02-18	--
10	SHEDULE OF FINISHES	AF/KKD/STP/SCH/18	1/1	10-07-24	--
11	FLOW SCHEMATIC FOR SEWAGE TREATMENT PLANT OF CAP 300 KLD MBBR BASED	APEX/CEAFSHL/STP/E/M/02/18	STP-FS-01R	10-07-24	-
12	FOUNDATION PLAN	APEX/CEAFSHL/STP/STR/02/18	1/13	26-02-18	--
13	RCC WALL AND SLAB PLAN AT PLINTH LVL	APEX/CEAFSHL/STP/STR/02/18	2/13	26-02-18	08-10-24
14	RCC WALL PLAN AT +2700 LVL	APEX/CEAFSHL/STP/STR/02/18	3/13	26-02-18	20-11-24
15	PLAN AT ROOF LVL AND SHEDULES	APEX/CEAFSHL/STP/STR/02/18	4/13	26-02-18	08-10-24
16	SECTION A-A AND B-B	APEX/CEAFSHL/STP/STR/02/18	5/13	26-02-18	--
17	SECTION 1-1, 2-2 AND 3-3	APEX/CEAFSHL/STP/STR/02/18	6/13	26-02-18	--
18	STAIRCASE DETAILS	APEX/CEAFSHL/STP/STR/02/18	7/13	26-02-18	--
19	HORIZONTAL REINFORCEMENT PLAN FOUNDATION TO +2700 LVL	APEX/CEAFSHL/STP/STR/02/18	8/13	26-02-18	--
20	HORIZONTAL REINFORCEMENT PLAN FROM +2700 LVL TO ROOF LVL	APEX/CEAFSHL/STP/STR/02/18	9/13	26-02-18	--
21	DETAILS OF PUMP FOUNDATION	APEX/CEAFSHL/STP/STR/02/18	10/13	26-02-18	--
22	RC DETAIL OF SEWAGE PUMP HOUSE AND SUMP PIT	APEX/CEAFSHL/STP/STR/02/18	11/13	26-02-18	--
23	GENERAL NOTES AND TYPICAL DETAILS-1	APEX/CEAFSHL/STP/STR/02/18	12/13	26-02-18	--
24	TYPICAL DETAILS-2	APEX/CEAFSHL/STP/STR/02/18	13/13	26-02-18	--

PARTICULAR SPECIFICATIONS (CONTD…)

Note :

(i) Drawing for site plan/ layout plan shall be referred in the office of CE(AF) Shillong Zone/ GE (P) (AF) Kalaikunda any working day during working hours.

(ii) The date of drg / Rev date shall be applicable as actually indicated in the drawing even though if erroneously not indicated in this list of drawings.

(iii) If any TD drawing which is not reflected in the list of drawings mentioned above but mentioned in working building drawings, shall be referred as per latest TD drawings followed in this HQ. These TD drawings shall be referred in the office of HQ CE (AF) Shillong Zone/ GE (P) (AF) Kalaikunda during working hours of mentioned offices.

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